Dynamic treatment effects in Education

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Abstract

Lifetime welfare calculations often rely on policy effects on educational attainment measured in a study's final observation window—for example, effects on college completion by age 25. But a higher completion rate by age 25 could reflect more people eventually graduating, earlier graduation among those who would have graduated anyway, or both. This paper develops a framework to assess the welfare implications of this uncertainty. I document the timing information in existing studies and model possible continuations of observed education effects. Two deterministic simulations show how different education paths can match the same observed effect yet imply different lifetime earnings values, and how additional observations can narrow this range. I apply the framework to 38 policy calculations, 37 with existing welfare benchmarks, reconstruct ranges for the marginal value of public funds (MVPF), and propose an ordering rule based on those ranges. Under the specified path and valuation assumptions, the same education evidence can support conflicting welfare judgments even when the estimated effects are treated as known. My contribution provides a framework for assessing the robustness of welfare comparisons to unobserved education paths and quantifying how estimates from each additional observation window can narrow uncertainty.

1 Introduction

A policy that raises bachelor's degree attainment by age 25 can create a permanent attainment gain, a temporary advantage from earlier graduation, or both. These possibilities have different economic consequences. Earlier completion can generate earnings before the comparison group catches up, while a permanent attainment gain can affect income and tax revenue over a much longer period. Educational costs also depend on when additional schooling occurs. An effect estimated at a particular age therefore answers an important causal question, but does not by itself determine the policy's lifetime value. This distinction matters when education estimates inform long-term earnings projections and comparisons of public spending, as in Hendren and Sprung-Keyser (2020).

This paper asks whether education effects measured within a finite window can rule out conflicting lifetime welfare judgments for the same policy. I take selected causal estimates from existing studies as inputs and construct education paths under explicit outcome conversions and age mappings. Retained reference inputs are matched exactly; additional point estimates are matched exactly when possible and otherwise fit by

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constrained minimum distance, with residuals disclosed. Within each reconstruction, I hold these inputs, the external comparison profile, path restrictions, and the valuation rule fixed. I then compare earnings net present values (NPVs) and the marginal value of public funds (MVPF), which relates beneficiaries' willingness to pay to net fiscal cost. The replacement passes each new education path through the retained income, tax, and education-cost functions. Source transfers remain fixed. Education costs follow recoverable source schedules; cases without these schedules retain their stated non-income amounts. The question is whether different paths can place beneficiary welfare above or below public cost, or change whether a policy finances itself. A comparison with the original welfare calculation can additionally reflect changes in conversion or valuation; the comparison among paths within a reconstruction holds those choices fixed.

How much the reported education effects reveal about lifetime attainment depends on when education is measured. I document this timing information in a review of 459 studies that combines a search of 12 economics and education journals over 2000–2025 with additional welfare-source papers and an author manuscript. Among the 190 studies reporting bachelor's-or-higher outcomes, 132 report one distinct estimation time or window for that outcome. More directly relevant to welfare extrapolation, 18 of the 31 studies reporting education-based lifetime or career valuation provide only one window for each valued education series. These counts describe the assembled sample rather than all education research. The original authors also discuss dynamics and extrapolation limits. Cohodes and Goodman (2014) interpret persistence and completion patterns as evidence of dropout rather than pure delay, while Angrist et al. (2022) and Schoonover et al. (2026) qualify their projected returns and discuss education timing. Their evidence informs the reconstruction without determining the unobserved continuation.

To connect the observed education effects to their possible continuation, I represent educational attainment for the same target population by two cumulative age profiles: $H_c^0(a)$ without the policy and $H_c^1(a)$ with it. Their difference is the education effect at each age. Both attainment levels are nondecreasing, but the difference can widen, narrow, or reverse sign. Reported effects constrain this difference at the source's observation ages, over its age windows, or through changes between observations, using the specified conversions. For the numerical exercises, I use logistic continuations anchored at the last unaffected age and restrict their levels, acquisition rates, and empirical support. Census–ACS profiles supply the comparison curves and calibration restrictions, matched to policy geography and birth cohorts in the applications. These population profiles are external calibrations, not counterfactuals identified by the source study's research design. The paths exactly match the retained reference inputs. Additional selected estimates may have nonzero best-fit residuals; neither unselected source results nor every added point is asserted to be matched exactly.

Two deterministic simulations then examine how education effects and their measurement ages restrict possible lifetime earnings values. The first compares permanent effects on eventual schooling with temporary effects generated by changes in acquisition speed or timing, while retaining the same age-25 measurement rule. In each of the three illustrated scenarios, the measured effect is compatible with both positive and negative education NPVs under a common valuation rule. Some compatible paths reverse the initial education advantage at later ages, so the alternatives extend beyond different durations of a positive effect. These reversals are possibilities within the specified path family. The second exercise holds the generating path fixed and varies the observation ages. In the illustrated comparisons, adding an effect estimate at one additional age can substantially restrict the compatible paths and narrow the resulting NPV range. The reduction depends on when that additional effect is measured. Even when attainment eventually converges, earnings forgone during an earlier delay need not be recovered.

Applying the model to existing policy evaluations, I reconstruct MVPF ranges for 38 policy calculations and propose a rule for ordering the 37 with an original welfare valuation (L1). The reconstructions build on Hendren and Sprung-Keyser (2020) or the original authors' valuations, replacing education-related earnings and fiscal returns with values implied by compatible education paths. Among these 37 calculations, 30 initial reconstructions (L2) admit positive-cost MVPFs on both sides of one, and 24 admit both beneficiary losses with public costs and beneficiary gains with self-financing.[1] Single-parameter comparisons show that permanent attainment changes often yield valuations close to L1, whereas matching the same education input through temporary changes in acquisition can produce substantially different lifetime earnings and MVPFs. Additional comparable observations narrow some ranges by excluding continuations allowed by the initial inputs; twelve calculations with L1 support this richer reconstruction (L3), including converted inputs with explicit timing mappings. Nine of these twelve L3 applications select one best-fitting path; three retain exact-match path sets. The cannabis education application also uses a best-fit L3. The ordering uses L3 where available and L2 otherwise, prioritizing higher lower endpoints and then higher upper endpoints among the paths found. Separately, 1,000 specified input repetitions per calculation examine how imprecision in education or accounting inputs changes the paths and their welfare values. These repetitions distinguish archived source draws from constructed perturbations. They address uncertainty in the inputs without supplying observations of the unmeasured education path.

The main contribution is a systematic analysis of how extrapolating finite-window education effects changes lifetime welfare judgments. Representing attainment with and without a policy as H^1 and H^0, I construct paths that match the same education inputs and show how permanent and temporary effects differ in their implications for earnings and fiscal returns. This makes explicit which parts of a welfare conclusion follow from the selected education evidence and which depend on continuation assumptions. Research on time to degree and dynamic educational choices establishes the importance of when schooling is acquired (Bound et al., 2012; Bárány et al., 2023; Heckman et al., 2016). Work on long-term extrapolation studies how short-term observations can inform later outcomes under additional restrictions (Athey et al., 2026; Obradović, 2026; Fan et al., 2026). I bring this question to the education-based valuations in Hendren and Sprung-Keyser (2020), using the literature review to document the available information and the simulations and policy reconstructions to assess its welfare consequences. Separating uncertainty about continuation from imprecision in estimated inputs also complements research on inference for calibrated parameters and welfare metrics (Cocci and Plagborg-Møller, 2025; Vohra, 2026).

Two verified education paths that satisfy the same restrictions but imply opposing welfare conclusions establish that those restrictions do not resolve the comparison. The sensitivity exercises assess whether such conflicting conclusions remain under alternative calibration, valuation, and search specifications. They cover all 38 calculations and use additional education observations where available. Across the common specifications, 16 calculations continue to admit beneficiary losses with public costs and beneficiary gains with self-financing. These checks support the existence of conflicting judgments under several stated assumptions, but do not validate every assumption or exhaust the feasible paths. A search that finds only favorable paths cannot establish that every compatible path is favorable. The reported extrema are candidate values, no probabilities are assigned to paths, and input-repetition summaries do not establish confidence coverage.

[1] Eleven tax calculations require an explicit policy-unit conversion, Early Head Start and Tennessee Promise use disclosed public-data income reconstructions, and algorithmic placement adds a long-run earnings channel; Section 4.1 explains these adjustments. Several calculations share a source study. The remaining application values only the education channel of cannabis legalization and has no whole-policy L1.

The comparison of alternative education paths also relates to research on lifetime earnings measurement, long-term prediction, and policy choice with incomplete information. Haider and Solon (2006) show that the relationship between current and lifetime earnings varies with age, while Bhuller et al. (2017) document how additional schooling changes the age profile of earnings. These studies examine earnings over the life cycle; the reconstruction here also varies the education path entering the valuation. To forecast beyond experimental follow-up, García et al. (2020) combine experimental and nonexperimental data with economic models to estimate lifetime program benefits and costs. Chen and Ritzwoller (2023) develop efficient estimators combining short-term experimental and longer-term observational outcomes, and Imbens et al. (2025) use the sequential structure of short-term outcomes to address persistent confounding in such data combinations. These approaches clarify how additional data and identifying restrictions support long-term estimates. The present analysis examines which welfare judgments remain unresolved when selected education observations leave room for different attainment continuations. It also relates to sufficient-statistics welfare analysis, which connects reduced-form estimates to welfare through economic restrictions (Chetty, 2009; Kleven, 2021). Projecting lifetime earnings and fiscal returns from an education effect similarly requires a restriction on its continuation. Finally, Manski (2011) studies policy choice when treatment responses are only partly known and no probability distribution is assigned to that uncertainty. This distinction between what the evidence determines and what a decision rule adds also guides the interpretation of the policy ordering in this paper.

The remaining sections move from the available education evidence to path construction, welfare applications, and their interpretation. Section 2 presents the motivating cases and literature statistics, and Section 3 develops the model and simulations. Sections 4 and 5 present the three main applications and the full policy collection. Section 6 discusses assumptions, robustness, and policy implications, and Section 7 concludes.

2 From Observed Education Effects to Lifetime Valuation

Assessing the lifetime value of education policies requires connecting effects estimated within finite observation windows to education paths over the life cycle. Figure 1 shows that educational attainment continues to rise after age 25. Among the 1985–1987 birth cohorts, bachelor's-or-higher attainment (BA+) rises from 29.34 percent at 25 to 40.80 percent at 35; approximate schooling rises from 13.30 to 13.97 years. The profiles use the full-population American Community Survey (ACS) extract for 2000–2022, with fixed cohort weights.[2] In Figure A1, the gains in degree attainment and schooling between ages 25 and 35 are larger for the 1985–1987 cohorts than for the 1975–1977 cohorts.

To document the time information used in higher-education research and its role in longer-run interpretation and valuation, I review the studies described in Appendix A. The sample contains 459 papers: 447 from a search of 12 economics journals over 2000–2025, 11 additional welfare-source papers, and one author manuscript. Among 190 papers with BA+ outcomes, 132 report one distinct BA+ estimation time or window (69.5 percent); one window can span several years.[3] Separately, 31 papers report education-based lifetime or career valuation. Among these, 18 report one age, follow-up, or stage-specific window per valued

[2]Each cohort contributes ages 16–35 with equal weight at every age. Appendix A.5 gives sample, person-weighting, and education-crosswalk details.

[3]The journal-only share is 130 of 186 BA+ papers (69.9 percent). Coding distinguishes attained age, individual follow-up, and policy event time.

education series, and 12 report multiple outcome times.[4] The following three college-aid studies illustrate the calculations and the dynamic evidence accompanying them.

Cohodes and Goodman (2014) estimate effects of Massachusetts' Adams Scholarship on college choice, persistence, and completion. Scholarship eligibility lowers the probability of earning a bachelor's degree within six years of high-school graduation by 2.5 percentage points near the eligibility threshold. The authors interpret the persistence and graduation patterns as evidence of dropout rather than pure delay, and use ACS earnings differences to illustrate lifetime losses for students induced to switch colleges. Hendren and Sprung-Keyser (2020) construct a policy-average MVPF from the scholarship's enrollment and completion effects, which together imply a 0.018-year reduction in schooling. They project the associated earnings and tax changes, then account for transfers and private and public education costs to obtain an MVPF of 0.72.

Dynarski (2003) examines college attendance after the termination of Social Security student benefits. Eligibility raises the probability of ever attending college full time by 21.9 percentage points by age 23 and 22.4 points by age 28 under the conservative attrition specification. The paper explicitly compares the two ages to distinguish accelerated attendance from lasting gains. The enrollment alternative in Hendren and Sprung-Keyser (2020) values the age-28 estimate using two schooling years per induced entrant and reports an MVPF of 4.87. This supplies a direct comparison between the observation retained for welfare and an additional observation of the same outcome at an earlier age.

Schoonover et al. (2026) estimate the effects of Tennessee Promise in overlapping two-year age groups from 18–19 through 23–24. At ages 23–24, the estimated increase in bachelor's-or-higher attainment is 2.7 percentage points, statistically significant at the 10% level under permutation inference. The authors combine the education effects with Tennessee-specific returns to schooling to project lifetime income and tax revenue. Their valuation yields positive willingness to pay and negative net public cost, classifying the program as self-financing with an infinite MVPF. They acknowledge that outcomes observed in early adulthood cannot fully distinguish lasting attainment gains from changes in the timing of schooling and labor-market entry.

These studies use different education outcomes and populations, and their authors already examine graduation dynamics or qualify the lifetime projections. The question pursued here is how much those projections can vary when the observed education evidence is retained but its unobserved continuation changes. The next section holds observed education effects fixed and varies age paths within stated economic restrictions. Applying the same valuation rule to each path isolates the role of those paths in the calculation.

3 Education Paths and Lifetime Valuation

A finite observation window leaves the evolution of educational attainment partly unobserved, even when the effects within that window are measured without error. To separate this information problem from sampling uncertainty, I hold a comparison path fixed and construct treated paths consistent with stipulated education effects. In each of the three displayed scenarios, the same age-25 effect admits paths with positive and negative education-channel net present values (NPV) under a common valuation rule. Adding the corresponding age-35 effect reduces the width of the numerical candidate range by 92.3–94.2 percent. These comparisons treat the observed effects as exact and hold the baseline and path restrictions fixed. The first experiment varies the education process while observing it at age 25; the second holds each generating process fixed and varies

[4] One further paper values institutional degree flows in policy event time. Appendix A.3 records the valued outcomes and their observation clocks.

the observation ages. Together, they show which valuation conclusions follow from the observed effects and which depend on the unobserved schooling path.

3.1 Education Paths, the Baseline, and Path Restrictions

An education effect measured at a particular age describes the attainment difference at that age, while leaving its earlier and later evolution unresolved. For a fixed population in cohort c, let $H_c^0(a)$ and $H_c^1(a)$ denote the mean of a specified educational attainment outcome at age a without and with the policy, respectively. The outcome may record cumulative college entry, degree attainment, or completed years of schooling.[5] The numerical illustration below uses years of schooling. For any of these cumulative outcomes, I distinguish the policy effect on the attainment stock from its age-a increment:

$$\Delta H_c(a) \equiv H_c^1(a) - H_c^0(a), \qquad \Delta h_c(a) \equiv \Delta H_c(a) - \Delta H_c(a-1). \tag{1}$$

Cumulative attainment satisfies $H_c^j(a) - H_c^j(a-1) \geq 0$ for $j \in \{0, 1\}$ in a fixed population. This restriction does not imply monotonicity of the treatment effect $\Delta H_c(a)$, because the two potential attainment paths can accumulate at different rates. The increment $\Delta h_c(a)$ may therefore be positive, negative, or zero, according to whether attainment accumulates faster, more slowly, or at the same rate under treatment. In particular, $\Delta h_c(a) < 0$ indicates a decline in the treatment–comparison attainment gap; neither level path need decline. Appendix B.1 gives the detailed definitions, and Figure 2 illustrates the two levels, their age-specific gaps, and the change in the gap between adjacent ages.

I represent the treated path with a three-parameter logistic continuation anchored to the comparison level. For a given starting age a_0, write $\tau_c = a_0 - 1$ and $S_c = H_c^0(\tau_c)$. With $\theta_c = (\Delta K_c, \Delta b_c, \Delta d_c)$, define $L_c(a; \theta_c) = \Lambda[(b_c + \Delta b_c)(a - d_c - \Delta d_c)]$, where $\Lambda(x) = (1 + e^{-x})^{-1}$. The treated path is

$$H_c^1(a; \theta_c, a_0) = \begin{cases} H_c^0(a), & a \leq \tau_c, \\ S_c + (K_c + \Delta K_c - S_c)\, \dfrac{L_c(a; \theta_c) - L_c(\tau_c; \theta_c)}{1 - L_c(\tau_c; \theta_c)}, & a > \tau_c \end{cases}. \tag{2}$$

The normalization makes the continuation pass through S_c at τ_c and approach $K_c + \Delta K_c$ at older ages. Thus, the first affected observation reflects acquisition since a common starting point, rather than a level difference imported when switching curves. The parameter K is measured in attainment units, b in inverse age-years, and d in age-years. They govern the plateau and the speed and timing of accumulation conditional on the anchor. I call an education effect permanent when it changes eventual attainment, so that $\lim_{a\to\infty} \Delta H_c(a) = \Delta K_c \neq 0$. An effect is temporary when the attainment gap eventually vanishes. In the single-parameter exercises, a nonzero K change is permanent, whereas b or d changes with K fixed are temporary.[6] Holding K, b, and the common anchor fixed, a higher d delays subsequent accumulation: attainment is lower at each finite age after the anchor, and any given level between the starting value and plateau is reached later. The delay varies across attainment levels, so an increase in d does not imply an equally sized delay at every point on the path. These are population-path parameters rather than individual graduation dates. Figure B1 illustrates their roles under the normalization; Appendix B.2 gives the derivatives.

[5] For binary attainment, $H_c^j(a)$ is a population share and $100\Delta H_c(a)$ is a percentage-point effect. Schooling levels and effects are measured in years. Current enrollment is not cumulative and need not rise with age.

[6] Temporary describes eventual convergence, not disappearance within the valuation horizon. Such an effect may persist beyond age 65 and can have a nonzero lifetime NPV. A permanent effect can also involve changes in acquisition speed or timing.

The choice of a_0 reflects when the policy can first affect the specified attainment outcome for the fixed target population. I assume no policy effect or anticipation through τ_c, which justifies the common starting level in Equation (2). This compares the same population under two policy states; it does not require the observed treatment and comparison samples to have identical pre-policy means. In the simulation, the hypothetical policy first affects the population at age 18, making age 17 the anchor. The anchor is therefore determined by exposure timing, not by the age at which the researcher happens to observe an effect.[7]

For the years-of-schooling simulation, $H_c^0(a)$ is the equally weighted average of fitted ACS profiles for the 1985–1987 birth cohorts, using the full-population data at ages 16–35 introduced in Figure 1. Here c indexes the synthetic three-cohort group. Each cohort is fitted separately, and the fixed one-third weights preserve its contribution at every age; values above age 35 are extrapolated.[8] I vary H_c^1 while holding this comparison curve fixed. Appendix C, Table C1, and Figure C1 document the construction and fit.

The admissible path class combines the logistic specification with restrictions on attainment levels and annual increments. Both experiments set the first possible effect age to $a_0 = 18$, value earnings through age 65, and extend education paths through age 72. I require completed schooling under treatment to be nondecreasing, to lie between zero and 25 years, and to increase by no more than one schooling year between adjacent ages, including the transition from age 17 to age 18. The total-parameter domain is $K \in [13.3370, 15.1395]$, $b \in [0.12, 0.65]$, and $d \in [8, 30]$.[9] Let C collect these maintained conditions. They define which paths enter the comparison; satisfying them does not establish that a particular path is the response to an actual policy.

3.2 What a Single Observed Effect Leaves Unresolved

An education estimate restricts a feature of $\Delta H_c(\cdot)$ rather than revealing the entire path. Let the observation operator $\mathcal{M}$ map a candidate path into the effects represented by a study:

$$\mathbf{m} = \mathcal{M}\{\Delta H_c(\cdot)\}. \tag{3}$$

For attainment measured at one age, the relevant entry of $\mathbf{m}$ is $\Delta H_c(a)$. An average effect over an age window $\mathcal{A}$ instead enters as $\sum_{a\in\mathcal{A}} w_a \Delta H_c(a)$, using the source's weights.[10] An effect on attainment acquired between ages a_1 and a_2 restricts $\Delta H_c(a_2) - \Delta H_c(a_1)$. These restrictions preserve the source study's outcome definition and time scale.[11] The source-specific mappings for credits, enrollment, and credentials enter the applications in Section 4. For the fixed comparison path and target population, the compatible path set is

$$\mathcal{P}(\mathbf{m}; a_0, C) = \left\{H_c^1(\cdot;\theta_c, a_0) \in C : \mathcal{M}\left\{H_c^1(\cdot;\theta_c, a_0) - H_c^0(\cdot)\right\} = \mathbf{m}\right\}. \tag{4}$$

An additional observed effect changes the moment restrictions; changing the parameter domain or the path conditions changes C. With one or two exact age-specific effects, the matching equations generally leave two

[7] Observation at 25 does not justify anchoring at 24. The anchor must precede possible effects and anticipation. If exposure ages differ, use a common unaffected age or cohort-specific anchors before aggregating with fixed weights.

[8] The reference triple averages parameters, whereas H_c^0 averages fitted curves. Their anchored continuations differ by at most 0.000226 schooling years over ages 18–65; zero parameter increments need not imply an exactly zero effect.

[9] The plateau limits are weighted first and 99th percentiles of a historical state–cohort fit bank. Transporting this support to the full-population baseline is a calibration assumption (Appendix C.3). Speed, timing, and annual-increment limits are design restrictions. Binary outcomes require share-consistent bounds instead.

[10] An exact 0.1-year effect at 25 imposes $\Delta H_c(25) = 0.1$. An equally weighted two-percentage-point BA effect at ages 23–24 imposes $[\Delta H_c(23) + \Delta H_c(24)]/2 = 0.02$, not a restriction at the midpoint.

[11] Six years after college entry need not mean age 24. Converting follow-up into age requires entry-age information or source-specific averaging. Policy event time also differs from individual follow-up.

or one free parameter dimensions. Appendix B.3 gives the minimum-distance formulation and the algebra used to solve these equations directly. I impose no probability distribution over compatible paths or their parameters. The resulting NPV and MVPF candidate ranges describe conditional possibilities, without assigning probabilities to a positive NPV or an MVPF above one.[12]

The first experiment fixes the observation age at 25 and varies the education process generating that observation. Holding two generating parameters at their reference values, I vary the third over its entire conditional feasible interval. Each scenario supplies its own age-25 effect. I then set aside knowledge of the generating parameters and search the common three-parameter class for paths matching that effect. Only one generating parameter varies at a time, but all three candidate parameters may vary: the observer is not assumed to know which parameter the policy changed. Appendix B.6 specifies the finite grids used to span the intervals.

An education effect observed at age 25 leaves open whether the effect is permanent or temporary. Panel A of Figure 3 compares a permanent increase in completed schooling, a temporary gain from faster accumulation, and a temporary loss from delayed accumulation followed by catch-up, from left to right.[13] The orange curves describe these stipulated responses. In the middle column, the early advantage largely disappears as the comparison population catches up; in the right column, the treated population initially falls behind and subsequently catches up. The shaded regions summarize the alternative paths found when only the age-25 effect is observed. Within each column, every candidate reproduces that effect and satisfies the same maintained restrictions, yet the subsequent education outcomes can differ. For both positive early effects, the candidates include persistent gains and later reversals of the attainment advantage. For the negative early effect, they include persistent shortfalls, catch-up, and a later attainment advantage. These are possible evolutions of population means; they do not identify particular individuals' graduation or return-to-school decisions. Observing the gap at one age therefore leaves uncertainty about both eventual schooling and how long an advantage or disadvantage lasts. Under the valuation rule introduced next, these differences change the duration of earnings gains or losses and the timing of schooling costs.

3.3 From Education Paths to Net Present Value

Education paths that reproduce the same observed effects can imply different economic values under a common earnings and cost schedule. For college policies without observed earnings effects, Hendren and Sprung-Keyser (2020) project income changes from education outcomes using returns estimated by Zimmerman (2014). The superscripts point and path distinguish the benchmark and its path-based extension. I write both values as functions of the schooling-effect path ΔH_c, holding the observation age a^* fixed. The benchmark uses only the observed gap $\Delta H_c(a^*)$ in completed schooling years per member of the target population.[14] Let $Y_0(a)$ be annual comparison earnings in a common dollar year and r_t^{HSK} the proportional earnings effect per additional schooling year in period t after policy onset. The dollar premium is $\pi_{\mathrm{YS}}^{\mathrm{HSK}}(a) = Y_0(a) r_{a-a_0+1}^{\mathrm{HSK}}$. With first possible effect age a_0, terminal valuation age A, and discount rate ρ, write the common discount

[12] Grid spacing does not assign path probabilities. Here the stipulated effects have no sampling error, so candidate ranges are not confidence intervals. Finite search also does not certify global bounds; Appendix B.7 distinguishes these issues.

[13] Only the named parameter moves to its conditional-interval midpoint. Relative to $(13.9222, 0.2575, 12.5649)$, the increments are $(0.3160, 0, 0)$, $(0, 0.0030, 0)$, and $(0, 0, 6.4351)$. The rule is independent of valuations and does not imply equal-sized policy shocks. Table B1 reports all scenario parameters.

[14] Appendix B.4 gives schooling conversions and the treatment of age-window averages.

factor as $D_\rho(a; a_0) = (1+\rho)^{-(a-a_0)}$. The single-effect benchmark is

$$\mathrm{NPV}_c^{\mathrm{point}}(\Delta H_c; a_0, A, \rho) = \sum_{a=a_0}^{A} D_\rho(a; a_0)\pi_{\mathrm{YS}}^{\mathrm{HSK}}(a)\Delta H_c(a^*). \tag{5}$$

The observation age a^* stays fixed as the valuation age a runs from a_0 to A. The benchmark therefore assigns the same NPV to all paths with the same observed gap. The retained schedule assigns negative returns in the first seven periods and positive returns thereafter.[15] Thus, a single observed education difference generates an income profile that varies with age through earnings, returns, and discounting. Its value depends on using that same education input in periods for which the education path is unobserved.

I extend this benchmark by assigning its signed earnings response to the age at which each part of the attainment gap is acquired. Define the signed completion increment

$$q_c(\tau) = \Delta H_c(\tau) - \Delta H_c(\tau - 1). \tag{6}$$

These increments need not be positive and are not probabilities of individual graduation. They record the difference between the two population paths' acquisition at each age. The net annual earnings effect and its present value are

$$\Delta y_c(a) = Y_0(a) \sum_{\tau} R(a, \tau) q_c(\tau), \tag{7}$$

$$\mathrm{NPV}_c^{\mathrm{path}} = \sum_{a=a_0}^{A} D_\rho(a; a_0)\Delta y_c(a). \tag{8}$$

For the common simulation, $R(a, \tau) = r^- \mathbf{1}\{\tau - 7 \le a < \tau\} + r^+ \mathbf{1}\{a \ge \tau\}$, evaluated only at income ages $a \ge a_0$. Here $r^- = -0.0667064939863901$ and $r^+ = 815/7241$. A single completion increment of size m at 25 reproduces the benchmark's negative response at ages 18–24 and positive response from 25 onward. Two increments sum two shifted responses; a smooth path sums increments across all completion ages. This is a maintained transport of an earnings-response template across completion ages. It does not assert that Hendren and Sprung-Keyser observed everyone graduating at 25 or that each additional schooling year entails seven actual years of enrollment.

Early earnings losses and later gains are already in $\Delta y_c(a)$. No separate opportunity-cost charge is subtracted, and annual changes in completed schooling are not set equal to contemporaneous enrollment. Linearity permits signed increments, including comparison-group catch-up. It also assumes that the same marginal response can be transported across the completion ages and sizes considered, without additional interactions or general-equilibrium effects. These assumptions concern valuation rather than the identification of the source education estimates.

The first experiment uses $a^* = 25$, income ages 18–65, a 3 percent discount rate to age 18, and 2000 dollars. The common baseline remains the fitted 1985–1987 cohorts. Education paths extend through 72 because income at 65 can include a pre-completion response associated with acquisition up to seven years later. The terminal attainment statistic remains the gap at 65. All restrictions, income prices, response coefficients, and discounting are held fixed across candidate paths. Appendix B.4 gives the equivalent stock

[15] The schedule uses approximately −0.0667 in periods 1–7 and 0.1126 thereafter, following the Hendren and Sprung-Keyser replication's conversion of Zimmerman's estimates. It is an illustrative benchmark; their policy accounts also use other education inputs and observed earnings.

formula and the continuous-age interpretation. Policy accounts then retain their own income, tax, transfer, and education-cost conventions.

Applying the common path valuation to the candidates in the first experiment produces both positive and negative values in all three displayed scenarios. Table 1 separates each scenario's single-effect benchmark, its known generating-path value, and the values of other paths matching the observed effect. The age-25 candidate NPV ranges, in thousands of 2000 dollars, are $[-3.53,\ 44.80]$ for the plateau change, $[-15.43,\ 42.58]$ for the speed change, and $[-26.41,\ 39.18]$ for the timing change. Finding paths with opposite signs establishes that the stipulated education evidence and maintained restrictions do not determine the NPV sign. The generating-path value exceeds the single-effect benchmark in the plateau and timing examples and is below it in the speed example, so the direction of that comparison is not uniform. Table B2 summarizes the full scans; the complete scenario results remain in the replication files.

The annual bars in Figure 4 separate the post-completion and pre-completion terms of the earnings response for the single-effect benchmark and the full attainment path. The illustration combines a higher plateau with changes in acquisition speed and timing; Figure B2 shows the corresponding levels.[16] Its gap is 0.364 years at 25 and 0.263 at 65. The single-effect and full-path NPVs are approximately \$15,390 and \$13,986. Blue bars show the post-completion earnings premium, orange bars the pre-completion earnings term, and black dots their sum. This is a decomposition of the maintained response template, not separate estimates of wage returns and opportunity costs.

The distinction between permanent and temporary effects can also be examined with the education inputs used in the policy applications. Figure B3 in the appendix compares each reference earnings NPV with the value obtained by matching its education input through a plateau, speed, or timing change alone. Table B3 reports the corresponding normalized NPVs and parameter changes, including calculations without a comparable pair. The other two parameter families remain at their calibrated baseline values. The K specification produces a permanent effect; the b and d specifications produce temporary effects as the paths eventually converge. Both axes use each account's monetary units and the same reference-expenditure normalization. These comparisons retain the applications' own valuation schedules, rather than imposing the simulation's common schedule on every policy. Appendix B.5 explains the construction, unit bridges, and differing matched samples. Differences from the reference projection can reflect both continuation and the source-to-path earnings mapping; they are not estimates of which mechanism actually occurred.

3.4 What Additional Observation Ages Reveal

Additional observations can distinguish paths that fit the same early effect, but the information they provide depends on when schooling is measured. The second experiment holds each generating path fixed and changes the observation ages, retaining the same baseline, path restrictions, and valuation rule. I use five positions spanning each parameter's conditional feasible interval and examine every single age and every pair of distinct ages from 18 through 35. The path remains fixed as observations change; its effects at different ages need not be equal. Each added observation restricts the same underlying path set. Appendix B.6 gives the full design and the numerical procedure for preserving this nesting.

Adding an age-35 effect sharply narrows the NPV ranges for the same three midpoint scenarios. Panel B of Figure 3 retains the generating parameters, baseline, restrictions, valuation rule, and axis scales of Panel A.

[16] The joint increments are $(\Delta K_c, \Delta b_c, \Delta d_c) = (0.2625, 0.0300, -0.4500)$. This illustration satisfies the common path restrictions but is outside the two experiment grids.

The candidate NPV ranges in Table 1 become $[10.61,\ 14.33]$, $[-2.92,\ 0.46]$, and $[-8.74,\ -3.97]$ thousand dollars, reductions in width of 92.3, 94.2, and 92.7 percent. The second observation matters because a single schooling gap does not determine how quickly it emerged or what follows it. Timing also matters for value: a year acquired earlier can produce earnings gains over more years but also moves forgone earnings closer to the discount base. In the timing example, the age-35 gap is only -0.043 schooling years, yet the candidates found have negative NPVs: near-convergence in later schooling need not erase the value lost during earlier delays. The candidates found for the plateau example are positive, whereas the speed example still admits both signs. Its generating speed lies close to the reference value under the one-year acquisition cap, so the stipulated effect is small. These findings do not prove sign identification: the search is finite, and one of the five timing scenarios also admits both signs after adding age 35.

The location of the additional observation matters alongside its availability. Table 2 compares selected designs within each scenario before summarizing across scenarios.[17] Across the three parameter groups, adding age 20 to age 25 reduces the NPV width by a median 33.5–68.1 percent, whereas adding age 24 reduces it by 38.2–74.9 percent. Adding age 30 yields median reductions of 76.0–88.3 percent. The age-24 pair is more informative than the age-20 pair in these summaries despite the shorter distance between observations. It restricts the curve near the first observation, while later ages constrain how much of the schooling gap persists. Neither comparison supplies a general rule about nearby and distant observations: the ranking depends on the logistic family, anchor, and support. Figure B4 summarizes all observation combinations, showing both interior-scenario medians and maximum widths across all five scenarios; the individual-scenario results remain in the replication files.

Finite-window evidence can therefore leave economically consequential differences among admissible education paths, while observations at other ages can narrow the differences found. The exercise conditions on the fitted baseline, logistic family, parameter domain, and valuation calibration. It does not evaluate source-estimate sampling error or sensitivity to alternative schooling functions and earnings-response mappings. The policy applications in Section 4 add source-specific outcome definitions, timing bridges, observed mechanisms, and full welfare accounting.

4 From Education Paths to Policy Welfare

Calculating the earnings NPV of an education path is only one step toward evaluating a policy: the welfare comparison also depends on transfers, private education payments, and the net cost to the government. Building on the marginal value of public funds (MVPF) framework of Hendren and Sprung-Keyser (2020), I incorporate alternative education paths into beneficiaries' willingness to pay and the government's fiscal account. The broader exercise reconstructs education-related welfare calculations for 38 policy cases, whose results are brought together in Table 3 and Section 5. This section examines three cases introduced in Section 2: the Adams Scholarship, Social Security student benefits, and Tennessee Promise.[18] These cases illustrate how the reconstruction procedure operates and how alternative education paths can alter welfare conclusions. Under the reference specification, the MVPF is finite and less than one for the Adams Scholarship, finite and greater than one for the Social Security enrollment alternative, and classified as self-financing for Tennessee

[17] Reductions are calculated within scenarios relative to age 25 alone, then summarized by the median at the 25, 50, and 75 percent interval positions. Largest widths and sign counts include both boundaries.

[18] The collection table normalizes both monetary components by fixed reference program costs; this section uses source dollars. MVPF is unchanged.

Promise. The discussion relates the candidate paths to each study's dynamic evidence and mechanisms, distinguishing the observed education effects from the assumptions needed to project lifetime benefits and fiscal costs.

4.1 Welfare Accounting and the Education-Path Replacement

The marginal value of public funds (MVPF) measures beneficiaries' willingness to pay for a policy change per dollar of its net cost to the government. Following Hendren and Sprung-Keyser (2020), let WTP_i denote willingness to pay and N_i net fiscal cost for policy contrast i. Both are present values for the same population and policy unit. For a positive net cost,

$$\text{MVPF}_i = \frac{\text{WTP}_i}{N_i}, \qquad N_i > 0. \tag{9}$$

Willingness to pay is the monetary value of the policy to its beneficiaries under the maintained welfare assumptions. Net fiscal cost includes the initial expenditure and subsequent fiscal effects of behavioral responses; it is not simply the program's budget. For example, higher earnings can increase private disposable income while also reducing net public cost through higher taxes and lower means-tested transfers. A policy with positive willingness to pay and nonpositive net fiscal cost is classified as self-financing and assigned an infinite MVPF.[19]

The education-policy accounts can be organized into earnings-related components and the remaining benefits and costs. Use the superscript ref for L1 and write

$$\text{WTP}_i^{\text{ref}} = B_i^{\text{ref}} + \text{NPV}_i^{\text{after,ref}} - C_i^{\text{priv,ref}}, \tag{10}$$

where B_i^{ref} contains the value of program transfers and other private benefits included in the source account, $\text{NPV}_i^{\text{after,ref}}$ is the disposable-income component generated by the education-related earnings projection, and $C_i^{\text{priv,ref}}$ is the associated private education payment. The corresponding public account is

$$N_i^{\text{ref}} = G_i^{\text{ref}} + C_i^{\text{pub,ref}} - \text{NPV}_i^{\text{tax,ref}}. \tag{11}$$

Here G_i^{ref} contains direct program expenditure and other net fiscal costs, $C_i^{\text{pub,ref}}$ is additional public education expenditure not already included in G_i^{ref}, and $\text{NPV}_i^{\text{tax,ref}}$ is the earnings-related fiscal return. The superscript tax includes changes in transfers when the source uses a combined tax-and-transfer rate. All components are signed policy-induced changes: a reduction in tuition payments raises willingness to pay, whereas lost tax revenue raises net cost.[20]

I assemble the applications from education-related entries in the Policy Impacts Library and their underlying studies, focusing on calculations that project lifetime earnings and fiscal returns from education effects measured within finite windows. Of the 38 policy calculations, 37 have an original welfare calculation, denoted L1. Most use the reconstruction by Hendren and Sprung-Keyser (2020), as in Adams; STBF and

[19] Self-financing is a welfare convention, not the arithmetic ratio at negative cost. If both components are nonpositive, I report them without an assigned MVPF. Hendren and Sprung-Keyser also apply the marginal framework to finite policy contrasts.

[20] Components vary by policy, and transfers must not be counted twice. Scholarship benefits need not equal expenditure dollar for dollar: Hendren and Sprung-Keyser distinguish induced students from those whose behavior is unchanged. Appendix D.1 gives accounting conventions.

Tennessee Promise use their original authors' welfare calculations (Angrist et al., 2022; Schoonover et al., 2026). The remaining application values the education channel of cannabis legalization.[21]

The 38 calculations share an outer net-income account while retaining different source mappings. Thirty-three use Hendren and Sprung-Keyser programs, three use independent authors' forecasts, and two are explicitly limited extensions. I first replay the source monetary components, including earnings, taxes, education payments, WTP, and net cost. I then replace the education inputs, holding the source mapping fixed where it is recoverable. STBF retains its nonlinear credential–experience function. Early Head Start and Tennessee Promise use public-data reconstructions because their full author income inputs are unavailable; their reported L1 values remain separate comparison benchmarks. No income multiplier is chosen to force these reconstructions to equal a reported total. Eleven tax cases retain the fixed finite-policy normalization in Appendix E.4.1.

Let $\mathcal{P}_i^j$ contain the relevant schooling, credential, or time-enrolled paths. The common earnings object is

$$\Delta y_i(a) = F_i(a; \mathcal{P}_i^1, z_i) - F_i(a; \mathcal{P}_i^0, z_i), \tag{12}$$

where z_i contains the retained source prices, response coefficients, and population conventions. The Hendren and Sprung-Keyser extension uses signed attainment increments with source-specific response templates; Social Security indexes its retained enrollment response to entry age. An author's nonlinear forecast instead receives the separate education inputs it originally used. In either case early losses enter this net-income effect once, without an additional opportunity-cost deduction. With ordinary age discounting, $E_i = \sum_a D_i(a)\Delta y_i(a)$; where the source discounts response phases from different bases, those phase-specific contributions are retained before aggregation (Appendix D.2).

Write T_i for the fiscal earnings component, P_i for private education payments, and C_i for public education costs. For accounts that value disposable earnings, the component replacement is

$$\mathrm{WTP}_i^{\mathrm{path}} = \mathrm{WTP}_i^{\mathrm{ref}} + (E_i^{\mathrm{path}} - E_i^{\mathrm{ref}}) - (T_i^{\mathrm{path}} - T_i^{\mathrm{ref}}) - (P_i^{\mathrm{path}} - P_i^{\mathrm{ref}}), \tag{13}$$

$$N_i^{\mathrm{path}} = N_i^{\mathrm{ref}} + (C_i^{\mathrm{path}} - C_i^{\mathrm{ref}}) - (T_i^{\mathrm{path}} - T_i^{\mathrm{ref}}). \tag{14}$$

The old component is removed before its replacement is added. In those source-account cases, source unit prices and the source completion allocation also transport private and public education costs across batch timing. That additional cost-timing bridge is stated explicitly; it is not inferred from the attainment data. Independent author cases retain their specified non-income components. Early Head Start retains its source convention of gross earnings in WTP, and algorithmic placement retains source WTP while extending only the fiscal earnings channel. These exceptions are not forced into a disposable-income formula. Figure D1 displays paired WTP and net cost, so each ratio comes from the same feasible path.

The comparison curves H_i^0 and empirical restrictions come from the Census–ACS calibration library, matched to each application's geographic target and birth cohorts. Adams uses Massachusetts, retaining all races, sexes, and birthplaces in the external population data. Social Security combines a source age-23 control level with a transported national entry shape. Tennessee requires an additional adjustment because its education effects are conditional on high school graduation: I match that denominator, the source's prior-year state definition, and its latest-window control means. These adjustments improve correspondence to the reported outcomes without recovering the source design's untreated potential path. Where a comparable

[21]The 20 September 2026 library snapshot and Appendix E supply 23 source-study families. Several tax contrasts can come from one study. Inclusion requires reconstructable education and accounting inputs; the collection is not a census of education policies.

control level is unavailable, a statewide population mean remains an external calibration rather than a measured mean for selected scholarship applicants.[22]

L1 comprises the original WTP, net fiscal cost, and MVPF calculation. L2 reconstructs the education inputs retained for L1; L3 preserves L2 and adds separately reported source windows or dynamics. For accounts based on Hendren and Sprung-Keyser (2020), both reconstructions retain their outcome-conversion rules; author-reported accounts use the case-specific conversions above. Additional information need not be a directly estimated schooling-years effect: an earlier enrollment estimate can enter through the retained duration conversion, provided its placement on the schooling path is stated explicitly. The same coefficient may already enter the reference total; L3 then retains its separate timing rather than introducing a previously unused estimate. Both reconstructions hold the comparison curve, empirical support, welfare account, and L2 constraints fixed.[23]

The common path-construction rule is constrained minimum distance:

$$\widehat{\Theta}_{i,L} = \underset{\theta_i \in \Theta_{i,L}}{\arg\min} \left[\widehat{\mathbf{m}}_{i,L} - \mathbf{g}_{i,L}(\theta_i)\right]' W_{i,L} \left[\widehat{\mathbf{m}}_{i,L} - \mathbf{g}_{i,L}(\theta_i)\right], \qquad L = 2, 3 \tag{15}$$

Here $\widehat{\mathbf{m}}_{i,L}$ contains the fixed source point estimates, and $\mathbf{g}_{i,L}$ applies their age, cohort, and outcome operators to a candidate path. The set $\Theta_{i,L}$ imposes the anchored logistic form, calibrated support, education-level bounds, the one-year annual schooling-acquisition cap, and credential ordering where relevant. L3 retains the L2 equalities and adds source observations to the fit. $W_{i,L}$ is fixed in advance; the ten minimum-distance L3 applications use the identity matrix in their common within-case units, without standard-error bands. If zero residual is attainable, all exact matches minimize the same criterion. Otherwise the retained paths minimize the discrepancies subject to the unchanged L2 restrictions, and their residuals are reported. The numerical search retains distinct minimizing solutions found, not every approximately fitting path. Input uncertainty is studied separately. Table E2 identifies the ten best-fit applications, for which the search currently retains one path each; this is a conditional model selection result, not identification of the true path.

For Tennessee, the retained non-income components yield WTP = $w_0 + (1 - t)E$ and $N = n_0 - tE$. At positive cost, MVPF exceeds one exactly when $E > n_0 - w_0$. The Adams and Social Security accounts additionally transport education payments, so their welfare cannot be summarized by a fixed threshold in earnings alone. Welfare comparisons across groups also depend on social welfare weights (Hendren and Sprung-Keyser, 2020). All ranges refer to verified numerical candidates rather than certified bounds on the continuous feasible set.

4.2 Adams Scholarship

Cohodes and Goodman (2014) use test-score eligibility thresholds to estimate the effects of Massachusetts' Adams Scholarship on college choice, persistence, and completion. The estimates used here concern the 2005–2006 high-school graduating classes. Eligibility shifts students toward participating public colleges. Within two years of high school, two-year enrollment rises by 0.7 percentage points and four-year enrollment changes little. At six years, bachelor's completion falls by 2.5 percentage points, while the two-year-completion

[22]The library covers 50 states, DC, and the nation. Multi-state targets use fixed weights; identification control states need not belong to the target population. Table E1 and Appendix D.5 document matching and support.

[23]Appendix E.7 records the observation windows, conversions, and any temporal or population mapping. Institutional policy lags and conditional college-year outcomes require such mappings; they are not relabeled as individual follow-up. Splitting a single entry observation into assumed study years alone does not create L3, as in Tennessee Pell and Georgia HOPE.

effect is approximately zero. The authors interpret similar declines in persistence and graduation as evidence favoring dropout over pure delay within follow-up.

Hendren and Sprung-Keyser (2020) convert these enrollment and completion effects into −0.018 schooling-equivalent years per eligible student, using assumed durations for each enrollment and completion margin. Their earnings projection gives an NPV of −$1,204.04. Combining this loss with scholarship benefits, private education payments, public costs, and tax-and-transfer effects yields willingness to pay of $1,364.23 and net cost of $1,896.35, in 2004 dollars. The L1 MVPF is 0.72. This calculation concerns scholarship eligibility rather than the separate effect among students induced to switch colleges.

I retain the source conversion, signed earnings response, tax mapping, and education-cost unit schedules. A single-batch replay reproduces the reference monetary components. Replacing that batch by signed path increments changes earnings and the timing of source private and public education costs. The account is $W = w_{\text{fixed}} + E - T - P$ and $N = n_{\text{fixed}} + C - T$, with fixed noneducation components and source-derived cost schedules. It therefore does not reduce to a fixed threshold in earnings alone. Appendix D.6 states the cost-timing bridge and the retained 2004-dollar units.

The paths use the Massachusetts ACS profile for the 1987 birth cohort, anchored at 17 so that education can respond from age 18. L2 imposes the converted six-year total at 24. L3 adds three source windows from the original study's Table 4, columns (3) and (5): immediate entry, entry within two years, and completion within four years. Applying the same conversion gives targets of 0.026, 0.007, and −0.011 schooling years at assigned ages 18, 20, and 22, respectively (Figure 5). These age placements are temporal allocations of converted estimates, not directly observed schooling stocks. L3 keeps the age-24 input exact and fits the additional point targets by equal-weight minimum distance. Appendix D.6 gives the arithmetic; Appendix E.9 reports sensitivity to calibration and valuation.

The L2 candidates leave the welfare comparison unresolved. The four-input L3 fit instead selects an MVPF of 0.63, with positive willingness to pay below positive net public cost. Its value depends on the added source estimates, their age allocation, and the fitting criterion: the retained path does not pass through all four converted targets exactly. The resulting singleton is therefore not a claim that the education path has been identified. The original authors' dropout interpretation concerns the observed persistence and completion history. A modeled path that later catches up is a conditional continuation beyond that history, not contrary evidence about the observed mechanism.

4.3 Social Security Student Benefits

Dynarski (2003) studies the withdrawal of Social Security student benefits in 1982. Before termination, eligible children of deceased, disabled, or retired workers could continue receiving benefits while attending college full time. The empirical comparison uses the change in college attendance among children with deceased fathers relative to children with living fathers, before and after termination. I retain the positive benefit-availability contrast: the education coefficients measure the gain from eligibility, rather than the negative effect of withdrawing aid.

The outcome is whether a person has ever attended college full time by a stated age. It is cumulative entry, not enrollment at that age or degree completion. Table 3 reports an effect of 21.9 percentage points by age 23 and 22.4 points by age 28. The latter uses the paper's conservative imputation for attrition; "lower bound" labels that specification, not a confidence limit. The respective standard errors are 10.2 and 10.6 points. Dynarski explicitly compares these ages to assess whether aid merely accelerates college attendance.

The similar estimates suggest a lasting attendance gain during observed follow-up, although their imprecision limits stronger conclusions.

For L1, I use the enrollment alternative reported by Hendren and Sprung-Keyser (2020), rather than their main completed-schooling specification. It combines the age-28 attendance effect with an assumed two years of schooling per induced entrant. At 3 percent discounting, reproducing this account gives earnings NPV of \$21,163.75, willingness to pay of \$23,072.74, and net public cost of \$4,739.23, all in 2000 dollars per eligible person. Their ratio is 4.868, which reproduces the reported MVPF of 4.87. The source-reported 95 percent interval is $[1.16, 32.23]$. Their main schooling specification instead uses 0.679 years and reports 4.86; its input and welfare account are not combined with the enrollment specification here.

Let $H^j(a)$ denote cumulative full-time attendance under policy state j. With an unaffected anchor at 17, L2 imposes $\Delta H(28) = 0.224$. L3 retains this equality and adds $\Delta H(23) = 0.219$. Both are source observation ages, so the second restriction supplies a separate follow-up without allocating an assumed study duration across intermediate ages. The two point inputs are matched exactly. Three logistic parameters can still vary along a one-dimensional matching set; observing two effects need not select a single path.

The comparison curve is calibrated to the source's age-23 attendance mean of 0.352 for the post-termination deceased-father group. Its age shape is transported from the national ACS 1985–1987 some-college profiles, which cover young adulthood. This is an external proxy for the age pattern of full-time entry, not an estimated untreated trajectory for the original DID population. I retain donor-based speed and timing restrictions and require the treated probability to lie between zero and one. I do not use cross-population variation in attainment levels to cap this selected population's policy response. Appendix D.7 explains why the historical cohort's backward-extrapolated entry profile and the earlier plateau restriction are unsuitable for this specification. The calibration and restrictions remain identical in L2 and L3.

The income mapping follows the source event: entry. Each signed increment of the attendance-effect path receives the two-year schooling conversion and the earnings response used by Hendren and Sprung-Keyser, with seven early response years from that entry age and the later response thereafter. A single batch entering at 18 reproduces the source negative response at ages 18–24 and positive response from 25. Seven years describe the earnings template, not literal years spent in college. The source education-cost bundle is transported from the same entry age, while the fixed program components remain unchanged. Early losses enter net income once; there is no additional opportunity-cost deduction.

Figure 6 displays the comparison curve and treated-path ranges. At fixed inputs, L2 yields earnings NPVs of 17.09–21.59 thousand dollars and MVPFs of 3.79–4.99. Adding the age-23 observation narrows the earnings range to 18.71–19.87 thousand dollars and the MVPF range to 4.21–4.52, a 74.4 percent contraction in its searched width. The retained candidates have positive WTP and positive net cost, with MVPF above one. The additional observation tightens the valuation within this specification; it does not overturn the source's interpretation of the observed attendance effects or validate the transported income and path assumptions. Unlike the L2 span, the narrower L3 span lies below the reference 4.87. The reference single-entry batch need not belong to the anchored logistic class or match the additional age-23 observation.

4.4 Tennessee Promise

Schoonover et al. (2026) study the expansion of tuition-free community college from a Tennessee county program in 2009 to a statewide program in 2015. Their attainment analysis uses 2000–2024 ACS data for high-school graduates aged 18–24, with exposure defined by eligibility at age 17. They estimate effects for

overlapping age pairs from 18–19 through 23–24, controlling for state and survey year. At ages 23–24, some college or higher rises by 3.5 percentage points, associate's or higher by 4.0 points, and bachelor's or higher by 2.7 points. The authors examine earlier windows and acknowledge that some participants may complete degrees after age 24.

The authors use Tennessee-specific education returns to project lifetime income within the framework of Hendren and Sprung-Keyser (2020). Private benefits include tuition relief for students who would have enrolled anyway and after-tax earnings for induced students. Scholarship and public education expenditure are offset by projected tax receipts. With a 26 percent tax rate and 3 percent discounting, they report willingness to pay of \$72,837 and net public cost of $-\$10,053$, giving a self-financing L1 classification. The reported tax gain of \$24,845 implies a gross earnings NPV of $E^{\text{ref}} = 24{,}845/0.26 = 95{,}557.69$ under this fiscal mapping.

The public author materials do not supply a complete age-by-credential earnings table. I therefore reconstruct Tennessee earnings from public ACS data and identify the result as a reconstruction, not an exact replication. The schedule uses 2021 dollars, two pre-completion study years for SC/AA and four for BA, zero earnings while studying, and the corresponding credential-versus-HS earnings difference after completion. Signed exclusive-credential increments aggregate these responses. Cumulative SC+, AA+, and BA+ shares are differenced into exclusive categories. BA+ is priced at the BA schedule without an extra graduate premium. These are additional bridges.

To connect the population attainment effects to the participant welfare unit, the main normalization combines the reported \$2,711 grant, the private-benefit remainder of \$2,124.31, and the 5.4-percentage-point entry effect. It implies a participant share of 0.249524 under the stated entry-effect bridge; this is an account-implied normalization, not observed take-up or an income-matching factor. With fixed components in the adopted 2021-dollar convention, $\text{WTP} = 2{,}124.31 + 0.74E$ and $N = 14{,}792 - 0.26E$. The source component sum differs from its printed program total by one dollar. Applying the public schedule to the nested endpoint inputs gives \$69,261.65, compared with the author's implied \$95,557.69; the difference is retained. Appendix D.8 documents the sample, timing, and unit assumptions.

L2 matches all three latest-window effects using anchored logistic paths for the 1992–1998 birth cohorts, with fixed equal cohort weights. The Tennessee ACS reference matches the high-school-graduate denominator, prior-year residence definition, and reported control levels. Both policy states share the age-17 anchor, and credential levels remain nested at every age (Figure 7). State-calibrated support and a restriction based on remaining completion in earlier cohorts limit the continuations; the main bachelor's plateau cannot exceed 45.4 percent. These are maintained restrictions on an external reference, not recovered untreated predictions from the source regressions.[24]

The latest-window effects alone admit earnings NPVs from -68.71 to 203.21 thousand dollars. L3 uses all eighteen Table 2 point estimates: three credentials observed in six overlapping age windows. It preserves the three latest-window effects exactly and minimizes the equal-weight squared discrepancies for the fifteen earlier coefficients. The search retains one best-fitting joint path, with earnings NPV 113.94 thousand dollars and a self-financing MVPF under the reconstructed account. Its root mean squared residual across eighteen inputs is 0.353 percentage points, with a maximum absolute residual of 0.667 points. Table D2 reports every fitted value and residual. Figure 7 focuses on ages 17–25. The source point estimates are fixed inputs here; their uncertainty is not used to define acceptance bands.

[24]Table D1 and Appendix D.8 document donor cohorts, plateau extrapolation, level scaling, and age weights.

At positive net cost, this reconstructed account has MVPF above one if $E > \$12,667.69$, while positive-benefit self-financing requires $E \geq \$56,892.31$. The comparison with the author's L1 depends on public earnings reconstruction, participant normalization, and price conventions as well as education timing. Comparisons among candidate paths hold those choices fixed. The source cautions about returns and completion after age 24 remain relevant.

Input uncertainty is examined separately from continuation at fixed education effects. Table 4 summarizes 1,000 prespecified input repetitions per application. Adams and Social Security propagate linked source inputs, with disclosed dependence assumptions for added observations. Tennessee uses researcher-constructed perturbations under stated dependence and scale choices. Tennessee's normal perturbations are not draws from the source permutation distribution. Appendix F documents these differences and unsuccessful searches; the endpoint summaries do not establish confidence coverage for the continuous path class.

The cases show how continuation and its restrictions enter distinct welfare questions: the assumptions needed to locate Adams' converted schooling loss in time, the value of a second observed attendance age for Social Security, and the fiscal classification of Tennessee Promise. Within each path comparison, the income and cost mappings remain fixed; the education paths entering those mappings change. Section 5 asks how frequently comparable fixed-input paths leave these judgments unresolved across the policy collection.

5 Welfare Judgments across Policies and Populations

The welfare implications of an education effect depend on its persistence, the ages at which attainment changes, and uncertainty in the education inputs. This section asks whether the observed information determines which side of MVPF one a policy lies on, whether it can finance itself, and how these conclusions change when additional comparable observations restrict the path. The collection contains 38 finite-path calculations, including 37 with L1. The tax-input conversions and account adjustments described in Section 4.1 remain identified within the full comparison. Single-parameter comparisons distinguish permanent effects on eventual attainment from temporary effects on acquisition speed or timing. Input repetitions examine a separate source of variation under each case's specified uncertainty scheme. Appendix G compares men and women within seven policies, holding terminal education effects and valuation prices fixed.

5.1 Continuation Values and Policy Comparisons

Figure 8 places the L2 ranges and available L3 exact-match ranges or best-fit values beside the reference MVPFs, ordered by descending L1. All 38 calculations use finite attainment paths. Among the 37 with L1, 30 admit positive-cost paths on both sides of MVPF one at fixed L2 inputs. Of these, 17 belong to the 24-account group displayed separately from the finite tax and special accounts. Taking the education coefficients as known therefore need not determine whether beneficiary welfare exceeds one dollar per dollar of net public cost.[25]

The uncertainty can concern the signs of the monetary components, rather than only the size of their ratio. In 24 of the 37 calculations with L1, including 12 in the 24-account group, one compatible path gives negative willingness to pay and positive public cost, whereas another gives positive willingness to pay and

[25] Several tax contrasts come from one study, so the calculation count is not a count of independent studies. The 24-account group spans 20 source-study families, with threshold pairs in 15. MVPF one weights beneficiary dollars and public resources equally; welfare priorities can also depend on distributional weights and financing alternatives (Hendren and Sprung-Keyser, 2020).

nonpositive public cost. These paired outcomes follow from different projected earnings and tax receipts under the same education restrictions. Fixed transfers remain unchanged. Where recovered, source private and public education-cost schedules are evaluated on each path, so these costs can also change. A gain that persists into high-earning ages can benefit recipients and repay the public expenditure; an initial gap that disappears or reverses can produce a different conclusion. These are possible aggregate continuations, not estimates of individual dropout or return-to-school behavior.

A large MVPF need not represent a large dollar benefit. Numerical checks find positive-cost approaches to zero net cost with a positive limiting WTP in 24 of the 37 calculations, and with a negative limiting WTP in 16. The resulting ratio diverges while both monetary components remain finite. Figure 8 therefore separates positive-cost ratio spans from self-financing states. The span does not assert that every intermediate ratio is attained, and a finite searched endpoint does not certify the boundary of the continuous feasible set.[26]

Figure 9 applies the same lower-endpoint ordering to all 37 calculations with a reference MVPF. It first compares lower endpoints and then upper endpoints, both descending, using L3 where available and L2 otherwise. For a best-fit singleton the endpoints coincide; its rank conditions on the fitting rule as well as the path restrictions. Finite endpoints are rounded to two decimals, and equal endpoint pairs share a rank. Some tax calculations retain unbounded positive-cost ratio spans although attainment and both monetary components are bounded. The divergence comes from finite paths approaching zero net cost. No path probabilities are used to break ties.[27] Both axes increase from small to large ranks, so a point below the diagonal has higher priority under this rule than under L1. The ordering expresses a preference over conditional endpoint summaries; it is not a social-welfare or certified worst-case ranking.

Twelve calculations with L1 admit L3 under the common rule for additional source windows or dynamics. Replacing one calculation's L2 by L3 improves its displayed priority in ten cases and leaves it unchanged in two, holding the others fixed. This is a consequence of the endpoint-ordering rule, not evidence that information raises true welfare. Additional restrictions or the best-fit criterion can exclude favorable as well as adverse continuations. For the nine best-fit L3 applications with L1, the displayed contraction includes selection by the maintained loss function. The added restrictions preserve the source windows under explicit outcome and age mappings. For example, Adams combines enrollment and completion coefficients, while Wisconsin retains their joint cohort–follow-up weights. Narrowing therefore depends on these mappings as well as on the additional observations. Appendix E.2 records the inputs and conversions; Figures E1–E8 display H^0 and H^1 for all 38 calculations. No probability distribution is assigned to the paths.

Table 3 reports willingness to pay and net cost alongside MVPF. The policy-type columns in the table and Figure 8 distinguish 14 direct student-aid calculations, 17 education tax-benefit calculations, four tuition or public-college-funding calculations, one college-assessment and course-placement intervention, one early-childhood intervention, and one education channel of another policy. These types describe the intervention, whereas the panels distinguish accounting specifications. Several calculations share a source study, so the category counts are not independent-study counts. Panel A contains the 24 retained comparison cases, including Tennessee's disclosed public-data reconstruction. Panel B reports the eleven finite tax reconstructions, whose source marginal coefficients require the stated finite-policy conversion. Their WTP and net-cost extrema are finite, even when their ratios are not. Panel C distinguishes the public-data Early

[26]Appendix E.3 documents complete paired paths, constraint checks, and one-sided ratio sequences. A large finite ratio or a rounded zero cost is insufficient evidence of an infinite limit.

[27]The tax calculations transport local attendance and accounting responses over a fixed finite schedule change calibrated to one point-estimate credit dollar. Appendix E.4.1 states this additional assumption.

Head Start account, the algorithmic-placement extension, and the cannabis education channel. Monetary components in Panels A and B use a fixed reference program-cost divisor within each calculation; Panel C uses native thousands. Every MVPF combines the two components of the same path. Very large normalized amounts indicate that the allowed future response can dominate a small reference expenditure; they do not establish that such a response is likely. The main finding is the existence of conflicting welfare judgments within a fixed reconstruction, conditional on both its path restrictions and its input conversion.

5.2 Permanent and Temporary Education Effects

The same finite-window education input can support a permanent or a temporary effect, with different implications for lifetime welfare. I examine this distinction by holding two parameter families at their calibrated comparison values and solving for a change in the third that matches the retained L2 education input. A nonzero change in K produces a permanent effect by altering eventual attainment. Changing b or d with K fixed produces a temporary effect: the population acquires education faster or more slowly, but the attainment gap eventually vanishes. The observation operator, empirical support, and welfare account remain unchanged across these three exercises. This makes the continuation assumption explicit without treating the observed input as evidence that one particular parameter changed. Appendix B.5 gives the matching procedure and its relation to the main path construction.[28]

Interpreting the observed education input as a permanent effect tends to preserve the reference welfare comparison. In Panel A of Figure 10, the finite MVPF pairs follow an upward pattern broadly aligned with the 45-degree equality line. Changing K leaves a permanent difference in eventual attainment, so the education gain or loss continues to affect projected earnings beyond the observed window. This resembles a reference calculation that uses the measured education effect to project sustained earnings gains or losses. The alignment is consistent with interpreting the reference valuations in this collection as assigning a permanent effect to the short-window education input. Shared education inputs and related return calibrations contribute to this resemblance, so it does not independently establish that the education effect is permanent.[29]

Temporary effects produce a different welfare comparison. In Panels B and C, several finite reconstructed MVPFs cluster near one across a wide range of reference values, rather than following the near-diagonal pattern in Panel A. With eventual attainment fixed, changes in b or d match the education input through faster or slower acquisition, and the attainment gap eventually fades. Figure B3 shows that these temporary effects still generate dispersed earnings NPVs, even though several corresponding MVPFs lie close to one. This pattern reflects how net earnings and education-dependent costs enter the retained welfare accounts. When earnings NPV is small relative to the retained transfers and other non-earnings components, those components largely determine willingness to pay relative to net public cost. Several accounts consequently remain near MVPF one even as their earnings NPVs differ.[30] Temporary effects therefore retain economic value; their influence on MVPF depends on their duration and on the scale and composition of the welfare account under

[28]The exercise preserves the individual components of each fitted H^0, so a zero change reproduces that baseline exactly. K and d change additively; b changes by a common proportion. STBF permits four outcome-specific changes and Tennessee Promise permits three credential-specific changes in the active family, subject to joint restrictions. This is a diagnostic rather than a decomposition of the main candidate ranges.

[29]Panel A contains 22 finite pairs, seven calculations that move from a finite reference MVPF to self-financing, and seven that are self-financing in both accounts. The diagonal denotes equality on the common asinh scale; it is not an estimated regression line.

[30]Among finite pairs, 7 of 23 temporary speed results and 7 of 15 temporary timing results lie between 0.75 and 1.25. Holding the other components fixed, $W = w_0 + (1 - t)E$ and $N = n_0 - tE$ approach the ratio w_0/n_0 as net earnings NPV E approaches zero,

the specified valuation rule. The contrast with Panel A separates two explanations of the same education input: a permanent change in eventual attainment and a temporary change in when education is acquired. A lifetime welfare calculation must distinguish them through additional evidence or an explicit continuation assumption. Greater precision within the observed window does not by itself resolve that distinction.

Table 5 gives each reference MVPF, its single-parameter reconstruction, and the required parameter change. It also records calculations without a matching solution. The matched samples differ, which limits comparisons of the plotted patterns. Holding two parameters fixed can leave no solution within the empirical support; one, eight, and seventeen calculations have no matched plateau, speed, or timing solution, respectively. Their absence is reported rather than interpreted as a zero response. None of the single-family solutions exactly reproduces all additional point inputs where L3 is available. Additional observations can therefore challenge a one-parameter explanation even when the reduced input is matched. This diagnostic is stricter than merely attaining the best-fit criterion used for fitted L3 applications. The results do not establish that the original lifetime projections are incorrect or that their authors ignored dynamics. They show how the welfare interpretation of a retained short-window input depends on the continuation assigned to it, while the source studies' observed dynamics and mechanism evidence remain relevant to choosing among those continuations.

5.3 From Uncertain Inputs to Paths and Welfare

Uncertainty in an estimated education effect carries through to the education paths that match it and to their welfare values. This section examines that propagation, together with sensitivity to selected accounting inputs, using the specified input repetitions. Figure 11 displays point-input values in dark colors and input-repetition endpoint summaries in lighter colors. Its Type column repeats the policy categories, separately from the bracketed input-method labels. Its L1 panel reports available source 95 percent MVPF intervals. The L2 and L3 panels instead summarize the existing 1,000 prespecified repetitions per calculation, keeping the ACS calibration and specified conversion and valuation rules fixed. The source income-response and education-cost functions stay fixed in form. The 33 calculations based on Hendren and Sprung-Keyser propagate their paired accounting inputs, including earnings-return draws where available. Their education perturbations can still be conditional or constructed, as identified below. The eleven finite tax displacements remain fixed at their point normalizations. STBF holds its recovered author-model coefficients fixed; Early Head Start and Tennessee hold their public wage schedules and cost or participation conventions fixed. Each summary runs from the 2.5th percentile of candidate minima to the 97.5th percentile of candidate maxima. The methods are intentionally named separately: propagation of archived joint source/accounting draws, constructed normal or correlated perturbations, and exercises varying economic inputs while holding education fixed.[31]

Among the 37 calculations with L1, 37 L2 summaries cross MVPF one under their specified repetition schemes. WTP summaries cross zero in 34 calculations and net-cost summaries cross zero in 33; 33 have an infinite upper MVPF endpoint under the self-financing convention or a verified positive-cost limit. Within the 24-account group, the corresponding counts are 24, 22, 20, and 20. These counts describe how the reconstructed endpoints respond to the inputs being varied. They do not measure the chance that a policy is

provided $n_0 > 0$. Path-dependent education costs can also change this limiting comparison. That ratio need not equal one. This accounting relationship does not imply general insensitivity to temporary effects: the ratio can change sharply near zero net cost.

[31] Appendix F.1 defines the order-statistic summaries and the inputs varied. These procedures do not share a sampling law or establish confidence coverage for all compatible paths. No probability distribution is assigned to paths within a repetition.

harmful or self-financing. The monetary components remain informative when the ratio becomes unstable near zero net cost.[32]

Table 4 identifies each repetition method, endpoint summary, and usable count. Across the collection, all 38,000 scheduled reduced-information repetitions were evaluated; 38,000 yielded verified paths and 38,000 yielded a defined MVPF. All thirteen L3 specifications now propagate their complete input vectors through 1,000 paired repetitions, giving 12,980 verified outcomes. L2 uses the same retained-input draw as L3. The original correlation groups are preserved and extended to added source windows under explicit assumptions, rather than treating an unavailable source covariance as known.[33] Tennessee's constructed normal shocks perturb and refit all eighteen point targets. Two displayed zero standard errors receive a 0.0005 reporting-precision scale. This is an imposed sensitivity convention, not a recovered standard error or a draw from the source permutation distribution.

Path uncertainty remains even if the original education effect is known exactly: different paths can match that effect and imply different welfare values. Estimation uncertainty arises because the original effect is itself estimated imprecisely. Allowing the input to vary changes the compatible paths and, in turn, the welfare range. The two sources can operate together, since each possible input may still admit several continuations. More precise estimates within the same window address estimation uncertainty; comparable observations at additional ages can help resolve path uncertainty. The repetitions reported here trace this propagation under the specified source-draw or input-sensitivity scheme.

6 Discussion

Education effects observed within a finite window can support different lifetime welfare judgments, depending on assumptions about persistence, attainment timing, and valuation. I take the source studies' causal identification as given; the additional assumptions link their estimates to education paths for the target population. Conversions must preserve policy units and observation windows, while assumed schooling durations cannot constitute new observations. The external comparison profile must also be relevant to the affected population. Matching Census–ACS geography and birth cohorts improves comparability, but population averages may not describe scholarship applicants or marginal students. Historical cohorts' unobserved early attainment is reconstructed using other cohorts' shapes. The common anchor imposes no effect or anticipation before the specified exposure age. Fit to ACS profiles cannot establish these population and exposure assumptions.[34]

Path and valuation assumptions further determine the comparisons. Logistic curves may omit interrupted schooling or reentry, and historical variation need not bound a policy response. The single-parameter exercises distinguish permanent from temporary explanations without identifying which applies. Earnings returns may also depend on completion age (Bárány et al., 2023). The policy reconstructions value the difference between the same income function under treated and comparison education paths. For accounts based on Hendren and Sprung-Keyser, signed attainment increments move the source negative-then-positive response template across

[32] Source intervals and reconstructed summaries can vary different inputs and use different dependence assumptions. Their widths are not a controlled comparison of inferential methods.

[33] Appendix F.2 records these extensions. Unsuccessful searches are retained without replacement or a claim of infeasibility; usable counts exclude them. Cannabis' six-year-only repetitions are diagnostic, while its primary result uses the three-window L3.

[34] Appendix D.5 records age support and donor shapes. The selected literature and policy collections also limit the generalizability of the reported frequencies.

the corresponding completion ages, or entry ages for the Social Security enrollment alternative. This assumes that the template transports across ages and adds across signed increments; it does not identify individual extra time enrolled from completed attainment. STBF retains a nonlinear credential–experience model, conditional on recovered sufficient statistics and the subgroup bridge. Early Head Start and Tennessee use disclosed public-data reconstructions where the source income tables are unavailable, so their L1 comparisons change valuation inputs as well as continuation. Recoverable source education-cost schedules move with the paths; STBF, Early Head Start, and Tennessee retain their stated non-income amounts. Appendix E.9 varies calibration, valuation, and search at fixed source inputs. Using L3 where available, positive-cost MVPFs cross one in 21 of 38 calculations, compared with 20 under narrower support. All seven common specifications admit negative-WTP, positive-cost paths alongside positive-WTP, nonpositive-cost paths in 16 calculations. These conditional contrasts neither validate all assumptions nor exclude unsearched paths; wider support preserves feasible paths but can change the selected minimum-distance fit. The ten fitted L3 results report nonzero residuals and one best-fitting path each. This selection is conditional on the logistic family and fixed weights; it is not proof that the underlying educational response or its welfare value is point identified.

Interpreting the welfare ranges also requires distinguishing input uncertainty from the rule used to rank policies. Different repetition schemes and incomplete covariance information prevent a common sampling interpretation, although all repetitions hold calibration and the path family fixed. Endpoint percentiles do not establish confidence coverage (Imbens and Manski, 2004), and more repetitions can reduce simulation error without resolving these limits or assigning probabilities to paths. The lower-endpoint ordering gives weight to adverse continuations among the paths found. Its ranks depend on search coverage, calibration, and information available for L3; they are neither expected-welfare nor certified worst-case rankings. Joint reporting of WTP and net cost distinguishes large benefits from ratios driven by near-zero costs, while self-financing and paths without an assigned ratio require separate interpretation. Spending recommendations further require distributional weights, a financing comparison, and assumptions about policy expansion. To assess the underlying extrapolations, later education outcomes withheld from path construction could be used to compare logistic predictions with constant effects and flexible monotone alternatives. Observed earnings could test valuation, while comparable age windows and joint coefficient covariances would improve reconstruction and inference. Such evidence can distinguish permanent from temporary effects with different welfare implications.

7 Conclusion

An education effect measured within a finite window leaves two economically distinct questions open: how much of the effect persists and when the remaining attainment occurs. This paper makes those questions explicit by constructing education paths that match retained reference restrictions, fitting additional point observations under the stated criterion, and passing each path through a specified valuation rule. The simulations show how additional observations can distinguish continuations that agree at an earlier age. Single-parameter comparisons separate permanent effects on eventual attainment from temporary effects on acquisition speed or timing. Temporary effects still carry lifetime earnings consequences, even when the attainment gap eventually vanishes. In the policy applications, verified paths often support different welfare judgments even when the retained education estimates are treated as known. At fixed inputs, 30 of the 37

calculations with L1 admit positive-cost MVPFs on both sides of one. More precise estimation within the same window would not remove these particular continuation differences.

The contribution is a way to assess which welfare judgments remain dependent on unobserved education paths. Compatible paths with different classifications establish that dependence under the stated calibration and valuation rules; they do not establish either path's probability. Input repetitions answer a separate question about imprecision in the retained estimates. The distinction helps identify whether further progress requires more precise inputs, additional observations of attainment, or stronger evidence on the valuation relationship.

The results favor reporting the observed education window and the assumed continuation separately. When additional outcomes are available, their contribution depends on whether they constrain attainment at a new age for a comparable population. Reporting willingness to pay and net fiscal cost alongside MVPF also distinguishes changes in benefits from ratios driven by costs near zero. Longer follow-up, richer education histories, and observed earnings can each improve the assessment, but they answer different questions. The useful standard is whether the available evidence distinguishes the paths that matter for the welfare comparison.

References

Abraham, K. G. and Clark, M. A. (2006). Financial aid and students' college decisions: Evidence from the District of Columbia Tuition Assistance Grant Program. *Journal of Human Resources*, 41(3):578–610.

Acton, R. (2021). Effects of reduced community college tuition on college choices and degree completion. *Education Finance and Policy*, 16(3):388–417.

Angrist, J. D., Autor, D. H., and Pallais, A. (2022). Marginal effects of merit aid for low-income students. *The Quarterly Journal of Economics*, 137(2):1039–1090.

Athey, S., Chetty, R., Imbens, G. W., and Kang, H. (2026). The surrogate index: Combining short-term proxies to estimate long-term treatment effects more rapidly and precisely. *Review of Economic Studies*, 93(4):2284–2312.

Bárány, Z. L., Buchinsky, M., and Corblet, P. (2023). Late bloomers: The aggregate implications of getting education later in life. NBER Working Paper 31874, National Bureau of Economic Research.

Bartik, T. J., Hershbein, B., and Lachowska, M. (2021). The effects of the Kalamazoo Promise Scholarship on college enrollment and completion. *Journal of Human Resources*, 56(1):269–310.

Bergman, P., Kopko, E., and Rodríguez, J. (2026). A seven-college experiment using algorithms to track students: Impacts and implications for equity and fairness. Author manuscript, June 2026 version.

Bettinger, E. (2004). How financial aid affects persistence. In Hoxby, C. M., editor, *College Choices: The Economics of Where to Go, When to Go, and How to Pay for It*, pages 207–238. University of Chicago Press, Chicago.

Bhuller, M., Mogstad, M., and Salvanes, K. G. (2017). Life-cycle earnings, education premiums, and internal rates of return. *Journal of Labor Economics*, 35(4):993–1030.

Bound, J., Lovenheim, M. F., and Turner, S. (2012). Increasing time to baccalaureate degree in the United States. *Education Finance and Policy*, 7(4):375–424.

Bruce, D. J. and Carruthers, C. K. (2014). Jackpot? The impact of lottery scholarships on enrollment in Tennessee. *Journal of Urban Economics*, 81:30–44.

Bulman, G. B. and Hoxby, C. M. (2015). The returns to the federal tax credits for higher education. *Tax Policy and the Economy*, 29(1):13–88.

Carruthers, C. K. and Welch, J. G. (2019). Not whether, but where? Pell grants and college choices. *Journal of Public Economics*, 172:1–19.

Castleman, B. L. and Long, B. T. (2016). Looking beyond enrollment: The causal effect of need-based grants on college access, persistence, and graduation. *Journal of Labor Economics*, 34(4):1023–1073.

Chen, J. and Ritzwoller, D. M. (2023). Semiparametric estimation of long-term treatment effects. *Journal of Econometrics*, 237(2):105545.

Chetty, R. (2009). Sufficient statistics for welfare analysis: A bridge between structural and reduced-form methods. *Annual Review of Economics*, 1(1):451–488.

Cocci, M. D. and Plagborg-Møller, M. (2025). Standard errors for calibrated parameters. *Review of Economic Studies*, 92(5):2952–2978.

Cohodes, S. R. and Goodman, J. S. (2014). Merit aid, college quality, and college completion: Massachusetts' Adams Scholarship as an in-kind subsidy. *American Economic Journal: Applied Economics*, 6(4):251–285.

Cornwell, C., Mustard, D. B., and Sridhar, D. J. (2006). The enrollment effects of merit-based financial aid: Evidence from Georgia's HOPE program. *Journal of Labor Economics*, 24(4):761–786.

Deming, D. J. and Walters, C. R. (2017). The impact of price caps and spending cuts on U.S. postsecondary attainment. NBER Working Paper 23736, National Bureau of Economic Research.

Denning, J. T. (2017). College on the cheap: Consequences of community college tuition reductions. *American Economic Journal: Economic Policy*, 9(2):155–188.

Dynarski, S. M. (2003). Does aid matter? Measuring the effect of student aid on college attendance and completion. *American Economic Review*, 93(1):279–288.

Fan, Y., Manzanares, C. A., Park, H., and Qi, Y. (2026). A sensitivity analysis of the surrogate index approach for estimating long-term treatment effects. Working paper, arXiv:2603.00580v1.

García, J. L., Heckman, J. J., Leaf, D. E., and Prados, M. J. (2020). Quantifying the life-cycle benefits of an influential early-childhood program. *Journal of Political Economy*, 128(7):2502–2541.

Goldrick-Rab, S., Kelchen, R., Harris, D. N., and Benson, J. (2016). Reducing income inequality in educational attainment: Experimental evidence on the impact of financial aid on college completion. *American Journal of Sociology*, 121(6):1762–1817.

Haider, S. and Solon, G. (2006). Life-cycle variation in the association between current and lifetime earnings. *American Economic Review*, 96(4):1308–1320.

Heckman, J. J., Humphries, J. E., and Veramendi, G. (2016). Dynamic treatment effects. *Journal of Econometrics*, 191(2):276–292.

Hendren, N. and Sprung-Keyser, B. (2020). A unified welfare analysis of government policies. *The Quarterly Journal of Economics*, 135(3):1209–1318.

Hess, A. J. (2025). *Think of the Children: The Role of Early Childhood Interventions in Long-Term Human Capital Formation*. PhD thesis, University of California, Los Angeles.

Hoxby, C. M. and Bulman, G. B. (2016). The effects of the tax deduction for postsecondary tuition: Implications for structuring tax-based aid. *Economics of Education Review*, 51:23–60.

Imbens, G., Kallus, N., Mao, X., and Wang, Y. (2025). Long-term causal inference under persistent confounding via data combination. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(2):362–388.

Imbens, G. W. and Manski, C. F. (2004). Confidence intervals for partially identified parameters. *Econometrica*, 72(6):1845–1857.

Kleven, H. J. (2021). Sufficient statistics revisited. *Annual Review of Economics*, 13(1):515–538.

Li, A. (2026). Higher education or just higher? The unintended effects of recreational marijuana legalization. Author manuscript, version supplied September 2026.

Long, B. T. (2004). The impact of federal tax credits for higher education expenses. In Hoxby, C. M., editor, *College Choices: The Economics of Where to Go, When to Go, and How to Pay for It*, pages 101–168. University of Chicago Press, Chicago.

Manski, C. F. (2011). Choosing treatment policies under ambiguity. *Annual Review of Economics*, 3(1):25–49.

Marx, B. M. and Turner, L. J. (2018). Borrowing trouble? Human capital investment with opt-in costs and implications for the effectiveness of grant aid. *American Economic Journal: Applied Economics*, 10(2):163–201.

Obradović, F. (2026). Identification of long-term treatment effects via temporal links, observational, and experimental data. Working paper, revised March 2026, arXiv:2411.04380v2.

Schoonover, P., Attridge, J., Carruthers, C. K., and Welch, J. G. (2026). A promise worth keeping? Impacts of free community college on degrees and earnings. NBER Working Paper 35226, National Bureau of Economic Research.

Seftor, N. S. and Turner, S. E. (2002). Back to school: Federal student aid policy and adult college enrollment. *Journal of Human Resources*, 37(2):336–352.

Turner, N. (2011). The effect of tax-based federal student aid on college enrollment. *National Tax Journal*, 64(3):839–861.

Vohra, V. (2026). Inference for welfare metrics. *Journal of Public Economics*, 256:105603.

Zimmerman, S. D. (2014). The returns to college admission for academically marginal students. *Journal of Labor Economics*, 32(4):711–754.

Table 1. Education valuation under one and two observed ages

Generating process	$\Delta H_c(25)$ (years)	Single-effect NPV	Generating-path NPV	Candidate NPV range	
				Age 25	Ages 25 and 35
Permanent (K)	0.265	11.20	14.08	[−3.53, 44.80]	[10.61, 14.33]
Temporary (b)	0.014	0.60	0.23	[−15.43, 42.58]	[−2.92, 0.46]
Temporary (d)	−0.402	−16.96	−6.07	[−26.41, 39.18]	[−8.74, −3.97]

Notes: NPVs are thousands of 2000 dollars per person. Each generating scenario changes only the named parameter; the other two increments are zero. The increments are $(0.3160, 0, 0)$, $(0, 0.0030, 0)$, and $(0, 0, 6.4351)$, respectively. $\Delta H_c(25)$ is the implied schooling effect. The benchmark uses that effect; the generating-path value uses the full path. Permanent and temporary describe the generating path. All three parameters may vary among candidates matching the stated ages. Ranges summarize verified candidates, not confidence intervals or certified global bounds.

Table 2. How additional observation ages change the NPV range

Observed ages	Median width ($1,000)	Largest width ($1,000)	Reduction (percent)	Both signs found (of 5 scenarios)
Panel A. Plateau (ΔK)				
25	48.34	65.92	0.0	3/5
20, 25	20.78	23.32	57.0	1/5
24, 25	16.39	18.40	66.1	1/5
25, 30	7.63	8.65	84.2	0/5
25, 35	3.72	4.31	92.3	0/5
Panel B. Speed (Δb)				
25	58.01	72.12	0.0	4/5
20, 25	18.50	22.71	68.1	2/5
24, 25	14.57	21.10	74.9	2/5
25, 30	6.81	12.76	88.3	1/5
25, 35	3.38	7.60	94.2	1/5
Panel C. Timing (Δd)				
25	65.59	91.01	0.0	5/5
20, 25	49.39	49.51	33.5	4/5
24, 25	40.51	47.01	38.2	4/5
25, 30	15.76	38.76	76.0	4/5
25, 35	4.77	13.54	92.7	1/5

Notes: Each panel varies only the named generating parameter; the other two increments are zero. Candidate parameters may all vary. Median widths and percentage reductions use the three interior interval positions (25, 50, and 75 percent). Reductions are calculated within each scenario relative to age 25 alone before taking the median. Largest widths and sign counts use all five positions, including the boundaries. Both signs found requires verified NPV candidates above \$0.01 and below −\$0.01. A zero count does not establish that such paths are absent from the continuous feasible set. Dollar units are 2000 dollars per person; the ranges are conditional numerical candidates.

Table 3. Reference welfare values, continuation ranges, and L3 best fits

Policy	Type	MVPF			WTP			Net fiscal cost		
		L1	L2	L3	L1	L2	L3	L1	L2	L3
Panel A. Primary comparisons: WTP and net cost per reference program dollar										
Tuition deduction: single, lower threshold	Tax	$+\infty^{s}$	$[-\infty, -7.40]^{s,\dagger}$	–	5.38	[−345.81, 583.82]	–	−5.10	[−93.18, 46.75]	–
Tennessee Promise^{r}	Aid	$+\infty^{s}$	$[-1.49, +\infty]^{s}$	$+\infty^{s}$	4.92	[−3.29, 10.31]	5.84	−0.68	[−2.57, 2.21]	−1.00
Tuition deduction: single, upper threshold	Tax	$+\infty^{s}$	$[-\infty, -6.33]^{s,\dagger}$	–	1.00	[−236.30, 372.31]	–	−1.13	[−60.20, 37.33]	–
Texas CC tuition	Funding	349.51	$[9.70, +\infty]^{s}$	$+\infty^{s}$	9.31	[5.87, 24.13]	10.87	0.03	[−1.63, 0.61]	−0.08
Michigan CC tuition	Funding	29.46	$[-4.44, +\infty]^{s}$	$[9.03, +\infty]^{s}$	7.80	[−35.00, 96.71]	[5.99, 14.42]	0.26	[−13.56, 7.90]	[−0.59, 0.66]
DC tuition assistance	Aid	22.98	$[-1.60, +\infty]^{s}$	–	7.62	[−2.62, 40.96]	–	0.33	[−2.89, 1.64]	–
HOPE tax credit	Tax	12.58	$[-5.09, +\infty]^{s}$	$+\infty^{s}$	5.23	[−19.03, 59.43]	29.30	0.42	[−6.58, 3.74]	−2.60
Florida access grant	Aid	7.42	[0.57, 22.32]	18.09	7.40	[0.67, 31.22]	23.02	1.00	[1.16, 1.40]	1.27
Social Security student benefits	Aid	4.87	[3.79, 4.99]	[4.21, 4.52]	3.03	[2.54, 3.08]	[2.74, 2.88]	0.62	[0.62, 0.67]	[0.64, 0.65]
College appropriations: spending	Funding	4.00	$[-5.43, +\infty]^{s}$	16.52	3.03	[−18.53, 42.71]	6.53	0.76	[−4.03, 3.41]	0.40
Georgia HOPE	Aid	4.00	$[-3.19, +\infty]^{s}$	–	3.12	[−5.43, 23.20]	–	0.78	[−1.15, 1.70]	–
Ohio Pell grants	Aid	2.49	[−11.75, 7.53]	–	2.88	[−5.34, 27.29]	–	1.16	[0.45, 3.62]	–
Adult Pell introduction	Aid	2.18	[1.95, 4.63]	–	2.71	[1.61, 13.45]	–	1.25	[0.83, 2.91]	–
Kalamazoo Promise	Aid	1.93	[0.65, 4.53]	3.77	1.76	[0.63, 4.12]	3.48	0.91	[0.91, 0.97]	0.92
Tennessee HOPE	Aid	1.86	[−75.64, 21.66]	–	1.89	[−29.05, 44.95]	–	1.02	[0.38, 2.08]	–
STBF scholarship	Aid	1.75	[1.25, 4.01]	–	1.52	[1.19, 2.50]	–	0.87	[0.62, 0.95]	–
Wisconsin Scholars Grant	Aid	1.43	[−1.94, 7.58]	1.96	1.42	[−2.13, 6.60]	1.94	0.99	[0.87, 1.10]	0.99
CUNY Pell grants	Aid	1.39	$[-\infty, 14.28]^{\dagger}$	[−3.17, 2.34]	1.42	[−27.52, 39.88]	[−2.69, 2.54]	1.02	[−0.14, 2.79]	[0.85, 1.09]
College appropriations: tuition	Funding	1.02	$[-8.86, +\infty]^{s}$	−1.97	1.02	[−22.45, 38.90]	−2.41	1.00	[−1.42, 2.53]	1.22
Tennessee Pell eligibility	Aid	0.85	$[-\infty, 2.70]^{\dagger}$	–	0.79	[−18.13, 27.17]	–	0.93	[−5.34, 10.06]	–
Tuition deduction: joint, upper threshold	Tax	0.77	$[-3.92, +\infty]^{s}$	–	1.00	[−120.89, 191.27]	–	1.29	[−44.89, 30.83]	–
Adams Scholarship	Aid	0.72	$[0.48, +\infty]^{s,\dagger}$	0.63	1.21	[−10.19, 6.55]	1.47	1.68	[−18.52, 13.39]	2.34
Tuition deduction: joint, lower threshold	Tax	−0.02	$[-4.35, +\infty]^{s}$	–	−0.03	[−143.15, 215.89]	–	1.38	[−46.06, 32.89]	–

Continued on next page

Table 3 (continued)

Policy	Type	MVPF			WTP			Net fiscal cost		
		L1	L2	L3	L1	L2	L3	L1	L2	L3
HOPE and Lifetime Learning credits	Tax	−8.81	[−∞, −9.59]^{s,†}	–	−42.82	[−199.55, 199.21]	–	4.86	[−19.57, 20.78]	–
Panel B. Finite tax reconstructions: per reference program dollar										
HOPE: joint, lower kink^{f}	Tax	+∞^{s}	[−∞, +∞]^{s,†}	–	18.41	[−21,373.75, 40,380.60]	–	−3.15	[−4,694.42, 2,529.55]	–
AOTC: dependent single, lower kink^{f}	Tax	+∞^{s}	[−∞, +∞]^{s,†}	–	23.39	[−40,937.11, 64,893.24]	–	−1.61	[−6,442.87, 3,927.52]	–
AOTC: joint, lower kink^{f}	Tax	+∞^{s}	[−∞, +∞]^{s}	–	9.96	[−53,630.62, 85,312.41]	–	−1.37	[−10,116.74, 6,349.58]	–
HOPE: independent single, lower kink^{f}	Tax	18.86	[−∞, +∞]^{s,†}	–	5.91	[−53,768.73, 101,434.66]	–	0.31	[−17,374.29, 9,427.43]	–
HOPE: dependent single, upper kink^{f}	Tax	11.83	[−∞, +∞]^{s,†}	–	4.59	[−17,357.02, 32,615.58]	–	0.39	[−3,366.47, 1,855.80]	–
AOTC: simulated instrument^{f}	Tax	10.05	[−∞, +∞]^{s,†}	–	5.36	[−82,841.87, 145,584.75]	–	0.53	[−16,160.55, 9,118.53]	–
AOTC: independent single, lower kink^{f}	Tax	6.75	[−∞, +∞]^{s}	–	2.45	[−179,550.49, 284,893.86]	–	0.36	[−56,780.46, 36,085.36]	–
HOPE: joint, upper kink^{f}	Tax	2.37	[−∞, +∞]^{s,†}	–	8.21	[−23,368.86, 44,169.47]	–	3.47	[−4,875.76, 2,696.42]	–
AOTC: dependent single, upper kink^{f}	Tax	−0.02	[−∞, +∞]^{s,†}	–	−0.02	[−42,932.08, 67,915.87]	–	1.12	[−7,231.21, 4,415.06]	–
HOPE: dependent single, lower kink^{f}	Tax	−1.64	[−∞, +∞]^{s,†}	–	−1.91	[−16,274.53, 30,497.00]	–	1.16	[−2,764.33, 1,548.51]	–
AOTC: joint, upper kink^{f}	Tax	−1.77	[−∞, +∞]^{s,†}	–	−7.63	[−55,169.57, 87,656.11]	–	4.31	[−10,682.18, 6,711.54]	–
Panel C. Reconstructed or extended accounts and education channel: native thousands										
Algorithmic college placement^{x}	Placement	+∞^{s}	[0.08, +∞]^{s}	–	0.22	0.22	–	−0.09	[−5.55, 2.82]	–
Early Head Start^{r}	Childhood	0.35	[−1.37, +∞]^{s}	–	3.91	[−37.51, 114.60]	–	11.22	[−1.10, 27.35]	–
Cannabis legalization: education channel	Other	–	–	−55.85	–	–	−2.50	–	–	0.04

Policy types: Aid, direct student financial aid; Tax, education tax benefits; Funding, tuition and public-college funding; Placement, college assessment and course placement; Childhood, early childhood services; Other, the education channel of a noneducation policy. Types describe interventions, whereas panels distinguish accounting specifications. *Notes:* L1 reproduces source values. A single number denotes coincident endpoints; the ten minimum-distance L3 cases each retain one best-fitting path. Brackets denote distinct fixed-input endpoints, including limiting values, not confidence intervals or certified global bounds. WTP and net-cost endpoints need not use the same path; each positive-cost MVPF uses paired components from one complete path. Infinite range endpoints retain verified one-sided limits from finite schooling paths. Superscript s on an MVPF entry records a separate self-financing state: positive WTP and nonpositive net cost. A dagger marks additional undefined-ratio paths. Superscripts f, r, and x identify the tax conversion, public-data reconstruction, and earnings extension. Panel A divides both monetary components by one fixed reference program-cost component per case. Panel B uses the same bounded logistic family. Source attendance and monetary inputs are transported to a fixed finite schedule change calibrated to one point-estimate credit dollar, retaining the two-year schooling conversion. The conversion and its age placement are assumptions. The first-stage point normalization stays fixed across repetitions. The AOTC simulated-instrument reconstruction also corrects the direct-credit unit while retaining the reported L1. Panel C uses thousands of native dollars. Early Head Start uses unscaled public ACS income and a funded-slot cost proxy, in 2000 dollars. Its source price year is unresolved. Tennessee uses public ACS income and an account-implied participant bridge, under a disclosed 2021-dollar convention. Algorithmic placement uses the verified June 2026 source components, but its reconstruction extends fiscal earnings only, keeping source WTP fixed. Their monetary differences cannot be attributed only to continuation. Cannabis has no whole-policy L1 and uses its three-horizon L3 as the primary result; its last-horizon-only L2 remains in the appendix data. A dash denotes an unavailable reference value or an inapplicable layer; missing L3 means no eligible additional source-window restriction. NF denotes an unsuccessful search. Duration allocations alone do not qualify as L3. Appendix E.4 documents units, normalization, and account-version qualifications.

Table 4. Reported intervals and conditional input-repetition endpoint summaries

Policy	L1: reported 95%	L2: endpoints	n_2	L3: endpoints	n_3	Scope
HOPE: joint, lower kink	[−3.51, +∞]	[−∞, +∞]	1000	–	–	J
Tuition deduction: single, lower threshold	[−∞, +∞]	[−∞, +∞]	1000	–	–	J
AOTC: dependent single, lower kink	[−0.84, +∞]	[−∞, +∞]	1000	–	–	J
AOTC: joint, lower kink	[−5.96, +∞]	[−∞, +∞]	1000	–	–	J
Tennessee Promise	[+∞, +∞]	[−1.62, +∞]	1000	[+∞, +∞]	1000	P
Algorithmic college placement	–	[0.07, +∞]	1000	–	–	P
Tuition deduction: single, upper threshold	[−∞, +∞]	[−∞, +∞]	1000	–	–	J
Texas CC tuition	[0.93, +∞]	[−1.81, +∞]	1000	[1.14, +∞]	1000	J
Michigan CC tuition	[−2.33, +∞]	[−24.02, +∞]	1000	[−3.74, +∞]	1000	J
DC tuition assistance	[1.12, +∞]	[−3.93, +∞]	1000	–	–	X
HOPE: independent single, lower kink	[−3.03, +∞]	[−∞, +∞]	1000	–	–	J
HOPE tax credit	[−29.15, +∞]	[−97.46, +∞]	1000	[−0.04, +∞]	1000	J
HOPE: dependent single, upper kink	[−5.36, +∞]	[−∞, +∞]	1000	–	–	J
AOTC: simulated instrument	[−18.36, +∞]	[−∞, +∞]	1000	–	–	P
Florida access grant	[0.69, +∞]	[−14.48, +∞]	1000	[0.53, +∞]	1000	J
AOTC: independent single, lower kink	[−1.61, +∞]	[−∞, +∞]	1000	–	–	J
Social Security student benefits	[1.16, 32.23]	[0.72, 27.30]	1000	[1.01, 21.68]	980*	J
College appropriations: spending	[1.25, 20.44]	[−1,925.69, +∞]	1000	[1.23, +∞]	1000	J
Georgia HOPE	[0.46, 22.38]	[−4.11, +∞]	1000	–	–	J
Ohio Pell grants	[0.82, 5.54]	[−∞, +∞]	1000	–	–	J
HOPE: joint, upper kink	[−2.22, +∞]	[−∞, +∞]	1000	–	–	J
Adult Pell introduction	[0.71, 6.11]	[−5.15, 102.12]	1000	–	–	J
Kalamazoo Promise	[0.97, 5.61]	[−0.65, 19.65]	1000	[0.78, 15.51]	1000	J
Tennessee HOPE	[0.92, 5.08]	[−∞, +∞]	1000	–	–	J
STBF scholarship	–	[0.94, 4.14]	1000	–	–	P
Wisconsin Scholars Grant	[0.96, 2.18]	[−2.84, +∞]	1000	[−0.04, 6.29]	1000	J
CUNY Pell grants	[−2.95, 12.88]	[−∞, +∞]	1000	[−∞, +∞]	922*	J
College appropriations: tuition	[−1.11, 5.13]	[−∞, +∞]	1000	[−4.48, 11.82]	997*	J
Tennessee Pell eligibility	[−1.59, 3.57]	[−∞, +∞]	1000	–	–	J
Tuition deduction: joint, upper threshold	[−1.83, 57.84]	[−4.87, +∞]	1000	–	–	J
Adams Scholarship	[−0.92, 3.05]	[−∞, +∞]	1000	[−8.38, 2.73]	931*	J
Early Head Start	–	[−2.20, +∞]	1000	–	–	P
AOTC: dependent single, upper kink	[−2.25, +∞]	[−∞, +∞]	1000	–	–	J
Tuition deduction: joint, lower threshold	[−2.25, 5.52]	[−4.67, +∞]	1000	–	–	J

Continued on next page

Table 4 (continued)

Policy	L1: reported 95%	L2: endpoints	n_2	L3: endpoints	n_3	Scope
HOPE: dependent single, lower kink	$[-6.45, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
AOTC: joint, upper kink	$[-17.06, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
HOPE and Lifetime Learning credits	$[-\infty, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
Cannabis legalization: education channel	–	–	–	$[-57.03, -54.19]$	1000	R

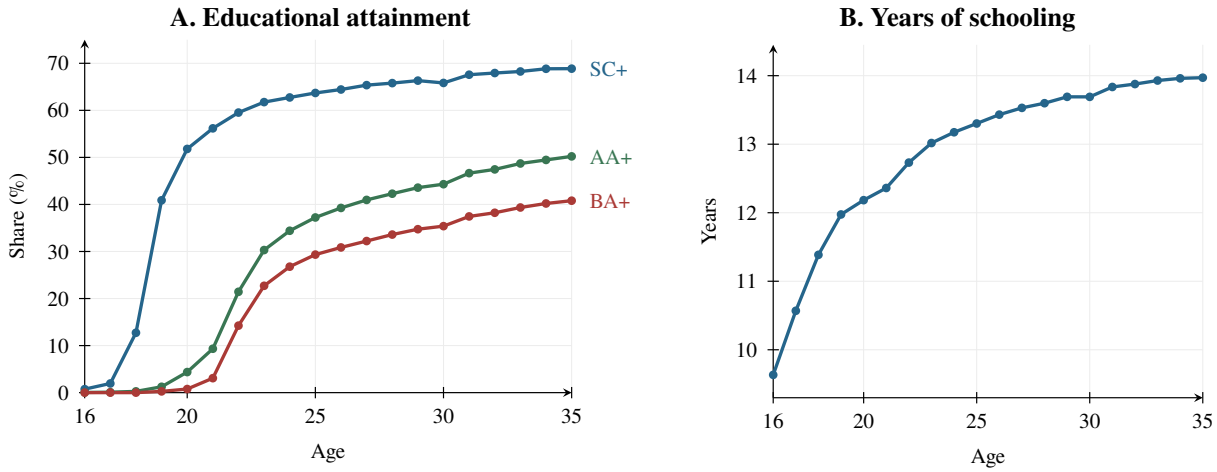
Notes: L1 reports available source 95 percent intervals. L2/L3 give the 2.5th percentile of candidate minima and the 97.5th percentile of maxima under 1,000 paired input repetitions. Each of the thirteen L3 schemes includes its complete input vector. Counts are repetitions with verified paths and a defined ratio; asterisks denote fewer than 1,000, and dashes denote no admitted layer or source interval. These summaries do not establish confidence coverage. Infinite endpoints include self-financing or verified positive-cost limits. J: archived joint source/accounting draws, extended to added inputs; P: constructed education perturbations; X: economic inputs with education fixed; R: synthetic correlated shocks. All schemes condition on the comparison profile, calibration, and stated dependence. Source intervals can vary different inputs. Appendix F gives details. Cannabis shows only its primary three-window L3. Policy types and superscripts follow Figure 8.

Table 5. Reference and single-parameter reconstructed MVPFs

	Reference	Permanent (K)		Temporary (b)		Temporary (d)	
Policy	MVPF	ΔK	MVPF	Δb (%)	MVPF	Δd	MVPF
Florida grant	7.42	0.323	10.54	—	—	—	—
DC tuition aid	22.98	0.637	SF	9.56	10.38	—	—
Adams	0.72	−0.0230	0.72	−1.03	0.81	0.334	0.85
STBF	1.75	Joint	1.86	—	—	—	—
Wisconsin grant	1.43	0.120	1.78	4.76	1.07	—	—
Kalamazoo	1.93	0.375	2.25	—	—	—	—
Michigan CC	29.46	0.0330	47.45	3.12	3.18	—	—
CUNY Pell	1.39	0.0164	1.70	0.526	1.11	−0.147	1.07
Ohio Pell	2.49	0.224	4.70	9.17	1.89	—	—
Tennessee HOPE	1.86	0.0435	2.79	1.55	1.12	−0.428	1.03
Tennessee Pell	0.85	−0.0142	0.66	−0.548	0.97	0.150	0.99
Georgia HOPE	4.00	0.236	12.63	8.73	1.96	—	—
Social Security	4.87	22.4	4.51	—	—	—	—
Appropriations: spending	4.00	0.0841	8.45	3.22	1.73	—	—
Early Head Start	0.35	—	—	—	—	—	—
Texas CC	349.51	0.248	SF	—	—	—	—
Adult Pell	2.18	0.0601	3.98	—	—	—	—
Appropriations: tuition	1.02	8.41×10^{-4}	1.04	0.0318	1.01	−0.0106	1.00
HOPE credit	12.58	0.144	SF	5.46	3.55	—	—
HOPE ind. single: lower	18.86	1.96×10^{-4}	SF	0.00946	83.88	−0.00414	46.41
HOPE dep. single: lower	−1.64	-2.72×10^{-4}	−6.21	−0.0131	−1.89	0.00575	−2.22
HOPE dep. single: upper	11.83	3.15×10^{-4}	SF	0.0152	5.50	−0.00665	4.61
HOPE joint: lower	SF	0.00124	SF	0.0599	SF	−0.0263	SF
HOPE joint: upper	2.37	4.74×10^{-4}	22.04	0.0229	1.83	−0.0100	1.74
Tuition ded. single: lower	SF	0.0379	SF	1.05	SF	−0.257	SF
Tuition ded. joint: lower	−0.02	−0.0227	−1.49	−0.631	0.42	0.149	0.63
HOPE / Lifetime Learning	−8.81	−0.0794	−10.75	−4.90	−13.68	—	—
AOTC single: upper	−0.02	-4.22×10^{-5}	−1.20	−0.00136	0.68	3.37×10^{-4}	0.81
AOTC single: lower	SF	9.72×10^{-4}	SF	0.0314	11.45	−0.00777	6.15
AOTC joint: upper	−1.77	-2.79×10^{-4}	−3.54	−0.00901	−0.47	0.00223	−0.17
AOTC joint: lower	SF	2.98×10^{-4}	SF	0.00962	SF	−0.00238	SF
AOTC ind. single: lower	6.75	2.01×10^{-5}	SF	6.51×10^{-4}	3.15	-1.61×10^{-4}	2.43
AOTC instrument	10.05	9.39×10^{-5}	SF	0.00390	3.79	−0.00136	3.02
Tennessee Promise	SF	Joint	SF	—	—	—	—
College placement	SF	0.0415	SF	3.41	SF	—	—
Tuition ded. single: upper	SF	0.000	SF	0.00	SF	0.000	SF
Tuition ded. joint: upper	0.77	0.000	0.77	0.00	0.77	0.000	0.77

Notes: Each pair varies the named parameter family; the other two remain at H^0 values. K changes eventual attainment; b and d effects eventually vanish. ΔK is in schooling years, except percentage points for Social Security and Early Head Start; Δd is in years and Δb is the proportional speed change in percent. STBF jointly changes time enrolled and three credential plateaus; Tennessee jointly changes three credential plateaus. Their outcome-specific changes are saved in the matched-solution data. No single-family solution matches every additional point input exactly; — denotes no admissible L2 match. SF denotes self-financing ($W > 0$, $N \leq 0$), represented by $+\infty$ in Figure 10. The table includes all 37 reference calculations; cannabis has no whole-policy reference. Appendix B.5 defines the matching procedure. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds.

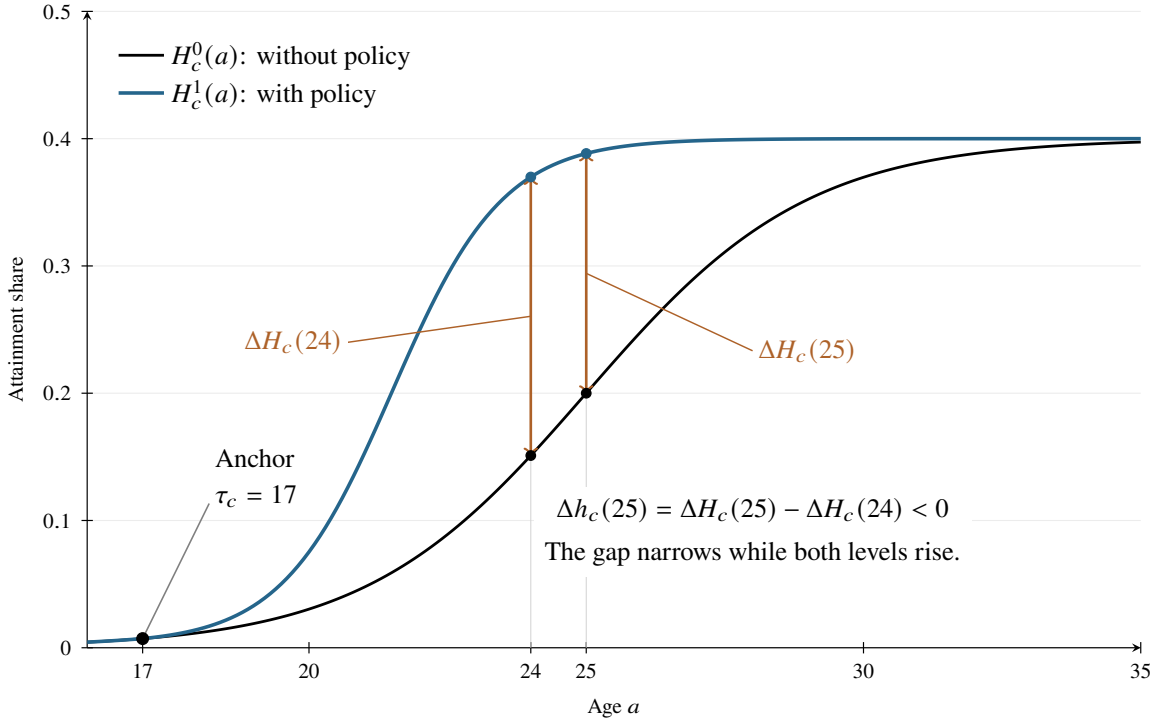
Figure 1. Education profiles for a fixed set of birth cohorts



Notes: ACS 2000–2022, birth cohorts 1985–1987, ages 16–35; all sexes, birthplaces, and racial and ethnic groups are included. Person weights apply within cohort–age cells; cohorts receive fixed one-third weights. Repeated cross sections do not fix composition within birth cohorts. SC+ denotes some college or higher, an attendance proxy; AA+ and BA+ denote cumulative degree thresholds. Schooling years use the EDUCD crosswalk. Markers show annual means; lines connect adjacent ages. Appendix A.5 documents these descriptive profiles.

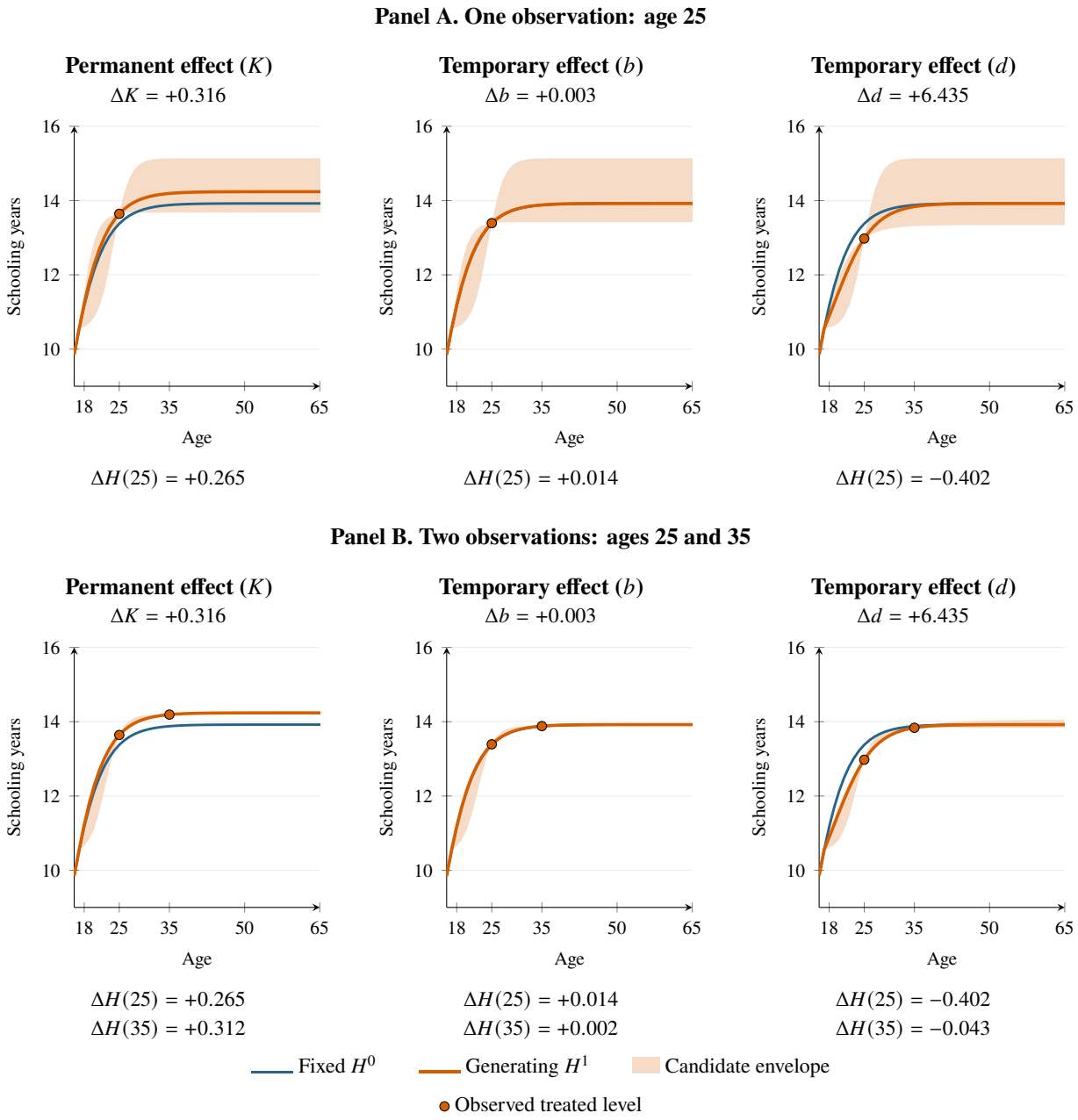
Source: Author's calculations from ACS microdata.

Figure 2. Educational attainment, treatment effects, and the pre-policy anchor



Notes: The hypothetical reference is $H_c^0(a) = 0.4\Lambda[0.5(a-25)]$. The treated path uses Equation (2), $(K, b, d) = (0.4, 1, 21.5)$, and the same age-17 level. Arrows mark attainment gaps at 24 and 25. Both levels rise while the gap narrows, giving $\Delta h_c(25) < 0$; both plateaus equal 0.4. These are illustrative attainment shares, not estimated effects.

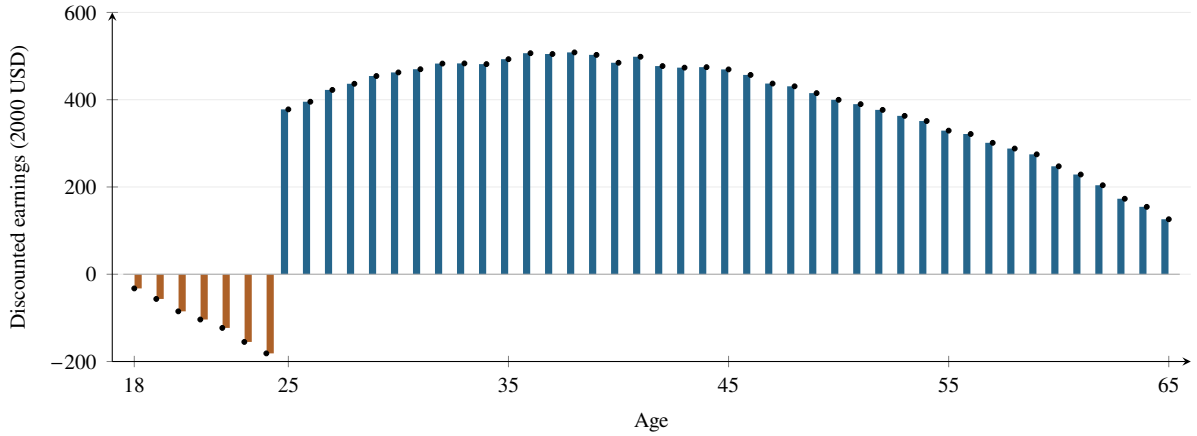
Figure 3. Compatible schooling paths with one and two observed ages



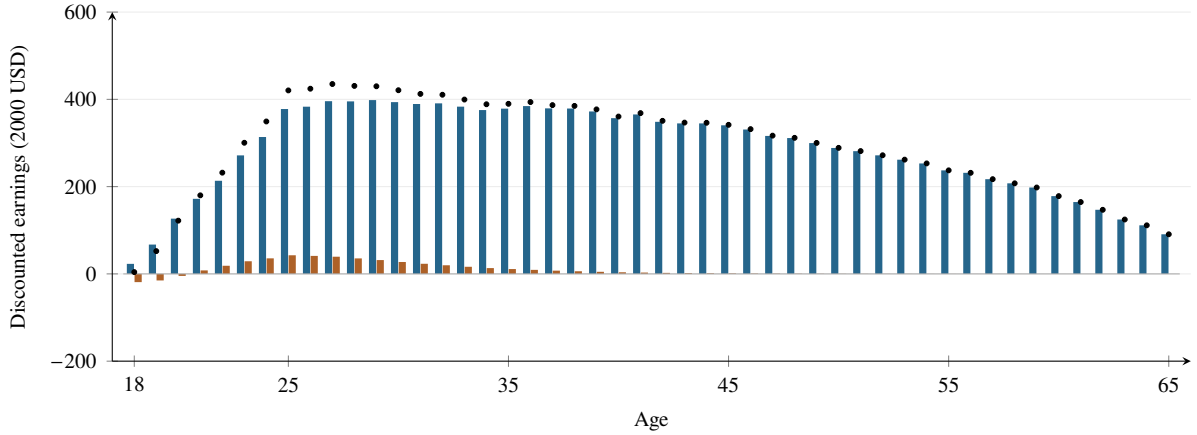
Notes: Panel A matches the age-25 effect; Panel B additionally matches the age-35 effect for the same generating process in each column. Columns vary one generating parameter at a time to the midpoint of its conditional feasible interval; the other two increments are zero. Permanent and temporary refer to the generating path, not all paths in its envelope. Blue lines show the fixed comparison path, and orange-red lines the generating path. Matching shading is the pointwise envelope of verified candidate paths, whose three parameters may all vary. Dots locate the treated levels implied by the observed effects and the fixed baseline. Every treated continuation is anchored to the comparison level at age 17. All six plots use identical age and schooling scales. Valuation results are reported in Table 1. Envelope edges need not form a single feasible path. No probability distribution is imposed on candidate paths, so the shading carries no probability interpretation. The envelopes are not confidence bands or certified global envelopes. Baseline values above age 35 are extrapolated.

Figure 4. Single-effect and full-path valuation with a permanent schooling gain

A. Single completion batch at age 25 Net earnings NPV: $15,390



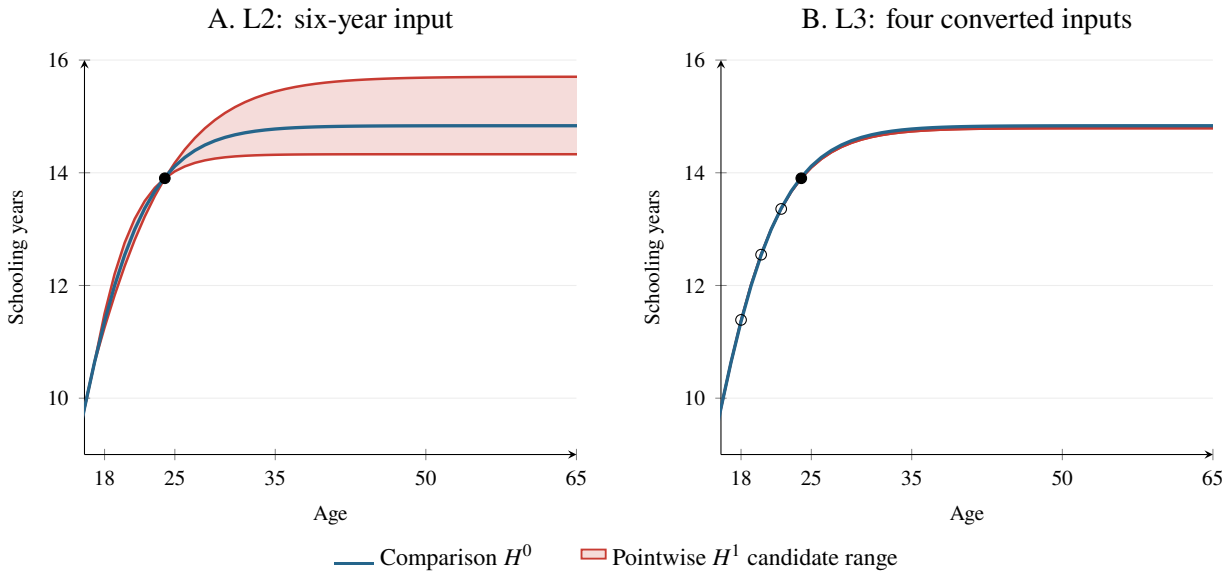
B. Completion batches along the full path Net earnings NPV: $13,986



Post-completion earnings Pre-completion earnings • Net earnings

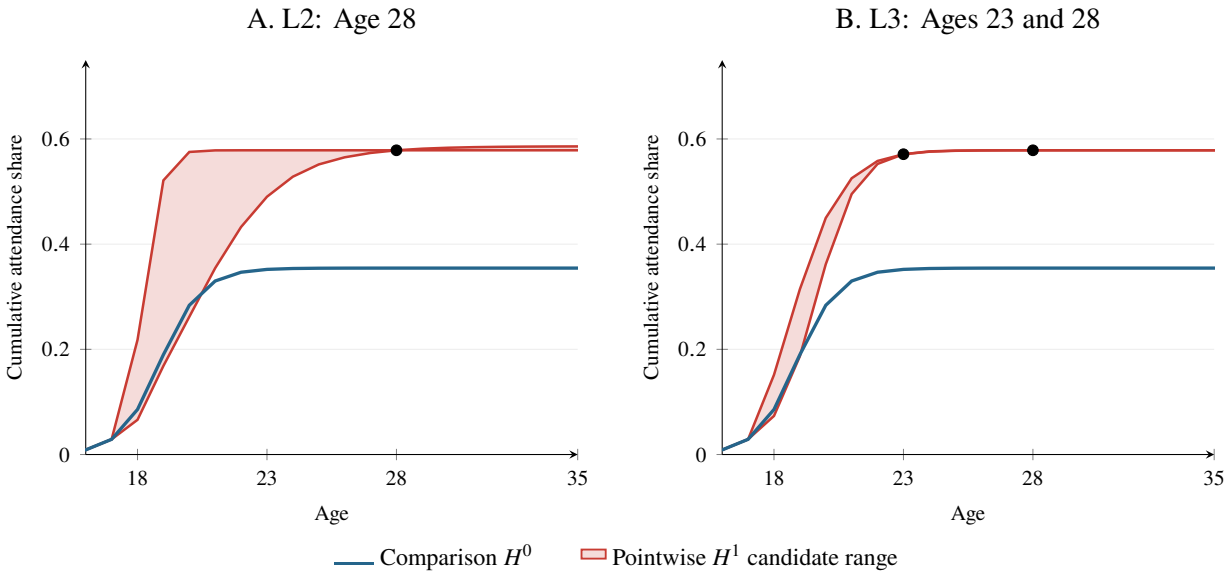
Notes: Both panels use $(\Delta K_c, \Delta b_c, \Delta d_c) = (0.2625, 0.0300, -0.4500)$, with schooling gaps of 0.364 years at 25 and 0.263 at 65. Figure B2 shows the level paths. Blue and orange bars separate the post-completion and pre-completion terms of the same earnings-response template; black dots give their sum. Panel A assigns the age-25 effect to one completion batch at 25; Panel B uses the full attainment path. An orange bar can be positive when the comparison population catches up. These are model components, not separately estimated wage returns and opportunity costs; no extra cost is deducted. Both panels cover ages 18–65 in 2000 dollars, discounted at 3 percent to age 18. Education paths extend to 72 to preserve the seven-year pre-completion window. This illustration is outside the two experiment grids.

Figure 5. Adams: adding three earlier converted education inputs



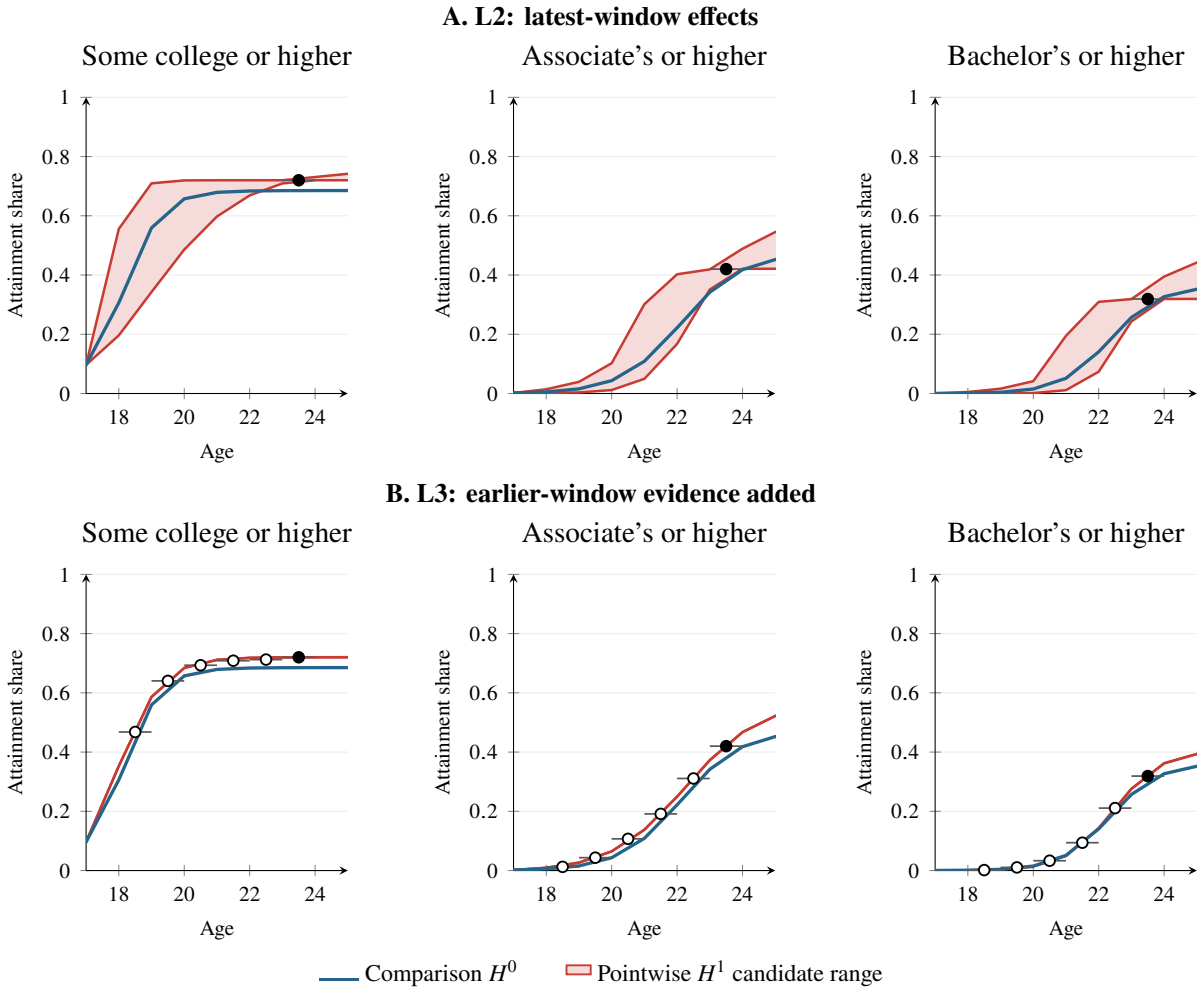
Notes: L2 matches −0.018 schooling-equivalent years at HS+6 (model age 24). Panel B fits the additional converted targets 0.026, 0.007, and −0.011 at ages 18, 20, and 22 while preserving the age-24 equality. The anchor is 17. Open markers indicate fitted targets; the filled marker indicates the exact restriction. Blue is H^0; red boundaries and shading give the pointwise H^1 candidate range. Its edges may combine different paths and are not confidence limits. Scales are common. Appendix D.6 documents the calibration and allocation.

Figure 6. Social Security: adding the earlier college-attendance observation



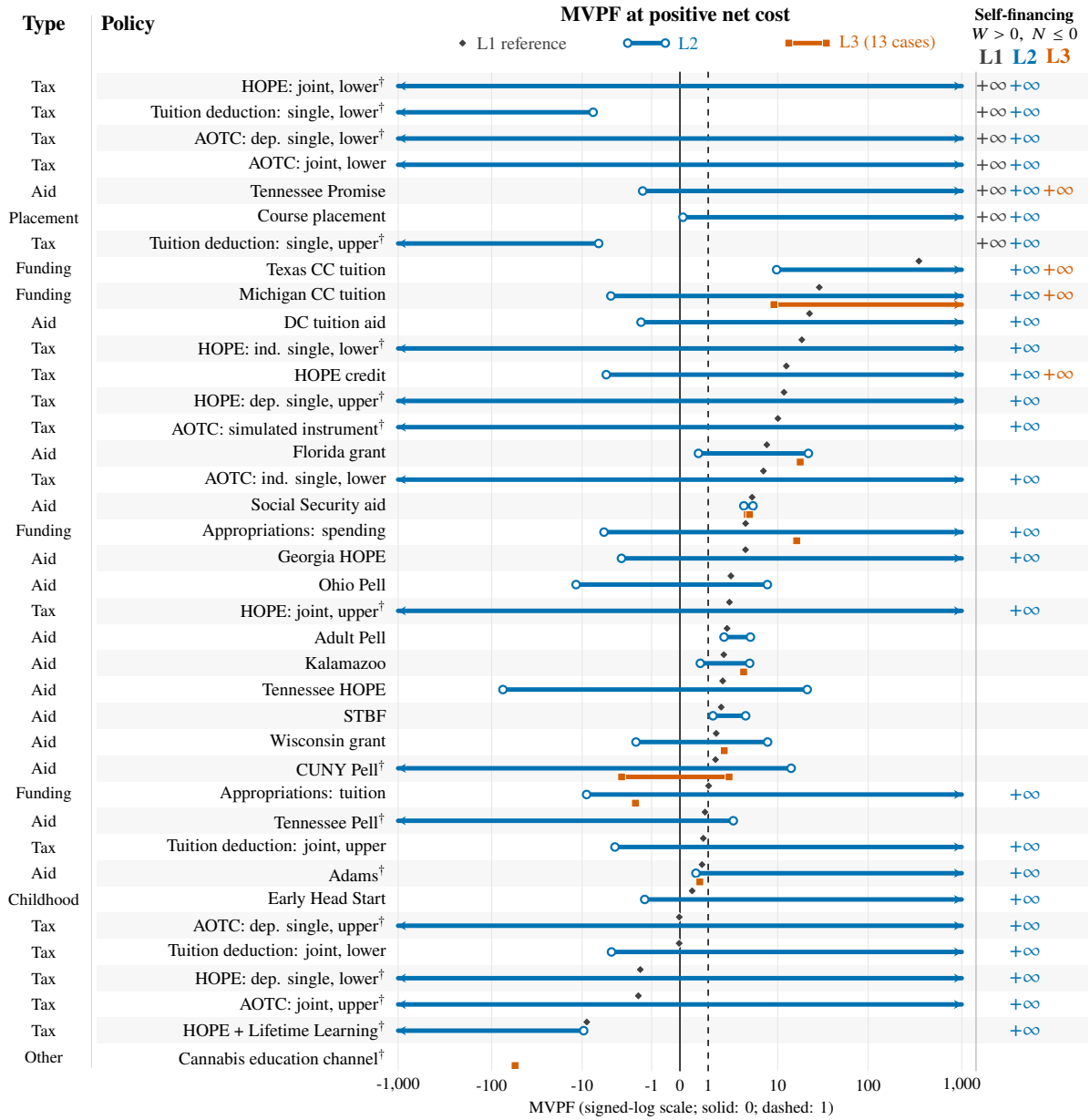
Notes: L2 matches the 0.224 effect on ever attending college full time by age 28. L3 also matches the 0.219 effect by age 23, keeping the age-28 equality. Filled markers show H^0 plus the exact source effects. Both panels use the same age-17 anchor, source age-23 control level, transported ACS entry shape, and probability and shape restrictions. The figure shows ages 16–35 for readability; valuation uses ages 18–65. The boundaries may combine different paths and are not confidence limits. Appendix D.7 documents the outcome proxy and calibration.

Figure 7. Tennessee Promise: L2 attainment ranges and the L3 fitted path



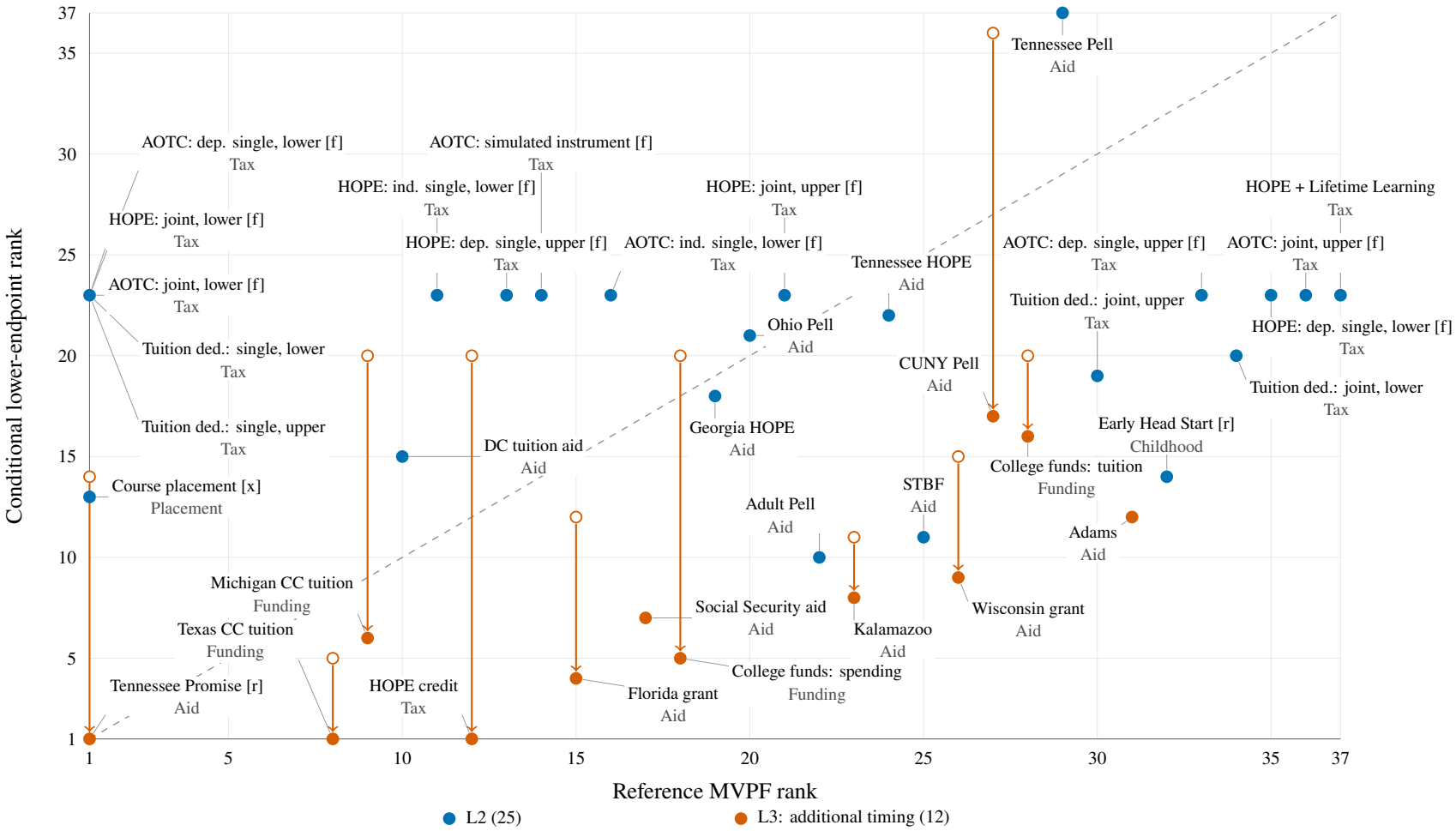
Notes: Blue is H^0; red boundaries and shading give the pointwise H^1 range among joint credential candidates. Filled markers show the exact 23–24 inputs; open markers show the fifteen earlier-window targets fitted in L3 from Schoonover et al. (2026). Marker heights equal the window mean of H^0 plus the reported effect. Horizontal bars span the source age windows; their midpoints locate windows, not single-age restrictions. Both panels use the same baseline, age-17 anchor, support, completion caps, and scales. Envelope edges may combine different paths across ages and credentials; they are not confidence bands. Appendix D.8 documents the restrictions; the appendix atlas shows ages through 65.

Figure 8. Education continuation ranges, ordered by the reference MVPF



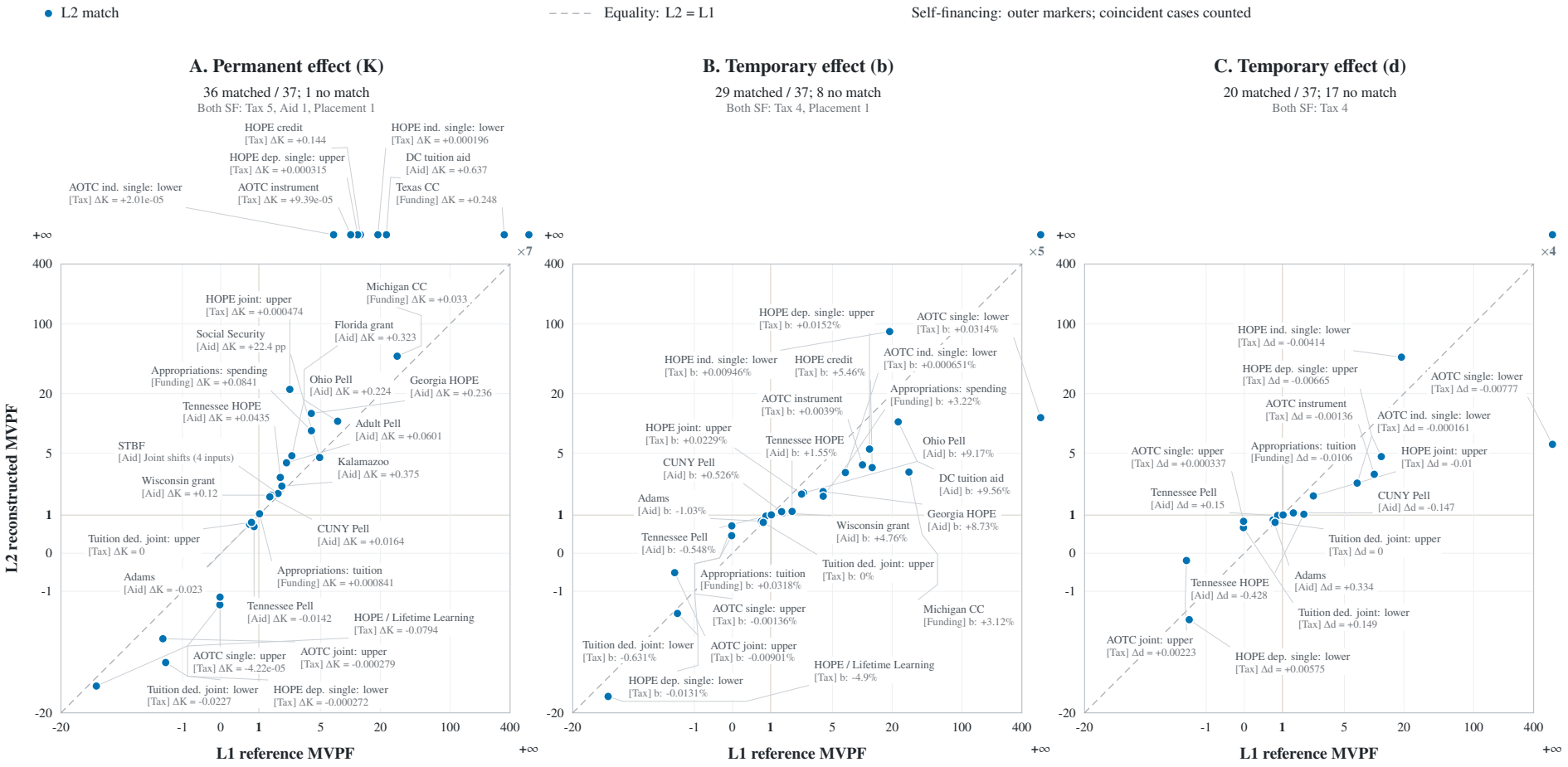
Notes: Rows follow descending L1. Types: Aid, direct student aid; Tax, education tax benefits; Funding, tuition and public-college funding; Placement, assessment and course placement; Childhood, early-childhood services; Other, another policy's education channel. Gray diamonds denote L1; blue circles and red squares bound positive-cost L2 and L3 spans. Separate +∞ marks denote self-financing ($W > 0$, $N \leq 0$); arrows include audited positive-cost ratio limits. Gaps do not establish infeasible intermediate ratios. Daggers mark undefined-ratio paths. Missing L3 means no added restriction is admitted under the retained mapping, not no source dynamics. Superscripts f, r, and x identify tax-unit conversion, public-data reconstruction, and a fiscal-only earnings extension. Cannabis has no whole-policy L1 and shows primary L3. Ten L3 cases have coincident endpoints at one best-fitting path; the other three retain exact-match ranges. These summaries are not confidence intervals or certified bounds. Table 3 gives values. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds.

Figure 9. L1 and candidate-based lower-endpoint ranks for 37 policy calculations



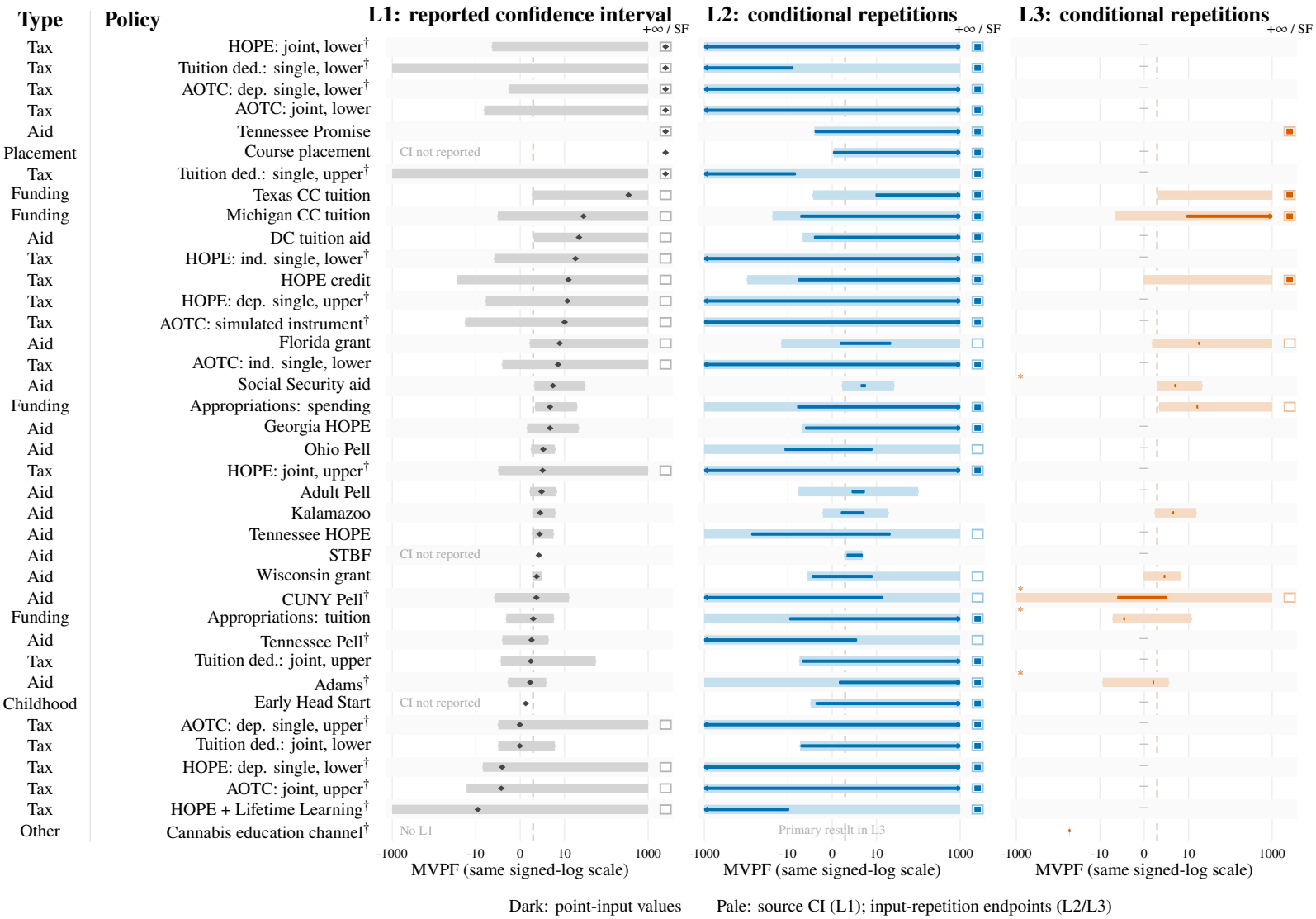
Notes: Higher lower endpoints rank first, followed by higher upper endpoints; admitted L3 replaces L2. Finite endpoints are rounded to two decimals, and equal pairs share a competition rank. Labels give policy names and Figure 8 types; [f], [r], and [x] mark tax conversion, public-data reconstruction, and a fiscal-only earnings extension. Coincident endpoint pairs share a marker. Arrows replace only the indicated policy's L2 by L3. Thirteen selected specifications contain undefined-ratio paths; those paths, not the calculations, are omitted. Cannabis lacks L1. Table E3 gives all ranks. This is a candidate-based display rule, not a social-welfare or certified worst-case ordering.

Figure 10. Reference and reconstructed MVPFs under permanent and temporary effects



Notes: Blue points match L2. None of these single-family solutions also matches every additional point input exactly; the main L3 fits remain separate. Bracketed labels give the policy types defined in Figure 8. Finite axes have identical asinh scales and square dimensions. Outer markers indicate self-financing: $W > 0$, $N \leq 0$. Top markers preserve finite L1, right markers preserve finite L2, and corner counts identify coincident double-self-financing cases; their policy types appear below the panel counts. Labels give ΔK in schooling years for YS or percentage points for credentials (STBF jointly shifts time and credential inputs), proportional b changes in percent, and Δd in years. The other two parameter families stay fixed. Permanent effects change eventual attainment; temporary effects eventually vanish but can retain lifetime value. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds. Appendix B.5 gives the method and case-level details.

Figure 11. Point-input ranges, source intervals, and conditional input repetitions



Notes: Policy types follow Figure 8. Dark marks show fixed-input results; pale bands show reported L1 intervals or conditional L2/L3 endpoint summaries. Filled +∞ marks denote self-financing; open squares denote infinite summary endpoints. Arrows extend beyond the axis; the guide marks one. Daggers mark undefined ratios; asterisks mark fewer than 1,000 usable repetitions. All thirteen L3 specifications include their full input vector. Method codes: J, registered source/accounting draws; P, constructed education perturbations; X, economic inputs with education fixed; R, synthetic correlated shocks. Tennessee's normal perturbations are not source permutation draws. Superscripts follow Figure 8. Table 4 gives endpoints and counts. Reconstructed summaries have no demonstrated coverage for all continuations.

A Literature Review and Descriptive Education Profiles

This appendix documents the two sources of descriptive evidence in Section 2: a review of when higher-education effects are measured and used in longer-run valuation, and ACS profiles of educational attainment across ages and birth cohorts. The literature review counts distinct papers; the policy applications elsewhere in the manuscript count policy–estimand calculations. For each reviewed paper, the accompanying records identify the source version, education outcomes, observation windows, relevant passages and exhibits, and inclusion decision. The review takes the original estimates as inputs and does not reassess each study's causal identification.

A.1 Search sample and inclusion

The journal search covers 2000–2025 in the *American Economic Review*, *Quarterly Journal of Economics*, *Journal of Political Economy*, *Review of Economic Studies*, *Review of Economics and Statistics*, *AEJ: Applied Economics*, *AEJ: Economic Policy*, *Journal of Human Resources*, *Journal of Labor Economics*, *Journal of Public Economics*, *Economics of Education Review*, and *Education Finance and Policy*. The 674 journal-search records are the supplied, preliminarily screened higher-education records in the full-text manifest, not all articles published in those journals. Full-text inclusion and duplicate checks retain 447 for analysis. The register preserves each supplied record and its inclusion decision; the exercise is a documented review of this selected search sample, not a census of education research. Eleven additional studies underlying welfare calculations and one unpublished author manuscript on marijuana legalization bring the retained sample to 459 papers. After deduplication of the additions against the journal search, the combined register contains 686 records. The remaining 227 records include exclusions and duplicates.

Inclusion requires an original estimate relating a policy, intervention, or external exposure to a higher-education outcome observed within a finite window. The education estimate can be secondary to the paper's main analysis. Studies reporting only test scores, aspirations, earnings, or high-school outcomes are excluded, as are purely descriptive comparisons, reviews, and editorials. The review imposes no fixed maximum follow-up length: an education outcome observed many years after exposure still provides finite-window evidence. Working and published versions of the same study count once, while distinct studies of one program can count separately. The resulting sample is selected. In particular, the additional welfare-source papers were included for their relevance to valuation, so their frequency cannot be interpreted as the prevalence of valuation in the wider literature.

A.2 Education Outcomes, Estimation Times, and Endpoints

The review distinguishes college entry (CE), associate-degree outcomes (AA), and bachelor's-degree or higher outcomes (BA+). CE requires first entry or ever entering college; current enrollment or persistence alone does not qualify. AA preserves the source study's distinction between an associate's degree and at least an associate's degree. The overall category also includes credits, completed schooling, graduate entry, and other qualifying education measures. A paper can contribute to several outcome groups, so the group counts cannot be added to obtain a total number of papers.

The timing count records distinct reported effect-estimation times or pooled windows within a comparable series. Effects reported at years 3 and 6 count as two times; effects reported separately at years 3, 4, 5, and 6 count as four. A single coefficient pooled over years 3–6 counts as one window. Alternative specifications,

treatment arms, and subgroups at the same horizon do not add times. The coding retains the time origin, reported horizons, selected series, and supporting table or figure. Independent samples or unmatched time origins are not combined into a single longitudinal sequence. The overall and outcome-specific counts can differ because they summarize different education evidence within the same paper.

Figure A2 and Table A1, Panel A, summarize these counts. All 459 papers have an overall count: 249 report one time or window and 210 report more than one. CE includes 237 eligible papers, of which 236 have a coded count; the remaining paper is excluded only from that panel's percentage denominator. AA has 40 coded papers and BA+ has 190. Among the BA+ papers, 132 report one time or window, or 69.5 percent. A single window can span several years, so these statistics describe the number of reported estimates over time rather than the length of follow-up or the duration of a policy effect.

An observation endpoint is recorded separately from the timing count and assigned to a defined clock. Attained age dates the person's education outcome. Individual follow-up measures elapsed time since an origin such as random assignment, high-school exit, or college entry. Policy or cohort event time dates an observation relative to a reform or exposed cohort and need not track the same people. These clocks are kept separate throughout. A six-year completion outcome is not assigned age 24 without evidence on the relevant entry age; academic terms are converted to years only when the source calendar supports that conversion.

Every paper receives one overall endpoint classification, reported in Table A1, Panel B. An *exact numerical endpoint* supplies a value on at least one defined clock (366 papers). An *interval or bound* restricts the endpoint without supplying a unique value (38). A *stage with unaligned timing* identifies an education stage or observation window without a common numerical endpoint (46). An *unresolved endpoint* has insufficient or conflicting information about the cutoff (9). All four categories remain in the literature sample. For example, an admission-stage effect can have one identifiable estimation window even when the paper does not report elapsed time between assignment and admission.

Figure A3 relates the number of reported times or windows to numerical endpoints for all education outcomes; Figure A4 presents the same comparison separately for CE, AA, and BA+. The overall panels contain 104 age pairs, 258 follow-up pairs, and 29 event-time pairs, representing 366 distinct papers. These counts overlap because a paper can supply more than one clock. The other 93 papers remain in the sample and in the endpoint classification, without an imputed coordinate. Bubble area is proportional to the number of papers at each coordinate. The displayed Spearman correlations are computed from the individual paper records, including ties, rather than treating each bubble as one observation. Correlations are shown only when a panel contains at least ten papers and both coordinates vary. The figures describe reporting patterns, not causal relationships or changes in treatment effects. In the overall panels, counts and endpoints can summarize different education outcomes within a paper; the outcome-specific panels narrow that comparison to the recorded education category.

A.3 Interpretation and Education-Based Lifetime Valuation

The review separates statements about finality from arguments about the adequacy of the observation window. A *finality or permanence statement* discusses whether an education difference is final or persistent, preserving the source authors' qualifications (18 papers). A *completion or window-adequacy argument only* explains why education is expected to be complete or follow-up sufficiently long, without explicitly asserting a permanent treatment effect (22 papers). The second category is assigned only when no qualifying finality statement is

recorded. The combined count of 40 therefore measures the presence of these arguments, not the number of studies establishing permanent effects.

Education-based lifetime or career valuation requires a quantitative projection from education evidence over a lifetime or working career. It includes projected earnings, fiscal effects, benefits relative to costs, and MVPF calculations; a complete welfare calculation is not required. Qualitative references to education returns, lifetime calculations unrelated to education estimates, and education costs totaled over a few years do not qualify. Thirty-one papers meet this definition. One also belongs to an interpretation category, so 70 papers satisfy at least one of these criteria and 389 have neither detected in the reviewed material (Table A2, Panel A). The latter classification does not imply that a paper contains no discussion of long-run outcomes.

Panel B of Table A2 links the 31 valuation papers to the education outcomes entering their calculations. This is a narrower question than the paper-level reporting counts above. It asks whether the valued education outcome is observed at one or several ages or individual follow-up times; it does not count earnings observations, different educational outcomes at different deadlines, or successive exposed cohorts as additional observations of one attainment path. When several outcomes enter a valuation, the classification uses the richest comparable series among those outcomes. A paper can therefore supply several inputs to a calculation while reporting only one window for each. For example, credits measured in one academic year and transfer measured in the next supply different education margins, whereas bachelor's completion reported at years 4, 5, and 6 supplies repeated observations of a degree-attainment margin. The table records available education information, not a claim that every reported time enters the authors' valuation formula.

Eighteen papers report one age, follow-up, or stage-specific window per valued education series; 12 report multiple outcome times. The remaining paper values annual institutional degree awards indexed by spending-shock time, which does not supply an individual completion deadline. Papers in the first group may still report policy-event dynamics across successive cohorts. Five of the six age-based papers supply an exact upper age, and 17 of the 20 follow-up papers supply an exact last deadline. Their endpoint spans are shown within each clock, without pooling ages and elapsed years or imputing missing cutoffs. The other follow-up records use a pooled 6–8-year window or upper bounds; the age record without an exact endpoint and the four stage-specific records remain in the table. A pooled adult-age window is not a lifetime education trajectory, and a multiple-time classification does not imply that final attainment is observed.

The accompanying 31-paper ledger records the valued outcome, reported times, time origin, endpoint qualifications, source version, and supporting pages or exhibits. It also preserves event-time information separately. These linked classifications supplement the 459-paper records rather than overwriting their broader definitions. They describe the selected valuation papers and should not be read as evidence that the original authors ignored dynamics or that their valuations are necessarily incorrect.

A.4 Source Versions and Replication Records

The review counts use recorded source versions, while the main text cites published articles where available. STBF record 628 uses the September 2020 working paper; the application uses Angrist et al. (2022), whose six-year degree results and graduation dynamics appear in Table III and Figure III. STBF applies the framework of Hendren and Sprung-Keyser (2020) but is not one of their earlier policy reconstructions. Version differences remain explicit in the evidence ledger.

For Adams, record 10 uses the May 2013 working paper and its roughly 30-percentage-point, $300,000 illustration for marginal college switchers. The published Cohodes and Goodman (2014) use about 27

percentage points and $270,000. Neither illustration is the 2.5-percentage-point eligibility effect or the separate policy-average MVPF in Hendren and Sprung-Keyser (2020). Tennessee record 684 uses the May 2026 version of Schoonover et al. (2026); the main BA-or-higher outcome at ages 23–24 is distinct from exact BA and the alternative ages 22–24 specification. Appendix D documents these welfare accounts, their inputs, and their authors' discussion of dynamics and extrapolation.

Restricting the review to the journal sample leaves 130 one-window papers among 186 coded BA+ papers, or 69.9 percent, compared with 132 of 190 papers, or 69.5 percent, in the combined sample. Education-based lifetime or career valuation appears in 28 of 447 journal papers and 31 of 459 papers overall. These comparisons show how the reported counts change when the additional sources are excluded. The replication files contain the paper register, timing and interpretation decisions, endpoint classifications, evidence locations, source versions, and coordinates underlying the figures. The exhibit code aggregates those recorded judgments without recoding the papers. Bubble plots use the September 21 endpoint review, which revisited the 120 previously unplotted papers and retained the earlier reviewed coordinates for the others.

A.5 ACS sample and construction of the age profiles

Figure 1 uses ACS microdata selected within survey years 2000–2022, birth cohorts 1985–1987, and ages 16–35. Cohort is indexed by survey year minus reported integer age. The selected observations span 2001–2022, and every cohort contributes all 20 ages: 1985 over 2001–2020, 1986 over 2002–2021, and 1987 over 2003–2022. The resulting sample contains 1,978,860 respondent records from all 50 states and the District of Columbia. There are no restrictions on birthplace, race, Hispanic origin, sex, or marijuana policy status. Positive person weights and valid detailed educational attainment (EDUCD) are required; neither condition excludes a record within this selection. All 60 cohort–age cells are observed.

Person weights are applied within cohort–age cells, and the three cohorts receive equal weights at every age. Let S_{ca} be the respondents in cohort c at age a, w_i the IPUMS person weight, and H_{ik} respondent i's value of education measure k. For a group g containing three birth cohorts C_g, the profile is

$$\overline{H}_{gk}(a) = \frac{1}{3} \sum_{c \in C_g} \frac{\sum_{i \in S_{ca}} w_i H_{ik}}{\sum_{i \in S_{ca}} w_i}. \tag{16}$$

For Figure 1, $C_g = \{1985, 1986, 1987\}$. Fixed cohort weights prevent changes in the mixture of birth cohorts across ages. The underlying ACS observations are repeated cross sections, so the profiles need not hold immigration, survival, survey coverage, or other within-cohort composition constant. They describe the education of residents observed at each age, including people born outside the United States.

In the ACS profiles, SC+ is some-college-or-higher attainment, used as a proxy for college attendance; it is not current enrollment or a directly observed date of first entry. AA+ and BA+ denote cumulative degree thresholds. The EDUCD crosswalk assigns 12 schooling years to high-school completion, 12.5 to less than one year of college, 13 to at least one year of college without a degree, 14 to an associate's degree, 16 to a bachelor's degree, 18 to a master's degree, 19 to a professional degree, and 20 to a doctorate. Lower-grade categories use the reported grade or the documented midpoint. YS is therefore an approximation based on attainment categories. The replication files provide the crosswalk, cohort–age means, respondent counts, weight sums, survey years, and source information. Using a fitted ACS profile as an untreated potential-outcome path in a policy application requires a separate calibration assumption.

A.6 Education profiles across birth cohorts

Figure A1 compares three groups of birth cohorts: 1975–1977, 1980–1982, and 1985–1987. The horizontal axis spans ages 16–35, but each group includes only ages observed for all three cohorts within the 2000–2022 survey window. The groups contribute ages 25–35, 20–35, and 16–35, respectively. Missing earlier ages remain unplotted, preserving the fixed cohort composition without extrapolation. The plotted sample contains 4,626,027 respondent records in 141 cohort–age cells and uses the same population definition, education crosswalk, person weights, and fixed one-third cohort weights as Figure 1.

Each curve shows the level of educational attainment at each observed age. Degree and attendance measures are displayed as population percentages; schooling is displayed in years. All three groups are observed at ages 25 and 35, allowing their subsequent education gains to be compared using the differences between these endpoints. These descriptive profiles are distinct from the policy effect ΔH used elsewhere in the paper. The accompanying cohort-growth data retain both endpoint levels and their differences.

Across the three groups, the age-25-to-35 gains in BA+ are 5.86, 9.73, and 11.46 percentage points. The corresponding AA+ gains are 7.30, 11.02, and 12.99 points, and approximate schooling increases by 0.424, 0.634, and 0.669 years. All three increments rise successively across these groups. The college-attendance proxy follows a different pattern, with gains of 4.79, 8.39, and 5.17 percentage points. Thus the pattern concerns later degree attainment and schooling accumulation, rather than a uniform increase across every education measure. These level differences do not measure the fraction of individual degree holders who completed after age 25, and the comparison does not isolate a causal effect of birth year or economic conditions.

B Education-Path Framework and Simulation

The Section 3 exercise asks which treated years-of-schooling paths remain compatible with stipulated age-specific effects and how their education-channel values differ under a common rule. It is a deterministic compatibility calculation. This appendix defines the observations and path restrictions, derives the matching and valuation formulas, and reports the experimental design, numerical checks, and complete results. The comparison path is the equal-cohort mean of the three fitted ACS profiles documented in Appendix C. Averaging the three cohort parameters does not make that comparison path itself a single logistic curve.

B.1 Schooling Stocks, Flows, and Observations

For a specified cumulative education outcome, $H_c^j(a)$ is its population mean at age a under policy state j. It is a share for a binary attainment outcome and is measured in schooling years for the years-of-schooling simulation. Its first difference records acquisition between adjacent ages. The effect $\Delta H_c(a)$ is the difference between the two stocks, whereas $\Delta h_c(a)$ is the difference between their acquisition flows, as defined in Equation (1). Even if both levels are nondecreasing, either effect can have either sign. For example, a treated group can acquire more schooling early and less later while retaining a positive stock difference throughout. Figure 2 illustrates these definitions with hypothetical degree-attainment shares. Both level paths increase, but the gap falls between ages 24 and 25, so $\Delta h_c(25) < 0$. These curves explain the notation and are not estimated policy effects or the ACS calibration. In the numerical schooling-year simulation, the first possible

effect age is $a_0 = 18$: the model sets $\Delta H_c(a) = 0$ before age 18 but allows $\Delta H_c(18)$ to differ from zero. The first flow is consequently $\Delta h_c(18) = \Delta H_c(18)$, relative to the zero gap at age 17.

The observation operator must retain what the source actually measures. An effect on attainment at a specified age enters as

$$\mathcal{M}_{a_*}(\Delta H_c) = \Delta H_c(a_*). \tag{17}$$

A population-weighted window average instead enters as

$$\mathcal{M}_{\mathcal{A}}(\Delta H_c) = \sum_{a \in \mathcal{A}} w_a \Delta H_c(a), \tag{18}$$

where the source's weights sum to one over $\mathcal{A}$. An effect on acquisition between two ages restricts the change in the stock gap:

$$\mathcal{M}^{\text{flow}}_{a_1,a_2}(\Delta H_c) = \Delta H_c(a_2) - \Delta H_c(a_1). \tag{19}$$

Observing two stock effects separately imposes two constraints, unlike observing only their difference. A completion rate within six years of entry is not automatically a stock measured at age 24: an attained-age mapping requires information on entry ages and the outcome definition. The experiments below use exact stock effects at one or two attained ages and require no such age bridge. They hold the target population fixed across those observations. For each compatible candidate I retain the terminal effect $\Delta H_c(65)$, the annual sum $\sum_{a=18}^{65} \Delta H_c(a)$, and the education NPV in Equation (30). The annual sum has schooling-year–age-year units and is distinct from additional completed schooling at the terminal age.

B.2 Logistic Parameters and Feasible Scenario Intervals

Write $p = (K, b, d) = (K_c + \Delta K_c, b_c + \Delta b_c, d_c + \Delta d_c)$ for the total treated-path parameters, and let $S = H^0_c(\tau_c)$. Define $L(a) = \Lambda[b(a-d)]$ and $F(a) = [L(a) - L(\tau_c)]/[1 - L(\tau_c)]$. The continuation in Equation (2) is $H^1_c(a) = S + (K - S)F(a)$ after τ_c. Holding the anchor fixed, its plateau derivative is $\partial H^1_c(a)/\partial K = F(a)$. For either shape parameter $v \in \{b, d\}$, the derivative is

$$\frac{\partial H^1_c(a)}{\partial v} = (K - S)\frac{[1 - L(\tau_c)]\,\partial_v L(a) - [1 - L(a)]\,\partial_v L(\tau_c)}{[1 - L(\tau_c)]^2}. \tag{20}$$

Here $\partial_b L(a) = (a - d)L(a)[1 - L(a)]$ and $\partial_d L(a) = -bL(a)[1 - L(a)]$. In particular, $\partial_d H^1_c(a) = -(K - S)b[1 - L(a)]F(a) < 0$ after the anchor when $K > S$ and $b > 0$. Figure B1 varies each total parameter with a common starting level. K changes the eventual plateau and scales the remaining acquisition $K - S$; b changes its speed; and d changes its timing subject to the common start. A nonzero K change thus produces a permanent education effect, whereas changes in b or d alone produce temporary effects with the same eventual attainment. These labels describe the generating path, not every candidate compatible with its observed effect. The negative derivative with respect to d means that a higher d postpones attainment while preserving the anchor and eventual plateau. Because normalization is affine in $L(a)$, the post-anchor continuation has its inflection and maximum instantaneous acquisition rate at $a = d$ when $d > \tau_c$. If $d \leq \tau_c$, that peak lies outside the continuation domain, and its acquisition rate declines after the anchor. The reference value $d = 12.564854$ is below the simulation's age-17 anchor, so it is not a post-policy age of peak acquisition. None of these parameters is an individual time to degree.

An equivalent interpretation uses the age at which the path reaches a specified attainment level. Let a_q satisfy $H_c^1(a_q) = S + q(K - S)$, where $q \in (0, 1)$ is the fraction of remaining attainment accumulated after the anchor. The normalized continuation implies

$$a_q = \tau_c + \frac{1}{b} \log\left(\frac{1 + q \exp[b(d - \tau_c)]}{1 - q}\right). \tag{21}$$

Its derivative with respect to d is $\partial a_q / \partial d = q \exp[b(d-\tau_c)]/\{1+q \exp[b(d-\tau_c)]\}$, which lies strictly between zero and one. Holding K, b, and the anchor fixed, a higher d therefore increases the age required to reach each such level. The delay depends on q and is smaller than the change in d; it is not a uniform horizontal shift. Here q refers to progress along the population attainment path, not the fraction of individuals who graduate. The age at which half of the remaining attainment has accumulated is $a_{1/2} = \tau_c + b^{-1} \log\{2 + \exp[b(d - \tau_c)]\}$, which generally differs from d.

The anchor implements an assumption about exposure in a fixed population. In the age-based experiment, individuals are unaffected through age 17, and a policy may first alter their measured attainment at age 18. The post-policy continuation therefore passes through the comparison level at 17, rather than attaching a curve whose value already differs there. This eliminates a mechanical level discontinuity at the transition; it does not require a zero effect at the first affected annual observation or equal derivatives at the anchor. In applications, eligibility, announcement, and exposure timing determine which age can serve as the last unaffected observation. An earlier anticipatory response would invalidate a later anchor. Equality refers to two potential outcomes for the same target population, not to equality of raw means in two selected samples. A common anchor for people first exposed at different ages requires a genuinely unaffected common age or an explicit cohort-specific construction before aggregation.

Each candidate is constrained to $0 \le H_c^1(a) \le 25$ schooling years at every annual age from 18 through 72. Over the same ages, its annual acquisition satisfies

$$0 \le H_c^1(a) - H_c^1(a - 1) \le 1. \tag{22}$$

The common-anchor construction imposes $H_c^1(a) = H_c^0(a)$ through age 17, so the first increment is the increase from that shared level along the normalized continuation. The one-year cap is a maintained bound on annual acquisition in the harmonized schooling measure, not a threshold estimated from causal policy responses. It restricts the difference between adjacent annual schooling levels, including the first post-anchor transition; it is not a bound on the instantaneous derivative or on the schooling treatment effect. The fixed comparison curve also satisfies this annual bound. Nondecreasing attainment restricts the levels and does not require a nondecreasing treatment effect. The parameter domain ensures $K > S$ and $b > 0$. Membership in this class establishes the stated mathematical restrictions, while the plausibility of its exposure and schooling assumptions remains a separate economic question.

For each coordinate $j \in \{K, b, d\}$, I fix the other two at their reference values and intersect its domain with all path restrictions. The reference triple p^0 is the mean of the fitted cohort parameters documented in Appendix C. The conditional feasible intervals are $I_K = [13.337015, 15.139521]$ for the plateau, $I_b = [0.120000, 0.401078]$ for speed, and $I_d = [8.000000, 30.000000]$ for timing. A scenario at position u in $I_j = [L_j, U_j]$ sets $p_j = L_j + u(U_j - L_j)$, leaving the other two parameters at p^0. The midpoint scenarios use $u = 1/2$ and increments 0.3160 schooling years, 0.0030 inverse age-years, and 6.4351 age-years. The timing midpoint consequently represents later accumulation. The annual-acquisition cap binds at the upper end of the speed interval, whose midpoint lies close to the reference speed and therefore implies a

small education effect. This rule selects locations within the conditional feasible intervals without equalizing observed effects or selecting on NPV. The grid is a design for examining the maintained model, not an estimate of a typical policy response.

B.3 Exact Matching and Numerical Search

I use the constrained minimum-distance rule in equation (15). For a simulated cohort, the same criterion can be written directly as

$$\widehat{\Theta}_c(\mathbf{m}, a_0) = \arg\min_{\theta_c \in \Theta_c} [\mathbf{m} - \mathbf{g}_c(\theta_c; a_0)]' W [\mathbf{m} - \mathbf{g}_c(\theta_c; a_0)] \tag{23}$$

where $\mathbf{g}_c$ predicts the retained moments and Θ_c imposes the maintained path restrictions. W is a fixed positive-definite matrix. For the exact simulation inputs, the minimum is zero and every such matrix gives the same zero-residual set. With three path parameters and one or two post-anchor effects, the minimizer need not be unique. The simulation solves these equations directly and retains verified exact matches. The ten policy L3 specifications without a verified exact match instead report constrained best fits and their nonzero residuals. This criterion is not a claim of covariance-weighted efficient GMM. The anchor is built into the candidate path and is not another observed policy effect.

Let $S = H_c^0(\tau_c)$, let (K, b, d) denote total treated-path parameters, and define $F(a; b, d) = [\Lambda\{b(a-d)\} - \Lambda\{b(\tau_c - d)\}]/[1 - \Lambda\{b(\tau_c - d)\}]$. For one effect at $a_* > \tau_c$, the implied treated level is $y = H_c^0(a_*) + m_{a_*}$. Conditional on (b, d), exact matching gives

$$K = S + \frac{y - S}{F(a_*; b, d)}. \tag{24}$$

I evaluate a 401×401 grid in (b, d), recover K, and retain candidates satisfying the level, increment, and parameter restrictions. For each output metric, local constrained optimization then searches for smaller and larger values from up to two separated feasible starting candidates, using scaled coordinates, at most 100 iterations, and an objective tolerance of 10^{-11}. The observed moment remains imposed algebraically.

For two effects at $\tau_c < a_1 < a_2$, let $y_j = H_c^0(a_j) + m_{a_j}$, $z_j = y_j - S$, $R = z_2/z_1$, and $t_j = \exp[-b(a_j - \tau_c)]$. Since $F(a_j; b, d) = (1 - t_j)/(1 + qt_j)$ for $q = \exp[b(d - \tau_c)]$, define $r = R(1 - t_1)/(1 - t_2)$. Solving the two matching equations gives $q = (r - 1)/(t_1 - rt_2)$. The implied timing parameter is

$$d = \tau_c + \frac{\log q}{b}, \tag{25}$$

and the plateau is

$$K = S + z_1 \frac{1 + qt_1}{1 - t_1}. \tag{26}$$

The search is one-dimensional in b. I use 1,601 grid values, refine detected feasible-interval boundaries by root finding, and refine candidate metric extrema locally within feasible intervals. A candidate must have $q > 0$, finite parameters, and satisfy all maintained restrictions. The registered generating paths have $0 < z_1 < z_2$, so the formulas are not used for degenerate observations at the anchor or with equal treated levels. I include the stipulated generating path in each library to verify nonemptiness. This is a deterministic compatibility exercise, not an estimator-performance experiment.

A candidate is accepted only if its largest moment residual is at most 10^{-8} schooling years and every path or parameter restriction holds to tolerance 10^{-9}. If a local optimizer stops just outside a boundary, I move

its proposal a small fraction toward its verified starting point, recover K from the exact matching equation, and check the conditions again; the feasibility tolerance is not relaxed. Independently feasible outputs are retained even when the optimizer does not report convergence, because these outputs supply candidate paths rather than certified extrema. Libraries are deduplicated using parameter values rounded to 12 decimal places. Every verified two-age candidate is also supplied to both component single-age libraries, after which their metric endpoints and pointwise path envelopes are recomputed. This preserves numerical nesting without treating the finite search as a proof of completeness.

B.4 A Unified Net-Earnings Response

The simulation uses annual comparison earnings $Y_0(a)$, income ages 18–65, and discount factors $D(a) = 1.03^{-(a-18)}$. Amounts are in 2000 dollars. With $m = \Delta H(25)$ and the retained signed return schedule from Hendren and Sprung-Keyser, the single-effect benchmark is

$$E^{\text{point}} = m \sum_{a=18}^{65} D(a)Y_0(a) \left[r^- \mathbf{1}\{18 \le a < 25\} + r^+ \mathbf{1}\{a \ge 25\} \right]. \tag{27}$$

The full-precision coefficients are $r^- = -0.0667064939863901$ and $r^+ = 815/7241 = 0.1125535147$. They retain the benchmark conversion of early and later earnings effects associated with the education estimate (Zimmerman, 2014; Hendren and Sprung-Keyser, 2020). This single-batch representation is an equivalent accounting construction, not a claim that the source observes a universal graduation age.

For an arbitrary attainment gap, define $q(\tau) = \Delta H(\tau) - \Delta H(\tau - 1)$ and retain the common pre-policy value $\Delta H(17) = 0$. The linear extension is

$$R(a, \tau) = r^- \mathbf{1}\{\tau - 7 \le a < \tau\} + r^+ \mathbf{1}\{a \ge \tau\}, \tag{28}$$

$$\Delta y(a) = Y_0(a) \sum_{\tau=18}^{72} R(a, \tau) q(\tau), \qquad 18 \le a \le 65, \tag{29}$$

$$E^{\text{path}} = \sum_{a=18}^{65} D(a) \Delta y(a). \tag{30}$$

The policy-onset floor truncates pre-completion responses before age 18. Acquisition after age 65 can affect earnings before 65 through the seven-year window, which is why the education support extends to 72. No earnings after 65 are valued. The annual acquisition cap is imposed on the full modeled education path; the reported terminal gap remains $\Delta H(65)$.

Summing the increments gives the equivalent expression

$$\Delta y(a) = Y_0(a) \left[r^+ \Delta H(a) + r^- \{\Delta H(a + 7) - \Delta H(a)\} \right]. \tag{31}$$

The program checks present values both through the increment sum and through this stock expression. A step of size m at 25 recovers Equation (27) exactly. For two steps at 23 and 25, the response is the sum of their respective shifted templates. Signed increments also admit delayed acquisition and catch-up: a negative increment is a difference between population acquisition flows, not a negative graduation probability. A seven-year negative-response window per schooling-equivalent increment is an extrapolation template, not seven additional years physically spent in school.

The continuous-age analogue is the signed Stieltjes integral

$$\Delta y(a) = Y_0(a) \int R(a,\tau)\, d\Delta H(\tau). \tag{32}$$

When ΔH is differentiable, $d\Delta H(\tau) = \Delta H'(\tau)d\tau$; when it has jumps, the same expression includes the corresponding discrete batches. The calculations use annual sums, not numerical estimates of a continuous-time structural model. The finite path family does not identify the kernel from education observations. Its use assumes linear transport across acquisition ages and maintains the earnings profile, response coefficients, and policy-onset convention.

Both early losses and later gains enter one net-earnings curve. No additional contemporaneous schooling-flow charge or separately calibrated cost-share parameter enters the current specification. In particular, the method does not impose equality between completed-schooling increments and extra enrollment time. Tuition, public education resources, transfers, and fiscal responses remain separate welfare-account items, because they are not all contained in earnings.

Figure 4 evaluates the joint-change illustration $(\Delta K, \Delta b, \Delta d) = (0.2625, 0.0300, -0.4500)$ using the same template for the single-batch benchmark and the full path. Its maximum annual schooling increment is 0.766 years. The two NPVs are \$15,390 and \$13,986. The bars display the two discounted terms in Equation (31): blue gives $D(a)Y_0(a)r^+\Delta H(a)$ and orange gives $D(a)Y_0(a)r^-\{\Delta H(a+7) - \Delta H(a)\}$. Panel A uses the step path $\Delta H(a) = m\mathbf{1}\{a \geq 25\}$. Black dots sum the two terms at each age, and their annual sum gives NPV. The orange term can become positive when the attainment gap narrows as the comparison population catches up. This accounting decomposition introduces no additional opportunity-cost deduction or separately identified labor-supply mechanism.

The policy collection shares this outer net-income accounting, while retaining source-specific inner mappings. Cases based on Hendren and Sprung-Keyser retain their source response phases and monetary conventions; STBF retains its nonlinear credential–experience model; Early Head Start and Tennessee use explicitly identified public-data reconstructions when the author inputs cannot be recovered. Appendix D documents these distinctions and source-reproduction checks.

B.5 Single-Parameter Explanations of Policy Education Inputs

The policy exercise asks whether an observed education input can be matched by a permanent effect through K alone or a temporary effect through b or d alone. It differs from the deterministic experiments in Section 3: the policy input is held fixed, and the active parameter is solved for rather than selected to generate a new input. Each calculation retains its geographic and cohort calibration, observation operator, policy unit, and valuation account. A weighted stock input and a difference between two schooling stocks are both admissible inputs to this exercise. Neither is replaced by an effect at a convenient single age. The eleven finite tax calculations retain their policy-dose conversion and their stated stock or acquisition-window operators.

I preserve every fitted component of the comparison path. Let j index those components within policy i, with fixed weights ω_{ij} and baseline parameters $z^0_{ij} = (K^0_{ij}, \log b^0_{ij}, d^0_{ij})$. For parameter family $q \in \{K, b, d\}$, the experiment uses

$$z^{1,q}_{ij}(t) = z^0_{ij} + te_q, \tag{33}$$

where e_q selects the corresponding coordinate. A plateau or timing experiment adds the same t to every component's K or d. A speed experiment multiplies each component's b by $\exp(t)$, so its labels report

$100[\exp(t) - 1]$ percent. The other two parameter families, component weights, and pre-policy anchors remain fixed. Applying the anchored continuation component by component and then averaging reproduces H_i^0 exactly at $t = 0$. This construction preserves baseline heterogeneity. A nonzero K shift changes the limiting attainment level and is therefore permanent. Changing b or d with each component's K fixed leaves the limiting aggregate gap at zero and is therefore temporary, although the gap can persist beyond the valuation horizon. It is a separate diagnostic and need not be a coordinate slice of the main search's common-parameter specification.

Let $\Delta H_i^q(\cdot; t)$ denote the resulting aggregate education effect. The matching calculation retains values in

$$\mathcal{T}_{iq} = \left\{ t : \; H_i^{1,q}(\cdot; t) \in C_i, \quad \mathcal{M}_i\{\Delta H_i^q(\cdot; t)\} = \mathbf{m}_i^{\mathrm{L2}} \right\}. \tag{34}$$

Here C_i includes the level and annual-acquisition restrictions, parameter bounds, timing support, and calibrated covariance ellipsoid. The ellipsoid is applied to the weighted parameter center in $(K, \log b, d)$ coordinates; Social Security retains its shape-only ellipse and probability bounds from Appendix D.7. Holding two coordinates fixed can substantially shorten the remaining interval relative to that coordinate's marginal bounds. The numerical routine scans this conditional interval, brackets sign changes, and checks local minima of the absolute matching residual for tangencies. Reported solutions have input residuals below 10^{-8} in the input's units. No input standard errors are added to obtain a match. Missing solutions remain missing rather than being assigned zero or an unrestricted parameter value.

STBF has four jointly matched inputs: time enrolled, any college, AA+, and BA+. Its active family permits one shift for each outcome, subject to credential nesting and feasibility of the recovered subgroup mean inputs. These shifts have different units and are not summed. Only the plateau family yields a verified joint match. Tennessee Promise has three education inputs. Its family experiment permits a separate change in the active parameter for SC+, AA+, and BA+, holding the other two parameters of each credential path fixed. The three latest-window effects must be matched jointly, and the paths must satisfy credential ordering and the late-completion restrictions. Thus, its plateau point represents three credential-specific plateau changes, reported in percentage points, rather than one schooling-year shift. Only this family has a jointly admissible solution. Each matched path is also checked against the additional L3 point inputs, without introducing standard-error bands. No single-family solution exactly matches that full added set in the current specification. This check does not replace the constrained fit in fitted main L3 applications.

Figure B3 compares the reference earnings NPV with the earnings NPV of each matching path. Each saved path value is already expressed in its current account units. A paired dollar comparison is made only when the monetary and discount-base units can be reconciled with L1. Both coordinates are then divided by the same positive reference program expenditure. This normalization permits a comparison within each policy without mixing its currency years or policy doses. The reference earnings component is recovered from the original account; it is not inferred from the MVPF ratio. Algorithmic placement has no source lifetime-earnings projection and is excluded from this comparison, rather than assigned a reference NPV of zero. The source monetary years for Early Head Start and Tennessee cannot be established sufficiently for a matched dollar comparison, so their reconstructed amounts are not plotted against L1 dollars. Their dimensionless MVPF comparisons remain available. Cannabis has no whole-policy L1. The paired samples contain 34, 28, and 20 calculations for K, b, and d, respectively. Table B3 reports the normalized values and parameter changes for all 37 reference calculations, including unpaired cases. A self-financing welfare account can still have a finite earnings NPV and remains in these plots.

The income comparisons retain each application's valuation rule, including its signed early-income response or author income function. They therefore differ from the common synthetic-cohort calibration used in the Section 3 simulations. Differences from a source earnings projection may reflect both the education continuation and the valuation mapping. For example, the Adams reference earnings loss is about $1,204 in the account's 2004-dollar units. Matching its converted −0.018 schooling-year input with plateau, speed, or timing changes gives losses of about $1,617, $417, and $372. The latter two temporary-effect paths converge to the comparison plateau, reducing the duration of the earnings loss; the permanent-effect path retains a lower plateau. This comparison does not establish which path occurred or overturn the source's evidence of dropout during follow-up.

The matched samples differ across parameter families because holding two parameters fixed can leave no admissible solution for the remaining parameter. This restriction combines the observation operator with the conditional parameter support; an absent point is not a zero effect and does not rule out timing responses outside the specified family. The replication materials retain the parameter sweeps, matching checks, earnings-unit comparisons, and exclusions.

B.6 Two Experiments and Summary Measures

The first experiment changes the generating process while fixing the observed age at 25. For each parameter j, I use $u = 0, 0.01, \ldots, 1$ in the scenario rule from Appendix B.2 and additionally retain its reference value, yielding 102 registered scenarios per parameter and 306 rows. The two parameters not varied in a generating scenario remain at their reference values. I then discard that knowledge of the generating mechanism and search the common three-parameter class for paths matching the scenario's age-25 effect. The three reference rows describe the same generating path and are retained within each mechanism's reporting grid; they are not three independent pieces of evidence. The complete finite grids remain in the replication results; they do not exhaust the continuum of real-valued parameters.

The second experiment holds a generating process fixed and changes the observed ages. It uses five interval positions, $u \in \{0, 0.25, 0.5, 0.75, 1\}$, for each parameter, giving the 15 scenarios in Table B1. For each scenario the design set contains all 18 single ages from 18 through 35 and all $\binom{18}{2} = 153$ distinct age pairs, producing $15 \times 171 = 2,565$ scenario–design comparisons. The age-25 results for these 15 scenarios are shared with the first experiment, leaving 2,856 distinct scenario–design identifiers across the two output files. The three midpoint scenarios in Figure 3 are unchanged between Panels A and B. A fixed generating process implies age-specific effects that generally differ across ages; it does not mean that the same numerical treatment effect is imposed at every age. Every verified pair candidate is also inserted into both component single-age libraries. This ensures that their numerical summaries respect the nesting implied by the observation restrictions.

For a scenario s and observation design D, let $\widehat{\mathcal{P}}_s(D)$ denote its finite library of verified candidates. For a candidate treated path h, write $\mathrm{NPV}_c^{\mathrm{path}}(h - H_c^0)$ for the value of its associated education-effect path under Equation (30), holding a_0, A, and ρ fixed. Its smallest NPV is $\widehat{\mathrm{NPV}}_s^{-}(D) = \min_{h \in \widehat{\mathcal{P}}_s(D)} \mathrm{NPV}_c^{\mathrm{path}}(h - H_c^0)$, and its largest is $\widehat{\mathrm{NPV}}_s^{+}(D) = \max_{h \in \widehat{\mathcal{P}}_s(D)} \mathrm{NPV}_c^{\mathrm{path}}(h - H_c^0)$. The reported width is

$$W_s(D) = \widehat{\mathrm{NPV}}_s^{+}(D) - \widehat{\mathrm{NPV}}_s^{-}(D). \tag{35}$$

For comparisons with observing age 25 alone, the proportional width reduction is $1 - W_s(D)/W_s(\{25\})$, calculated within each scenario before summarizing. A zero benchmark width would make that proportional

measure undefined rather than justify recording a zero reduction. Median widths and median proportional reductions use the three interior positions, $u = 0.25, 0.5, 0.75$. Largest widths and sign counts use all five positions, keeping the two boundary cases visible. A pair with a positive candidate above \$0.01 and a negative candidate below −\$0.01 is recorded as admitting both NPV signs. The cents-level threshold avoids treating numerical zero as a sign reversal. These are equal-weight summaries of the experimental scenarios, not estimates of the frequency of mechanisms among real policies.

B.7 Numerical Checks and Interpretation

All 306 first-experiment rows and 2,565 second-experiment rows were completed. The largest observed-moment residual among retained candidates was 3.55×10^{-15} schooling years, and the most negative recorded constraint margin was -9.99×10^{-10}, within the declared numerical tolerance. The audit checked 2,295 pair designs for nesting in their component single-age libraries; the largest recorded containment discrepancy was 1.06×10^{-10} in the numerical metric comparisons. An independent calculation from the stored endpoint parameters recomputed 17,136 metric-endpoint records using the normalized logistic difference directly. Its largest NPV difference was 2.36×10^{-10} dollars, and its largest terminal-schooling and cumulative-gap differences were 1.60×10^{-14} schooling years and 1.99×10^{-13} schooling-year–age units, respectively. Some endpoint records correspond to the same path because multiple metrics or designs can select it.

The local optimizer made 6,708 recorded constrained-refinement calls: 4,558 converged and passed feasibility checks, 1,690 did not report convergence but supplied independently verified feasible candidates, and 460 outputs were rejected. Boundary repair toward a verified seed, as described in Appendix B.3, was used for 1,100 accepted outputs without relaxing the tolerances. Feasible outputs contribute witnesses and candidate extrema regardless of their stopping flag; they are not certified optima. The current diagnostics do not include a separate grid-refinement pilot for this earnings-response specification, so the recorded feasibility and arithmetic checks do not establish grid stability or search completeness. No sampling bootstrap or Monte Carlo coverage calculation is part of these two experiments.

The numerical checks establish that saved candidates meet the maintained restrictions, not that those restrictions hold for real policies. ACS fit is informative within observed ages, while extrapolation above age 35, population transport, the logistic family, empirical support, and the flow-to-cost mapping remain assumptions. Finding opposite-sign values establishes conditional ambiguity; finding only one sign does not certify sign identification. Finite-bank envelopes can switch candidate identity across ages, so their edges need not be feasible paths. No probabilities or confidence levels are attached to them. Alternative families, calibrations, and noisy observations require separate analyses.

B.8 Results across the Complete Experimental Grids

Table B2 summarizes all 102 age-25 scenarios per parameter. The complete 306-row file retains the generating parameter, observed effect, terminal schooling and cumulative-gap ranges, NPV endpoints, and generating-path NPV. Endpoints for different metrics can belong to different paths. Median NPV widths are \$48,620, \$58,266, and \$65,536 for the plateau, speed, and timing scans; 56, 91, and 102 scenarios, respectively, admit both signs. These counts summarize the chosen grid and assign no probabilities to policy mechanisms.

Table 2 reports selected observation designs, and Figure B4 covers every single age and age pair. Median widths use the three interior generating scenarios; maximum widths use all five, preserving the most permissive boundary or interior result. Diagonal cells represent one age and upper-triangle cells two distinct ages. All panels share one dollar-width scale. Supplementary Figures S1–S3 report the individual-scenario heatmaps. All 2,565 scenario–design rows remain in the replication material, allowing the aggregate patterns to be checked against every generating path.

The two panels of Figure 3 use the same midpoint scenarios and common axes, so their visual difference comes from the additional age-35 constraint and the accompanying candidate search. For these three scenarios the NPV-width reductions are 92.3–94.2 percent. The candidates found for the plateau example have positive NPVs, those for the timing example have negative NPVs, and those for the speed example still span both signs. The speed midpoint is close to the reference speed within its cap-restricted feasible interval. These midpoint results do not describe every registered scenario: the additional age-35 observation admits both signs in one of the five speed scenarios and one of the five timing scenarios. Other age pairs need not produce the same reduction. Within the registered interior scenarios, adding age 24 to age 25 gives larger median proportional reductions than adding age 20 in each parameter group, despite the smaller age separation. The result concerns where observations constrain this restricted curve family, and should not be generalized to an optimal sampling design under noisy estimates without a separate analysis.

C Comparison Schooling Paths and Empirical Support

C.1 Construction of the Comparison Curve

For the schooling-year exercise in Section 3, I hold fixed a comparison path formed from three ACS birth-cohort fits. Let $\widehat{H}_{0q}(a)$ be the fitted mean completed schooling profile for birth cohort q. The reference path is

$$H_c^0(a) = \frac{1}{3} \sum_{q=1985}^{1987} \widehat{H}_{0q}(a). \tag{36}$$

The subscript c in the simulation therefore refers to this synthetic target group, not to one of the three birth cohorts. The ACS figure in Figure 1 shows descriptive attainment at observed ages; Equation (36) instead averages fitted cohort curves and is used as a fixed input to the path search. Neither object is an estimated policy counterfactual for a specific case in Section 4.

The 60 person-weighted cohort–age means and their population weights come from the sample and education crosswalk documented in Appendix A.5. All three cohorts contribute every age from 16 through 35. The following fit uses these cell means, with no additional demographic restriction.

Let $\overline{YS}_{qa}$ denote the person-weighted mean in cell (q, a) and ω_{qa} its population weight normalized to sum to one over the 20 ages within cohort q. I estimate each cohort's parameters separately by nonlinear weighted least squares:

$$(\widehat{K}_q, \widehat{b}_q, \widehat{d}_q) = \underset{K,b,d}{\arg\min} \sum_{a=16}^{35} \omega_{qa} \left[\overline{YS}_{qa} - K\Lambda\{b(a-d)\}\right]^2. \tag{37}$$

The fitting bounds are $K \in [0.01, 25]$, $b \in [0.001, 5]$, and $d \in [6, 75]$. Five starting values lead to the same retained objective in each cohort; each fit allows at most 3,000 function evaluations with numerical tolerances

of 10^{-13}. These fitting bounds differ from the narrower treated-path search domain in Section 3. Table C1 reports the cohort sizes, parameter estimates, and weighted root mean squared errors. I then average fitted curves with fixed one-third weights, rather than allowing cohort shares to change with age.

C.2 Observed Ages and Extrapolation

Each selected cohort contributes all ages 16–35, so age 25 and age 35 are both within every cohort's observed support. Figure C1 compares the mean fitted curve with the mean of the observed cohort–age schooling measures over that common range. Cohort-specific weighted root mean squared errors are 0.112–0.115 schooling years, and the largest absolute cell residual is 0.327 years. This is an in-sample descriptive fit diagnostic. Every comparison-path value above age 35 is extrapolated. The age-17 fitted value is the shared level at which candidate continuations are anchored, under the assumption that the hypothetical policy first affects measured attainment at $a_0 = 18$. Fixing cohort weights removes changes in the included birth-cohort shares; it does not turn repeated cross sections into a panel or establish the profile as the untreated path for a policy population. The simulation conditions on the fitted curve rather than propagating its sampling or extrapolation uncertainty.

C.3 Parameter Reference and Restrictions

The reference triple used to express $\theta_c = (\Delta K_c, \Delta b_c, \Delta d_c)$ is the arithmetic mean of the three fitted cohort parameter triples. Its values are approximately $(13.922226, 0.257500, 12.564854)$. Its logistic curve is not generally equal to the mean of three logistic curves in Equation (36); their largest difference over ages 16–65 is 0.000412 schooling years. After anchoring the reference-parameter continuation at the common age-17 level, its largest discrepancy from H_c^0 over ages 18–65 is 0.000226 schooling years. Thus, the zero-increment reference exactly shares the anchor but does not impose an exactly zero effect at later ages.

The plateau restriction $K \in [13.3370, 15.1395]$ is the weighted first-to-99th percentile interval of a historical bank of 716 state-by-birth-cohort logistic fits. The bank covers the 50 states and the District of Columbia and birth cohorts 1980–1994, using ACS observations from 2000–2023. Its source construction retains U.S.-born non-Hispanic White, non-Hispanic Black, and Hispanic respondents, valid education records, positive person weights, and recorded male or female sex. State–cohort profiles are fitted by population-weighted nonlinear least squares. Retained fits have at least 14 age cells spanning at least 13 years, begin by age 20, reach at least age 29, converge, and lie inside the fitting bounds.[35] The interval restricts candidate eventual schooling to a broad central range of historical fitted profiles. Its transfer to the current full-population, 1985–1987 baseline is a maintained calibration assumption: the populations and survey windows differ, and the interval is neither a confidence interval for that baseline nor validation of policy-induced endpoints.

The speed and timing domains, $b \in [0.12, 0.65]$ and $d \in [8, 30]$, and the one-year annual acquisition cap are experimental design restrictions rather than estimated quantiles from the current cohort sample. The simulation does not impose the historical bank's timing-index bounds or joint parameter ellipsoid. Appendix B.6 describes the parameter sweeps within these restrictions. The reported widths are conditional

[35] Plateau weights sum person weights over each profile; quantiles interpolate cumulative normalized weights. The resulting limits are 13.337015059499 and 15.139521154274. Observations span ages 16–43 collectively, so plateaus remain fitted asymptotes rather than observed lifetime attainment.

on this domain, the baseline, and the valuation calibration; sensitivity to a population-matched support bank, alternative bounds, or other path families has not been evaluated.

D Policy Welfare Accounting and Education-Path Replacement

The welfare reconstruction replaces specified education-related components of a source account while retaining its policy contrast and the remaining accounting assumptions. This appendix defines the units and component boundaries, maps an education path into private income and fiscal returns, and states the extra information needed to let other education costs vary. The formulas describe the accounting procedure; they do not establish that a candidate path is the true policy response or that a finite search has recovered all compatible paths.

D.1 Policy Units and the Reference Ledger

Index the policy contrast by i. L1 reports $\text{WTP}_i^{\text{ref}}$, N_i^{ref}, and their associated MVPF, using the components in Equations (10) and (11). Each entry uses the same target population, treatment contrast, currency year, and discount base. Values may be stated per eligible person, per recipient, or per specified change in aid, but all components within an account must use the same unit. Multiplying every component by the same positive scale leaves the MVPF unchanged; mixing components from different populations or treatment margins does not. An intention-to-treat effect and an effect for students induced to switch institutions therefore require a documented conversion before they can enter one account. Monetary values from different cases also require conversion to common dollar units before their levels, rather than their dimensionless ratios, are compared.

The terms B_i^{ref} and G_i^{ref} collect the private and public components outside the replaced earnings projection and the separately recorded education payments. They are not necessarily pure cash transfers and statutory program costs. Depending on the source, they may include other measured private benefits, effects on public spending, or fiscal receipts. The private payment $C_i^{\text{priv,ref}}$ and public cost $C_i^{\text{pub,ref}}$ must follow the source's tuition and subsidy convention. For example, if the benefit of a grant is recorded separately in B_i^{ref}, the same grant cannot also be deducted from the private tuition charge to create a second benefit. Public education costs must likewise exclude payments already included in G_i^{ref}. Where the source nets private tuition receipts from institutional expenditure, that convention is retained. The original component sum, not a generic cost-per-degree formula, determines the reference values.

Interpreting income changes as willingness to pay retains the welfare assumptions of the source calculation. In their college-admission example, Hendren and Sprung-Keyser (2020) treat the earnings gain as a return to human capital rather than compensation for greater effort, and value the income change net of taxes and education expenses. A benefit requiring additional work effort, an unmeasured consumption value of college, or an effect on other people's earnings would require a corresponding welfare adjustment. Scholarship transfers also depend on whether the recipient would have attended without the award. These assumptions are not consequences of the logistic path specification. For the three main applications, the reference ledger must follow the Adams and Social Security enrollment calculations in Hendren and Sprung-Keyser (2020) and the Tennessee Promise calculation in Schoonover et al. (2026).

The accounting check is to recover the source's willingness to pay and net cost by summing its recorded components before substituting any path. Feeding the original replaced components back into Equations (13) and (14) must then return those same totals. This is an arithmetic identity check, not evidence that the original

continuation belongs to the candidate logistic family. If a source does not separately report a component needed for the subtraction, recovering it requires a documented calculation or an explicit additional calibration. A missing component cannot be silently set to zero. The case records report original education inputs and timing in Table D3, followed by the reference and replacement components in Table D4.

D.2 From Attainment to Disposable Income and Fiscal Returns

The observation operator in Equation (3) determines what a candidate education path must reproduce. A stock effect at one age, an average over an age window, and an effect on schooling acquired during an interval constrain different features of the path. Converting credits or credentials into schooling years is a separate step and must preserve the source's treatment margin. In particular, a change in the probability of earning a bachelor's degree is not automatically four additional years per person: the appropriate conversion depends on schooling already counted through enrollment, partial completion, and other credentials. The anchor must precede policy exposure or anticipation for the target population. Neither age 17 nor the simulated 1985–1987 comparison group is imposed on every application merely because it is used in Section 3.

Let $e_i^s(a)$ be the annual pretax earnings effect in account $s \in \{\text{ref}, \text{path}\}$, and let $\mathcal{A}_i$ be the set of ages included in the replaced projection. The discount factor is $D_i(a) = D_{\rho_i}(a; a_{d,i})$, with source-specific rate ρ_i and discount-base age $a_{d,i}$. That base need not equal the first age at which attainment can change. All annual earnings effects are expressed in the same currency year, and their gross present value is

$$\text{NPV}_i^{\text{earn},s} = \sum_{a \in \mathcal{A}_i} D_i(a) e_i^s(a). \tag{38}$$

For schooling-year inputs from Hendren and Sprung-Keyser, the path extension uses

$$e_i^{\text{path}}(a) = Y_{0i}(a) \sum_{\tau} R_i(a, \tau) \big[\Delta H_i(\tau) - \Delta H_i(\tau - 1)\big]. \tag{39}$$

The source programs determine each phase's effective return, eligible-person conversion, age range, monetary year, discount convention, and tax rate. A single batch at the source reference completion age reproduces earnings PV, fiscal PV, private and public education costs, WTP, and net cost separately. The 33 accounts based on Hendren and Sprung-Keyser are checked component by component. Social Security now uses the separately reproduced enrollment alternative, indexed to entry as described below; its source arithmetic is reconstructed for all 1,000 paired input repetitions. Source code and reported L1 vintages remain distinct: for example, the Tennessee Pell program yields 0.8437 while the retained library L1 is 0.85.

The common template has seven negative-response years followed by a positive response, but source exceptions are retained. Ohio Pell has six early response years and different early and later discount bases. The independent-single tax cases retain their later source income clock, and adult Pell retains its adult entry clock and 1973-dollar account. Kalamazoo's released program overlaps a positive response beginning at 18 with a negative response at 18–24 and discounts the positive block from age 11. This code convention is reproduced and transported with its own phase timing; it is not silently replaced by the illustrative post-25 template. Georgia HOPE retains its sector mixture, including its zero community-college early component.

When source phases b use different discount bases, the implemented present value is $E_i = \sum_a \sum_b D_{ib}(a) e_{ib}(a)$, and fiscal value is computed with the same phase's retained tax rule. The nominal net-income curve is $\Delta y_i(a) = \sum_b e_{ib}(a)$; the unified discounted curve is $\Delta y_i^{PV}(a) = \sum_b D_{ib}(a) e_{ib}(a)$. Its age sum equals the implemented NPV. A single common $D_i(a)$ is used only when it reproduces the source

phase conventions. The source-block export records the distinctions, avoiding an apparent source match created solely by a final scaling factor.

Independent forecasts instead use the original function's input structure when recoverable. STBF's F_i is nonlinear in credential shares and potential experience. Early Head Start uses the public BA–HS schedule described below; Tennessee uses signed exclusive-credential batches. No case receives an extra schooling-flow opportunity-cost charge on top of its net-income response. Negative portions of that response are earnings losses, not separately identified pure costs of time in school.

With a retained linear tax-and-transfer schedule, let $\vartheta_i(a)$ denote the fraction of an earnings change that accrues to the government through taxes and reduced transfers. The disposable-income and fiscal components are

$$\mathrm{NPV}_i^{\mathrm{after},s} = \sum_{a\in\mathcal{A}_i} D_i(a)\big[1-\vartheta_i(a)\big]e_i^s(a) \tag{40}$$

and

$$\mathrm{NPV}_i^{\mathrm{tax},s} = \sum_{a\in\mathcal{A}_i} D_i(a)\vartheta_i(a)e_i^s(a). \tag{41}$$

The two components sum to Equation (38). Negative earnings effects reduce private income and, under a positive rate, reduce fiscal revenue. The same schedule is used in the reference and path calculations when the comparison is intended to hold taxation fixed. If the source uses different short- and long-run rates, the age-specific rates remain in the sum; a single average rate generally cannot reproduce both components for every candidate path. Nonlinear tax rules or separate transfer responses require their source-specific mapping rather than these linear formulas. In particular, applying a nonlinear tax schedule to mean earnings need not recover mean taxes without distributional information.

Holding the education-payment components fixed, when $\vartheta_i(a) = \bar{\vartheta}_i$ is constant, the income bookkeeping is especially transparent. Writing $\Delta E_i = \mathrm{NPV}_i^{\mathrm{earn,path}} - \mathrm{NPV}_i^{\mathrm{earn,ref}}$, the income replacement changes willingness to pay by $(1-\bar{\vartheta}_i)\Delta E_i$ and net fiscal cost by $-\bar{\vartheta}_i\Delta E_i$. The private gain and public revenue gain partition the same pretax earnings change. Where education payments vary across paths, their changes also enter WTP and net cost. This conditional special case explains the signs in Equations (13) and (14); it does not impose one tax rate across policies. If the original income forecast and the candidate forecast use different return or opportunity-cost rules, their difference incorporates that accounting change as well as the education continuation. Separating the two effects requires evaluating a specified reference path under the same candidate valuation rule, when such a reference path can be recovered from the source.

D.3 Education Payments, Resource Costs, and the Scope of Replacement

Source education-cost functions are preserved through an explicit timing bridge where they are recoverable. For each account based on Hendren and Sprung-Keyser, I separate fixed noneducation components from the source private-payment and public-education amounts. The source schooling quantity and completion allocation identify their per-unit schedules. Applying those schedules to signed completion batches (entry batches for the Social Security enrollment alternative) reproduces the original education costs at the source reference batch, while allowing payments to vary with path timing and quantity. This transports the source institution mix and marginal unit costs; it does not infer actual enrollment, sector switching, or aid eligibility from attainment alone. Where a source aggregate combines enrollment and credential margins, holding that composition fixed per schooling equivalent is an additional restriction.

STBF, Early Head Start, and Tennessee Promise retain their stated non-income amounts while replacing income. This isolates a conditional income-channel comparison; it does not let every alternative path change grant duration, tuition, or public instruction resources. Algorithmic placement retains original WTP and adds only fiscal earnings. Cannabis prices its education-resource flow separately from its signed income response. These scopes are recorded by case rather than described as one common fixed residual for all 38.

With fixed conversion rules, prices, and valuation coefficients, the account can also be collected into a ratio of affine functions of path summaries:

$$\mathrm{MVPF}_i(\Delta H_i) = \frac{\alpha_i + \mathbf{w}_i^\top \mathbf{z}_i(\Delta H_i)}{\beta_i + \mathbf{n}_i^\top \mathbf{z}_i(\Delta H_i)}, \qquad N_i(\Delta H_i) > 0. \tag{42}$$

The constants α_i and β_i collect retained components; $\mathbf{z}_i$ contains the specified discounted earnings and education-cost summaries. For the replacement in Equations (13) and (14), take

$$\mathbf{z}_i = \left(\mathrm{NPV}_i^{\mathrm{after,path}}, \mathrm{NPV}_i^{\mathrm{tax,path}}, C_i^{\mathrm{priv,path}}, C_i^{\mathrm{pub,path}}\right)^\top,$$

with $\mathbf{w}_i = (1, 0, -1, 0)^\top$ and $\mathbf{n}_i = (0, -1, 0, 1)^\top$. The constants are $\alpha_i = \mathrm{WTP}_i^{\mathrm{ref}} - \mathrm{NPV}_i^{\mathrm{after,ref}} + C_i^{\mathrm{priv,ref}}$ and $\beta_i = N_i^{\mathrm{ref}} + \mathrm{NPV}_i^{\mathrm{tax,ref}} - C_i^{\mathrm{pub,ref}}$. Thus both income and education payments are removed at their reference values before their path values enter the account. If a payment is held fixed, its two terms cancel. This representation does not imply linearity in the logistic parameters or impose fixed coefficients on every education-policy valuation. Returns that depend on education expansion, endogenous prices, or additional behavioral channels would require further modeling.

D.4 Comparing Paths Under a Common Welfare Account

L1 is the original welfare calculation. L2 imposes the education moments selected from that calculation and searches the maintained continuation class. L3 adds comparable source information while retaining the L2 moments and all other conditions. Write $\widehat{\mathcal{D}}_{i\ell}$ for the finite set of verified education-effect paths found for policy i at information layer $\ell \in \{2, 3\}$. The joint welfare values are

$$\widehat{\mathcal{V}}_{i\ell} = \left\{ \left(\mathrm{WTP}_i^{\mathrm{path}}(\Delta H_i), N_i^{\mathrm{path}}(\Delta H_i) \right) : \Delta H_i \in \widehat{\mathcal{D}}_{i\ell} \right\}. \tag{43}$$

The numerator and denominator of each MVPF come from the same member of this set. Table D5 gives selected complete paths and their paired welfare components. Pairing the largest willingness to pay with the smallest net cost from different paths can create a ratio that no candidate attains. Likewise, the reference MVPF falling inside a candidate range does not establish that the reference continuation is a member of the path family. A later observation refines the admissible set only when it concerns the same population and outcome under a compatible time mapping. Changing the sample, functional form, or cost calibration defines an alternative specification rather than a nested information addition.

For positive net costs, $\mathrm{MVPF}_i > 1$ if and only if $\mathrm{WTP}_i - N_i > 0$. Finding two compatible, positive-cost paths on opposite sides of this threshold establishes ambiguity conditional on the maintained restrictions and account. Finding candidates on only one side does not establish that all compatible paths lie there. The self-financing region requires positive willingness to pay and nonpositive net cost and is reported separately. Ratios can also become arbitrarily large in magnitude along a sequence of positive-cost paths whose costs approach zero; an education effect crossing zero is not sufficient to establish such a sequence. A large finite ratio or a rounded zero denominator is therefore not a proof of an infinite bound. The analysis assigns no

probabilities to candidate paths, and any sampling analysis of the source estimates remains distinct from this continuation exercise.

The threshold at one compares a dollar of measured private willingness to pay with a dollar of net public cost. It is not a universal policy-adoption rule. As Hendren and Sprung-Keyser (2020) show, a budget-neutral comparison of policies affecting different people also depends on the social weights attached to their beneficiaries and on the financing alternative. The cross-policy comparisons in Section 5 retain that distinction. Within a policy, the accounting exercise asks the narrower question of whether the same education evidence supports different threshold or fiscal classifications under a fixed valuation rule.

D.5 Geographic References, Source Control Levels, and Empirical Support

The geographic library uses the unrestricted 2000–2022 IPUMS extract: no race, sex, or birthplace restriction is imposed. It contains the 50 states, the District of Columbia, and a national aggregate. Person weights form outcome means within state–birth-cohort–age cells; birth cohort is survey year minus integer age. The 2000 component is Census sample 200001, followed by annual ACS samples in 2001–2022. The sample inventory is retained with the replication files. The full library supplies years of schooling and some-college-or-higher (SC+), associate's-or-higher, bachelor's-or-higher, and postgraduate outcomes. Case-specific geographic and cohort weights are fixed across ages, so changing the observed age does not change those chosen weights. This does not turn repeated cross sections into individual histories or eliminate migration and other within-cell composition changes. The library is an external description of education profiles, not an additional source of policy identification.

For each geographic cell and outcome, I fit a logistic profile to person-weighted age means, using observed ages 16–35 when the coverage supports estimation. State shapes are regularized toward their national counterparts. A cohort with insufficient young-adult support instead borrows a national shape and uses its available observations to calibrate the level. The data therefore involve both forward extrapolation for young cohorts and backward reconstruction for historical cohorts. The 1947 cohort used for Adult Pell is calibrated at ages 53–65, and the old library entry profile for the 1963 Social Security cohort at ages 37–59. Both library profiles borrow the 1978 national shape, itself fitted at ages 22–35. Social Security now uses the source-level and observed-youth-shape calibration below instead of that backward-extrapolated entry profile. Neither construction supplies observed early-life histories for the historical cohort. Table E1 reports actual support and fit modes; the accompanying registry lists every state–cohort–outcome cell, donor, and level-calibration range. These records also distinguish Tennessee's subsequent source-population and control-level adjustments. The library contains 10,920 fits across 52 geographic units, 35 birth cohorts, and six outcome definitions.

A published control mean is used only when its population, outcome, and clock match the effect under study. In Tennessee, the effects concern high school graduates and residence one year before interview. I therefore construct the corresponding Tennessee profiles using migration information and rescale each outcome to its reported age-23–24 control mean. The common cohort weights remain fixed. This operation supplies an external level and shape; it does not recover the regression's conditional untreated prediction, and conditioning on high school graduation is itself part of the source population definition. Adams uses a full-population schooling profile because the source degree means do not identify total completed schooling. STBF uses separate external time and credential timing indexes, while its income function retains recovered source mean inputs. Its 3.93-year enrollment mean is not converted to total schooling by adding twelve. The distinction between source-matched levels and external population proxies is explicit in the case registry.

The empirical support is constructed before searching policy paths. Let $x = (K, \log b, d)^\top$. For each outcome I retain fitted profiles for the 1983–1987 donor cohorts in the 51 state-level regions. A raw logistic profile with its own level at the anchor can be written exactly in the anchored normalized form, using the same parameter triple; a separate post-anchor refit is therefore unnecessary. Such refits are saved as diagnostics because many schooling fits have weakly determined timing parameters at the numerical boundary. The main covariance averages the relevant state's within-donor covariance and the pooled-state covariance with equal weights. National targets use the pooled covariance; multi-state targets average the state covariances with the fixed target weights. The support center is the weighted parameter location of the target cohort's fitted reference, and the restriction is

$$(x - \bar{x}_i)^\top \Sigma_i^{-1} (x - \bar{x}_i) \leq 11.34486673. \tag{44}$$

The cutoff retains the earlier 99-percent chi-square scale only as an empirical calibration convention. It is not a confidence region for a treatment response and assigns no probability to candidate paths. Centering historical dispersion on a different population or cohort is a maintained transport restriction. The wider sensitivity doubles Σ_i and retains every previously verified main candidate. The Social Security entry specification restricts the two shape coordinates using their marginal covariance and imposes logical probability bounds on the plateau instead, as documented below.

Timing restrictions refer to fractions of the increase after the anchor, rather than to the unanchored curve's half-plateau age. For anchor τ, the age at fraction q of the remaining logistic increase is

$$a_q(b, d; \tau) = \tau + \frac{1}{b} \log\left\{ \frac{1 + q \exp[b(d - \tau)]}{1 - q} \right\}. \tag{45}$$

For $q = 0.1$ and 0.9, pooled donor deviations between the 0.5th and 99.5th percentiles are transported around the corresponding target reference. The wider sensitivity multiplies these deviations by $\sqrt{2}$. Schooling paths remain nondecreasing and acquire at most one year per annual step after the anchor; probability paths stay between zero and one. These restrictions describe the admitted economic continuations. They are not validated causal restrictions merely because they are informed by ACS profiles.

For L2, a deterministic Sobol shape grid and a 97×97 regular grid cover $(\log b, d)$. Conditional on a shape pair, the retained moment determines K algebraically. Three exact-match L3 cases additionally solve a second moment equation by bracketing and root refinement. The other ten L3 cases minimize the fixed equal-weight point discrepancies in equation (15), preserving L2 exactly. The main scalar fits use up to 64 diverse feasible starts and Tennessee uses 42 joint starts, supplemented by separate credential fits; constrained sequential quadratic programming is followed by an interior least-squares polish only when it passes every original restriction. Distinct minimizing basins are retained; repeated optimizer outputs near the same minimum are not interpreted as a welfare interval. All ten point searches currently retain one best-fitting path with nonzero additional-input residuals. Table E2 summarizes fit quality and Table E4 reports the schooling-input residuals. Tennessee's complete residuals appear in Table D2. Annual paths, the L2 equalities, credential ordering, and welfare accounts are independently reconstructed. These checks establish the validity of the saved solutions, not certified global optimality or uniqueness.

D.6 Adams: Education Conversion and Additional Source Timing

The Adams reconstruction retains the eligibility RD contrast for the 2005–2006 graduating classes and the source program used by Hendren and Sprung-Keyser (2020). The source observes college entry immediately

after high school and within two years, persistence in the fourth year, and graduation within four and six years. The retained conversion assigns one year to additional two-year enrollment, two years to additional four-year enrollment, half a year to two-year completion, and one further year to bachelor's completion. The corresponding multipliers are $(1, 2, 0.5, 1)$, giving $0.007 + 2(0) + 0.5(0) - 0.025 = -0.018$ schooling-equivalent years per eligible student. The baseline is the full-population Massachusetts profile for birth cohort 1987, with age 17 as the last unaffected observation. L2 imposes

$$\Delta H_{\text{Adams}}(24) = -0.018. \tag{46}$$

L3 preserves this endpoint and adds the original study's Table 4, columns (3) and (5), immediate-entry, two-year-entry, and four-year-completion results. The conversion gives

$$\Delta H_{\text{Adams}}(18) = 2(0.009) + 0.008 = 0.026, \tag{47}$$

$$\Delta H_{\text{Adams}}(20) = 2(0) + 0.007 = 0.007, \tag{48}$$

$$\Delta H_{\text{Adams}}(22) = 0.007 - 0.017 + 0.5(-0.002) = -0.011. \tag{49}$$

These equalities define the additional targets, rather than exact restrictions on the selected fit. The age-24 input remains exact; equal-weight least squares selects the remaining shape. Retaining the 0.007 enrollment contribution at ages 22 and 24 follows the source's additive conversion, which values education acquired even without a degree. Completion adds incremental schooling equivalents and is not counted again as a full four- or two-year program. The assigned ages and timing of these equivalents are explicit reconstruction assumptions. The source's local RD population also differs from all Massachusetts residents, so its external baseline remains a calibration.

The search retains 1 best-fitting L3 path within the 5,916 L2 candidates. Tables E2 and E4 report the nonzero discrepancies. The four observed windows restrict the fit more than the retained six-year input, but neither the added information nor the imposed loss function identifies an actual individual mechanism. The original authors' dropout evidence during follow-up remains relevant when interpreting possible later catch-up. Input repetitions preserve the original four-year and two-year error groups and extend them to the added entry and graduation coefficients; this extension is specified in Appendix F.

The source program yields earnings NPV $-\$1,204.04$, WTP \$1,364.23, and net cost \$1,896.35 in 2004 dollars. The new single-batch replay reproduces these components individually. It retains the original phase-specific earnings and fiscal rules and transports the source private and public education-cost schedules with the signed schooling-equivalent batches. The fixed noneducation parts are

$$w_{\text{fixed}} = W^{\text{ref}} - E^{\text{ref}} + T^{\text{ref}} + P^{\text{ref}}, \tag{50}$$

$$n_{\text{fixed}} = N^{\text{ref}} + T^{\text{ref}} - C^{\text{ref}}. \tag{51}$$

For a candidate path, $W = w_{\text{fixed}} + E - T - P$ and $N = n_{\text{fixed}} + C - T$. Private payments and public education costs can therefore vary with the candidate under the source cost-timing bridge. There is no fixed threshold in E alone. The source institution mix per schooling equivalent remains a maintained transport assumption when the total converted education effect is negative. All kernels are already in 2004 dollars; no second 188.9/172.2 conversion is applied.

The fitted L3 earnings NPV is -2.74 thousand dollars, with WTP 1.66 and net public cost 2.64 thousand dollars. The paired MVPF is 0.63. These values use the same signed completion-response and education-cost mappings as L2. They describe the selected fit at fixed education targets, rather than a confidence interval or an exhaustive feasible-path bound.

D.7 Social Security: Cumulative Entry and the Enrollment-Based Account

The source inputs are the full-time-attendance column of Table 3 in Dynarski (2003): 0.219 (SE 0.102) by age 23 and 0.224 (SE 0.106) by age 28. The second is the attrition-imputation lower-bound specification. The alternative age-28 estimates, 0.248 and 0.256, change the treatment of missing outcomes and are not additional observation ages. The enrollment alternative in Hendren and Sprung-Keyser uses 0.224 and two schooling years per induced entry. L2 therefore matches $\Delta H(28) = 0.224$, and L3 also matches $\Delta H(23) = 0.219$. The 0.679-year schooling input belongs to their separate completed-schooling specification and is excluded from this attendance-path calculation.

The age-23 baseline level is 0.352 from Table 1's post-termination deceased-father group. This sample mean is not the untreated prediction from the controlled DID regression. I use it as a disclosed calibration level and transport the equally weighted national 1985–1987 ACS some-college age shapes, rescaling their pooled age-23 mean to 0.352. Birth cohort 1963 remains the application index; the shape-donor cohorts are not relabeled as the source sample. Both paths coincide through age 17. Full-time attendance and ACS some-college attainment are related but distinct outcomes, so the shape transport remains an additional assumption.

This entry specification uses a different calibration from the superseded schooling specification. The library's 1963 entry curve has only ages 37–59 and borrows the 1978 shape, whose youngest observations are at 22. Backward extrapolation gives an implausibly diffuse entry profile for this application. Its calibrated support permits an age-28 effect no larger than approximately 0.078, below the reported 0.224. Moreover, cross-population plateau dispersion is not a logical bound on a large aid response among a selected population. I instead impose $0 \le K \le 1$ and calibrate only $(\log b, d)$ to the existing 1983–1987 state-donor covariance. The shape ellipse uses cutoff 9.21034, the 99-percent two-dimensional chi-square scale as a calibration convention, without a probability interpretation. The anchored 10th- and 90th-percent completion-time restrictions use the same donor-deviation rule as other cases. The reference shape, covariance, and timing limits are fixed before matching the treated effects and shared by L2 and L3. Sensitivity varies the shape covariance and timing spreads, while retaining the probability bounds. Comparisons with the superseded YS results would consequently change the outcome, source account, and calibration, not just the observation window.

For entry increments $q(\tau) = \Delta H(\tau) - \Delta H(\tau - 1)$, the net earnings response is

$$\Delta y(a) = 2 \sum_{\tau} q(\tau) \left[r_{-} Y_{-}(a) \mathbf{1}\{\tau \le a < \tau + 7\} + r_{+} Y_{+}(a) \mathbf{1}\{a \ge \tau + 7\} \right], \tag{52}$$

with source returns $r_{-} = -0.066706494$ and $r_{+} = 0.112553515$, source age-income bases Y_{-}, Y_{+}, income ages 18–65, and discounting at 3 percent to age 18. The corresponding tax rates are 0.003 and 0.199 for the early and later response blocks. This entry-indexed form is algebraically the same signed-batch construction as equation (39), with an outcome-specific response kernel. Here the event is entry and the factor two is retained schooling per induced entrant; completion-indexed kernels in the other schooling applications use their own source event. Negative q reverses the batch response and is not interpreted as negative numbers of individual entrants.

Let $d_2 = 1 + 1/1.03$, $B = 6700$, $s_0 = 0.352$, $m = 0.224$, $p = 2968.006431$, and $c = 6466.027832$. The source fixed private component is $w_0 = B s_0 d_2 = 4648.108738$, and the fixed public program component is $n_0 = B(s_0 + m) d_2 = 7605.996208$. Calculations retain the archived full precision of m. The entry-indexed

private-payment and public-education bundles are

$$P = \sum_{\tau} q(\tau) 1.03^{-(\tau-18)} [-2(B-p)], \qquad C = \sum_{\tau} q(\tau) 1.03^{-(\tau-18)} [(c-p)d_2]. \tag{53}$$

These retain the source's distinct private-benefit and public-cost duration conventions rather than forcing both into one generic tuition schedule. The account is $W = w_0 + E - T - P$ and $N = n_0 + C - T$. For a reference entry batch $q(18) = m$, the separate components are $E = 21163.754221$, $T = 4411.054305$, $P = -1671.933171$, and $C = 1544.291593$. They recover $W = 23072.741825$, $N = 4739.233496$, and MVPF 4.868454328. No terminal scaling aligns the result to the source ratio. Across candidate paths the original fixed program amount is retained; the exercise does not let alternative entry timing change statutory eligibility or grant duration independently.

The source marginal SEs are reserved for the separate 1,000 input repetitions. Each repetition retains the released attendance draw and its paired early and later return draws. The added age-23 coefficient receives the same standardized shock as age 28, with its own 0.102 SE. This imposes perfect cross-window dependence because the joint covariance was not recovered; it is not an author bootstrap of both ages. Source program components are reconstructed for each input repetition, while the external baseline, two-year conversion, shape restrictions, and response transport stay fixed. All input IDs, including negative education draws and unsuccessful searches, are retained. Appendix F reports conditional endpoint summaries rather than coverage guarantees.

D.8 Tennessee Promise: Matched Credential Paths and Earlier Windows

The Tennessee source uses 2000–2024 ACS data and overlapping age groups among high school graduates. Exposure is the share of a cohort eligible at age 17, with geographic assignment based on state of residence one year earlier. The regressions include state and survey-year fixed effects. The reconstruction uses fixed equal weights across the 1992–1998 birth cohorts, which reach age 17 during the county-to-statewide expansion in 2009–2015. These are the model cohorts, not a claim that the source regressions use only those birth years. Each outcome's Tennessee age shape is fitted to the corresponding external population and rescaled so that its age-23–24 mean equals the source control mean: 0.685 for some college or higher, 0.380 for associate's or higher, and 0.292 for bachelor's or higher. The treated curves share their outcome-specific age-17 reference levels. This reproduces selected population levels, not a fitted untreated DID trajectory; the control mean may itself combine treatment states, covariates, and cohorts differently from the regression contrast.

For $j \in \{C, A, B\}$, the retained source effects are $m_j \in \{0.035, 0.040, 0.027\}$. I impose

$$\frac{\Delta H_j(23) + \Delta H_j(24)}{2} = m_j. \tag{54}$$

Equal age weights approximate the unavailable regression operator. At every annual age, both reference and treated levels satisfy $H_B^s(a) \le H_A^s(a) \le H_C^s(a)$. Each credential has its own anchored parameter triple and state-centered support. The constituent shape pools are combined with a deterministic Sobol design, supplemented with combinations near earnings extremes and earlier-window fits. The main bank contains 97,863 complete joint candidates. Each reported welfare pair uses one such joint candidate; separate credential extrema are never combined as if they were one feasible path.

The additional late-completion restriction is relative to those who have not yet attained the outcome. Let $\bar{H}_{jc}$ be a donor cohort's mean level at ages 23–24. Define the historical completion fraction

$$R_j = \max_{c \in \{1983,\ldots,1987\}} \frac{K_{jc} - \bar{H}_{jc}}{1 - \bar{H}_{jc}}. \tag{55}$$

If necessary, the main allowance also includes the fitted target reference's own fraction. The candidate restriction is

$$K_j^1 \le \bar{H}_j^1 + (1 - \bar{H}_j^1) R_j. \tag{56}$$

The donor profiles use the same Tennessee high-school-graduate denominator. This rule avoids treating a statewide attainment share as a universal ceiling for a selected population. It still assumes that remaining completion among the treated can be bounded by the chosen donor experience, whose plateau is itself extrapolated. The main bachelor's ceiling is 0.4537. Table D1 gives the other quantities; Appendix E.9 separately varies this allowance and the parameter support, so the two restrictions are not changed together.

L3 compares predicted adjacent-age averages with all eighteen point estimates in Table 2, retaining the three latest-window effects exactly. The fifteen earlier coefficients enter equal-weight least squares in probability units. The source reports clustered SEs and uses policy permutations for inference; neither the SEs nor the graphical permutation intervals form acceptance bands in this fixed-input exercise. The best-fitting joint path has eighteen-input RMSE 0.003525 and maximum absolute residual 0.006673 in probability units. Table D2 records all targets, predictions, and residuals. The same ACS reference, cohort weights, logistic family, support, ordering restrictions, late-completion caps, and income account apply in L2 and L3. The retained L3 path is also supplied to the L2 bank.

The source welfare account applies Tennessee-specific education returns to project earnings and taxes. It assumes 2.3 years of Promise participation and five years to a bachelor's degree. Direct scholarship spending averages \$2,760 before discounting; the source reports total public cost before tax offsets of \$14,793. Tuition relief for students who would have enrolled without the program and projected after-tax earnings for induced students enter willingness to pay. The reported totals are \$72,837 for willingness to pay, $-\$10,053$ for net public cost, and \$24,845 for additional taxes. The authors qualify the use of average education returns for induced students and the public-expenditure assumptions. The reconstruction uses a participant-based account with the unit bridge and price-year convention stated below.

The author age-by-credential income table could not be recovered. Public ACS 2011–2021 data supply reconstructed mean wages for employed, out-of-school Tennessee residents under the documented prior-residence sample, in 2021 dollars. They are not author-supplied income inputs. Write $Y_j(a)$ for earnings in exclusive category $j \in \{SC, AA, BA\}$ and $Y_{HS}(a)$ for high-school earnings. The completion kernel is $-Y_{HS}(a)$ during d_j study years before completion and $Y_j(a) - Y_{HS}(a)$ from completion onward, with $(d_{SC}, d_{AA}, d_{BA}) = (2, 2, 4)$. It is zero outside these periods and before policy onset. This approximates zero earnings while studying; the source's 2.3-year Promise duration and five-year BA assumptions are not claimed to have been recovered as annual income paths.

Let $q_j(\tau)$ be signed increments of the exclusive effects $(\Delta C - \Delta A, \Delta A - \Delta B, \Delta B)$. The reconstructed account is

$$E = \frac{1}{p} \sum_{a=18}^{64} 1.03^{-(a-18)} \sum_j \sum_\tau K_j(a, \tau) q_j(\tau). \tag{57}$$

The SC+ path records some college or higher; BA+ is priced at BA earnings without a graduate premium. Education paths extend to 68 for the four-year pre-completion window. Negative exclusive increments are retained as differences between two credential distributions.

The unit bridge is explicit. With grant $g = 2{,}711$, private remainder $w_0 = 2{,}124.307692$, and entry effect 0.054, set $u = w_0/g$ and $p = 0.054/(1-u) = 0.2495243215$. Under the maintained interpretation, u is the uninduced participant share and p maps population effects into participant units. This is account-implied, not observed take-up. It does not select p to match earnings PV. The earnings scale is one. The source-rounded exclusive endpoint exercise gives \$59,881.13; the nested endpoint vector used in the path model gives \$69,261.65. Both differ from the reported implied \$95,557.69, and neither is rescaled to that total. Fixed source non-income amounts are assigned the 2021-dollar convention to match reconstructed wages; their exact original price vintage is unavailable. Consequently the L1–L2 comparison includes price and unit assumptions in addition to continuation.

The private remainder is obtained by removing the reference after-tax earnings contribution from total willingness to pay:

$$w_0 = 72{,}837 - 0.74\left(\frac{24{,}845}{0.26}\right) \simeq 2{,}124.31. \tag{58}$$

Thus, \$2,124.31 is the private transfer and tuition component left after the earnings contribution is removed from the reported total. It is recovered from the account, not an additional estimate of a policy effect. Adding the projected tax receipts back to net public cost gives

$$n_0 = -10{,}053 + 24{,}845 = 14{,}792. \tag{59}$$

This is the reconstructed pretax fiscal cost, containing scholarships and education spending. It differs by one dollar from the source's displayed \$14,793 total because the reported components do not sum exactly at their published precision. I retain the reported net-cost and tax totals for an internally consistent reconstruction.

The candidate account is therefore WTP $= 2{,}124.31 + 0.74E$ and $N = 14{,}792 - 0.26E$, using unrounded remainders in the calculations. Equivalently, it starts from the reported willingness to pay and net cost and replaces the private and fiscal shares of E^{ref} with those of E. Returning E^{ref} recovers the source totals. Fiscal break-even occurs at $E = n_0/0.26 = 56{,}892.31$; with positive willingness to pay, net cost at or below zero yields the self-financing classification. Grant duration, tuition, and public education resources remain in the fixed remainders. Their invariance across paths is a restriction on this comparison, not a prediction about how enrollment costs respond to each education continuation.

Among the L2 candidates, willingness to pay ranges from -48.72 to 152.50 thousand dollars and net cost from -38.04 to 32.66 thousand dollars. The L3 best fit gives willingness to pay of 86.44 thousand dollars and net cost of -14.83 thousand dollars. Its positive benefit and negative net cost imply self-financing. Figure 7 displays L2 attainment ranges and the L3 fitted path alongside H^0. Appendix E.9 refits this point criterion under alternative calibration and valuation assumptions. Pixel-width sensitivity is inapplicable to the point-estimate criterion.

D.9 Source-Input Repetitions

Table 4 reports 1,000 prespecified repetitions per application, holding calibration and path restrictions fixed. STBF uses constructed normal perturbations; Adams uses linked education and accounting draws. Tennessee

perturbs the eighteen point coefficients using displayed clustered SEs and refits the same constrained minimum-distance criterion in each repetition. This does not recover the source permutation distribution. Verified and unsuccessful draw counts are reported in Appendix F; no unsuccessful input draw is replaced. Appendix F defines the quantiles and explains the statistical interpretation, including Tennessee's SE convention.

D.10 STBF: The Author Credential–Experience Income Function

The source is the deposited main and appendix workbooks and their accompanying earnings programs for the 2012–2014 four-year strata (Angrist et al., 2022). The four pooled controlled effects at six years are $m_T = 0.3647777438$, $m_C = 0.00715823844$, $m_A = 0.05182238668$, and $m_B = 0.08133502305$. Their marginal SEs are 0.04131259769, 0.002961894264, 0.015011851676, and 0.015532993712. Time enrolled is measured in years, and the other inputs are cumulative credential shares. Each outcome has a separate anchored path, with fixed external weights over cohorts and graduation ages:

$$\mathcal{M}_o\{\Delta H_o\} = m_o, \qquad o \in \{T, C, A, B\}. \tag{60}$$

The 2012–2014 graduating cohorts receive equal weights. Graduation ages 17, 18, and 19 receive weights 0.065207, 0.575880, and 0.358913, placing the six-year inputs at ages 23–25 and their unaffected anchors at 16–18. These are external calibration weights, not verified trial entry-age frequencies. The three credential levels satisfy $H^s_C \geq H^s_A \geq H^s_B$ at every age. The time-enrolled effect uses the external schooling-shaped family as a timing index, not as an observed level of enrollment or completed schooling. All four same-horizon estimates enter L2; they do not create a second follow-up date or an L3 specification.

For each of four race–sex groups g, the source Poisson mean-input forecast can be written

$$F_g(a; \Delta p_g, \Delta T_g) = \exp\left[\alpha_g + (B_g + \beta_g' \Delta p_g(a))\mathbf{1}\{x_g(a) > 0\} + f_g(x_g(a))\right], \tag{61}$$

$$x_g(a) = \max\{a - 18 - \bar{T}_g - \Delta T_g(a), 0\}. \tag{62}$$

Here f_g is the retained source quartic with its original scaling, $\Delta p_g = (\Delta C_g - \Delta A_g, \Delta A_g - \Delta B_g, \Delta B_g)$ contains exclusive SC, AA, and BA changes, and $B_g = \beta_g' \bar{p}_g$. The forecast evaluates mean inputs; it is not an estimate of the expectation of this nonlinear function over an unobserved individual distribution. Group-specific coefficients, 3 percent discounting to age 18, income ages 18–65, and 2019-dollar conventions are retained. No Hendren and Sprung-Keyser return coefficient is substituted into this function.

The source's 3 and 5 percent income PVs identify the sufficient statistics $(\bar{T}_g, B_g)$ conditional on its coefficients and education effects. In source group order, the recovered pairs are $(1.90624931, 0.28136372)$, $(2.01871376, 0.32275122)$, $(1.56582623, 0.23536598)$, and $(1.62036888, 0.21935042)$. Individual baseline credential means remain underidentified. Their feasible affine sets are retained, and each candidate must permit one fixed baseline share vector per group that remains valid at all ages. No arbitrary feasibility witness is presented as an author mean. Conditional on the source control total of 1,924 and the exported aggregate PVs, integer weights are proportional to $(522, 798, 242, 362)$; this recovery assumption is recorded separately from directly supplied coefficients.

Pooled and subgroup controlled effects are not exactly aggregation-consistent. The main bridge transports each pooled timing index to group g according to $\Delta H_{go}(a) = m_{go}\Delta H_o(a)/m_o$, holding the source effect ratios fixed. Thus the pooled paths are timing indexes, not claims to equal the weighted subgroup means at every age. Reconstructing the original fixed effects recovers the source group PVs and the workbook aggregate

of \$21,129.2291. The published text instead reports \$21,150; both are retained, and no multiplicative correction is made. Current path income is $\sum_g w_g[F_g(a;\Delta p_g,\Delta T_g) - F_g(a;0,0)]$. WTP is $32{,}250 + 0.8E$ and net cost is $32{,}250 - 0.2E$.

The 1,000 conditional repetitions multiply all four marginal SEs by the same registered normal draw. This imposes perfect dependence rather than recovering an author joint bootstrap. Group coefficients, sufficient statistics, effect ratios, and external ACS calibration remain fixed. The complete eight-dimensional joint path search is repeated for each input draw.

E Full Policy Collection and Case Registry

This appendix documents the 38 policy calculations in Section 5: their source inputs, calibration, welfare comparisons, and ordering rule. Separate profiles describe the 35 cases outside the main applications. Appendix F treats input uncertainty; the replication materials supply full-precision observation operators and complete paths.

E.1 Cases, Source Families, and Calibration

Table E1 lists all 38 calculations. The collection retains 34 earlier calculations and adds algorithmic college placement, two further tuition-deduction contrasts, and the cannabis education account. Several calculations can come from one paper. The source-account cases represent 22 research families, with one further family for the author cannabis study. The grouping combines related Bulman–Hoxby versions and tax-schedule calculations, the spending and tuition calculations associated with Deming–Walters, and the Kalamazoo Promise research versions. The source-authors string is retained separately. A family-level witness count records whether at least one member supplies the specified paths; it is not an estimate of a policy-population frequency.

The geographic registry selects the source's treated population rather than combining it mechanically with control states. Birth cohorts and mixture weights remain fixed across candidates. Geographic matching improves comparability, but an ACS mean is not an identified untreated trajectory for a source study's selected population. The source-control status records that distinction, including Tennessee's high-school-graduate denominator and control-level calibration. Education conversion, age mapping, the anchored logistic continuation, empirical support, and transport of welfare prices remain maintained assumptions. Reproduction of a source coefficient or account is distinct from validation of those assumptions.

The accompanying replication materials record each moment's outcome, reconstruction, target, available standard error, and nonzero age–cohort weights, together with the comparison profiles, anchors, valuation coefficients, and sources. A converted coefficient is not relabeled as observed completed schooling. In particular, current enrollment and accumulated education are different outcomes; the source-specific bridge remains part of each reconstruction. Full-precision inputs permit the matching equation to be reproduced without inferring an age from a figure or replacing a window by its midpoint.

E.2 Education Inputs and Observation Windows

These profiles document the 35 calculations outside the three main applications in Appendices D.6, D.7, and D.8. Each identifies the original estimate and window, its conversion into a path input, and the welfare

comparison in Tables 3 and 4. Unless stated otherwise, L1 and its education conversion are from Hendren and Sprung-Keyser (2020) and the associated Policy Impacts calculation. Here ΔH measures schooling-equivalent years except for Early Head Start, where it measures BA+ attainment, and STBF, which jointly uses time enrolled and three credential outcomes. A probability change of 0.01 is one percentage point. Displayed inputs are rounded; calculations retain their recorded precision. Outcome conversion and age placement are separate assumptions: dividing credits by 24 does not determine their observation age. Table E1 reports comparison populations; Figures E1–E8 plot the corresponding H^0/H^1 ranges by policy.

L2 preserves its converted input exactly. Michigan, CUNY, and Social Security student benefits also admit exact L3 matches. The other ten L3 applications keep the L2 equality and fit additional point targets by constrained equal-weight minimum distance, without standard-error bands or precision weights. Their fitted values are conditional selections, not uniquely identified policy effects. L3 uses additional source windows or dynamics under the stated conversion; dividing a single observed enrollment effect across assumed schooling years does not qualify. Repetition summaries below use 1,000 paired inputs per layer and report the 2.5th percentile of minima and 97.5th percentile of maxima, conditional on verified paths and defined ratios. Original correlation groups are preserved and explicitly extended to added coefficients as described in Appendix F.2. These summaries are not validated confidence regions. Reported L1 intervals, where available, retain their original inferential meaning and may vary different inputs.

Florida Student Access Grant

Castleman and Long (2016) use a regression discontinuity design to estimate how Florida Student Access Grant eligibility affects enrollment, credits, and completion. Outcomes extend from the first semester through seven years after high school. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 7.42. The education input is the 4.366-credit effect on four-year cumulative credits in the original study's Table 4, column (5). Dividing by 24 gives 0.181917 schooling-equivalent years. With first exposure assigned to age 18 and the unaffected anchor at 17, L2 retains $\Delta H(22) = 0.181917$. L2 yields a searched positive-cost MVPF span of $[0.57, 22.32]$, crossing one; these endpoints bracket L1. L3 adds Table 4, columns (1)–(4): 0.355, 1.098, 2.666, and 3.847 credits. The same conversion gives fixed targets of 0.014792, 0.045750, 0.111083, and 0.160292 at assigned ages 18.5, 19, 20, and 21. These are modeled age placements of the source windows; the half-year target is evaluated on the continuous curve. Keeping L2 exact, the additional targets enter the constrained fit. L3 yields a positive-cost MVPF of 18.09, above one. This fitted value lies above L1; the reduction relative to L2 also reflects the fitting criterion. Figure E1, Florida access grant panel, shows age-65 gaps of $[-0.102, 1.067]$ under L2 and 0.720 years under L3; L2 permits eventual reversal. Paired input repetitions give L2 $[-14.48, +\infty]$; L3 $[0.53, +\infty]$. The reported L1 95 percent interval is $[0.69, +\infty]$.

District of Columbia Tuition Assistance Grant

Abraham and Clark (2006) examine whether the District of Columbia Tuition Assistance Grant increases college enrollment, comparing enrollment changes for DC graduates with those in comparison jurisdictions. The retained outcome is first-time freshman enrollment among recent high-school graduates, rather than subsequent completion. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 22.98. The original study's published Table 7a, Total row for all schools, final difference column, supplies the 0.089 enrollment effect used in the reference calculation. Its two-year duration assumption gives $2(0.089) = 0.178$ schooling-equivalent years. L2 places this converted total at age 20, imposing $\Delta H(20) = 0.178$ with an age-17 anchor.

Age 20 is an assumed endpoint of the duration conversion, not a directly observed schooling age for every student. L2 yields a searched positive-cost MVPF span of $[-1.60, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. No L3 is admitted: assigning 0.089 years to an earlier age would split the same assumed duration rather than add a distinct follow-up estimate. The source's enrollment comparisons therefore establish the input, but do not determine its subsequent persistence. Figure E1, DC tuition assistance panel, shows the L2 paths; the age-65 gap is $[-0.364, 1.268]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-3.93, +\infty]$. The reported L1 95 percent interval is $[1.12, +\infty]$.

STBF Scholarship

Angrist et al. (2022) study randomized Susan Thompson Buffett Foundation scholarship offers in Nebraska. The retained 2012–2014 four-year-college strata supply four six-year effects: time enrolled 0.3648 years, any-college attainment 0.7158 percentage points, AA+ 5.1822 points, and BA+ 8.1335 points. L2 matches all four jointly, using the fixed external cohort and entry-age mixture documented in Appendix D.10. Four outcomes at the same horizon do not create a second observation age, so no L3 is admitted. Time enrolled remains a separate timing index rather than completed schooling.

The author credential–experience income model, its four race–sex functions, recovered sufficient statistics, subgroup effect-ratio bridge, 20 percent tax rate, and $32,250 transfer are retained. Its source workbook earnings NPV is $21,129.23, compared with the printed $21,150; the printed L1 account gives WTP $49,170, net cost $28,020, and MVPF 1.75. This move to the additional-case collection changes neither the estimates nor the author model. Current L2 earnings NPVs range from 7.60 to 60.60 thousand 2019 dollars, with MVPFs of 1.25–4.01. Credential levels are nested, and source mean-input feasibility is checked over each complete joint path.

Figure E8 displays STBF's comparison levels H^0 and treated levels H^1, with the same level-range convention used for other applications. Its time-enrolled index uses the external schooling-shaped timing calibration; that plotted level is not the author's control mean of time enrolled. The author model retains its separate source means. Shaded edges can combine different complete candidates and are not confidence bands. The 1,000 constructed repetitions use a common standardized education shock across the four outcomes; Appendix D.10 records this imposed dependence and the unchanged income inputs.

Wisconsin Scholars Grant

Goldrick-Rab et al. (2016) randomize Wisconsin Scholars Grants across three entering cohorts and report semester credits, cumulative credits, and completion. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 1.43. The reference input combines Table 5's 0.9-credit third-year effect for cohort 1 with its pooled 2.1-credit effect for cohort 2 after three years and cohort 3 after two years. Weights of 0.314570 and 0.685430 and division by 24 yield 0.071772 schooling years. With age-17 anchors, L2 imposes $0.314570\Delta H_{1990}(21) + 0.342715\Delta H_{1991}(21) + 0.342715\Delta H_{1992}(20) = 0.071772$. Equal splitting of the pooled cohorts and entry at 18 are calibration assumptions; cohort 3's unobserved third year receives no weight. L2 yields a searched positive-cost MVPF span of $[-1.94, 7.58]$, crossing zero and one; these endpoints bracket L1. L3 separately retains both cumulative estimates and converts Table 5's annual credit contributions: $(0.5, 0.2, 0.1)/24$ for cohort 1, $(0.4, 1.0)/24$ for pooled cohorts 2–3, and $0.5/24$ for cohort 2's third year. The first-year contribution restricts $\Delta H(19) - \Delta H(17)$; later contributions restrict the next annual

differences. Reported semester and cumulative coefficients do not satisfy exact adding-up identities, so the fit also reflects this source aggregation. Keeping L2 exact, the additional targets enter the constrained fit. L3 yields a positive-cost MVPF of 1.96, above one. This fitted value lies above L1; the reduction relative to L2 also reflects the fitting criterion. Figure E1, Wisconsin Scholars Grant panel, shows age-65 gaps of $[-0.543, 1.011]$ under L2 and 0.145 years under L3; L2 permits eventual reversal. Paired input repetitions give L2 $[-2.84, +\infty]$; L3 $[-0.04, 6.29]$. The reported L1 95 percent interval is $[0.96, 2.18]$.

Kalamazoo Promise

Bartik et al. (2021) estimate how Kalamazoo Promise eligibility affects enrollment, attempted credits, and credentials. Their credit results cover two, three, and four years after high school; later degree outcomes use different follow-up samples. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 1.93. The original study's Appendix Table A.7, Panel C, column (1), supplies a fourth-year credit effect of 6.562, retained as 6.56 in the reference calculation. Dividing by 24 gives 0.273333 schooling-equivalent years. Assuming entry at 18, L2 imposes $\Delta H(22) = 0.273333$ and anchors the paths at 17. L2 yields a searched positive-cost MVPF span of $[0.65, 4.53]$, crossing one; these endpoints bracket L1. L3 adds the same table's two- and three-year credit effects, 3.236 and 4.309. Division by 24 gives targets $\Delta H(20) = 0.134833$ and $\Delta H(21) = 0.179542$, while the fourth-year L2 input remains exact. Treating these follow-ups as a common response requires comparability across the available cohorts. Enrollment and credentials are not additionally counted as attempted credits. Keeping L2 exact, the additional targets enter the constrained fit. L3 yields a positive-cost MVPF of 3.77, above one. This fitted value lies above L1; the reduction relative to L2 also reflects the fitting criterion. Figure E1, Kalamazoo Promise panel, shows age-65 gaps of $[-0.118, 1.059]$ under L2 and 0.834 years under L3; L2 permits eventual reversal. Paired input repetitions give L2 $[-0.65, 19.65]$; L3 $[0.78, 15.51]$. The reported L1 95 percent interval is $[0.97, 5.61]$.

Michigan community-college tuition

Acton (2021) estimates responses to a $1,000 reduction in Michigan community-college tuition. Enrollment within one year is observed for 2009–2011 graduates; credentials are measured in 2017, six to eight years after high school. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 29.46. The reference uses the original study's Table 4, Panel B, columns (1)–(4), enrollment effects $(0.036, -0.021, -0.004, -0.003)$, and Table 5, columns (4)–(6), credential effects $(-0.002, 0.003, 0.011)$. Schooling weights $(1, 1, 1, 2, 0.5, 1, 2)$ give 0.029 years. L2 places this total at representative age 25, $\Delta H(25) = 0.029$, with anchor 17. L2 yields a searched positive-cost MVPF span of $[-4.44, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. L3 separately uses the early enrollment contribution, $0.036 - 0.021 - 0.004 + 2(-0.003) = 0.005$. It imposes $[\Delta H(18) + \Delta H(19)]/2 = 0.005$, preserving L2. The equal-weight age window is an additional temporal allocation of the converted enrollment input, not a source estimate of completed schooling at either age. It uses an earlier source window rather than inventing a second observation from the same enrollment result. L3 yields a searched positive-cost MVPF span of $[9.03, +\infty]$ and separate self-financing paths, entirely above one among the paths found. Figure E1, Michigan CC tuition panel, shows age-65 gaps of $[-0.224, 0.568]$ under L2 and $[0.014, 0.063]$ years under L3; L2 permits eventual reversal. Paired input repetitions give L2 $[-24.02, +\infty]$; L3 $[-3.74, +\infty]$. The reported L1 95 percent interval is $[-2.33, +\infty]$.

CUNY Pell grants

Marx and Turner (2018) estimate how an additional $1,000 of Pell aid affects borrowing and academic attainment among CUNY entrants in 2007–2011. Their first-year-student results include current-year and three-year credit outcomes. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 1.39. The original study's Table 8, Panel C, column (2), gives the 0.223-credit three-year effect retained for lifetime valuation. Dividing by 24 gives 0.009292 schooling-equivalent years. With entry assigned to 18 and anchor 17, L2 imposes $\Delta H(21) = 0.009292$. L2 yields a searched positive-cost MVPF span of $[-\infty, 14.28]$, crossing zero and one; these endpoints bracket L1. L3 adds the first-year entrants' current-year effect, 0.212 credits, as $\Delta H(19) = 0.212/24 = 0.008833$. Both converted points are matched exactly. The first- and third-year regressions have 38,100 and 32,271 observations, so this construction assumes comparability of their available samples. The separate 0.492-credit estimate for returning students is excluded because it concerns a different risk set. L3 yields a searched positive-cost MVPF span of $[-3.17, 2.34]$, crossing zero and one. Figure E2, CUNY Pell grants panel, shows age-65 gaps of $[-0.701, 1.047]$ under L2 and $[-0.102, 0.040]$ years under L3; L2 permits eventual reversal. Paired input repetitions give L2 $[-\infty, +\infty]$; L3 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-2.95, 12.88]$.

Ohio Pell grants

Bettinger (2004) examines whether Pell aid prevents withdrawal between the 1999–2000 and 2000–2001 academic years in Ohio. The retained estimate concerns students already enrolled in four-year colleges. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 2.49. The original study's Table 5.3, column (4), reports that $1,000 in imputed Pell aid reduces withdrawal by 0.064. The reference reverses the sign and assigns one additional schooling year to persistence, obtaining 0.064 expected schooling years. L2 imposes $\Delta H(20) = 0.064$. The modeled first affected year begins at 19, so age 18 is the unaffected anchor. This timing reflects a persistence intervention for existing college students, rather than an initial-entry intervention. L2 yields a searched positive-cost MVPF span of $[-11.75, 7.53]$, crossing zero and one; these endpoints bracket L1. No L3 is admitted. Other columns change the sample or specification for the same transition and do not provide a later attainment observation for these students. Consequently, the positive short-window persistence effect leaves both its duration and later catch-up to the maintained path restrictions. Figure E2, Ohio Pell grants panel, shows the L2 paths; the age-65 gap is $[-0.284, 1.070]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[0.82, 5.54]$.

Tennessee HOPE scholarship

Bruce and Carruthers (2014) study Tennessee HOPE eligibility and post-high-school college choice among 2006–2009 graduates. Their discontinuity comparison observes enrollment choices rather than a common later degree-completion endpoint. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 1.86. The original study's Table 3 reports a -0.022 effect on two-year enrollment in the retained low-GPA eligibility comparison. The reference uses a one-year schooling contrast for the sector response and reverses its sign, giving 0.022 schooling-equivalent years. L2 imposes $\Delta H(20) = 0.022$, with first exposure assigned to 18 and anchor 17. The assigned endpoint locates the reference duration conversion; it is not an observed age common to all graduates. L2 yields a searched positive-cost MVPF span of $[-75.64, 21.66]$, crossing zero and one; these endpoints bracket L1. No additional comparable attainment window is admitted to L3. Sector-specific

choices and robustness specifications are not later observations, and the requirement to enroll within 16 months is an eligibility rule, not a common follow-up period. L2 therefore examines continuations of the retained sector-switching input without rescaling it into a recipient effect. Figure E2, Tennessee HOPE panel, shows the L2 paths; the age-65 gap is $[-0.684, 1.094]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[0.92, 5.08]$.

Tennessee Pell eligibility

Carruthers and Welch (2019) estimate initial-fall college-entry responses at the Pell eligibility cutoff among 2006–2009 Tennessee graduates. The original study's Table 4, column (1), distinguishes enrollment by sector and location. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 0.85. The retained four-year probability effects sum to $P_4 = -0.006236$. Two-year and grouped for-profit effects sum to $P_2 = 0.005190$. The reference assigns two schooling years to four-year entry and one to two-year entry, giving $2P_4 + P_2 = -0.007282$. L2 matches this total in the second post-high-school year, at ages 19–21 for graduation ages 17–19. Their weights are 0.065207, 0.575880, and 0.358913; graduation cohorts receive equal weights. Each component is anchored one year before graduation, at 16, 17, or 18. L2 yields a searched positive-cost MVPF span of $[-\infty, 2.70]$, crossing zero and one; these endpoints bracket L1. No L3 is admitted. The sum $P_4 + P_2 = -0.001046$ uses the same initial-entry coefficients; imposing it at an earlier age would allocate the assumed duration rather than add an observed window. The age and cohort weights are calibration assumptions, not additional causal estimates. Figure E2, Tennessee Pell eligibility panel, shows the L2 paths; the age-65 gap is $[-0.675, 1.034]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-1.59, 3.57]$.

Georgia HOPE scholarship

Cornwell et al. (2006) estimate Georgia HOPE effects on freshman enrollment in a 1988–1997 state-year panel. The outcome compares successive entering classes rather than following one class through completion. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 4.00. The published study's Table 3, SREB-control columns (4), (6), and (10), supplies log effects 0.086, 0.132, and −0.045 for public four-year, private four-year, and public two-year enrollment. Table 4, column (5), contributes 560 fewer out-of-state entrants. The reference applies $\exp(\beta) - 1$, sector counts, and recipient normalization, obtaining four- and two-year probability responses of 0.055988 and −0.014674. The duration conversion gives $2(0.055988) - 0.014674 = 0.097302$. L2 imposes this schooling gap at age 20, equally weighting 1975–1979 cohorts with anchor 17. L2 yields a searched positive-cost MVPF span of $[-3.19, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. No L3 is admitted: a first-year value of 0.041314 would recombine the same sector estimates, not introduce another attainment window. The continuation comparison therefore remains conditional on both the reference enrollment conversion and its representative age placement. Figure E2, Georgia HOPE panel, shows the L2 paths; the age-65 gap is $[-0.362, 1.155]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-4.11, +\infty]$. The reported L1 95 percent interval is $[0.46, 22.38]$.

College appropriations: spending

Deming and Walters (2017) estimate public-college enrollment responses to institutional spending using 1990–2013 institution-year data. Their current-year and next-three-year coefficients describe policy time, not individual follow-up. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 4.00. The retained elasticities are 0.304, 0.796, 0.845, and 0.830, corresponding to the original study's spending panel across four lag columns (Table 3, Panel A, in the available draft; Table A6 in the reference version). The reference divides their sum by 2.02 and multiplies by 1,000/29,856.23, giving 0.046013 schooling years. Assigning the four contributions to ages 18–21, L2 imposes $\Delta H(21) = 0.046013$ with anchor 17. L2 yields a searched positive-cost MVPF span of $[-5.43, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. L3 fits the separate annual targets $(0.005041, 0.013199, 0.014011, 0.013762)$ for $\Delta H(a) - \Delta H(a-1)$ at $a = 18, \ldots, 21$, preserving their exact total. This requires transporting institutional policy-time responses to a representative age path. All coefficients already enter L1's sum; L3 adds their sequence, rather than previously unused coefficient values. Keeping L2 exact, the additional targets enter the constrained fit. L3 yields a positive-cost MVPF of 16.52, above one. This fitted value lies above L1; the reduction relative to L2 also reflects the fitting criterion. Figure E2, College appropriations: spending panel, shows age-65 gaps of $[-0.504, 1.158]$ under L2 and 0.120 years under L3; L2 permits eventual reversal. Paired input repetitions give L2 $[-1{,}925.69, +\infty]$; L3 $[1.23, +\infty]$. The reported L1 95 percent interval is $[1.25, 20.44]$.

Early Head Start

Hess (2025) estimates how access to Early Head Start affects later educational attainment. The source pools adult outcomes in linked Census–ACS data and distinguishes population intent-to-treat effects from participant effects. Hess calculates an MVPF (L1) of 0.35. The original study's Table 1.2, full-exposure All/F row and College Grad ATT column, supplies a 0.168 participant effect on bachelor's attainment; the corresponding population effect is 0.012. The reconstruction uses 0.168 directly as a probability gap, without converting it into schooling years. L2 imposes a weighted $\Delta H_c(22) = 0.168$ across 1989–1999 cohorts. Age 22 is a representative endpoint, not a common source follow-up. Because exposure occurs in childhood, no college-age equality anchor is imposed. Both BA profiles remain zero through 19 under the credential model's structural restriction. L2 yields a searched positive-cost MVPF span of $[-1.37, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. No L3 is admitted: partial-exposure, sex-specific, and some-college estimates change the target or outcome. The reconstructed public ACS earnings and funded-slot cost account is an extension of the source valuation, with an unresolved original price-year comparison. Figure E7, Early Head Start panel, shows the L2 paths; the age-65 gap is $[-0.059, 0.271]$ probability units, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-2.20, +\infty]$.

Texas community-college tuition

Denning (2017) estimates how Texas community-college district access and lower tuition affect 1994–2005 high-school graduates. Enrollment is observed in the graduation year and each of the following six years; bachelor's completion is observed after six and eight years. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 349.51. The reference sums the original study's Table 6 graduation-year and Table 7 subsequent enrollment effects across community colleges and universities, giving 0.209450 schooling years. Online Appendix Table

A4 adds $2(0.0078 - 0.0033) = 0.009000$ for later bachelor's completion. L2 retains $\Delta H(26) = 0.218450$, with graduation assigned to 18 and anchor 17. L2 yields a searched positive-cost MVPF span of $[9.70, +\infty]$ and separate self-financing paths, entirely above one among the paths found; these endpoints bracket L1. L3 separately fits cumulative targets $(0.031550, 0.077550, 0.112750, 0.145650, 0.172150, 0.187250, 0.209450)$ at ages 18–24, keeping the age-26 total exact. These sums preserve the seven reported annual enrollment responses under the reference conversion. The later completion difference supplies the final increment; no year-seven enrollment response is imputed. Keeping L2 exact, the additional targets enter the constrained fit. L3 selects a self-financing path. Figure E3, Texas CC tuition panel, shows age-65 gaps of $[-0.004, 0.796]$ under L2 and 0.223 years under L3; L2 permits eventual reversal. Paired input repetitions give L2 $[-1.81, +\infty]$; L3 $[1.14, +\infty]$. The reported L1 95 percent interval is $[0.93, +\infty]$.

Adult Pell Grant introduction

Seftor and Turner (2002) estimate how the introduction of federal student aid changes current enrollment among adults ages 22–35. Their repeated-cross-section comparison does not follow individual entrants through later attainment. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 2.18. The original study's Table 2, final panel, columns (1) and (3), supplies enrollment effects of 0.015 for men and 0.014 for women. Weights proportional to 47,407 and 62,405 observations yield 0.0144317. The reference's two-year duration gives 0.028863421 schooling years. L2 imposes $\Delta H(30) = 0.028863421$ with anchor 27, representing exposure beginning at 28. The source's broad adult age range does not identify these specific model ages; they locate the assumed acquisition period in the representative adult path. L2 yields a searched positive-cost MVPF span of $[1.95, 4.63]$, entirely above one among the paths found; these endpoints bracket L1. No L3 is admitted. Sex and age contrasts concern different populations, and splitting the two-year duration would add a timing assumption without another follow-up. The reconstruction consequently holds the converted adult-enrollment input fixed while allowing its later education gap to vary. Figure E3, Adult Pell introduction panel, shows the L2 paths; the age-65 gap is $[-0.011, 0.206]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-5.15, 102.12]$. The reported L1 95 percent interval is $[0.71, 6.11]$.

College appropriations: tuition

Deming and Walters (2017) also estimate enrollment responses to tuition changes in institution-year data from 1990–2013. Four coefficients cover the current policy year and three subsequent years. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 1.02. The original tuition panel's four lag coefficients are −0.016, 0.066, 0.031, and −0.073 (the available draft's Table 3; the reference version's Table A6). The reference divides their sum by 2.02 and multiplies by 1,000/8,599.86, yielding 0.000460519 years under its released tuition-cut sign convention. L2 imposes $\Delta H(21) = 0.000460519$, with anchor 17 and modeled acquisition ages 18–21. L2 yields a searched positive-cost MVPF span of $[-8.86, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. L3 fits annual changes at those four ages to $(-0.000921, 0.003799, 0.001785, -0.004202)$ while retaining the exact total. The source coefficients already contribute to L1; their separate sequence supplies L3's extra restriction. Mapping institutional policy time into individual ages is additional to the inherited schooling conversion, and does not establish a longitudinal treatment effect for a fixed entering cohort. Keeping L2 exact, the additional targets enter the constrained fit. L3 yields a positive-cost MVPF of −1.97, implying beneficiary losses. This fitted value lies

below L1; the reduction relative to L2 also reflects the fitting criterion. Figure E3, College appropriations: tuition panel, shows age-65 gaps of $[-0.573, 1.052]$ under L2 and -0.091 years under L3; L2 permits eventual reversal. Paired input repetitions give L2 $[-\infty, +\infty]$; L3 $[-4.48, 11.82]$. The reported L1 95 percent interval is $[-1.11, 5.13]$.

HOPE tax credit

Turner (2011) estimates tax-aid effects on first-year full-time enrollment and second-year continuation among college-age youth. The second response concerns students who entered their first year in the preceding academic year. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 12.58. The original study's Table 4, columns (1)–(2), Subsidy row, provides probability responses of 0.00235 and 0.00210 per \$100. The reference combines a \$1,104 subsidy with one year for entry and 1.5 years for continuation: $(1{,}104/100)[0.00235 + 1.5(0.00210)] = 0.060720$. With anchor 17, L2 imposes $\Delta H(20.5) = 0.060720$. The fractional endpoint represents the assumed duration and is evaluated continuously. L2 yields a searched positive-cost MVPF span of $[-5.09, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. L3 adds $\Delta H(19) = 0.025944$ and $\Delta H(20) = 0.049128$. These respectively allocate the first-year contribution and one additional year of the continuation contribution; the remaining half-year completes L2's total. This construction preserves the reference conversion while adding an explicit age allocation and a bridge between the two source risk sets. It does not claim that the source directly estimates three schooling stocks. Keeping L2 exact, the additional targets enter the constrained fit. L3 selects a self-financing path. Figure E3, HOPE tax credit panel, shows age-65 gaps of $[-0.351, 1.020]$ under L2 and 0.441 years under L3; L2 permits eventual reversal. Paired input repetitions give L2 $[-97.46, +\infty]$; L3 $[-0.04, +\infty]$. The reported L1 95 percent interval is $[-29.15, +\infty]$.

HOPE credit: independent single filers, phase-out start

Bulman and Hoxby (2015) estimate the HOPE attendance response at the phase-out start for independent single filers. The observation is current attendance among ages 19–20 in 2007; it is not a later follow-up of one entering class. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 18.86. The original study's Table 14, relevant filer row and first Attendance column, supplies the signed slope coefficient 3×10^{-6} per income-margin dollar. Its matching Table 13 credit first stage is 0.1156229. A fixed schedule displacement normalized to one point-estimate credit dollar and the reference two-year duration give $2\beta/\varphi = 0.000051893$ schooling-equivalent years. With anchor 17, L2 imposes $\{[\Delta H_{1987}(21)-\Delta H_{1987}(19)]+[\Delta H_{1988}(20)-\Delta H_{1988}(18)]\}/2 = 0.000051893$. These assigned acquisition windows implement the duration conversion; they are distinct from the survey-age window. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. No L3 is admitted: the source reports one attendance window for this contrast, and splitting its assumed duration would not add an observation. Appendix E.4.1 separates the coefficient, fixed policy unit, and dollar account. Figure E3, HOPE: independent single, lower kink panel, shows the L2 paths; the age-65 gap is $[-0.475, 0.995]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-3.03, +\infty]$.

HOPE credit: dependent single filers, phase-out start

Bulman and Hoxby (2015) estimate the HOPE attendance response at the phase-out start for dependent single filers. The observation is current attendance among ages 19–20 in 2007; it is not a later follow-up of one entering class. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of −1.64. The original study's Table 14, relevant filer row and first Attendance column, supplies the signed slope coefficient -5×10^{-6} per income-margin dollar. Its matching Table 13 credit first stage is 0.1386706. A fixed schedule displacement normalized to one point-estimate credit dollar and the reference two-year duration give $2\beta/\varphi = -0.000072113$ schooling-equivalent years. With anchor 17, L2 imposes $\{[\Delta H_{1987}(21) - \Delta H_{1987}(19)] + [\Delta H_{1988}(20) - \Delta H_{1988}(18)]\}/2 = -0.000072113$. These assigned acquisition windows implement the duration conversion; they are distinct from the survey-age window. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. No L3 is admitted: the source reports one attendance window for this contrast, and splitting its assumed duration would not add an observation. Appendix E.4.1 separates the coefficient, fixed policy unit, and dollar account. Figure E3, HOPE: dependent single, lower kink panel, shows the L2 paths; the age-65 gap is $[-0.475, 0.995]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-6.45, +\infty]$.

HOPE credit: dependent single filers, phase-out end

Bulman and Hoxby (2015) estimate the HOPE attendance response at the phase-out end for dependent single filers. The observation is current attendance among ages 19–20 in 2007; it is not a later follow-up of one entering class. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 11.83. The original study's Table 14, relevant filer row and first Attendance column, supplies the signed slope coefficient 6×10^{-6} per income-margin dollar. Its matching Table 13 credit first stage is 0.1441341. A fixed schedule displacement normalized to one point-estimate credit dollar and the reference two-year duration give $2\beta/\varphi = 0.000083256$ schooling-equivalent years. With anchor 17, L2 imposes $\{[\Delta H_{1987}(21) - \Delta H_{1987}(19)] + [\Delta H_{1988}(20) - \Delta H_{1988}(18)]\}/2 = 0.000083256$. These assigned acquisition windows implement the duration conversion; they are distinct from the survey-age window. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. No L3 is admitted: the source reports one attendance window for this contrast, and splitting its assumed duration would not add an observation. Appendix E.4.1 separates the coefficient, fixed policy unit, and dollar account. Figure E4, HOPE: dependent single, upper kink panel, shows the L2 paths; the age-65 gap is $[-0.474, 0.995]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-5.36, +\infty]$.

HOPE credit: joint filers, phase-out start

Bulman and Hoxby (2015) estimate the HOPE attendance response at the phase-out start for joint filers. The observation is current attendance among ages 19–20 in 2007; it is not a later follow-up of one entering class. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of $+\infty$. The original study's Table 14, relevant filer row and first Attendance column, supplies the signed slope coefficient 12×10^{-6} per income-margin dollar. Its matching Table 13 credit first stage is 0.0731243. A fixed schedule displacement normalized to one point-estimate credit dollar and the reference two-year duration give $2\beta/\varphi = 0.000328208$ schooling-equivalent years.

With anchor 17, L2 imposes $\{[\Delta H_{1987}(21) - \Delta H_{1987}(19)] + [\Delta H_{1988}(20) - \Delta H_{1988}(18)]\}/2 = 0.000328208$. These assigned acquisition windows implement the duration conversion; they are distinct from the survey-age window. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; self-financing is also the L1 classification. No L3 is admitted: the source reports one attendance window for this contrast, and splitting its assumed duration would not add an observation. Appendix E.4.1 separates the coefficient, fixed policy unit, and dollar account. Figure E4, HOPE: joint, lower kink panel, shows the L2 paths; the age-65 gap is $[-0.474, 0.996]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-3.51, +\infty]$.

HOPE credit: joint filers, phase-out end

Bulman and Hoxby (2015) estimate the HOPE attendance response at the phase-out end for joint filers. The observation is current attendance among ages 19–20 in 2007; it is not a later follow-up of one entering class. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 2.37. The original study's Table 14, relevant filer row and first Attendance column, supplies the signed slope coefficient 4×10^{-6} per income-margin dollar. Its matching Table 13 credit first stage is 0.0638205. A fixed schedule displacement normalized to one point-estimate credit dollar and the reference two-year duration give $2\beta/\varphi = 0.000125352$ schooling-equivalent years. With anchor 17, L2 imposes $\{[\Delta H_{1987}(21) - \Delta H_{1987}(19)] + [\Delta H_{1988}(20) - \Delta H_{1988}(18)]\}/2 = 0.000125352$. These assigned acquisition windows implement the duration conversion; they are distinct from the survey-age window. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. No L3 is admitted: the source reports one attendance window for this contrast, and splitting its assumed duration would not add an observation. Appendix E.4.1 separates the coefficient, fixed policy unit, and dollar account. Figure E4, HOPE: joint, upper kink panel, shows the L2 paths; the age-65 gap is $[-0.474, 0.996]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-2.22, +\infty]$.

Tuition deduction: single filers, phase-start

Hoxby and Bulman (2016) estimate attendance responses around federal tuition-deduction thresholds. This contrast concerns single filers at the deduction's start. The retained outcome is attendance in the first full tax year after expected on-time high-school graduation. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of $+\infty$. The original study's Table 5, column (5), baseline specification, supplies the probability effect 0.005. The reference's two-year enrollment conversion gives 0.010 schooling-equivalent years. L2 imposes $[\Delta H(18) + \Delta H(19)]/2 = 0.010$ with anchor 17. Equal weights at ages 18 and 19 are an additional age mapping of the tax-year window; they are neither recovered source weights nor two independently estimated schooling effects. L2 yields a searched positive-cost MVPF span of $[-\infty, -7.40]$ and separate self-financing paths, entirely negative among the positive-cost paths; self-financing is also the L1 classification. Table 7 contains additional attendance windows around expected graduation. They are not admitted as nested L3 because their specification and population have not been reconciled with the retained Table 5 input under a common schooling conversion. The absence of L3 therefore does not mean the paper lacks dynamic results. Figure E4, Tuition deduction: single, lower threshold panel, shows the L2 paths; the age-65 gap is $[-0.832, 1.554]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-\infty, +\infty]$.

Tuition deduction: joint filers, phase-start

Hoxby and Bulman (2016) estimate attendance responses around federal tuition-deduction thresholds. This contrast concerns joint filers at the deduction's start. The retained outcome is attendance in the first full tax year after expected on-time high-school graduation. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of −0.02. The original study's Table 5, column (2), baseline specification, supplies the probability effect −0.003. The reference's two-year enrollment conversion gives −0.006 schooling-equivalent years. L2 imposes $[\Delta H(18) + \Delta H(19)]/2 = -0.006$ with anchor 17. Equal weights at ages 18 and 19 are an additional age mapping of the tax-year window; they are neither recovered source weights nor two independently estimated schooling effects. L2 yields a searched positive-cost MVPF span of $[-4.35, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. Table 7 contains additional attendance windows around expected graduation. They are not admitted as nested L3 because their specification and population have not been reconciled with the retained Table 5 input under a common schooling conversion. The absence of L3 therefore does not mean the paper lacks dynamic results. Figure E4, Tuition deduction: joint, lower threshold panel, shows the L2 paths; the age-65 gap is $[-0.877, 1.470]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-4.67, +\infty]$. The reported L1 95 percent interval is $[-2.25, 5.52]$.

HOPE and Lifetime Learning tax credits

Long (2004) studies the introduction of HOPE and Lifetime Learning credits using repeated cross sections. The retained result concerns current enrollment among respondents ages 18–24, rather than cumulative attainment at a single age. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of −8.81. The original study's Table 3.11, Panel A, column (2), reports an odds ratio of 0.9342; Table 3.10 supplies the baseline enrollment rate of 0.463. The reference approximation $(0.9342 - 1)0.463$ and a two-year duration give −0.060930810 schooling years. This is the inherited approximation, not the exact odds-to-probability transformation. L2 imposes $\Delta H(25) = -0.060930810$ with an age-17 anchor. Age 25 aligns the converted schooling input with the start of the reference account's long-run earnings returns. This is an assumed age mapping, not a directly observed age-25 schooling effect; subsequent attainment remains free to change. L2 yields a searched positive-cost MVPF span of $[-\infty, -9.59]$ and separate self-financing paths, entirely negative among the positive-cost paths; its positive-cost values fall below L1. No L3 is admitted. The age-18–19 subsample has a separate odds ratio, but its corresponding baseline enrollment probability has not been established. Reusing the broader sample's baseline would introduce an unverified conversion, so that subgroup is not turned into an additional schooling target. Figure E4, HOPE and Lifetime Learning credits panel, shows the L2 paths; the age-65 gap is $[-0.362, 0.495]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-\infty, +\infty]$.

AOTC: dependent single filers, phase-out end

Bulman and Hoxby (2015) estimate the AOTC attendance response at the phase-out end for dependent single filers. The observation is current attendance among ages 19–20 in 2011; it is not a later follow-up of one entering class. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of −0.02. The original study's Table 14, relevant filer row and first Attendance column, supplies the signed slope coefficient -2×10^{-6} per income-margin dollar. Its matching Table 13 credit first stage is 0.2164388. A fixed schedule displacement

normalized to one point-estimate credit dollar and the reference two-year duration give $2\beta/\varphi = -0.000018481$ schooling-equivalent years. With anchor 17, L2 imposes $[\Delta H_{1991}(20) + \Delta H_{1992}(19)]/2 = -0.000018481$. Equal age–cohort weights and placement of the duration-equivalent input at the observed ages are explicit reconstruction assumptions. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. No L3 is admitted: the source reports one attendance window for this contrast, and splitting its assumed duration would not add an observation. Appendix E.4.1 separates the coefficient, fixed policy unit, and dollar account. Figure E5, AOTC: dependent single, upper kink panel, shows the L2 paths; the age-65 gap is $[-0.805, 1.433]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-2.25, +\infty]$.

AOTC: dependent single filers, phase-out start

Bulman and Hoxby (2015) estimate the AOTC attendance response at the phase-out start for dependent single filers. The observation is current attendance among ages 19–20 in 2011; it is not a later follow-up of one entering class. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of $+\infty$. The original study's Table 14, relevant filer row and first Attendance column, supplies the signed slope coefficient 43×10^{-6} per income-margin dollar. Its matching Table 13 credit first stage is 0.2019312. A fixed schedule displacement normalized to one point-estimate credit dollar and the reference two-year duration give $2\beta/\varphi = 0.000425888$ schooling-equivalent years. With anchor 17, L2 imposes $[\Delta H_{1991}(20) + \Delta H_{1992}(19)]/2 = 0.000425888$. Equal age–cohort weights and placement of the duration-equivalent input at the observed ages are explicit reconstruction assumptions. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; self-financing is also the L1 classification. No L3 is admitted: the source reports one attendance window for this contrast, and splitting its assumed duration would not add an observation. Appendix E.4.1 separates the coefficient, fixed policy unit, and dollar account. Figure E5, AOTC: dependent single, lower kink panel, shows the L2 paths; the age-65 gap is $[-0.804, 1.434]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-0.84, +\infty]$.

AOTC: joint filers, phase-out end

Bulman and Hoxby (2015) estimate the AOTC attendance response at the phase-out end for joint filers. The observation is current attendance among ages 19–20 in 2011; it is not a later follow-up of one entering class. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of -1.77. The original study's Table 14, relevant filer row and first Attendance column, supplies the signed slope coefficient -7×10^{-6} per income-margin dollar. Its matching Table 13 credit first stage is 0.1146557. A fixed schedule displacement normalized to one point-estimate credit dollar and the reference two-year duration give $2\beta/\varphi = -0.000122105$ schooling-equivalent years. With anchor 17, L2 imposes $[\Delta H_{1991}(20) + \Delta H_{1992}(19)]/2 = -0.000122105$. Equal age–cohort weights and placement of the duration-equivalent input at the observed ages are explicit reconstruction assumptions. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. No L3 is admitted: the source reports one attendance window for this contrast, and splitting its assumed duration would not add an observation. Appendix E.4.1 separates the coefficient, fixed policy unit, and dollar account. Figure E5, AOTC: joint, upper

kink panel, shows the L2 paths; the age-65 gap is $[-0.805, 1.432]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-17.06, +\infty]$.

AOTC: joint filers, phase-out start

Bulman and Hoxby (2015) estimate the AOTC attendance response at the phase-out start for joint filers. The observation is current attendance among ages 19–20 in 2011; it is not a later follow-up of one entering class. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of $+\infty$. The original study's Table 14, relevant filer row and first Attendance column, supplies the signed slope coefficient 7×10^{-6} per income-margin dollar. Its matching Table 13 credit first stage is 0.1074026. A fixed schedule displacement normalized to one point-estimate credit dollar and the reference two-year duration give $2\beta/\varphi = 0.000130351$ schooling-equivalent years. With anchor 17, L2 imposes $[\Delta H_{1991}(20) + \Delta H_{1992}(19)]/2 = 0.000130351$. Equal age–cohort weights and placement of the duration-equivalent input at the observed ages are explicit reconstruction assumptions. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; self-financing is also the L1 classification. No L3 is admitted: the source reports one attendance window for this contrast, and splitting its assumed duration would not add an observation. Appendix E.4.1 separates the coefficient, fixed policy unit, and dollar account. Figure E5, AOTC: joint, lower kink panel, shows the L2 paths; the age-65 gap is $[-0.805, 1.433]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-5.96, +\infty]$.

AOTC: independent single filers, phase-out start

Bulman and Hoxby (2015) estimate the AOTC attendance response at the phase-out start for independent single filers. The observation is current attendance among ages 19–20 in 2011; it is not a later follow-up of one entering class. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 6.75. The original study's Table 14, relevant filer row and first Attendance column, supplies the signed slope coefficient 1×10^{-6} per income-margin dollar. Its matching Table 13 credit first stage is 0.2268326. A fixed schedule displacement normalized to one point-estimate credit dollar and the reference two-year duration give $2\beta/\varphi = 0.000008817$ schooling-equivalent years. With anchor 17, L2 imposes $[\Delta H_{1991}(20) + \Delta H_{1992}(19)]/2 = 0.000008817$. Equal age–cohort weights and placement of the duration-equivalent input at the observed ages are explicit reconstruction assumptions. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. No L3 is admitted: the source reports one attendance window for this contrast, and splitting its assumed duration would not add an observation. Appendix E.4.1 separates the coefficient, fixed policy unit, and dollar account. Figure E5, AOTC: independent single, lower kink panel, shows the L2 paths; the age-65 gap is $[-0.805, 1.433]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-1.61, +\infty]$.

AOTC: simulated instrument

Bulman and Hoxby (2015) use a simulated instrument to estimate how education tax credits affect attendance. The original study's Table 16 reports separate age-19 and age-22 responses over 2006–2011. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 10.05. The reference retains the age-19 coefficient, 1.4×10^{-5} per actual credit dollar, from the first Attendance column. Multiplying by the assumed two-year

duration gives 0.000028 schooling-equivalent years for a one-dollar policy change. L2 imposes the equally weighted $\Delta H_c(20) - \Delta H_c(18) = 0.000028$ restriction across 1987–1992 cohorts, with anchor 17. The positive age-18 gap is not mechanically set to zero. The accounting transfer is one dollar because this coefficient is already second stage; Appendix E.4.1 documents that unit correction. L2 yields a searched positive-cost MVPF span of $[-\infty, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. The age-22 coefficient concerns different cohorts, so no longitudinal L3 is admitted for the retained age-19 population. L1 remains the source MVPF; its comparison with L2 therefore includes the unit correction as well as the continuation model. Figure E5, AOTC: simulated instrument panel, shows the L2 paths; the age-65 gap is $[-0.586, 1.155]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-18.36, +\infty]$.

Algorithmic college placement

Bergman et al. (2026) randomize algorithmic rather than test-only course placement at seven New York community colleges. Entering cohorts have unequal follow-up through 2018–2019. Bergman and coauthors calculate an MVPF (L1) of $+\infty$. The June 2026 study's Table 3, column (3), reports 0.562 additional college credits for the pooled sample. The reconstruction uses $0.562/24 = 0.023417$ schooling-equivalent years. The authors' own MVPF uses placement benefits and costs while omitting uncertain lifetime earnings; the reconstruction adds the fiscal earnings channel but preserves their private valuation. L2 imposes $\Delta H(23) = 0.023417$ with anchor 20 and first possible exposure at 21. These are representative ages; the pooled coefficient is not a common two-year follow-up. L2 yields a searched positive-cost MVPF span of $[0.08, +\infty]$ and separate self-financing paths, crossing one; self-financing is also the L1 classification. No L3 is admitted. The 1.717-credit result for the more mature fall-2016 cohort changes the population relative to pooled L2, and first-term course-placement outcomes are not total earned-credit stocks. Thus, this remains an extension of the source welfare account, not a comparison that changes only the education continuation. Figure E6, Algorithmic college placement panel, shows the L2 paths; the age-65 gap is $[-0.305, 0.608]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[0.07, +\infty]$.

Tuition deduction: single filers, phase-end

Hoxby and Bulman (2016) estimate attendance responses around federal tuition-deduction thresholds. This contrast concerns single filers at the deduction's end. The retained outcome is attendance in the first full tax year after expected on-time high-school graduation. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of $+\infty$. The original study's Table 5, column (6), baseline specification, supplies the probability effect 0. The reference's two-year enrollment conversion gives 0 schooling-equivalent years. L2 imposes $[\Delta H(18) + \Delta H(19)]/2 = 0$ with anchor 17. Equal weights at ages 18 and 19 are an additional age mapping of the tax-year window; they are neither recovered source weights nor two independently estimated schooling effects. L2 yields a searched positive-cost MVPF span of $[-\infty, -6.33]$ and separate self-financing paths, entirely negative among the positive-cost paths; self-financing is also the L1 classification. Table 7 contains additional attendance windows around expected graduation. They are not admitted as nested L3 because their specification and population have not been reconciled with the retained Table 5 input under a common schooling conversion. The absence of L3 therefore does not mean the paper lacks dynamic results. Figure E6, Tuition deduction: single, upper threshold panel, shows the L2 paths; the age-65 gap is $[-0.860, 1.501]$ years,

allowing either sign of the late-life effect. Paired input repetitions give L2 $[-\infty, +\infty]$. The reported L1 95 percent interval is $[-\infty, +\infty]$.

Tuition deduction: joint filers, phase-end

Hoxby and Bulman (2016) estimate attendance responses around federal tuition-deduction thresholds. This contrast concerns joint filers at the deduction's end. The retained outcome is attendance in the first full tax year after expected on-time high-school graduation. Hendren and Sprung-Keyser (2020) calculate an MVPF (L1) of 0.77. The original study's Table 5, column (3), baseline specification, supplies the probability effect 0. The reference's two-year enrollment conversion gives 0 schooling-equivalent years. L2 imposes $[\Delta H(18) + \Delta H(19)]/2 = 0$ with anchor 17. Equal weights at ages 18 and 19 are an additional age mapping of the tax-year window; they are neither recovered source weights nor two independently estimated schooling effects. L2 yields a searched positive-cost MVPF span of $[-3.92, +\infty]$ and separate self-financing paths, crossing zero and one; these endpoints bracket L1. Table 7 contains additional attendance windows around expected graduation. They are not admitted as nested L3 because their specification and population have not been reconciled with the retained Table 5 input under a common schooling conversion. The absence of L3 therefore does not mean the paper lacks dynamic results. Figure E6, Tuition deduction: joint, upper threshold panel, shows the L2 paths; the age-65 gap is $[-0.860, 1.502]$ years, allowing either sign of the late-life effect. Paired input repetitions give L2 $[-4.87, +\infty]$. The reported L1 95 percent interval is $[-1.83, 57.84]$.

Cannabis legalization: education channel

Li (2026) estimates recreational-cannabis legalization effects on bachelor's completion among first-time, full-time entrants, using 2006–2017 entering cohorts and post-entry policy exposure. The original manuscript's Table 3 reports effects of -0.016, -0.011, and -0.018 on completion within four, five, and six years. There is no original MVPF (L1). The education-only account assigns two additional schooling years per bachelor's completion, yielding targets -0.032, -0.022, and -0.036. The published reconstruction is L3 only. It anchors at 17, assigns entry to 18, and places the targets at ages 22, 23, and 24, equally weighting 1988–1999 calibration cohorts. The age-24 target is exact; the earlier two enter constrained minimum distance. This common onset is a modeling assumption despite heterogeneous actual exposure after entry. All three comparable BA windows enter the primary reconstruction. The two-year-institution outcome concerns another entrant population and is excluded. A six-year-only L2 is retained solely as a computational diagnostic and is not treated as a second substantive result. L3 yields a positive-cost MVPF of -55.85, implying beneficiary losses. Figure E6, Cannabis legalization: education channel panel, shows the L3 paths; the age-65 gap is -0.096 years, preserving a negative late-life gap. Paired input repetitions give L3 $[-57.03, -54.19]$.

E.3 Welfare Witnesses and Information Layers

For finite-policy calculations, let $\mathcal{B}_{i\ell}$ be the saved finite bank for calculation i and layer ℓ. Crossing MVPF one at positive cost requires a path h^- with $N_i(h^-) > 0$ and $W_i(h^-) < N_i(h^-)$ and another path h^+ with $N_i(h^+) > 0$ and $W_i(h^+) > N_i(h^+)$. Both satisfy the same retained information and accounting restrictions. A harmful-costly versus self-financing comparison instead pairs $W_i(h^-) < 0$, $N_i(h^-) > 0$ with $W_i(h^+) > 0$, $N_i(h^+) \leq 0$. The replication materials identify each complete path and its paired welfare values; they do not

combine marginal extrema. Table 3 supplies the separate component extrema; they are descriptive summaries and cannot be combined into a ratio.

All main L3 applications take the selected source estimates as fixed point inputs and retain the L2 equalities. Michigan, CUNY, and Social Security student benefits admit verified exact matches. Florida, Adams, Wisconsin, Kalamazoo, public-college spending, tuition caps, Texas tuition, HOPE tax credit, Tennessee Promise, and the cannabis education channel use the same minimum-distance rule when the added point values cannot all be matched exactly by the retained family. Each of these ten searches currently returns one best-fitting path with nonzero residuals. The original source estimates are not changed to make the family fit. Tables E2, E4, and D2 disclose the discrepancies. Their zero-width fitted welfare summaries describe conditional selections, not absence of statistical uncertainty or certified point identification. Source standard errors enter only the separate input-uncertainty exercises.

A main L3 restriction must preserve a time window actually reported by the source while retaining the education-conversion rule used in L2. Wisconsin qualifies because Table 5 reports semester credit effects and cumulative totals. Its annual flows sum adjacent observed semesters under the retained credits-to-schooling conversion. The later-cohort third-year restriction uses cohort 2 alone because cohort 3 is not observed that long; no missing year is imputed. The entry-age mapping, cohort weights, and common response parameters are already maintained in L2. For the separate uncertainty sensitivity, annual-flow SE upper bounds sum marginal semester SEs where covariance is unavailable; these SEs do not weight or relax the main point fit. The circle marker therefore identifies added source timing, not a reconstruction free of calibration assumptions.

L3 may preserve converted source dynamics rather than directly measured annual schooling. The two college-appropriation calculations use four reported policy-lag responses, retaining the reference persistence and enrollment conversion and assigning the resulting sequence to modeled acquisition years. This requires transport from institutions to a representative population. Adams, Michigan, and HOPE tax credit likewise retain explicit timing or risk-set mappings. These assumptions are recorded alongside the source information; the L3 label does not establish their validity. Tennessee Pell and Georgia HOPE remain L2 only because their proposed extra years merely split a single sector-entry observation using assumed durations. They add no separately observed window or dynamic response.

Every retained main L3 row, including each best-fit selection, is included in L2, and extrema are recomputed from the saved rows. Component-width ratios therefore cannot exceed one apart from numerical tolerance. Nesting of recorded banks does not establish that either exhausts the continuous feasible set. Failure to verify a candidate remains a search outcome. A span between finite endpoints does not assert that every intermediate value has been verified. No distribution over candidates or over the continuous continuation class is specified.

E.4 Monetary Units and Reference Components

Table 3 uses a positive reference program-cost component as a display denominator wherever its provenance is established. The same denominator divides willingness to pay and net cost in L1 and both reconstructions of a calculation. It is not the path-dependent net cost, and this scaling leaves each MVPF unchanged. The component follows the native account, such as a scholarship's discounted transfer or a tax credit's direct revenue cost; it need not represent an identical policy dose across applications. In tax-schedule calculations, the original policy unit can be a small change in a credit or deduction. A large normalized component can therefore combine a broad continuation range with a small reference expenditure. It should not be read as a

large absolute dollar amount for an average program participant. The replication materials document the normalization, source, and monetary base for all 38 calculations. The underlying point-input ranges and their units are preserved separately from the rounded table. Supplementary Figure S4 displays the native-unit component spans.

The reference component pair is taken from the same source account used to identify L1, with unrounded ledger values retained when they match that record. Reported reference ratios are not silently replaced by ratios recomputed from rounded components. For Early Head Start, the library account gives $E^{\text{ref}} = 21{,}944$, $W^{\text{ref}} = 3{,}906$, and $N^{\text{ref}} = 11{,}216$. Its fiscal sum is $24{,}257 + 1{,}540 - 10{,}478 - 4{,}103 = 11{,}216$: program cost plus the community-college cost change, less university savings and additional taxes. The dissertation reports net cost of about 12,755. Its displayed calculation appears to add the 1,540 community-college term again after including it in the net education-cost change. I use the library ledger consistently and retain the dissertation amount as an alternative account, rather than mix their components. The source price year remains unresolved.

Early Head Start is a public-data reconstruction rather than a source-total alignment. The 2017–2019 national ACS provides age-specific BA-versus-HS/GED wages, including zero wages, with person weights and documented dollar adjustments. The reconstruction uses 2000 dollars and values the retained 0.168 ATT degree input through its age-specific path. Source age placement at 22 is a modeling convention, not an exact age identified by the reported adult outcome. No early signed coefficient is invented for this childhood intervention and no separate schooling-flow charge is added.

The program-cost proxy uses the available FY2001–2008 statutory EHS allocation per funded slot, valued for three years; it is \$28,158.29 in discounted 2000 dollars. It is an average funded-slot proxy rather than the experimental sample's measured marginal cost. The retained public-source college-price reconstruction gives a fixed private education payment of \$11,463.05 and a public college-cost change of $-\$5,680.05$. Thus $W = E - 11{,}463.05$ follows the source convention of gross earnings in WTP, and $N = 22{,}478.24 - 0.187E$. Using gross earnings in WTP while also valuing fiscal revenue is an inherited convention, not a universal private-disposable-income rule. The public college prices require an explicit 2021 publication-price convention before conversion to 2000 dollars because their exact source vintage is unavailable. A constant 0.168 degree gap gives \$78,683.28 and an age-22 step gives \$77,910.64; both differ from the reported \$21,944. No multiplier is used to conceal that difference. The alternative program-cost exercise uses the FY2001 slot proxy. Input repetitions condition on the fixed public wage and cost schedules.

Algorithmic college placement retains the June 2026 source components, $W = 220$ and $N = -90$. The reconstruction retains source WTP of 220 and extends only the fiscal account using the signed education-related income mapping. It does not add another private earnings benefit. It remains an explicitly marked extension in the full ordering, rather than evidence that continuation alone overturns an otherwise identical account. Cannabis is separate because it has no whole-policy reference and values only the education channel.

E.4.1 Marginal Tax Inputs in a Finite Policy Experiment

The eleven HOPE/AOTC reconstructions use the same finite, anchored logistic attainment levels as the other applications. Their source coefficients require an additional policy-unit conversion. Table 14 of Bulman and Hoxby (2015) reports changes in attendance–income slopes at tax-schedule kinks for ages 19–20, in 2007 for HOPE and 2011 for AOTC. Table 13 supplies the corresponding credit–income first stages. Let β_i be the attendance coefficient and $\varphi_i > 0$ the credit first stage, both signed in the reference policy direction. I fix a

schedule displacement $q_i = 1/\varphi_i$, calibrated to raise the credit by one dollar at the point estimates. I assume that the local attendance and accounting responses transport linearly over this displacement. This specifies a finite counterfactual; the source derivative alone does not identify its exact effect. Keeping the reference two-year enrollment-to-schooling conversion gives the input

$$m_i^{\mathrm{YS}} = 2\beta_i q_i = \frac{2\beta_i}{\varphi_i}. \tag{63}$$

For the Table 16 simulated-instrument estimate, the attendance coefficient is already per actual credit dollar, so $q_i = 1$ without another first-stage division. These conversions define schooling-equivalent inputs, not directly observed completed-schooling effects.

The source ages determine the cohort calibration; assumed duration is stated separately. Nationwide ACS profiles use the 1987–1988 birth cohorts for HOPE, 1991–1992 for the AOTC kinks, and 1987–1992 for the simulated-instrument input observed at age 19 during 2006–2011. Equal cohort-cell weights are maintained because the source regression weights have not been recovered. In the annual acquisition convention, current enrollment at age a contributes to schooling in years a and $a + 1$ under the two-year duration assumption. The converted total therefore restricts $\Delta H_c(a + 1) - \Delta H_c(a - 1)$. For HOPE, the joint restriction is

$$\tfrac{1}{2}\left[\Delta H_{i,1987}(21) - \Delta H_{i,1987}(19)\right] + \tfrac{1}{2}\left[\Delta H_{i,1988}(20) - \Delta H_{i,1988}(18)\right] = m_i^{\mathrm{YS}}. \tag{64}$$

The AOTC kink reconstruction instead uses the source-age average $[\Delta H_{i,1991}(20) + \Delta H_{i,1992}(19)]/2 = m_i^{\mathrm{YS}}$. This places the duration-equivalent total in the observed age cells rather than treating the two-year assumption as an observed schooling increment. It is a distinct temporal allocation, stated explicitly rather than attributed to the source regression. The simulated-instrument operator averages $\Delta H_c(20) - \Delta H_c(18)$ across its six age-19 cohorts. The second acquisition year is a duration allocation, not another source observation, and does not create L3. All tax paths assume no attainment effect through age 17, allowing first effects at age 18. This exposure restriction is maintained rather than estimated. Earlier placements at ages 21–23 or 25–27 are not treated as source observation windows. Independent-filer earnings schedules beginning at 25 remain valuation assumptions. The age-22 simulated-instrument result refers to a different cohort composition and remains outside the main L3 set.

The restored paths retain the current full-population ACS calibration, the transported joint parameter support, completion-timing restrictions, nondecreasing levels, and the one-schooling-year annual acquisition cap. A common finite treated parameter triple is used across each application's cohort components, with their separate baseline anchor levels. Thus both H^0 and H^1 are finite attainment paths throughout the comparison. For example, the HOPE independent-single contrast has $\beta_i = 3 \times 10^{-6}$ and $\varphi_i = 0.1156229$. Its fixed schedule displacement is about 8.649 income-margin units, and its two-year input is approximately 0.000051893 schooling years. This is a local-linear calibration of a one-dollar credit change, not the effect of adopting the whole tax program. Matching an ACS support restriction does not establish the plausibility of every policy response inside that support.

The monetary conversion follows the same fixed displacement. For the ten kink contrasts, the non-earnings components become $w_{0i}^F = q_i w_{0i}^{\mathrm{source}}$ and $n_{0i}^F = q_i n_{0i}^{\mathrm{source}}$. The per-schooling-year earnings and fiscal schedules retain their units and values. The account evaluates the finite cohort-average gap, $\overline{\Delta H_i}(a)$, as

$$W_i^F(h) = w_{0i}^F + \sum_a w_i(a)\overline{\Delta H_i}(a). \tag{65}$$

The corresponding net cost is $N_i^F(h) = n_{0i}^F + \sum_a n_i(a)\overline{\Delta H}_i(a)$. The reported L1 ratio and interval are unchanged. Its WTP, net cost, and program-cost display divisor receive the same normalization, so their normalized values are unchanged as well. For the simulated instrument, I retain the unit correction that sets the direct credit payment to one dollar rather than the first-stage amount of approximately 0.38. At each input repetition, the correction adds $1 - \text{tax_value}$ to net cost and $(1 - \text{tax_value}) \min(1 - \beta_i, 1)$ to the non-earnings WTP component. Its L1–L2 comparison therefore includes this accounting correction as well as continuation.

All 1,000 archived joint input IDs are retained. The point normalization q_i stays fixed across repetitions; education, tax-credit, cost, and earnings inputs still vary together according to their registered scheme. In particular, five negative first-stage draws are retained, rather than discarded or divided into the education coefficient. Their credit changes are not constrained to equal one dollar away from the point estimates. The simulated-instrument repetitions remain a constructed reference-input exercise, not draws from the source's sampling distribution. Converting monetary units from dollars to cents leaves the same physical policy experiment, path constraints, and MVPF unchanged. Changing the size of the finite policy experiment is instead a substantive sensitivity, because the ACS support restricts levels rather than responses per dollar.

A limited dose sensitivity repeats the finite-path screening for point credit changes of 0.50, 1, and 2, scaling the education input and non-earnings account components together. At all three doses, each tax calculation has verified negative-WTP, positive-cost paths and positive-WTP, nonpositive-cost paths. This checks the existence of these conflicting classifications on the tested grid; it does not establish invariant endpoints or welfare bounds across policy sizes. Currency-unit checks separately preserve the physical policy experiment and recover the same schooling restriction and MVPF.

The replication materials document the source coefficients, first stages, joint age–cohort operators, equation residuals, and complete finite paths. WTP and net-cost extrema are finite. Some MVPF endpoints nevertheless remain infinite because verified finite paths approach zero net cost from above with nonzero WTP. Those ratio limits are distinguished from self-financing states. Equal endpoint pairs retain tied ranks; no additional responsiveness cap or path probability is introduced to separate them.

E.5 Candidate-Based Lower-Endpoint Ordering

Figure 9 ranks all 37 calculations with a reference MVPF. Twelve use L3: Florida, Adams, Wisconsin, Kalamazoo, Michigan, CUNY Pell, Social Security student benefits, both college-appropriation calculations, HOPE tax credit, Texas CC tuition, and Tennessee Promise. The other twenty-five use L2, including the eleven finite tax reconstructions just defined. Early Head Start uses the public-data reconstructed account, and algorithmic placement retains its marked earnings-channel extension. Cannabis has no whole-policy reference. Table E3 reports every rank; Table E5 and the case profiles document the source windows; the restricted 24-case ordering remains in the data.

For each selected finite-path specification, let $\underline{M}_i$ and $\overline{M}_i$ be the lower and upper endpoints of defined MVPFs, including audited one-sided limits and the self-financing convention. Let r_2 round finite numbers to two decimals and leave infinities unchanged. The display orders policies lexicographically in ascending order of

$$\left(-r_2(\underline{M}_i), -r_2(\overline{M}_i)\right). \tag{66}$$

Equal endpoint pairs receive the same competition rank, one plus the number of strictly preferred calculations. Reference ranks compare the recorded L1 values, preserving ties. Repeating the full ordering with three- or four-decimal endpoint rounding leaves all ranks unchanged in the current results. This numerical stability

check does not establish global extrema: a newly discovered adverse path could change an ordering based on candidate minima.

An open marker replaces only that policy's selected L3 set by L2, holding other selected sets fixed. Ten of the twelve added-information cases improve under the rule and two remain unchanged. The arrows describe the rule's response to restrictions, not an increase in the policy's true welfare. Thirteen ranked calculations admit paths with nonpositive WTP and nonpositive net cost. Those unassigned-ratio paths, rather than the calculations containing them, are omitted from the ordering; their monetary components remain in the collection table. The rule assigns neither artificial finite values to infinity nor probabilities to paths, and it is not a social-welfare or certified maximin ranking.

E.6 Reported Timing and Its Use in Reconstruction

Table E5 distinguishes source timing coverage from imposed path restrictions. It counts distinct age or individual-follow-up windows for the relevant contrast and outcome family. An overlapping two-year age interval counts once; additional credentials, estimators, subgroups, and tax contrasts at that window do not create more horizons. Institutional policy lags and college-grade risk sets retain their own clocks. The count measures neither precision nor independent observations.

Among the 37 calculations with L1, 17 have one verified window, five have two, and 11 have at least three. Four use other clocks: Georgia's state-year freshman flows, the two Deming–Walters institution-level calculations, and Turner's college-grade and pooled enrollment outcomes. The published Tennessee HOPE paper now verifies one initial-college-choice window. Its 16-month eligibility deadline is not a common observed follow-up, and its alternative sectors and specifications do not constitute additional attained-education times. The unranked cannabis application reports completion at three horizons. All 38 calculations therefore receive a coverage record, and all 37 calculations with L1 enter the expanded ordering with their specification groups identified.

Several multiple-window studies still require a population, outcome, or specification bridge before their results constrain one path. Absence of L3 does not imply that a source lacks dynamic evidence. Nor does L3 imply that the reference valuation ignored an estimate: Texas can use separately reported timing while retaining coefficients already used in its aggregate conversion. Adams also enters L3 through its separately reported early entry input, conditional on the temporal allocation in Appendix D.6. The registry separates reference inputs, source windows, imposed restrictions, and their empirical or assumed origins, with source locations and hashes. It does not treat the number of windows as a causal determinant of ranking changes. Several policy contrasts also share a study family.

E.7 Source Comparability and Numerical Verification

Additional coefficients constrain the same path only if their population, policy contrast, outcome, and clock can be reconciled with L2. The profiles identify unresolved specification differences for tuition deductions, different age-cohort populations for the AOTC simulated instrument, and changed entering cohorts for algorithmic placement. These comparisons remain separate specifications. The published DC and Georgia papers now support the source coefficients retained in their reference conversions; a corresponding DC education SE is still unavailable for the current repetition scheme.

The main L3 fits retain Florida's first-semester and annual credit totals and Wisconsin's reported semester credit flows. All thirteen admitted L3 specifications have 1,000 scheduled joint input repetitions. Existing source dependence groups are preserved and explicitly extended to additional inputs where necessary; unavailable covariance is not treated as a reason to omit these repetitions. Appendix F distinguishes inherited groupings from the added dependence assumptions. Tennessee uses eighteen table point estimates; its earlier graphical interval screen does not enter the main fit. Three L3 specifications admit exact-match banks and ten use best-fit selections.

For finite-policy calculations, all saved point candidates are independently evaluated at annual ages through 72, or through 68 for Tennessee, while income is valued only over each registered earnings horizon, checking their anchors, nondecreasing attainment, credential nesting where required, encoded moments, empirical support, and paired welfare values. Every verified L3 candidate also enters L2. Searches supplement shape grids with multiple-start component optimization and equations for zero WTP, WTP equal to cost, zero net cost, and selected ratios. At a zero-cost root with nonzero WTP, decreasing positive-cost targets check the one-sided limit. These checks prevent a sparse grid from obscuring narrow positive-cost branches, but do not certify global finite endpoints or connectedness of the feasible set. The same checks apply to registered input repetitions. An unsuccessful search remains distinct from evidence of infeasibility.

The numerical checks use each calculation's current calibration and observation operator. In particular, Wisconsin's point paths and all 1,000 L2 repetitions use the joint cohort–follow-up operator in Appendix E.2. Its input sequence, external baseline, and welfare account are held fixed. The separate replication guide identifies the current records and archived specifications.

E.8 The Complete Path Display

Figures E1–E6 display comparison and treated schooling levels for the 34 single-schooling-equivalent calculations. Figure E7 covers Social Security's full-time-attendance path, Early Head Start's BA+ path, and Tennessee Promise's three joint credential paths. Figure E8 displays STBF's four joint level paths. Together they cover all 38 finite-policy calculations, including the three main applications. Black lines show H^0. Blue and red boundaries and shading show the pointwise minimum and maximum H^1 in the saved L2 and admitted L3 sets. These ranges describe attainment without selecting paths by their welfare values. Tennessee's L2 ranges and L3 fitted path satisfy credential nesting; the cannabis education account displays only its primary L3 best fit. For the ten selected fits, coincident upper and lower traces represent one retained solution.

The envelope edges can switch candidate identity across ages and credentials and need not themselves form feasible paths. They summarize the current finite search, not a probability distribution, sampling interval, or certified boundary of all possible continuations. Complete candidate paths and their paired welfare values remain in the replication records. Single-schooling panels share one age and schooling scale; credential panels share one age and attainment-share scale. STBF's schooling-shaped timing index is labeled separately from completed schooling. The source profiles in Appendix E.2 and the expanded operators distinguish observed follow-ups from their assigned ages. Neither a later crossing nor a widening modeled gap establishes an individual behavioral mechanism or shows that the source researchers overlooked dynamics.

E.9 Point-Input Sensitivity across All Applications

Figure E9 and Table E6 compare all 38 calculations under alternative calibration, valuation, and numerical-search choices. The selected information set is L3 for thirteen calculations, including the cannabis education channel, and L2 for the other twenty-five. Both layers remain in the accompanying data where available. These counts differ from the L2-only comparison of the 37 cases with L1.

E.9.1 Changes to Calibration, Valuation, and Search

Narrower and wider support multiply the empirical covariance by 0.5 and 2 and transported timing deviations by their square roots. Source moments, anchors, natural level bounds, and the one-year annual schooling-acquisition cap are retained. The geographic alternative uses national profiles and support for the same cohorts; cases already nationally calibrated repeat that baseline. Tennessee retains its high-school-graduate denominator and source-control matching, while its late-completion allowance is varied separately. These exercises assess sensitivity to population transport; they do not estimate confidence levels.

The income alternatives multiply each path's net income and associated tax effects by 0.75 or 1.25. Fixed transfers and education-resource schedules retain their original values. Negative and positive portions of income are scaled together, without adding or subtracting another opportunity cost. For STBF, the multiplier applies after evaluating the nonlinear author income function. For Early Head Start and Tennessee, it applies to their disclosed public-data reconstruction.

Every alternative re-evaluates the relevant path rule. L2 and exact-match L3 banks retain monetary-extremum searches; the ten minimum-distance L3 applications instead refit the fixed-point minimum-distance criterion before valuation. Their denser check uses more optimizer starts and the denser feasible seed bank. A wider support can change a selected minimum even though it contains the old feasible paths; the minimizer set itself need not expand. The denser scalar L2 grid uses 2^{16} shape pairs, with supplemented joint pools for the two joint models. All common specifications have verified results. Annual monetary accounts are independently checked. These finite searches do not certify global extrema or uniqueness and are separate from the 1,000 input repetitions.

E.9.2 Which Welfare Comparisons Survive?

Positive-cost MVPFs cross one in 21 of the 38 main selected sets. The counts are 20 under narrower support, 22 under wider support, 20 with national calibration, 21 at the 75 percent income scale, and 21 at the 125 percent scale. The denser search also gives 21. Main and denser searches both find self-financing paths in 24 selected sets. Ten main L3 entries are best-fit singletons, so their contraction reflects the fitting criterion as well as added observations. These counts describe found paths, not exhaustive classifications of the continuous feasible sets.

In the main specification, 19 sets contain both a negative-WTP, positive-cost path and a positive-WTP, nonpositive-cost path. 16 calculations retain both types in every common specification. The count is 17 under narrower support, 19 under wider support, 18 under national calibration, 17 at the lower income scale, and 20 at the higher scale. Several tax cases share a study, so these counts are not independent replications. Functional form, population transport, and source identification remain maintained.

E.9.3 Source-Specific Checks

STBF replaces the unrounded time-enrolled input by the printed 0.360 years, keeping all three credential inputs. A separate diagnostic sets the experience-delay channel to zero within the author income function while retaining the joint education restrictions. It does not remove a separately estimated cost. All its found positive-cost MVPFs exceed one across the common and source-specific checks. The main range is approximately $[1.25, 4.01]$; the denser search finds a slightly lower endpoint of 1.24, illustrating numerical-search sensitivity without changing this threshold comparison.

Adams uses four converted point inputs under the stated age allocation. Its main L3 fit has an MVPF of 0.63, with positive WTP and positive net cost. Common alternatives give fitted MVPFs from 0.40 to 0.88; all remain below one. Setting the negative portion of the completion-response template to zero changes income valuation without supplying an additional education observation. Private and public education-cost functions continue to move with the candidate path.

Social Security retains the source-control attendance level at age 23 and the transported 1985–1987 entry-profile shape. Its narrower and wider alternatives vary the shape ellipse and timing support, retaining the probability bound $0 \leq K \leq 1$; the national alternative repeats the national baseline. Each L3 path continues to match the age-23 and age-28 attendance coefficients exactly. All found positive-cost MVPFs exceed one in the common support and income-scale alternatives. This result conditions on transporting the modern ACS age shape and using the two-years-per-entrant account of Hendren and Sprung-Keyser; neither restriction is identified by the two attendance estimates.

Tennessee refits the eighteen point targets under alternative source-control-level matching, national calibration, and late-completion allowances, as well as the common support and income changes. Every retained L3 fit in these checks is self-financing under its corresponding public-data account. Graphical interval widths do not enter the point-estimate criterion. No income multiplier is chosen to force agreement with the published total; the separately documented participant-unit bridge remains fixed within each comparison.

The eleven finite tax experiments additionally use point credit changes of 0.50 and 2, scaling education inputs and fixed monetary components together. Early Head Start substitutes the FY2001 funded-slot program-cost proxy for the FY2001–2008 average, with the same public wage schedule. Cannabis varies education-cost sectors, state weights, and the negative portion of the income response. Its main tax base is net income, and every specification concerns education alone. Monetary components from different paths are never combined to form a ratio. One-at-a-time changes do not establish robustness to all simultaneous departures.

F Conditional Input Repetitions and Additional Information

The uncertainty exercise distinguishes variation across paths at fixed education inputs from variation caused by imprecisely estimated inputs. All 38 calculations use 1,000 prespecified input repetitions, including a six-year-only diagnostic for cannabis. Each of the thirteen admitted L3 specifications also uses 1,000 paired repetitions of its complete input vector. Comparison profiles, empirical support, outcome conversions, age mappings, and valuation functions remain fixed. Table 4 compares the resulting endpoint summaries with available reported L1 intervals. Table F1 reports WTP and net cost separately. Comparisons are within policy: source accounts can differ in population, monetary year, policy unit, and included uncertainty.

F.1 Conditional Endpoint Summaries

For repetition r, let $\underline{M}_{i\ell}^{(r)}$ and $\overline{M}_{i\ell}^{(r)}$ denote the lowest and highest MVPFs among verified paths for policy i and reconstruction ℓ. A fitted singleton has equal within-repetition endpoints. Numerical checks for positive-cost approaches to zero net cost remain separate from self-financing. The reported summary is

$$\left[q_{.025}\left(\{\underline{M}_{i\ell}^{(r)}\}_{r\in\mathcal{R}_{i\ell}}\right), q_{.975}\left(\{\overline{M}_{i\ell}^{(r)}\}_{r\in\mathcal{R}_{i\ell}}\right)\right], \tag{67}$$

where $\mathcal{R}_{i\ell}$ contains repetitions with verified paths and a defined ratio. Usable counts are reported relative to the original 1,000, without replacing unsuccessful searches. WTP and net-cost summaries use all repetitions with verified monetary components, including those without an assigned MVPF. Each ratio pairs components from the same path and repetition; separate component extrema are never combined. The fixed finite-policy normalization for tax calculations is preserved.[36]

These endpoint percentiles are conditional summaries, not demonstrated confidence regions for the continuous path class. Search may miss feasible paths or extrema, and uncertainty in external ACS calibration is not resampled. The input schemes also differ: the 33 Hendren and Sprung-Keyser accounts retain their paired source and accounting draws, including earnings-return inputs where available. STBF, Early Head Start, algorithmic placement, the simulated-instrument education input, and Tennessee use constructed perturbations under their stated scales and dependence. DC tuition assistance varies accounting inputs with its education effect fixed because a corresponding source SE is unavailable. More repetitions improve numerical precision under these schemes without establishing a common sampling interpretation.

Reported L1 MVPF intervals are available for 34 calculations. They use the frozen Policy Impacts Library snapshot of 20 September 2026, with Social Security using its enrollment-alternative interval $[1.16, 32.23]$; Tennessee's $[+\infty, +\infty]$ interval comes directly from Schoonover et al. (2026), Section 5.2.2. No comparable reported interval was recovered for STBF, Early Head Start, or algorithmic placement; cannabis has no original whole-policy MVPF. Source and reconstructed interval widths can reflect different inputs, accounts, dependence assumptions, and coverage targets.

F.2 Dependence across Education Inputs

I preserve the correlation groups, signs, marginal standard errors, and paired accounting inputs used in the Hendren and Sprung-Keyser replication. Their grouped-draw approach does not impose one correlation coefficient on every input: a common shock links coefficients within a group, while distinct groups have separate shocks. For L3 coefficients absent from the original account, I extend this grouping according to their outcome, sample, and observation window. These extensions are explicit assumptions about unreported covariance, rather than covariance matrices recovered from the source studies.

Florida's five cumulative-credit coefficients share the retained fourth-year shock. If $\widehat{m}_j$ and SE_j denote a coefficient and its marginal standard error, its repeated input is $\widehat{m}_j + SE_j Z^{(r)}$, with the same $Z^{(r)}$ at all five horizons. Kalamazoo's earlier cumulative credits, CUNY's first-year credits, and Social Security's age-23 full-time-attendance effect likewise share their respective retained endpoint shocks. Wisconsin preserves independent groups for cohort 1 and pooled cohorts 2–3; each added annual coefficient shares

[36] Quantiles use the inverse empirical distribution, without interpolating finite and infinite values. With 1,000 usable repetitions, the endpoints are the 25th minimum and 975th maximum order statistics. Positive-benefit, nonpositive-cost paths are self-financing; paths with both components nonpositive receive no assigned MVPF.

its corresponding group's shock. Annual standard-error scales formed from marginal semester SEs remain imposed upper-bound scales where their covariance is unavailable. Adams preserves the original four-year and two-year outcome groups and extends them to immediate entry and four-year completion. The other source-based L3 applications preserve their registered primitive draws through the updated conversion and age operators. L2 always uses the identical endpoint draw appearing in L3.

STBF retains its constructed joint education perturbations and recovered author income coefficients. Tennessee uses one common normal shock across the six age windows within each cumulative credential, with three independent credential groups. The eighteen point targets are perturbed and refitted, keeping the three perturbed latest-window inputs exact. Marginal scales use displayed clustered SEs; the two BA coefficients whose SEs print as zero retain a 0.0005 reporting-precision floor. This is an imposed scale, not a recovered SE. These draws do not recover the source's permutation distribution, and its warning about clustered-SE precision continues to apply. Cannabis uses the dependence extension described below.

Complete positive dependence within a group is a baseline assumption, not a guarantee of conservative welfare ranges. With equal marginal scales, it removes uncertainty from differences between coefficients; such differences can be central to timing. Therefore the summaries describe uncertainty under the specified grouping. Recovering joint source estimates would permit replacing these assumptions without changing the point-input construction.

F.3 Additional Information and Unsuccessful Searches

Thirteen calculations admit L3 and 25 do not. All thirteen have point-input candidates and complete prescribed repetition schemes. Michigan, CUNY, and Social Security retain exact additional restrictions; the other ten refit constrained equal-weight minimum distance while preserving their L2 inputs. Newly found L3 paths also enter the paired L2 set. An absent L3 therefore indicates that no additional source restriction is admitted under the retained conversion, not that the original study contains no dynamics. Appendix E.2 explains each decision.

All 38,000 reduced-information repetitions were evaluated; 38,000 yielded verified paths and 38,000 yielded a defined MVPF. Across the thirteen L3 specifications, 13,000 repetitions were evaluated and 12,980 yielded verified paths. The 20 unsuccessful L3 searches occur in the Social Security enrollment specification; its endpoint percentiles use the 980 verified repetitions. Unsuccessful searches remain recorded as numerical outcomes, without replacement or claims that no feasible path exists. Likewise, paths without an assigned ratio retain their monetary components. Tables 4 and F1 display the usable counts for these distinct objects.

F.4 Cannabis: An Education-Channel Welfare Account

The cannabis calculation values the education channel of the registered policy contrast without claiming to value legalization as a whole. Its retained bachelor's-completion effects at four, five, and six years after the source's reference event are -0.016, -0.011, and -0.018. A fixed two-schooling-years-per-degree bridge converts these to -0.032, -0.022, and -0.036 schooling-equivalent years. The registered age mapping places them at model ages 22, 23, and 24, with age 17 as the last unaffected observation; it is a maintained mapping rather than individually observed exposure ages. The primary reconstruction is L3 only. Its six-year input is retained exactly and the earlier two targets enter constrained minimum distance, without SE bands. The baseline uses the 14 adopting states and the 1988–1999 birth cohorts, with equal cohort weights and fixed

state weights from the age-18 population. These state weights are external calibration weights, not recovered weights from the source's causal estimator. Appendix E.9 includes an equal-state point-calibration sensitivity.

Let $E_i = \sum_a D_i(a)\Delta y_i(a)$ be net education-related earnings, including the signed pre-completion and post-completion response, and let F_i be the discounted schooling-equivalent acquisition flow priced by the education-resource schedule. Under the fixed effective tax rate, $T_i = t_i E_i$. The education account is

$$W_i^{\text{edu}} = E_i - T_i - p_i F_i, \tag{68}$$

$$N_i^{\text{edu}} = g_i F_i - T_i, \tag{69}$$

where p_i is private payment and g_i is public education-resource cost net of that payment. The main schedule uses $t_i = 0.189$, $p_i = 5{,}997.02$, and $g_i = 8{,}557.30$, in 2000 dollars. No separate forgone-work cost is deducted. Private payments and public resource costs use the 2015 public four-year-college inputs in the Hendren and Sprung-Keyser replication, aggregated by full-time-equivalent enrollment within states, deflated, and combined with the fixed geographic weights. Pricing a schooling-equivalent increment as one full-time-equivalent education year and treating average expenditure as marginal cost remain calibration assumptions. Attainment effects do not identify enrollment sector or actual time out of employment.

The paired components remain central. The main L3 search retains one best-fitting path, giving earnings NPV $-\$3,635.92$, WTP $-\$2,498.50$, and positive net education cost \$44.73. Its MVPF is -55.85. These values are paired components of the same fitted path. Alternative sector, state-weight, support, and income-response assumptions are refit separately in Appendix E.9. The complementary resource comparison remains $W_i^{\text{edu}} - N_i^{\text{edu}} = E_i - (p_i + g_i)F_i$, since the tax transfer cancels. A fitted singleton is conditional on the path family and target criterion, not evidence that the true education-channel value is known.

Other legalization components remain outside this account. For example, adding a specified direct noneducation cost C_i would give net cost $C_i + N_i^{\text{edu}}$, so, for a positive-benefit path, the largest such cost consistent with fiscal self-financing is $-N_i^{\text{edu}}$ and the net-benefit break-even cost is $W_i^{\text{edu}} - N_i^{\text{edu}}$. These are conditional accounting thresholds, not estimates of the actual policy budget. Cannabis excise revenue, health, crime, enforcement, and consumption value would require separate evidence before a total-policy MVPF could be reported. In particular, a fiscal surplus in the education account does not establish that legalization itself is self-financing.

The cannabis repetition results also have a narrower interpretation than a new microdata bootstrap. They extend the registered calibration using the existing 15-cluster synthetic Rademacher shock for the six-year effect, now shared across all three BA horizons and scaled by their marginal SEs. This extension imposes complete positive dependence across horizons. These are researcher input perturbations, not recovered cluster influence functions or a re-estimation of the earlier CBASE model. The resulting endpoint summaries in Tables 4 and F1 condition on that perturbation rule, the degree-to-schooling bridge, and the fixed education-cost schedule. Keeping this limitation explicit permits an education-related valuation while leaving unsupported noneducation components outside the calculation.

G Policy-Specific Male and Female Education Paths

This appendix uses published subgroup treatment effects to construct male and female continuations within seven policies. The purpose is to distinguish differences in the size of an education effect from differences in when education is acquired. No restriction presumes that women graduate later or face larger timing

losses. These degree-channel exercises preserve the original subgroup outcomes; they do not replace the pooled-policy education inputs or welfare accounts used elsewhere in the paper.

G.1 Source Effects and the Reference Populations

Table G1 records the retained point inputs and the resulting standardized education valuations. The college-aid exercises use reported degree outcomes. For Adams, Table 7, Panel D, column 5 of Cohodes and Goodman (2014) reports effects on graduation from any four-year college within four years of high school. The sample contains the 2005–2008 high-school classes, rather than the 2005–2006 classes used for the pooled six-year result. The source test does not reject equality of the male and female coefficients. For STBF, Online Appendix Figure A4 of Angrist et al. (2022) reports six-year BA effects for the four-year applicant strata, with both estimates rounded to 0.08. The reported year-six gender contrast is 0.001, with a standard error of 0.005. Equal displayed coefficients therefore do not establish identical underlying estimates. The figure also reports earlier degree outcomes, but its samples differ by year. I retain the six-year inputs and do not combine the earlier values into a sex-specific L3 without an additional sample-transport assumption.

For Wisconsin, Table 7 of Goldrick-Rab et al. (2016) reports the first cohort's administrative-sample four-year BA effect as 4.7 percentage points, with a treatment-by-male interaction of 0.0 points at the printed precision. Both subgroup inputs are therefore 0.047 in this conditional calculation. The gender level coefficient is not a treatment effect. This specification differs from the table's smaller financial-aid sample and from the pooled credit-based account used in the collection. The table does not supply the covariance needed to recover a male subgroup standard error. For Michigan community-college tuition, Table 7, Panel B, column 7 of Acton (2021) reports the effects of in-district residence on BA completion. These are not per-dollar tuition slopes. The degree sample contains the 2009–2011 high-school classes and excludes degrees obtained in high school; its outcomes are observed through 2017. The source test does not reject equality of the male and female BA effects. The enrollment columns use additional cohorts and do not supply an extra BA observation for the same population.

Early Head Start retains the full-exposure college-graduation ATT in Table 1.2 of Hess (2025): 0.130 for men and 0.212 for women, with source standard errors of 0.040 and 0.050. The corresponding ITT estimates are 0.009 and 0.015. The ATT inputs cannot be described as effects on the entire population exposed to county program availability. The source pools adult attainment; the reconstruction assigns age 22 as the completion endpoint and uses the main case's BA-channel calibration. I retain its national 1989–1999 cohort mixture and full-population denominator, rather than imposing the high-school-graduate restriction used for the college-aid references. Transporting a national age profile to the program's participants remains an external calibration assumption.

For Tennessee, Schoonover et al. (2026) report BA-or-higher effects of 0.012 for men and 0.041 for women at ages 23–24. The female estimate is statistically significant at the 10 percent level; the male estimate is not statistically significant. This contrast does not itself test equality between the two effects. Figure 4F supplies estimates and marginal permutation intervals for the earlier overlapping two-year age groups, 18–19 through 22–23. The reference population follows the main case's high-school-graduate denominator and prior-year Tennessee residence. The fixed 1992–1998 cohort mixture is retained, separately by sex. No pooled control mean is relabeled as a sex-specific control mean.

For Kalamazoo, Table 8 of Bartik et al. (2021) reports within-district IV effects on obtaining a BA/BS within six years after high-school graduation: 0.041 for men and 0.132 for women, with heteroskedasticity-

robust standard errors of 0.070 and 0.077. The source test does not reject equality of the sex-specific BA/BS effects. The corresponding pre-Promise eligible-group means are 0.295 and 0.304. This is not the alternative between-district estimand in the source's Table 9. It is also not the attempted-credit input used in the pooled Kalamazoo Promise account. The subgroup exercise retains the reported degree outcome directly.

All calibrations use the read-only ACS 2000–2022 extract, valid person weights and education codes, and all race and birthplace categories. Means are calculated within sex–cohort–age cells, retaining fixed cohort weights that are identical across sexes within a policy. The college-aid references use high-school graduates; Early Head Start uses the full population. Geography is current residence except for Tennessee's known prior-year residence. Massachusetts birth cohorts 1987–1990 approximate the Adams high-school classes under graduation at 18, mapping four-year completion to age 22. Wisconsin's 2008 entering cohort is approximated by birth cohort 1990 and entry at 18, also mapping its four-year result to age 22. These state profiles are proxies for the respective scholarship populations, not measured control paths at the RD cutoff or among grant-eligible college entrants.

STBF retains the pooled case's fixed entry-age mixture for the 2012–2014 cohorts and applies the six-year degree moment at the corresponding ages 23, 24, and 25. The same age and cohort weights apply to both sexes; subgroup-specific entry-age distributions were not recovered. Michigan's 2009–2011 high-school classes are approximated by birth cohorts 1991–1993. Completion through 2017 then enters a fixed equal-weight mixture at ages 26, 25, and 24, rather than a single midpoint. Kalamazoo's 2006–2008 post-policy graduating cohorts are approximated by birth cohorts 1988–1990 under graduation at 18. Six-year follow-up is mapped to age 24, and the state's sex-specific age shape is rescaled to the source subgroup mean at that age. This transports a pre-policy eligible-group level and is not an identified untreated path for the marginal IV population. Source age distributions would permit less restrictive mappings.

Logistic reference curves are fitted separately by cohort and sex using available means through age 35 and the national-shape shrinkage rule. Recent cohorts lacking later ages borrow a same-sex shape and fit their own level. The Adams, STBF, Wisconsin, Michigan, and Early Head Start calibrations use 96 successful reference and donor fits, with a maximum weighted root-mean-square error of approximately 7.5 percentage points. Supplementary Figures S5–S7 display all seven policies. The common-reference sensitivity below removes differences between the transported male and female profiles.

G.2 Continuation Restrictions and Earlier Source Information

For subgroup $s \in \{m, f\}$, the treated continuation follows the anchored Logistic specification in Section 3. Each component equals its reference curve through the retained degree-channel anchor, is nondecreasing, and lies in $[0, 1]$. The college-aid anchors use the calibrated entry ages. Early Head Start retains the main case's approximation of zero BA attainment through age 19; that is a degree-outcome approximation, not a claim that childhood treatment has no earlier effects. The subgroup moment restriction is

$$\mathcal{M}_i\{H_s^1 - H_s^0\} = \widehat{\delta}_{is}, \tag{70}$$

where the operator retains each policy's component ages and weights. Tennessee averages ages 23 and 24, while the other operators use the mappings described above. The subgroup coefficients are treated as fixed point inputs, including their published rounding. Sampling uncertainty is not eliminated by this conditioning. The pooled-policy repetitions do not become subgroup draws or confidence intervals.

The geographic joint-support covariance is shared across sexes and transported to each reference center, using the same state/pooled construction as the main reconstruction. A common residual-attainment cap limits the fraction of those without a BA at the observed window who can acquire one later. The cap takes the largest residual share among the two subgroup reference curves and the corresponding 1983–1987 sex-specific donor curves. Donor attainment is evaluated using the source operator's age weights. The cap is fixed before the candidate search and is identical for men and women within a policy. This avoids imposing a systematically more permissive continuation on either sex; it remains an empirical transport restriction, not a proven policy-response bound.

Tennessee's earlier subgroup intervals are recovered from the embedded image of Figure 4F. I retain the image coordinates and scale, round the recovered limits outward to 0.001, and add a 0.002 outward allowance for graphical resolution. The extraction includes marker pixels if they extend beyond a very short interval, so this screen is deliberately conservative relative to the digitized plot. Earlier model moments must fall inside all five such intervals. They are not imposed as exact point estimates, converted to normal standard errors, or interpreted jointly as a 95 percent confidence region. The author's textual last-window estimates remain the exact inputs. Both endpoint-only and screened results are saved. No other policy in this subgroup exercise receives a second degree moment by distributing one coefficient across assumed years or combining different outcomes or samples. This auxiliary subgroup interval sensitivity is separate from the pooled application, whose current L3 fits fixed Table 2 point estimates. The subgroup allowance concerns graphical uncertainty and does not establish joint coverage or supply the pooled L3 rule. These degree-channel exercises do not alter the pooled L3 classifications.

G.3 Separating Timing from the Terminal Effect

A larger early effect for women does not by itself show a greater delay loss. To separate timing from the eventual gain, I additionally impose the same terminal BA effect and valuation prices on the two sexes. Earlier attainment changes the timing of both negative and positive portions of the income response. I apply the Section 3 signed completion template with a fixed two-schooling-years-per-degree conversion for BA relative to some college. This is an explicit common-price standardization, not an estimate of observed schooling or the full credential mix for either sex:

$$V_{is}^{\mathrm{BA}} = 2 \sum_{a=18}^{65} D_{.03}(a;18) Y_0(a) \left[r^{+} \Delta H_{is}^{\mathrm{BA}}(a) + r^{-} \{ \Delta H_{is}^{\mathrm{BA}}(a+7) - \Delta H_{is}^{\mathrm{BA}}(a) \} \right]. \tag{71}$$

The coefficients are $r^{+} = 815/7241$ and $r^{-} = -0.066706494$. Paths extend through age 72 so that the seven pre-completion response years are included for earnings through age 65. The terminal comparison below still fixes the age-65 effect. No additional opportunity cost is subtracted. Prices are common across sexes and policies, in 2000 dollars discounted to age 18. This standardized channel excludes sex-specific wage returns, other credentials, transfers, and fiscal costs; it does not estimate a full sex-specific MVPF. Direct annual net-income summation agrees with the stock-kernel calculation.

The unrestricted point-input spans in Table G1 mix timing and eventual attainment. For the timing comparison, each subgroup must additionally satisfy

$$\Delta H_{im}^{\mathrm{BA}}(65) = \Delta H_{if}^{\mathrm{BA}}(65) = \eta, \tag{72}$$

for a fixed terminal-effect grid. Adams uses $\eta \in \{-0.15, -0.10, -0.05, 0\}$, allowing the same terminal reduction or zero effect after its negative four-year inputs. The other policies use $\eta \in \{0.02, 0.05, 0.10, 0.15\}$.

These grid points are comparisons, not quantiles of a path distribution. Table G2 reports every value and the number of verified paths for both sexes. The common-reference sensitivity uses the equally weighted mean of the two subgroup reference curves within each component, retaining the source effects, age weights, support construction, residual cap and terminal grid.

Female-minus-male NPV endpoints compare independent complete paths at the same η, common prices, and common exposure-age weights within the policy. Each endpoint retains a pair of parameterized path witnesses. A negative difference supplies a witness of lower female education NPV at that terminal effect. It would not establish that such a path occurred, that it is more likely, or that pregnancy or marriage caused it. Figure G1 summarizes the smallest and largest paired differences across each policy's tested grid. Its endpoints can come from different terminal effects, but each male–female pair shares one terminal effect. The figure does not rank policies or assign probabilities to the continuations.

Six policies have a verified equal-terminal comparison under separate reference curves at one or more grid values. Adams, STBF, Wisconsin, and Michigan community-college tuition each permit both signs of the NPV difference at an individual fixed terminal value. Early Head Start and Tennessee have positive differences in the saved comparisons. For Tennessee, when both terminal effects are five percentage points, the saved paths give women a standardized education NPV approximately \$46–\$277 higher than men's, in 2000 dollars. The common-reference sensitivity has the same direction. The positive comparisons describe the paths found and do not establish that every compatible continuation favors women. Wisconsin admits both sexes at a two-point terminal gain, while no female path is found at five points. The separate-reference Kalamazoo banks do not yield a common terminal value on the four-point grid. Under the common reference, a ten-point terminal gain is supported for both sexes, with higher female NPV in the saved comparisons. STBF and Wisconsin, which have identical rounded point inputs, permit both NPV orderings under the common reference. Source precision, population transport, finite search, age mapping, and common-price standardization limit conclusions about subgroup welfare.

These findings motivate examining subgroup dynamics without imposing a female-delay interpretation. Marriage, childbearing, work, and college access may influence education timing, but the reconstructed paths do not identify those mechanisms. Tennessee's authors themselves discuss delayed transitions in schooling, work, and family formation. Their evidence does not establish that marriage or childbirth caused women to delay graduation. Distinguishing that mechanism requires additional subgroup or longitudinal evidence.

Table A1. Literature coverage, estimation times, and observation endpoints

A. Reported effect-estimation times or windows

	Full sample				Journal sample	
Outcome	Eligible	Coded	One	One (%)	One/coded	One (%)
All education outcomes	459	459	249	54.2	244/447	54.6
College entry (CE)	237	236	193	81.8	189/231	81.8
Associate's degree (AA)	40	40	32	80.0	32/39	82.1
Bachelor's degree or higher (BA+)	190	190	132	69.5	130/186	69.9

B. Availability of a numerical observation endpoint

Endpoint classification	Papers	Percent
Exact numerical endpoint	366	79.7
Interval or one-sided bound	38	8.3
Stage identified; clocks not aligned	46	10.0
Endpoint missing or conflicting evidence	9	2.0
Total	459	100.0

Notes: Panel A counts papers within overlapping outcome groups. One time or pooled window is not one year of follow-up. One CE paper lacks a category-specific count. Journal results exclude 11 additional welfare-source papers and one author manuscript. Panel B assigns each of the 459 papers to one endpoint category. Attained age, individual follow-up, and policy/cohort event time remain distinct clocks. Coding rules and evidence records are described in Appendix A.2.

Table A2. Interpretation of finite-window evidence and lifetime valuation

A. Interpretation and valuation

Category	Full sample	Journal sample
Finality or permanence statements	18	17
Completion or window-adequacy arguments only	22	22
Either interpretation category	40	39
Education-based lifetime or career valuation	31	28
Both interpretation and valuation	1	1
At least one category	70	66
Neither detected	389	381

B. Timing of the valued education outcome in 31 papers

Outcome clock	One window	Multiple windows	Papers	Last endpoint Range (n)
Attained age	5	1	6	22–54 (5)
Individual follow-up	9	11	20	1.5–11 (17)
Education stage only	4	0	4	—
Institutional event time	—	—	1	5 (1)
Total	18	12	31	

Notes: Panel A uses 459 papers overall and 447 journal papers. The first two categories are mutually exclusive; valuation can overlap either. Neither detected refers to the reviewed version, not an absence of all long-run discussion. Panel B codes comparable times for the education outcome entering valuation, rather than all outcomes or successive exposed cohorts. One institutional-flow study is separate from individual follow-up. Endpoint ranges use only exact values, with their counts in parentheses; follow-up origins differ across papers. Bounds, pooled intervals, and unaligned stages remain in the sample. The linked 31-paper ledger records sources and coding decisions.

Table B1. Generating scenarios for the observation-design experiment

Scenario	K	b	d	$\Delta H_c(25)$	$\Delta H_c(35)$	NPV
K000	13.3370	0.2575	12.5649	-0.4904	-0.5777	-26.05
K025	13.7876	0.2575	12.5649	-0.1126	-0.1328	-5.99
K050	14.2383	0.2575	12.5649	+0.2652	+0.3121	+14.08
K075	14.6889	0.2575	12.5649	+0.6430	+0.7570	+34.14
K100	15.1395	0.2575	12.5649	+1.0209	+1.2018	+54.20
b000	13.9222	0.1200	12.5649	-1.1271	-0.5345	-25.33
b025	13.9222	0.1903	12.5649	-0.4167	-0.1116	-7.40
b050	13.9222	0.2605	12.5649	+0.0143	+0.0025	+0.23
b075	13.9222	0.3308	12.5649	+0.2553	+0.0323	+4.01
b100	13.9222	0.4011	12.5649	+0.3864	+0.0402	+6.11
d000	13.9222	0.2575	8.0000	+0.0787	+0.0072	+1.26
d025	13.9222	0.2575	13.5000	-0.0295	-0.0027	-0.46
d050	13.9222	0.2575	19.0000	-0.4017	-0.0429	-6.07
d075	13.9222	0.2575	24.5000	-1.2601	-0.1990	-19.24
d100	13.9222	0.2575	30.0000	-2.1886	-0.7111	-40.20

Notes: Each parameter uses the 0, 25, 50, 75, and 100 percent positions of its conditional feasible interval. Scenario names identify the varied parameter and interval position. Only that generating parameter varies; the other two equal the reference values. All three candidate parameters may vary during matching. NPV is in thousands of 2000 dollars per person, under the common signed completion-response rule. Every continuation shares the age-17 baseline level. The mean comparison curve differs slightly from the anchored continuation evaluated at the mean parameter vector.

Table B2. Complete age-25 scans: summary across generating parameters

Generating parameter	Parameter interval	Age-25 effect interval (YS)	Scenarios	Candidate NPV width			Both signs found
				Minimum	Median	Maximum	
K	$[13.337, 15.140]$	$[-0.490, 1.021]$	102	21.92	48.62	65.92	56/102
b	$[0.120, 0.401]$	$[-1.127, 0.386]$	102	43.83	58.27	72.12	91/102
d	$[8.000, 30.000]$	$[-2.189, 0.079]$	102	55.49	65.54	91.01	102/102

Notes: Each row summarizes 101 equally spaced positions in the conditional feasible interval plus the reference value. Only the indicated generating parameter changes; the other two stay at their reference values. All three candidate parameters are free when matching the resulting age-25 effect. Widths are in thousands of 2000 dollars per person. Both signs requires candidates below −$0.01 and above $0.01. Counts summarize a deterministic design and are not path probabilities. The complete 306-row file, including terminal effects, cumulative gaps, generating values, and parameterized extrema, is supplied in the accompanying replication materials.

Table B3. Reference and single-parameter reconstructed earnings NPVs

	Reference	Permanent (K)		Temporary (b)		Temporary (d)	
Policy	NPV	ΔK	NPV	Δb (%)	NPV	Δd	NPV
Florida grant	8.63	0.323	15.52	—	—	—	—
DC tuition aid	8.95	0.637	30.68	9.56	7.27	—	—
Adams	−1.07	−0.0230	−1.43	−1.03	−0.37	0.334	−0.33
STBF	0.66	Joint	0.74	—	—	—	—
Wisconsin grant	0.59	0.120	1.05	4.76	0.14	—	—
Kalamazoo	1.10	0.375	1.55	—	—	—	—
Michigan CC	9.45	0.0330	11.49	3.12	3.41	—	—
CUNY Pell	0.56	0.0164	1.04	0.526	0.17	−0.147	0.12
Ohio Pell	2.53	0.224	8.88	9.17	1.46	—	—
Tennessee HOPE	1.31	0.0435	2.77	1.55	0.35	−0.428	0.23
Tennessee Pell	−0.30	−0.0142	−0.62	−0.548	−0.08	0.150	−0.05
Georgia HOPE	2.81	0.236	7.03	8.73	1.20	—	—
Social Security	2.78	22.4	2.61	—	—	—	—
Appropriations: spending	2.70	0.0841	5.16	3.22	0.89	—	—
Early Head Start	n.a.	—	—	—	—	—	—
Texas CC	11.16	0.248	13.99	—	—	—	—
Adult Pell	2.47	0.0601	6.47	—	—	—	—
Appropriations: tuition	0.03	8.41×10^{-4}	0.05	0.0318	9.55×10^{-3}	−0.0106	6.55×10^{-3}
HOPE credit	5.95	0.144	14.27	5.46	2.91	—	—
HOPE ind. single: lower	6.44	1.96×10^{-4}	36.06	0.00946	8.04	−0.00414	8.21
HOPE dep. single: lower	−4.14	-2.72×10^{-4}	−16.49	−0.0131	−3.44	0.00575	−3.48
HOPE dep. single: upper	5.06	3.15×10^{-4}	20.18	0.0152	4.20	−0.00665	4.27
HOPE joint: lower	23.75	0.00124	94.72	0.0599	19.73	−0.0263	20.04
HOPE joint: upper	9.72	4.74×10^{-4}	38.76	0.0229	8.08	−0.0100	8.20
Tuition ded. single: lower	6.88	0.0379	27.53	1.05	3.22	−0.257	1.80
Tuition ded. joint: lower	−1.66	−0.0227	−6.62	−0.631	−0.79	0.149	−0.44
HOPE / Lifetime Learning	−57.97	−0.0794	−77.41	−4.90	−25.65	—	—
AOTC single: upper	−1.40	-4.22×10^{-5}	−3.42	−0.00136	−0.41	3.37×10^{-4}	−0.25
AOTC single: lower	31.04	9.72×10^{-4}	75.53	0.0314	9.02	−0.00777	5.54
AOTC joint: upper	−11.54	-2.79×10^{-4}	−28.09	−0.00901	−3.36	0.00223	−2.06
AOTC joint: lower	12.01	2.98×10^{-4}	29.24	0.00962	3.49	−0.00238	2.14
AOTC ind. single: lower	1.86	2.01×10^{-5}	6.37	6.51×10^{-4}	0.82	-1.61×10^{-4}	0.50
AOTC instrument	5.97	9.39×10^{-5}	21.26	0.00390	3.52	−0.00136	2.94
Tennessee Promise	6.46	Joint	n.c.	—	—	—	—
College placement	n.a.	0.0415	n.c.	3.41	n.c.	—	—
Tuition ded. single: upper	0.00	0.000	0.00	0.00	0.00	0.000	0.00
Tuition ded. joint: upper	0.00	0.000	0.00	0.00	0.00	0.000	0.00

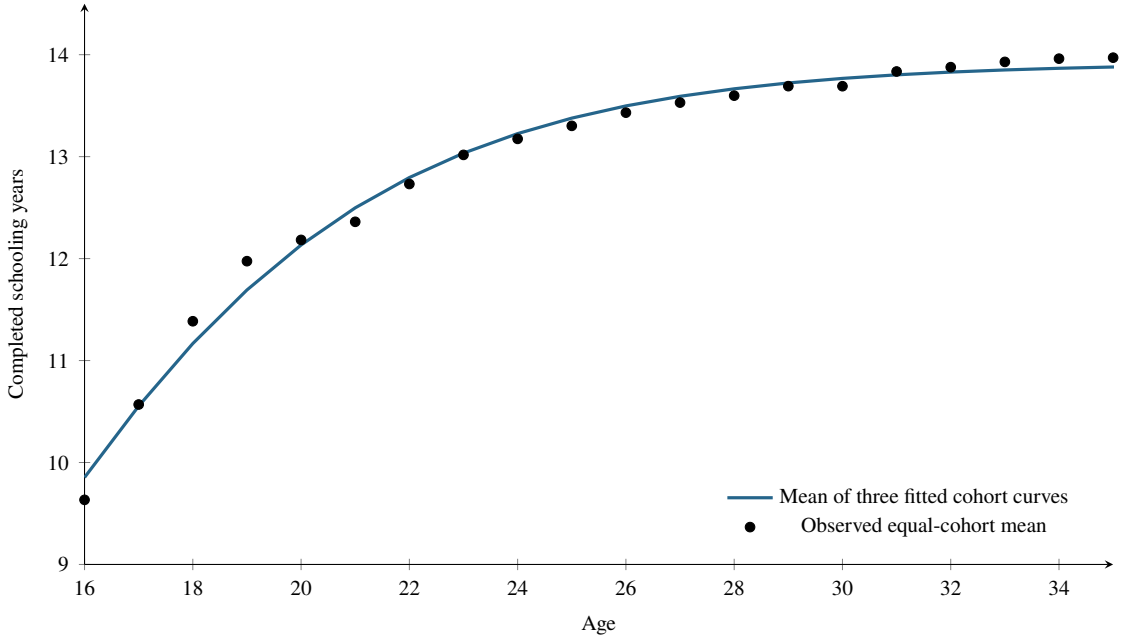
Notes: Each pair varies the named parameter family; the other two remain at H^0 values. K changes eventual attainment; b and d effects eventually vanish. ΔK is in schooling years, except percentage points for Social Security and Early Head Start; Δd is in years and Δb is the proportional speed change in percent. STBF jointly changes time enrolled and three credential plateaus; Tennessee jointly changes three credential plateaus. Their outcome-specific changes are saved in the matched-solution data. No single-family solution matches every additional point input exactly; — denotes no admissible L2 match. NPVs use comparable account units per reference expenditure. n.a.: unavailable reference projection or normalizer; n.c.: no comparable pair, including unresolved source monetary years for EHS and Tennessee. Figure B3 plots the comparable pairs. Appendix B.5 defines the matching procedure. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds.

Table C1. ACS cohort support and comparison-path fits

Birth cohort	Records	Ages	K	b	d	RMSE
1985	630,409	16–35	13.8926	0.2533	12.5107	0.112
1986	659,595	16–35	13.9037	0.2614	12.6170	0.113
1987	688,856	16–35	13.9704	0.2578	12.5669	0.115

Notes: The full-population ACS extract covers 2000–2022; the selected cohort–age cells occur in 2001–2022. Each cohort contributes all 20 ages. Fits minimize weighted squared differences from person-weighted cell means, using normalized cell population weights within each cohort. The comparison curve averages the three fitted curves with fixed one-third weights. RMSE is in schooling years. Parameter estimates and fit diagnostics are descriptive inputs, not causal policy effects.

Figure C1. Comparison-path fit: ACS birth cohorts 1985–1987



Notes: The graph covers observed ages 16–35 only. Each cohort is fitted to its person-weighted age means using normalized cell population weights. The three fitted curves and observed cell means receive fixed one-third cohort weights. This in-sample fit diagnostic does not validate extrapolation beyond age 35.

Table D1. Tennessee Promise: population matching and continuation limits

Quantity	Some college+	AA+	BA+
Source control mean, ages 23–24	0.6850	0.3800	0.2920
Matched treated mean, ages 23–24	0.7200	0.4200	0.3190
Remaining-completion fraction cap	0.1402	0.2635	0.1978
Implied treated plateau ceiling	0.7593	0.5728	0.4537
Largest saved treated plateau	0.7527	0.5728	0.4537

Notes: The high-school-graduate denominator and prior-year Tennessee residence follow the source population convention. Source control means match levels; cohort fits supply age shapes. Donor completion fractions bound the permitted later attainment under the stated transport assumption. All joint candidates satisfy credential nesting and the current source-window restrictions. The plateau ceiling is not an estimated causal bound.

Table D2. Tennessee Promise: L3 point targets and fitted residuals

Outcome	Age window	Target	Fitted	Residual	Exact L2?
SC+	18–19	3.500	3.677	+0.177	–
SC+	19–20	3.200	2.681	-0.519	–
SC+	20–21	2.500	2.951	+0.451	–
SC+	21–22	2.700	3.335	+0.635	–
SC+	22–23	2.800	3.467	+0.667	–
SC+	23–24	3.500	3.500	+0.000	Yes
AA+	18–19	0.200	0.756	+0.556	–
AA+	19–20	1.400	1.626	+0.226	–
AA+	20–21	3.100	2.518	-0.582	–
AA+	21–22	2.600	2.837	+0.237	–
AA+	22–23	2.900	2.928	+0.028	–
AA+	23–24	4.000	4.000	+0.000	Yes
BA+	18–19	-0.100	-0.027	+0.073	–
BA+	19–20	0.100	-0.076	-0.176	–
BA+	20–21	0.001	-0.141	-0.142	–
BA+	21–22	-0.200	0.028	+0.228	–
BA+	22–23	1.200	1.057	-0.143	–
BA+	23–24	2.700	2.700	+0.000	Yes

Notes: Entries are percentage points. Targets come from Table 2 of Schoonover et al. (2026), retaining the reported precision, including the 0.00001 probability coefficient for BA+ at ages 20–21. Predictions use the same adjacent-age averages and cohort weights. The three latest-window effects are exact. One joint path minimizes equal-weight squared discrepancies subject to the retained support, level, anchor, nesting, and late-completion restrictions. Its eighteen-target RMSE is 0.353 percentage points and maximum absolute residual is 0.667 points. No graphical permutation interval enters this fixed-input fit. Normal-input perturbations are reported separately and do not recover the source permutation distribution.

Table D3. Source education inputs, clocks, and reconstruction roles

Education input	Effect	SE	Clock / role
Adams: eligibility RD, 2005–2006 graduating classes			
Four-year enrollment, immediate	0.0090	0.0080	HS+0; L3 conversion
Two-year enrollment, immediate	0.0080	0.0050	HS+0; L3 conversion
Four-year enrollment, within two years	0.0000	0.0070	HS+2; conversion
Two-year enrollment, within two years	0.0070	0.0050	HS+2; conversion
Bachelor's completion, within four years	-0.0170	0.0100	HS+4; L3 conversion
Two-year completion, within four years	-0.0020	0.0030	HS+4; L3 conversion
Bachelor's completion, within six years	-0.0250	0.0090	HS+6; conversion
Two-year completion, within six years	0.0000	0.0040	HS+6; conversion
Schooling-equivalent input, age 18	0.0260	–	Assigned age; L3 target
Schooling-equivalent input, age 20	0.0070	–	Assigned age; L3 target
Schooling-equivalent input, age 22	-0.0110	–	Assigned age; L3 target
Schooling-equivalent input, age 24	-0.0180	–	Assigned age; exact L2
Social Security: benefit eligibility; full-time college attendance			
Ever attended by age 28	0.2240	0.1060	Exact L2; attrition lower bound
Ever attended by age 23	0.2190	0.1020	Exact additional L3
Tennessee Promise: age-window effects of program exposure			
Some college+, ages 18–19	0.03500	0.0040	L3 point target
Some college+, ages 19–20	0.03200	0.0040	L3 point target
Some college+, ages 20–21	0.02500	0.0040	L3 point target
Some college+, ages 21–22	0.02700	0.0030	L3 point target
Some college+, ages 22–23	0.02800	0.0040	L3 point target
Some college+, ages 23–24	0.03500	0.0040	L2
AA+, ages 18–19	0.00200	0.0010	L3 point target
AA+, ages 19–20	0.01400	0.0020	L3 point target
AA+, ages 20–21	0.03100	0.0030	L3 point target
AA+, ages 21–22	0.02600	0.0030	L3 point target
AA+, ages 22–23	0.02900	0.0040	L3 point target
AA+, ages 23–24	0.04000	0.0040	L2
BA+, ages 18–19	-0.00100	0.0000[a]	L3 point target
BA+, ages 19–20	0.00100	0.0000[a]	L3 point target
BA+, ages 20–21	0.00001	0.0010	L3 point target
BA+, ages 21–22	-0.00200	0.0020	L3 point target
BA+, ages 22–23	0.01200	0.0030	L3 point target
BA+, ages 23–24	0.02700	0.0040	L2

Notes: Degree effects are in probability units; 0.01 is one percentage point. Adams uses the original study's Table 4, columns (3) and (5), with the retained enrollment/completion multipliers. The four assigned ages are separate timing assumptions; derived-input uncertainty is generated from the grouped coefficient draws rather than a separately reported SE. Social Security uses Dynarski (2003), Table 3's full-time-attendance column. The age-28 lower-bound label refers to attrition imputation, not an uncertainty interval. STBF's separate joint inputs are documented in Appendix D.10. Tennessee coefficients and clustered SEs reproduce rounded Table 2 values; its L3 fits these point estimates without SE bands (Appendix D.8). [a] A displayed-zero SE receives a 0.0005 floor in the current 1,000 normal-input repetitions as an imposed reporting-precision scale. No joint confidence region is implied.

Table D4. Source welfare accounts and the path replacement

Component	Adams	Social Security	Tennessee Promise
Net earnings NPV	-1,204.04	21,163.75	95,557.69
Reference private nonincome component	2,329.97	6,320.04	2,124.31
Reference pretax public component	1,658.06	9,150.29	14,792.00
L1 WTP	1,364.23	23,072.74	72,837.00
L1 net cost	1,896.35	4,739.23	-10,053.00
L1 MVPF	0.72	4.87	$+\infty$
Dollar year	2004	2000	2021 convention

Notes: The table preserves each reference account. Social Security uses the Hendren and Sprung-Keyser enrollment alternative with two schooling years per induced entrant. Its early and later tax rates are 0.003 and 0.199; its source private and public education-payment bundles follow distinct duration conventions. The displayed nonincome components are their values at the source reference batch, not frozen residuals for every path. Adams also transports source education payments. Tennessee uses the public ACS reconstruction and participant normalization described in Appendix D.8; its implied source earnings NPV is retained separately from the reconstructed schedule. STBF remains in Appendix E.2, with its author model and reference account unchanged. Income responses include early earnings losses once.

Table D5. Verified joint paths and paired welfare components

Path	Outcome	K	b	d	$\Delta H(65)$	WTP	Net cost	MVPF
A1	YS	14.3287	0.3484	14.1078	-0.5059	7.27	15.10	0.48
A2	YS	15.7049	0.1847	12.7776	0.8692	-11.50	-20.90	–
A3	YS	14.7917	0.2537	12.7004	-0.0430	1.66	2.64	0.63
S1	Entry	0.5843	0.5849	19.9509	0.2298	19.34	5.10	3.79
S2	Entry	0.5785	2.7363	18.1979	0.2240	23.46	4.70	4.99
S3	Entry	0.5785	1.2779	19.6195	0.2240	20.81	4.95	4.21
S4	Entry	0.5785	1.0163	18.6110	0.2240	21.89	4.85	4.52
T1	SC+	0.7271	1.0647	19.0777	0.0419	-48.72	32.66	-1.49
	AA+	0.4241	2.2608	21.2173	-0.0479			
	BA+	0.3217	3.0000	21.6230	-0.0406			
T2	SC+	0.7200	3.0000	17.5874	0.0348	152.50	-38.04	$+\infty$
	AA+	0.5728	0.7691	22.1110	0.1008			
	BA+	0.4537	0.8964	22.4814	0.0914			
T3	SC+	0.7203	1.4437	17.8946	0.0351	86.44	-14.83	$+\infty$
	AA+	0.5687	0.8835	22.2619	0.0967			
	BA+	0.4057	1.3628	22.4467	0.0434			

Notes: WTP and net cost are thousands of case-specific dollars. A1–A2, S1–S2, and T1–T2 minimize/maximize L2 earnings NPV; A3 is the Adams minimum-distance fit to four converted inputs; its coincident extrema are shown once. S3–S4 minimize/maximize Social Security L3 earnings NPV. T3 is the Tennessee L3 best-fitting path, shown once because its extrema coincide. Neither selected fit supplies distinct bound witnesses. All rows with the same path label form one joint candidate. Social Security uses cumulative full-time attendance in probability units and the entry-indexed Hendren and Sprung-Keyser enrollment alternative. Its reference calibration is documented in Appendix D.7. Each witness passes the full joint restrictions; parameters are rounded only for display. Age 65 is extrapolated, and Tennessee earnings valuation ends at 64.

Table E1. Policy registry, calibration populations, and available L3 paths

Policy	ACS geography	Birth cohorts	Outcome	L3 paths
Florida access grant	Florida	1982	YS	1
DC tuition assistance	District of Columbia	1981	YS	–
Adams Scholarship	Massachusetts	1987	YS	1
STBF scholarship	Nebraska	1993–1997	T, C, A, B	–
Wisconsin Scholars Grant	Wisconsin	1990–1992	YS	1
Kalamazoo Promise	Michigan	1988	YS	1
Michigan CC tuition	Michigan	1992	YS	49
CUNY Pell grants	New York	1991	YS	50
Ohio Pell grants	Ohio	1981	YS	–
Tennessee HOPE	Tennessee	1989	YS	–
Tennessee Pell eligibility	Tennessee	1987–1992	YS	–
Georgia HOPE	Georgia	1975–1979	YS	–
Social Security student benefits	National	1963; shape 1985–1987	Entry	56
College appropriations: spending	National	1983	YS	1
Early Head Start	National	1989–1999	BA+	–
Texas CC tuition	Texas	1987	YS	1
Adult Pell introduction	National	1947	YS	–
College appropriations: tuition	National	1983	YS	1
HOPE tax credit	National	1981	YS	1
HOPE: independent single, lower kink	National	1987–1988	YS	–
HOPE: dependent single, lower kink	National	1987–1988	YS	–
HOPE: dependent single, upper kink	National	1987–1988	YS	–
HOPE: joint, lower kink	National	1987–1988	YS	–
HOPE: joint, upper kink	National	1987–1988	YS	–
Tuition deduction: single, lower threshold	National	1988	YS	–
Tuition deduction: joint, lower threshold	National	1988	YS	–
HOPE and Lifetime Learning credits	National	1980	YS	–
AOTC: dependent single, upper kink	National	1991–1992	YS	–
AOTC: dependent single, lower kink	National	1991–1992	YS	–
AOTC: joint, upper kink	National	1991–1992	YS	–
AOTC: joint, lower kink	National	1991–1992	YS	–
AOTC: independent single, lower kink	National	1991–1992	YS	–
AOTC: simulated instrument	National	1987–1992	YS	–
Tennessee Promise	Tennessee	1992–1998	CE, AA+, BA+	1
Algorithmic college placement	New York	1995	YS	–

Continued on next page

Table E1 (continued)

Policy	ACS geography	Birth cohorts	Outcome	L3 paths
Tuition deduction: single, upper threshold	National	1988	YS	–
Tuition deduction: joint, upper threshold	National	1988	YS	–
Cannabis legalization: education channel	RML 14 states	1988–1999	YS	1

Notes: All 38 calculations have numerical path banks; cannabis displays its primary L3. A dash means no additional timing restriction is encoded. L3 requires an additional reported source window under the retained conversion; synthetic duration allocations are excluded. No candidate means a main L3 search with no verified path; it is not a proof of infeasibility. YS denotes a source-specific schooling-equivalent input; STBF jointly uses a time-enrolled index and three cumulative credentials; Tennessee uses joint cumulative SC+, AA+, and BA+ paths. The underlying machine-readable registry additionally records source-study families, source windows, operator weights, source-control qualifications, and information-layer reasons. The main counts retain historical IDs.

Table E2. L3 fixed-point minimum-distance fits

Policy	Targets	RMSE	Max. error	NPV	MVPF
Florida grant	5	0.01735	0.02366	26.46	18.09
Adams Scholarship	4	0.00818	0.01417	-2.74	0.63
Wisconsin grant	9	0.01638	0.02664	8.09	1.96
Kalamazoo Promise	3	0.01164	0.01730	57.29	3.77
College spending	5	0.00097	0.00175	7.57	16.52
Texas tuition	8	0.00388	0.00744	13.12	$+\infty$
Tuition caps	5	0.00190	0.00247	-4.60	-1.97
HOPE tax credit	3	0.00053	0.00084	16.19	$+\infty$
Tennessee Promise	18	0.35251	0.66733	113.94	$+\infty$
Cannabis: education	3	0.00595	0.00914	-3.64	-55.85

Notes: All ten searches retain one best-fitting path with nonzero additional-input residuals and exact L2 constraints. RMSE includes every encoded target, including the exact retained moment(s); errors are schooling years except Tennessee, which is in percentage points. NPV is thousands of case-specific dollars; price years and policy units differ. The identity weight matrix is fixed within each case. These are conditional fitted values, not ranges from SE screens or certified unique/global solutions. Three other admitted L3 cases retain zero-residual path banks. Tables E4 and D2 report individual discrepancies.

Table E3. Conditional welfare ranks

Policy	L1	Own L2	Selected	Layer
HOPE: joint, lower kink [f]†	1	23	23	L2
Tuition deduction: single, lower threshold†	1	23	23	L2
AOTC: dependent single, lower kink [f]†	1	23	23	L2
AOTC: joint, lower kink [f]	1	23	23	L2
Tennessee Promise [r]	1	14	1	L3
Algorithmic college placement [x]	1	13	13	L2
Tuition deduction: single, upper threshold†	1	23	23	L2
Texas CC tuition	8	5	1	L3
Michigan CC tuition	9	20	6	L3
DC tuition assistance	10	15	15	L2
HOPE: independent single, lower kink [f]†	11	23	23	L2
HOPE tax credit	12	20	1	L3
HOPE: dependent single, upper kink [f]†	13	23	23	L2
AOTC: simulated instrument [f]†	14	23	23	L2
Florida access grant	15	12	4	L3
AOTC: independent single, lower kink [f]	16	23	23	L2
Social Security student benefits	17	7	7	L3
College appropriations: spending	18	20	5	L3
Georgia HOPE	19	18	18	L2
Ohio Pell grants	20	21	21	L2
HOPE: joint, upper kink [f]†	21	23	23	L2
Adult Pell introduction	22	10	10	L2
Kalamazoo Promise	23	11	8	L3
Tennessee HOPE	24	22	22	L2
STBF scholarship	25	11	11	L2
Wisconsin Scholars Grant	26	15	9	L3
CUNY Pell grants	27	36	17	L3
College appropriations: tuition	28	20	16	L3
Tennessee Pell eligibility†	29	37	37	L2
Tuition deduction: joint, upper threshold	30	19	19	L2
Adams Scholarship	31	12	12	L3
Early Head Start [r]	32	14	14	L2
AOTC: dependent single, upper kink [f]†	33	23	23	L2
Tuition deduction: joint, lower threshold	34	20	20	L2
HOPE: dependent single, lower kink [f]†	35	23	23	L2
AOTC: joint, upper kink [f]†	36	23	23	L2
HOPE and Lifetime Learning credits†	37	23	23	L2

Notes: All 37 calculations with a reference MVPF (L1) are retained. Own L2 replaces only that calculation's selected set by L2. Selected uses L3 where admitted and L2 otherwise. [f] marks a finite tax experiment with a local-linear input conversion, [r] a public-data reconstructed account, and [x] a fiscal-only earnings extension. † marks at least one unassigned-ratio path; those paths, not the calculations, are omitted from ranking. Finite endpoints are rounded to two decimals before lower-endpoint-first ordering, and tied endpoint pairs share a competition rank. Table 3 reports the endpoints; Table E5 and the case profiles document source observation windows. Cannabis has no whole-policy L1. The restricted 24-case ordering and rounding sensitivities remain in the replication materials. These are conditional display ranks, not social-welfare priorities or certified global bounds.

Table E4. L3 schooling-input targets, fitted values, and residuals

Policy	Source input	Target	Fitted	Residual	Exact L2?
Florida grant	Year 1 cumulative	0.045750	0.055306	+0.009556	–
	Year 2 cumulative	0.111083	0.093794	-0.017289	–
	Year 3 cumulative	0.160292	0.136635	-0.023657	–
	Year 4 cumulative	0.181917	0.181917	-0.000000	Yes
	First semester	0.014792	0.038354	+0.023562	–
Adams Scholarship	Immediate entry (age 18)	0.026000	0.011830	-0.014170	–
	Entry within two years (age 20)	0.007000	0.010579	+0.003579	–
	Four-year completion (age 22)	-0.011000	-0.003653	+0.007347	–
	Six-year completion (age 24)	-0.018000	-0.018000	-0.000000	Yes
Wisconsin grant	Cohort 1 cumulative	0.037500	0.064138	+0.026638	–
	Cohort 1: year 1 flow	0.020833	0.018002	-0.002832	–
	Cohort 1: year 2 flow	0.008333	0.022278	+0.013945	–
	Cohort 1: year 3 flow	0.004167	0.023858	+0.019692	–
	Cohorts 2–3: year 1 flow	0.016667	0.041935	+0.025268	–
	Cohorts 2–3: year 2 flow	0.041667	0.023689	-0.017978	–
	Cohort 2: year 3 flow	0.020833	0.019302	-0.001531	–
	Cohorts 2–3 cumulative	0.087500	0.075275	-0.012225	–
	Retained weighted aggregate	0.071772	0.071772	+0.000000	Yes
Kalamazoo Promise	Year 4 cumulative	0.273333	0.273333	-0.000000	Yes
	Year 2 cumulative	0.134833	0.124491	-0.010342	–
	Year 3 cumulative	0.179542	0.196841	+0.017299	–
College spending	Retained four-year sum	0.046013	0.046013	+0.000000	Yes
	Lag 0 contribution	0.005041	0.006115	+0.001075	–
	Lag 1 contribution	0.013199	0.011449	-0.001750	–
	Lag 2 contribution	0.014011	0.013972	-0.000039	–
	Lag 3 contribution	0.013762	0.014476	+0.000714	–
Texas tuition	Model age 18 cumulative	0.031550	0.038841	+0.007291	–
	Model age 19 cumulative	0.077550	0.078264	+0.000714	–
	Model age 20 cumulative	0.112750	0.114291	+0.001541	–
	Model age 21 cumulative	0.145650	0.144871	-0.000779	–
	Model age 22 cumulative	0.172150	0.169402	-0.002748	–
	Model age 23 cumulative	0.187250	0.188188	+0.000938	–
	Model age 24 cumulative	0.209450	0.202006	-0.007444	–
	Model age 26 cumulative	0.218450	0.218450	-0.000000	Yes
Tuition caps	Retained four-year sum	0.000461	0.000461	+0.000000	Yes
	Lag 0 contribution	-0.000921	0.001002	+0.001923	–
	Lag 1 contribution	0.003799	0.001332	-0.002467	–
	Lag 2 contribution	0.001785	0.000044	-0.001740	–
	Lag 3 contribution	-0.004202	-0.001918	+0.002284	–
HOPE tax credit	First-year entry (age 19)	0.060720	0.060720	-0.000000	Yes
	Continuation (age 20)	0.025944	0.026342	+0.000398	–
	Retained duration total (age 20.5)	0.049128	0.048292	-0.000836	–

L3 schooling-input residuals (continued)

Policy	Source input	Target	Fitted	Residual	Exact L2?
Cannabis: education	Four-year completion bridge	-0.032000	-0.027232	+0.004768	–
	Five-year completion bridge	-0.022000	-0.031136	-0.009136	–
	Six-year completion bridge	-0.036000	-0.036000	+0.000000	Yes

Notes: All entries are schooling-equivalent years. Residual equals fitted minus target. The retained L2 moments are exact to numerical precision; additional observations are fixed point targets, not intervals. Display rounding does not change the fitted inputs. The Wisconsin semester and cumulative estimates do not obey an established exact adding-up identity; their reported precision and cohort windows are preserved. The age, population, and conversion bridges are stated in Appendix E.2. No standard error is used to widen acceptance or weight these fits.

Table E5. Reported education timing and its use in reconstruction

Cases	Windows	Source timing evidence	Current reconstruction
Florida access grant	3+	First-semester and years 1–4 credits	Reported timing retained as L3
DC tuition assistance	1	First college entry within 12 months of HS	L2 only; no added timing constraint
Adams Scholarship	3+	Entry, persistence, and BA through year 6	L3: converted targets at ages 18, 20, 22 and 24
STBF scholarship	3+	Semester enrollment; BA in years 4–6	L2 only; no added timing constraint
Wisconsin Scholars Grant	3+	Semester credits; cumulative years 2/3	Reported timing retained as L3
Kalamazoo Promise	3+	Credits in years 2–4; degrees in years 4/6	Reported timing retained as L3
Michigan CC tuition	2	Entry within a year; later credentials	L3: early converted entry component
CUNY Pell grants	2	First-year and three-year credits	Reported timing retained as L3
Ohio Pell grants	1	Entry to next-year persistence	L2 only; no added timing constraint
Tennessee HOPE	1	First post-HS college enrollment	L2 only; no added timing constraint
Tennessee Pell eligibility	1	Initial-fall college-sector choice	L2 only; allocated timing excluded
Georgia HOPE	Other	State-year freshman flows	L2 only; allocated timing excluded
Social Security student benefits	2	Attainment at ages 23 and 28	Reported timing retained as L3
College appropriations	Other	Institution enrollment by policy lag	L3: lag responses under explicit time/population mapping
Early Head Start	1	Pooled adult attainment	L2 only; no added timing constraint
Texas CC tuition	3+	Annual enrollment at HS+0 through HS+6	Reported timing retained as L3
Adult Pell introduction	1	Pooled current adult enrollment	L2 only; no added timing constraint
HOPE tax credit	Other	College-grade and pooled enrollment	L3: first- and second-year inputs under stated age and risk-set mappings
HOPE/AOTC income kinks	1	Contemporaneous credit-kink response	L2 only; no added timing constraint
Tuition-deduction thresholds	3+	Years 0–4 from expected HS graduation	Different operator; L2 in main plot
HOPE and Lifetime Learning credits	1	Pooled current college enrollment	L2 only; no added timing constraint
AOTC: simulated instrument	2	Attendance at ages 19 and 22	Different operator; L2 in main plot
Tennessee Promise	3+	Six age windows, 18–19 to 23–24	Reported timing retained as L3
Algorithmic college placement	2	Initial term; pooled and oldest-cohort follow-up	L2 only; no added timing constraint
Cannabis legalization: education channel	3+	BA completion within 4, 5, 6 years	Reported timing retained as L3; not ranked

Notes: Windows count source ages or follow-up periods, not coefficients. Institutional lags and college-grade contrasts are listed as Other. L3 retains L2 and adds source windows or dynamics under stated outcome, timing, and population mappings. A duration-only split of one input is not L3. Cannabis lacks L1. The timing audit records source locations.

Table E6. Point-input sensitivity: numerical comparisons

Policy	Set	Main	Narrow	Wide	National	Earnings 75%	Earnings 125%	Denser
Direct student financial aid								
Florida access grant	L3	18.09	18.25	17.88	17.64	5.99	–^{S}	18.09
DC tuition assistance	L2	$[-1.60, > 10^6]^{S}$	$[6.67, > 10^6]^{S}$	$[-5.01, > 10^6]^{S}$	$[1.04, > 10^6]^{S}$	$[-1.30, > 10^6]^{S}$	$[-1.77, > 10^6]^{S}$	$[-1.60, > 10^6]^{S}$
Adams Scholarship	L3	0.63	0.62	0.63	0.63	0.88	0.40	0.63
STBF scholarship	L2	[1.25, 4.01]	[1.27, 2.78]	[1.24, 6.55]	[1.28, 5.69]	[1.18, 2.98]	[1.31, 5.43]	[1.24, 4.01]
Wisconsin Scholars Grant	L3	1.96	2.09	1.77	1.95	1.60	2.38	1.96
Kalamazoo Promise	L3	3.77	3.62	3.84	4.29	2.60	5.43	3.77
CUNY Pell grants	L3	[−3.17, 2.34]	[−2.01, 1.79]	[−5.19, 3.02]	[−2.96, 2.60]	[−3.01, 1.78]	[−3.26, 3.03]	[−3.17, 2.34]
Ohio Pell grants	L2	[−11.75, 7.53]	[−1.73, 7.08]	$[< -10^6, 7.83]$	[−8.76, 7.49]	$[< -10^6, 3.73]$	[−7.52, 19.58]	[−11.75, 7.53]
Tennessee HOPE	L2	[−75.64, 21.66]	[−34.06, 18.34]	[−371.25, 24.53]	[−77.66, 23.24]	$[< -10^6, 6.48]$	$[-15.01, > 10^6]^{S}$	[−75.64, 21.66]
Tennessee Pell eligibility	L2	$[< -10^6, 2.70]$	$[< -10^6, 2.66]$	$[< -10^6, 2.72]$	$[< -10^6, 2.70]$	$[< -10^6, 1.66]$	$[< -10^6, 4.25]$	$[< -10^6, 2.70]$
Georgia HOPE	L2	$[-3.19, > 10^6]^{S}$	$[-1.36, > 10^6]^{S}$	$[-4.78, > 10^6]^{S}$	$[-3.50, > 10^6]^{S}$	[−3.05, 56.42]	$[-3.28, > 10^6]^{S}$	$[-3.19, > 10^6]^{S}$
Social Security benefits	L3	[4.21, 4.52]	[4.27, 4.48]	[4.11, 4.56]	[4.21, 4.52]	[2.88, 3.04]	[6.23, 6.81]	[4.21, 4.52]
Adult Pell introduction	L2	[1.95, 4.63]	[2.87, 4.59]	[−0.50, 4.64]	[1.95, 4.63]	[1.64, 2.73]	[2.28, 8.20]	[1.95, 4.63]
Tennessee Promise	L3	–^{S}	–^{S}	–^{S}	–^{S}	–^{S}	–^{S}	–^{S}
Education tax benefits								
HOPE tax credit	L3	–^{S}	–^{S}	–^{S}	–^{S}	–^{S}	–^{S}	–^{S}
HOPE: indep. single, start	L2	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[-1000.00, > 10^6]^{S}$	$[-52.08, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$
HOPE: dep. single, start	L2	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[-2{,}697, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$
HOPE: dep. single, end	L2	$[< -10^6, > 10^6]^{S}$	$[-443.66, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[-220.34, > 10^6]^{S}$	$[-161.93, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$
HOPE: joint, start	L2	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[-15.87, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$
HOPE: joint, end	L2	$[< -10^6, > 10^6]^{S}$	$[-21.92, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, 1{,}000]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$
Tuition ded.: single, start	L2	$[< -10^6, -7.40]^{S}$	$[< -10^6, -7.65]^{S}$	$[< -10^6, -7.26]^{S}$	$[< -10^6, -7.40]^{S}$	$[< -10^6, -13.36]^{S}$	$[< -10^6, -5.99]^{S}$	$[< -10^6, -7.40]^{S}$
Tuition ded.: joint, start	L2	$[-4.35, > 10^6]^{S}$	$[-4.28, > 10^6]^{S}$	$[-4.41, > 10^6]^{S}$	$[-4.35, > 10^6]^{S}$	$[-4.38, > 10^6]^{S}$	$[-4.34, > 10^6]^{S}$	$[-4.35, > 10^6]^{S}$
HOPE and Lifetime Learning	L2	$[< -10^6, -9.59]^{S}$	$[< -10^6, -9.79]^{S}$	$[< -10^6, -9.33]^{S}$	$[< -10^6, -9.59]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, -7.02]^{S}$	$[< -10^6, -9.59]^{S}$
AOTC: single, end	L2	$[< -10^6, > 10^6]^{S}$	$[-116.46, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[-188.71, > 10^6]^{S}$	$[-95.11, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$
AOTC: single, start	L2	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[-1{,}240, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$
AOTC: joint, end	L2	$[< -10^6, > 10^6]^{S}$	$[-8.79, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[-12.62, > 10^6]^{S}$	$[-300.32, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$
AOTC: joint, start	L2	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, -13.39]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$
AOTC: indep. single, start	L2	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$
AOTC: simulated instrument	L2	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[< -10^6, > 10^6]^{S}$	$[-372.79, > 10^6]^{S}$	$[< -10^6, 36.36]^{S}$	$[< -10^6, > 10^6]^{S}$
Tuition ded.: single, end	L2	$[< -10^6, -6.33]^{S}$	$[< -10^6, -6.41]^{S}$	$[< -10^6, -6.27]^{S}$	$[< -10^6, -6.33]^{S}$	$[< -10^6, -8.17]^{S}$	$[< -10^6, -5.68]^{S}$	$[< -10^6, -6.33]^{S}$
Tuition ded.: joint, end	L2	$[-3.92, > 10^6]^{S}$	$[-3.84, > 10^6]^{S}$	$[-3.98, > 10^6]^{S}$	$[-3.92, > 10^6]^{S}$	$[-3.72, > 10^6]^{S}$	$[-4.03, > 10^6]^{S}$	$[-3.92, > 10^6]^{S}$
Tuition and public-college funding								
Michigan CC tuition	L3	$[9.03, > 10^6]^{S}$	$[16.60, > 10^6]^{S}$	$[3.81, > 10^6]^{S}$	$[3.89, > 10^6]^{S}$	[4.70, 26.12]	$[21.35, > 10^6]^{S}$	$[9.03, > 10^6]^{S}$
College spending	L3	16.52	25.71	9.27	16.52	6.78	189.51	16.52
Texas CC tuition	L3	–^{S}	–^{S}	–^{S}	–^{S}	14.33	–^{S}	–^{S}
College tuition caps	L3	−1.97	−1.36	−2.82	−1.97	−1.53	−2.27	−1.97
College assessment and course placement								
Algorithmic placement	L2	$[0.08, > 10^6]^{S}$	$[0.12, > 10^6]^{S}$	$[0.06, > 10^6]^{S}$	$[0.08, > 10^6]^{S}$	$[0.10, > 10^6]^{S}$	$[0.06, > 10^6]^{S}$	$[0.08, > 10^6]^{S}$
Early childhood services								
Early Head Start	L2	$[-1.37, > 10^6]^{S}$	[−0.48, 13.31]	$[-1.60, > 10^6]^{S}$	$[-1.37, > 10^6]^{S}$	[−1.19, 17.32]	$[-1.54, > 10^6]^{S}$	$[-1.37, > 10^6]^{S}$
Education channel of other policies								
Cannabis: education channel	L3	−55.85	−53.65	−58.94	−47.77	–	−14.94	−55.85

Notes: Brackets give the span of positive-cost ratios among verified candidates; equal displayed endpoints are printed once. Short policy names retain the distinctions in the policy registry. Superscript S marks self-financing separately. $> 10^6$ and $< -10^6$ indicate attained finite values beyond those display thresholds, not certified infinite bounds. = means that the main baseline is already national. Narrow and wide multiply the support covariance by 0.5 and 2 and timing deviations by their square roots. Earnings changes scale the earnings and associated fiscal channels; fixed transfers and education-resource components are held fixed. Each row uses the same point-input specification across columns.

Table E6. Point-input sensitivity: numerical comparisons (continued)

Policy	Specific change	Set	WTP (1,000)	Net cost (1,000)	Positive-cost MVPF
Adams Scholarship	Zero negative income-response template	L3	1.48	2.64	0.56
STBF scholarship	No author-model experience delay	L2	[43.52, 85.86]	[18.85, 29.43]	[1.48, 4.56]
STBF scholarship	Printed time-enrolled input (0.360)	L2	[38.39, 81.03]	[20.05, 30.71]	[1.25, 4.04]
Tennessee Promise	Pooled late-completion allowance	L3	86.44	−14.83	–[S]
Tennessee Promise	No source-control level scaling	L3	86.49	−14.85	–[S]
HOPE: indep. single, start	Credit dose: 0.50 dollars	L2	[−53.78, 101.42]	[−17.37, 9.43]	$[< -10^6, > 10^6]^S$
HOPE: indep. single, start	Credit dose: 2 dollars	L2	[−53.74, 101.47]	[−17.38, 9.42]	$[< -10^6, > 10^6]^S$
HOPE: dep. single, start	Credit dose: 0.50 dollars	L2	[−16.27, 30.50]	[−2.77, 1.55]	$[< -10^6, > 10^6]^S$
HOPE: dep. single, start	Credit dose: 2 dollars	L2	[−16.28, 30.48]	[−2.76, 1.55]	$[< -10^6, > 10^6]^S$
HOPE: dep. single, end	Credit dose: 0.50 dollars	L2	[−17.36, 32.61]	[−3.37, 1.86]	$[-83.48, > 10^6]^S$
HOPE: dep. single, end	Credit dose: 2 dollars	L2	[−17.34, 32.63]	[−3.37, 1.86]	$[-124.32, > 10^6]^S$
HOPE: joint, start	Credit dose: 0.50 dollars	L2	[−21.40, 40.34]	[−4.69, 2.53]	$[-38.45, > 10^6]^S$
HOPE: joint, start	Credit dose: 2 dollars	L2	[−21.31, 40.46]	[−4.70, 2.52]	$[-22.42, > 10^6]^S$
HOPE: joint, end	Credit dose: 0.50 dollars	L2	[−23.38, 44.15]	[−4.88, 2.70]	$[< -10^6, > 10^6]^S$
HOPE: joint, end	Credit dose: 2 dollars	L2	[−23.34, 44.20]	[−4.87, 2.70]	$[-42.16, > 10^6]^S$
AOTC: single, end	Credit dose: 0.50 dollars	L2	[−42.93, 67.92]	[−7.23, 4.41]	$[-387.50, > 10^6]^S$
AOTC: single, end	Credit dose: 2 dollars	L2	[−42.93, 67.91]	[−7.23, 4.42]	$[-112.99, > 10^6]^S$
AOTC: single, start	Credit dose: 0.50 dollars	L2	[−40.96, 64.86]	[−6.44, 3.93]	$[< -10^6, 64.57]^S$
AOTC: single, start	Credit dose: 2 dollars	L2	[−40.89, 64.97]	[−6.45, 3.92]	$[< -10^6, > 10^6]^S$
AOTC: joint, end	Credit dose: 0.50 dollars	L2	[−55.16, 87.67]	[−10.69, 6.71]	$[-20.42, > 10^6]^S$
AOTC: joint, end	Credit dose: 2 dollars	L2	[−55.18, 87.63]	[−10.68, 6.72]	$[-8.22, > 10^6]^S$
AOTC: joint, start	Credit dose: 0.50 dollars	L2	[−53.64, 85.30]	[−10.11, 6.35]	$[< -10^6, > 10^6]^S$
AOTC: joint, start	Credit dose: 2 dollars	L2	[−53.61, 85.34]	[−10.12, 6.35]	$[< -10^6, > 10^6]^S$
AOTC: indep. single, start	Credit dose: 0.50 dollars	L2	[−179.55, 284.89]	[−56.78, 36.09]	$[< -10^6, > 10^6]^S$
AOTC: indep. single, start	Credit dose: 2 dollars	L2	[−179.55, 284.90]	[−56.78, 36.09]	$[< -10^6, > 10^6]^S$
AOTC: simulated instrument	Credit dose: 0.50 dollars	L2	[−31.48, 55.32]	[−6.14, 3.47]	$[< -10^6, > 10^6]^S$
AOTC: simulated instrument	Credit dose: 2 dollars	L2	[−31.47, 55.33]	[−6.14, 3.46]	$[-19.32, > 10^6]^S$
Early Head Start	FY2001 funded-slot cost proxy	L2	[−37.51, 114.60]	[−0.52, 27.93]	$[-1.34, > 10^6]^S$
Cannabis: education channel	All-public-college costs	L3	−2.68	0.09	−30.29
Cannabis: education channel	Community-college costs	L3	−2.91	0.13	−22.12
Cannabis: education channel	Equal state weights	L3	−2.94	0.13	−21.99
Cannabis: education channel	Zero negative income-response template	L3	−2.71	0.09	−28.68

Notes: Equal displayed endpoints are printed once. Monetary components use each registered account unit; EHS and Tennessee are disclosed public-data reconstructions. Each component span is computed across paired paths; separate extrema must not be combined to form an MVPF. Tax-dose rows describe finite credit changes and can be small in absolute dollars. Adams L3 uses an explicit time allocation of the separately reported early entry input. The data file reports unrounded NPV, WTP, cost, ratios, and all L2/L3 results.

Table F1. Willingness to pay and net fiscal cost: point paths and input repetitions

Policy	Layer	WTP: point	WTP: repetitions	Net cost: point	Net cost: repetitions
Florida grant	L2	[0.60, 27.81]	[−9.49, 51.26]	[1.04, 1.25]	[−4.05, 7.08]
Florida grant	L3	[20.50, 20.50]	[2.55, 38.86]	[1.13, 1.13]	[−2.83, 5.43]
DC tuition aid	L2	[−3.08, 48.18]	[−10.07, 86.68]	[−3.40, 1.93]	[−12.14, 6.54]
Adams	L2	[−11.50, 7.40]	[−787.98, 730.47]	[−20.90, 15.11]	[−228.12, 219.90]
Adams	L3	[1.66, 1.66]	[−16.18, 21.72]	[2.64, 2.64]	[−2.92, 7.92]
STBF	L2	[38.33, 80.73]	[30.65, 82.05]	[20.13, 30.73]	[19.80, 32.61]
Wisconsin grant	L2	[−13.53, 42.02]	[−30.51, 70.02]	[5.54, 6.98]	[−1.28, 13.18]
Wisconsin grant	L3	[12.37, 12.37]	[−0.27, 31.07]	[6.29, 6.29]	[4.37, 8.35]
Kalamazoo	L2	[11.13, 72.60]	[−13.20, 118.03]	[16.04, 17.15]	[5.87, 26.88]
Kalamazoo	L3	[61.25, 61.25]	[17.13, 112.08]	[16.26, 16.26]	[6.92, 25.29]
Michigan CC tuition	L2	[−7.88, 21.76]	[−18.38, 42.64]	[−3.05, 1.78]	[−8.15, 4.05]
Michigan CC tuition	L3	[1.35, 3.24]	[−1.48, 7.72]	[−0.13, 0.15]	[−0.99, 0.60]
CUNY Pell	L2	[−27.52, 39.88]	[−52.36, 70.73]	[−0.14, 2.79]	[−7.07, 11.30]
CUNY Pell	L3	[−2.69, 2.54]	[−24.21, 25.42]	[0.85, 1.09]	[−1.79, 3.38]
Ohio Pell	L2	[−5.34, 27.29]	[−10.63, 48.18]	[0.45, 3.62]	[−1.50, 9.35]
Tennessee HOPE	L2	[−28.18, 43.61]	[−49.85, 76.69]	[0.37, 2.01]	[−6.75, 11.07]
Tennessee Pell	L2	[−25.65, 38.45]	[−51.60, 70.85]	[−7.56, 14.24]	[−56.66, 50.68]
Georgia HOPE	L2	[−10.03, 42.88]	[−22.65, 84.37]	[−2.13, 3.14]	[−11.96, 8.22]
Social Security aid	L2	[19.34, 23.46]	[4.66, 45.25]	[4.70, 5.10]	[1.48, 8.64]
Social Security aid	L3	[20.81, 21.89]	[6.77, 42.48]*	[4.85, 4.95]	[1.82, 8.63]*
Appropriations: spending	L2	[−19.38, 44.67]	[−36.11, 79.55]	[−4.21, 3.57]	[−12.35, 7.37]
Appropriations: spending	L3	[6.83, 6.83]	[1.68, 13.68]	[0.41, 0.41]	[−0.84, 1.49]
Early Head Start	L2	[−37.51, 114.60]	[−76.09, 114.87]	[−1.10, 27.35]	[−1.15, 34.50]
Texas CC tuition	L2	[5.88, 24.15]	[−2.81, 46.37]	[−1.63, 0.61]	[−6.55, 3.89]
Texas CC tuition	L3	[10.88, 10.88]	[1.99, 22.77]	[−0.08, −0.08]	[−2.37, 2.04]
Adult Pell	L2	[0.14, 1.18]	[−0.23, 2.31]	[0.07, 0.26]	[0.01, 0.51]
Appropriations: tuition	L2	[−22.46, 38.91]	[−42.92, 72.50]	[−1.42, 2.54]	[−11.24, 8.42]
Appropriations: tuition	L3	[−2.41, −2.41]	[−12.43, 5.32]	[1.22, 1.22]	[0.38, 2.99]
HOPE credit	L2	[−8.27, 25.82]	[−17.03, 48.24]	[−2.86, 1.62]	[−8.35, 3.73]
HOPE credit	L3	[12.73, 12.73]	[−0.08, 32.72]	[−1.13, −1.13]	[−5.17, 1.55]
HOPE: ind. single, lower	L2	[−53.77, 101.43]	[−97.27, 184.42]	[−17.37, 9.43]	[−38.00, 19.98]
HOPE: dep. single, lower	L2	[−16.27, 30.50]	[−30.46, 57.47]	[−2.76, 1.55]	[−9.38, 5.44]
HOPE: dep. single, upper	L2	[−17.36, 32.62]	[−32.35, 61.24]	[−3.37, 1.86]	[−10.39, 5.82]
HOPE: joint, lower	L2	[−21.37, 40.38]	[−39.52, 74.85]	[−4.69, 2.53]	[−12.63, 6.67]
HOPE: joint, upper	L2	[−23.37, 44.17]	[−42.83, 81.41]	[−4.88, 2.70]	[−13.73, 7.18]
Tuition deduction: single, lower	L2	[−31.21, 52.69]	[−60.65, 100.97]	[−8.41, 4.22]	[−20.55, 11.27]
Tuition deduction: joint, lower	L2	[−41.01, 61.84]	[−80.26, 122.56]	[−13.19, 9.42]	[−27.35, 18.53]
HOPE + Lifetime Learning	L2	[−9.75, 9.74]	[−18.77, 21.36]	[−0.96, 1.02]	[−3.34, 3.05]
AOTC: dep. single, upper	L2	[−42.93, 67.92]	[−78.48, 124.90]	[−7.23, 4.42]	[−20.14, 12.27]
AOTC: dep. single, lower	L2	[−40.94, 64.89]	[−74.94, 119.28]	[−6.44, 3.93]	[−18.85, 11.59]

Continued on next page

Table F1 (continued)

Policy	Layer	WTP: point	WTP: repetitions	Net cost: point	Net cost: repetitions
AOTC: joint, upper	L2	[−55.17, 87.66]	[−99.34, 158.36]	[−10.68, 6.71]	[−26.88, 16.38]
AOTC: joint, lower	L2	[−53.63, 85.31]	[−96.67, 154.28]	[−10.12, 6.35]	[−25.91, 15.77]
AOTC: ind. single, lower	L2	[−179.55, 284.89]	[−317.82, 506.61]	[−56.78, 36.09]	[−111.37, 68.42]
AOTC: simulated instrument	L2	[−31.48, 55.32]	[−57.80, 102.28]	[−6.14, 3.47]	[−16.84, 9.45]
Tennessee Promise	L2	[−48.72, 152.50]	[−58.19, 160.81]	[−38.04, 32.66]	[−40.97, 35.97]
Tennessee Promise	L3	[86.44, 86.44]	[70.60, 100.77]	[−14.83, −14.83]	[−19.92, −9.30]
Course placement	L2	[0.22, 0.22]	[0.22, 0.22]	[−5.55, 2.82]	[−6.11, 3.20]
Tuition deduction: single, upper	L2	[−34.67, 54.63]	[−67.26, 106.70]	[−8.83, 5.48]	[−20.99, 13.09]
Tuition deduction: joint, upper	L2	[−40.59, 64.21]	[−80.83, 128.11]	[−15.07, 10.35]	[−29.69, 19.53]
Cannabis education channel	L3	[−2.50, −2.50]	[−2.70, −2.28]	[0.04, 0.04]	[0.04, 0.05]

Notes: Components use thousands of each case-specific account unit; monetary years and policy doses differ. Point entries give candidate extrema. Repetition entries give the 2.5th percentile of minima and 97.5th percentile of maxima under 1,000 prescribed paired inputs. Stars mark fewer than 1,000 usable repetitions; Table 4 reports ratio counts. WTP and cost extrema may come from different paths and must not be divided to form an MVPF. All thirteen admitted L3 specifications have complete prescribed input schemes, with inherited and extended correlation groups documented in Appendix F.2. These conditional summaries do not establish confidence coverage or path probabilities. Cannabis reports its education-only L3.

Table G1. Published male and female degree effects and standardized education NPVs

Policy	Sex	Effect (pp)	SE (pp)	Paths	NPV span ($1,000)
Adams Scholarship	Men	-0.8	0.9	2,325	[−15.36, 1.21]
	Women	-1.4	1.0	1,913	[−16.26, −0.61]
STBF scholarship	Men	8.0	–	5,173	[4.22, 12.63]
	Women	8.0	–	5,232	[4.42, 11.95]
Wisconsin Scholars Grant	Men	4.7	–	2,145	[−14.34, 8.28]
	Women	4.7	–	2,108	[−12.21, 3.46]
Kalamazoo Promise	Men	4.1	7.0	5,642	[−3.64, 6.69]
	Women	13.2	7.7	5,436	[8.46, 16.53]
Michigan CC tuition	Men	1.7	1.0	7,737	[−1.42, 6.35]
	Women	1.9	1.1	7,040	[−2.22, 4.80]
Early Head Start	Men	13.0	4.0	1,063	[−5.47, 12.34]
	Women	21.2	5.0	946	[−1.17, 13.96]
Tennessee Promise	Men	1.2	–	2,888	[−2.13, 10.52]
	Women	4.1	–	902	[−1.71, 7.09]

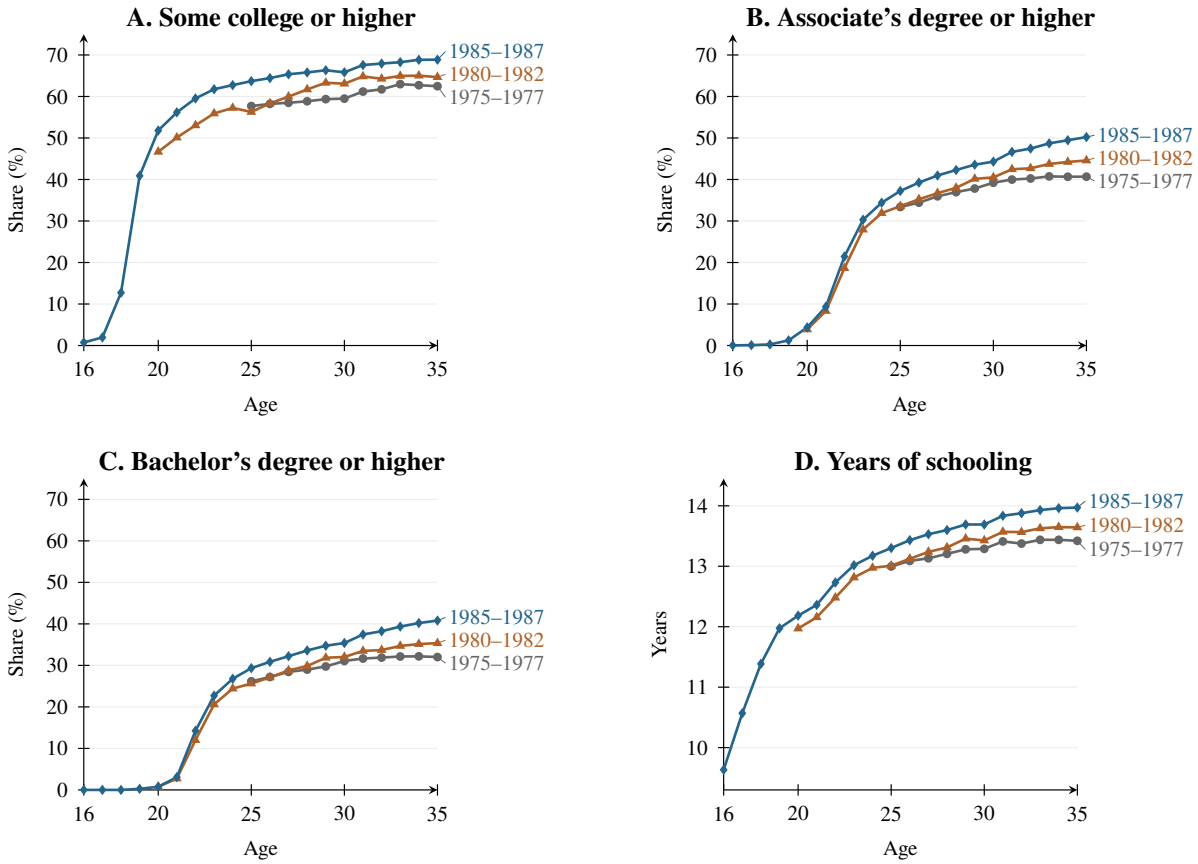
Notes: Effects are the source studies' subgroup point estimates, in percentage points. Source standard errors are reported where available; a dash means that a subgroup SE was not recovered. Tennessee retains a separate auxiliary subgroup interval sensitivity. STBF and Wisconsin use rounded published coefficients; their identical displayed effects do not establish exact equality. The NPV spans allow the terminal effects to vary and therefore combine timing with eventual attainment. They summarize verified finite candidate banks, with no probabilities assigned to paths. All accounts use a common two-schooling-year equivalent per BA increment and the Section 3 net-earnings response, in 2000 dollars discounted to age 18 at 3 percent. No other education changes, transfers or fiscal costs are included. These are education-channel values, not total-policy MVPFs. Source tables, populations, age mappings and estimands are described in Appendix G.1.

Table G2. Male and female education NPVs at the same terminal effect

		Sex-specific reference curves		Common reference curve	
Policy	η (pp)	Paths (M/F)	$V_f - V_m$ (\$)	Paths (M/F)	$V_f - V_m$ (\$)
Adams Scholarship	-15	32/64	[−449, −249]	43/42	[−101, 35]
	-10	62/64	[−539, −31]	44/46	[−252, 179]
	-5	93/54	[−675, 332]	69/75	[−468, 384]
	0	122/0	–	95/0	–
STBF scholarship	2	0/0	–	0/0	–
	5	71/72	[−585, 733]	72/72	[−664, 664]
	10	66/70	[−626, 559]	67/67	[−587, 587]
	15	0/0	–	0/0	–
Wisconsin Scholars Grant	2	74/83	[−95, 863]	84/84	[−425, 425]
	5	90/0	–	98/98	[−576, 576]
	10	0/0	–	0/0	–
	15	0/0	–	0/0	–
Kalamazoo Promise	2	25/0	–	24/0	–
	5	26/0	–	26/0	–
	10	0/19	–	25/20	[475, 1,279]
	15	0/18	–	0/19	–
Michigan CC tuition	2	63/65	[−602, 404]	63/64	[−474, 527]
	5	58/60	[−513, 317]	59/60	[−391, 442]
	10	0/0	–	0/0	–
	15	0/0	–	0/0	–
Early Head Start	2	45/43	[37, 208]	49/33	[210, 356]
	5	45/51	[35, 194]	50/44	[210, 347]
	10	41/59	[29, 231]	48/52	[217, 383]
	15	0/0	–	0/53	–
Tennessee Promise	2	21/11	[39, 320]	13/14	[58, 316]
	5	22/11	[46, 277]	13/16	[85, 312]
	10	21/0	–	0/17	–
	15	0/0	–	0/0	–

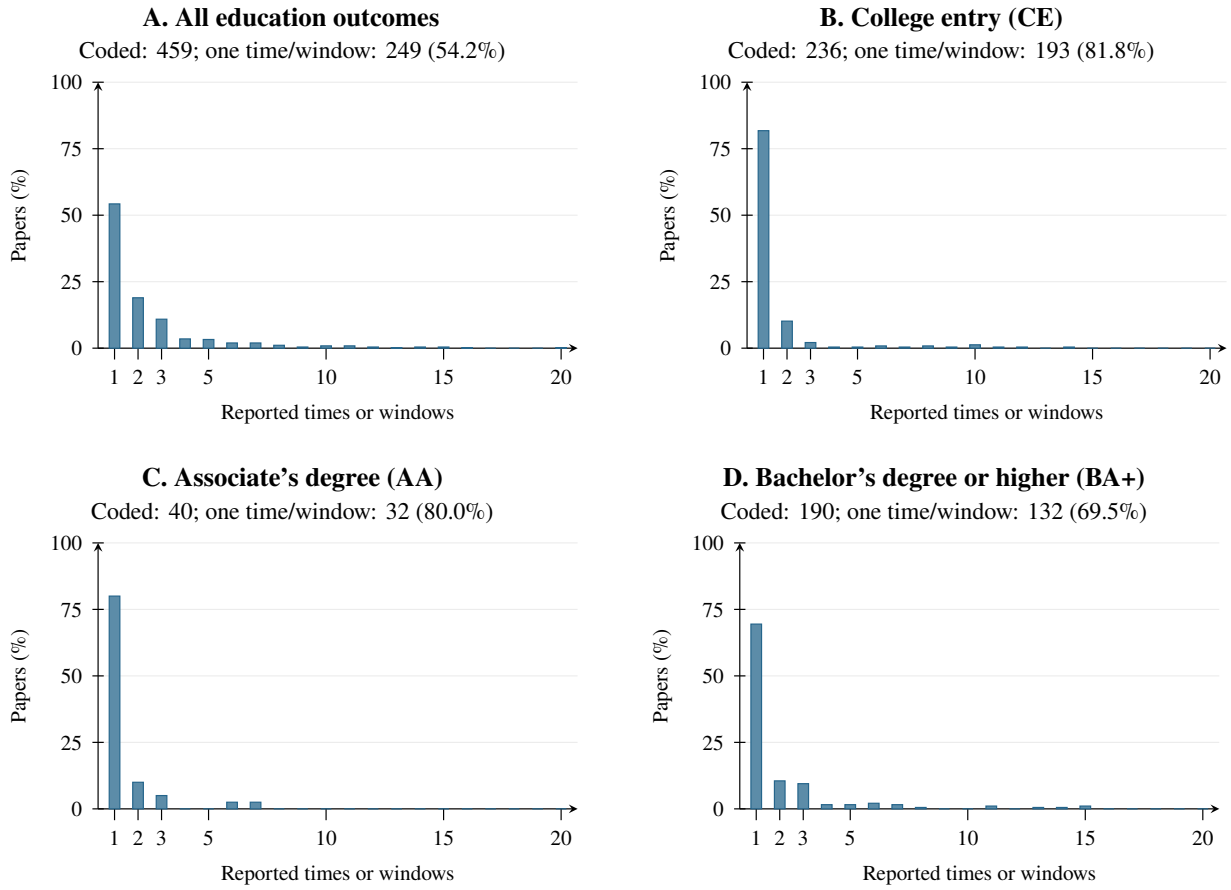
Notes: Each pair matches its own male and female source effects and the same age-65 BA effect η. Prices and exposure-age weights are identical across sexes within a policy. Negative η represents a common terminal reduction, as allowed for Adams. The common-reference sensitivity replaces both reference curves by their equally weighted mean, retaining the same inputs and grid. Entries count verified paths, not observations or repetitions. The displayed dollar span runs from the smallest to the largest female-minus-male NPV difference among the retained pairs. A dash denotes no verified pair at that grid value; finite search failure does not prove incompatibility. Spans are numerical summaries, not confidence intervals or certified global bounds. Tennessee also retains a separate auxiliary subgroup interval sensitivity. Policy populations and estimands differ, so these are within-policy comparisons rather than a cross-policy welfare ranking.

Figure A1. Education profiles across birth cohorts, ages 16–35



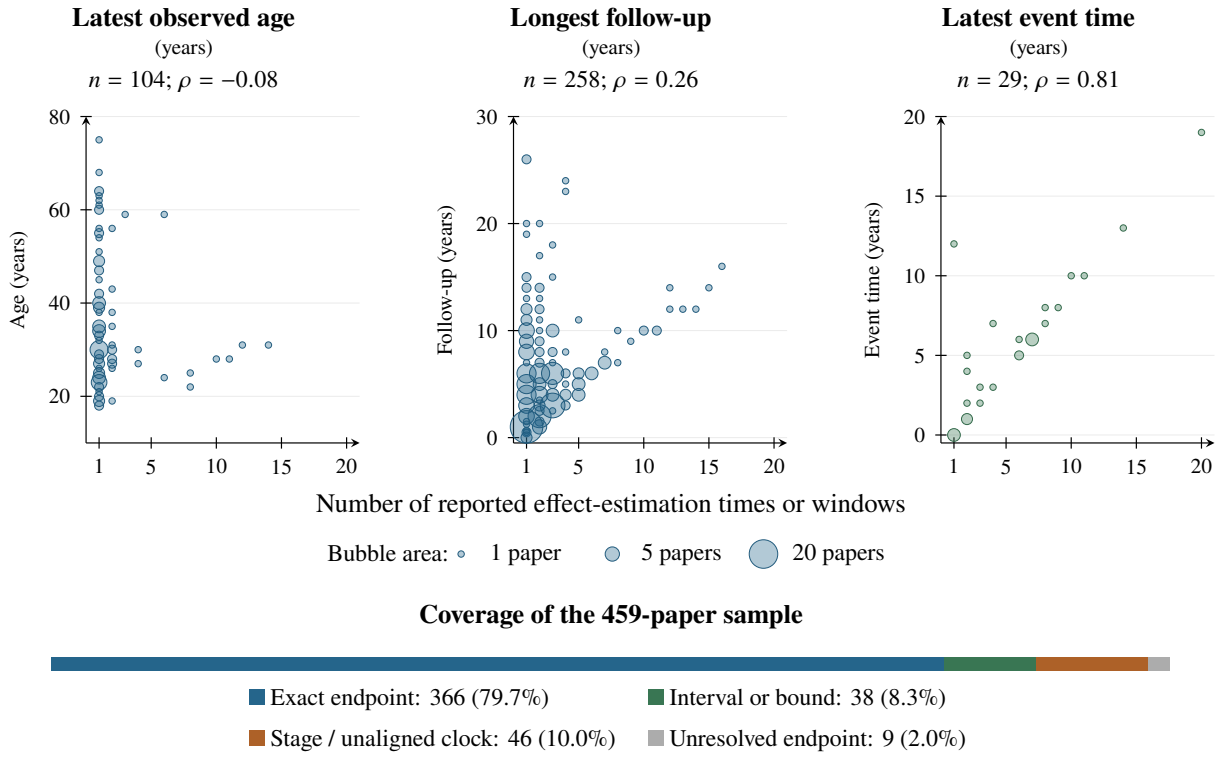
Notes: Labels at the right identify birth cohorts. Markers show observed annual education levels; lines connect adjacent observed ages. With fixed one-third cohort weights, ACS 2000–2022 supports ages 25–35 for 1975–1977, ages 20–35 for 1980–1982, and ages 16–35 for 1985–1987. Missing earlier ages remain unplotted. Person weights apply within cohort–age cells; all sexes, birthplaces, and racial and ethnic groups are included. Attainment panels share percentage scales; schooling uses years. SC+ is some college or higher; AA+ and BA+ are cumulative thresholds. Appendix A.6 gives construction details.

Figure A2. Number of reported effect-estimation times or windows



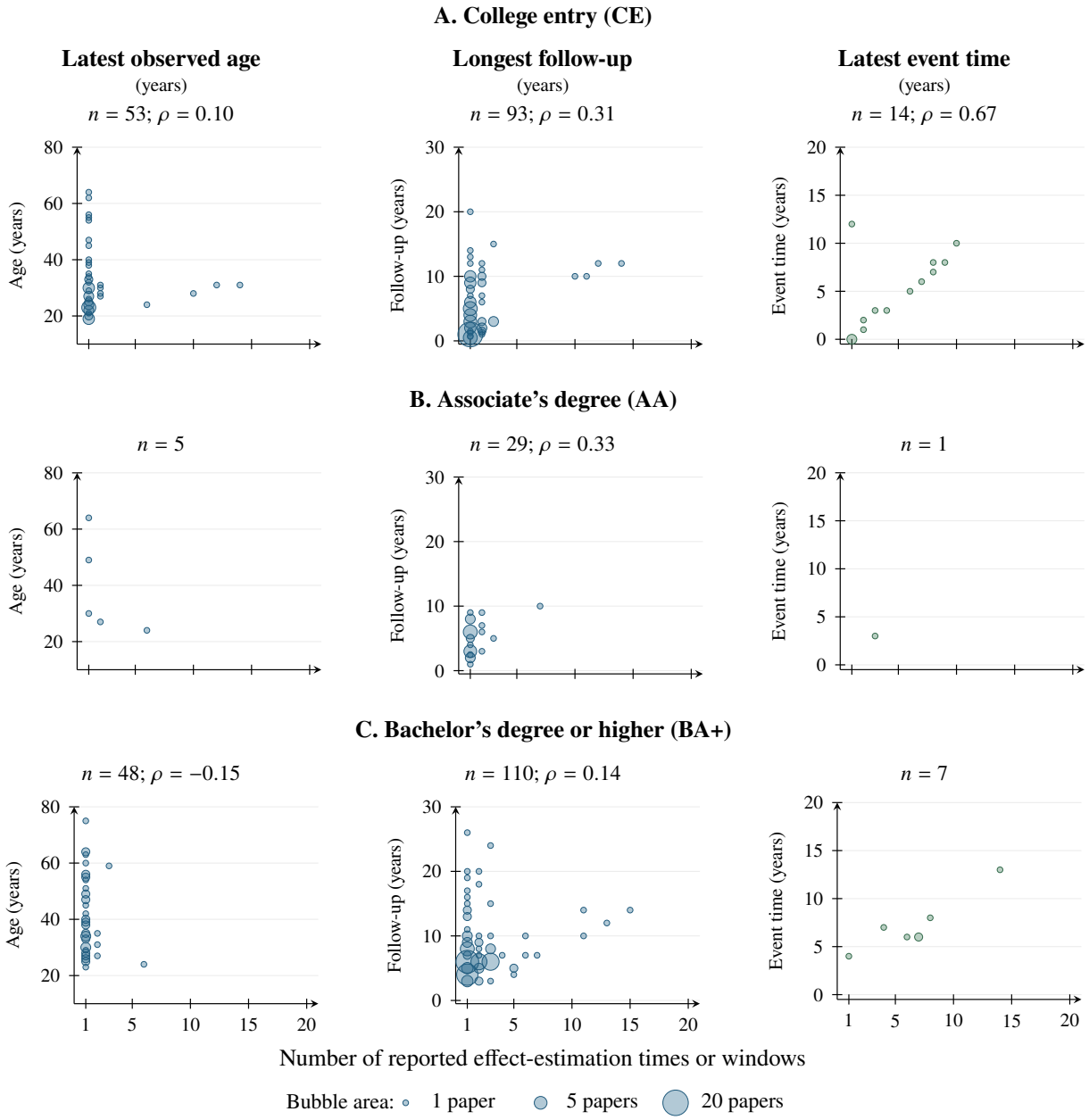
Notes: Percentages use papers with a usable count in each overlapping outcome group; one CE paper is excluded. A point or pooled window counts once; separately reported horizons count separately. Counts do not measure follow-up length or effect persistence. Axes are common. Table A1 gives counts; Appendix A.2 gives coding rules.

Figure A3. Estimation times and observation endpoints: all education outcomes



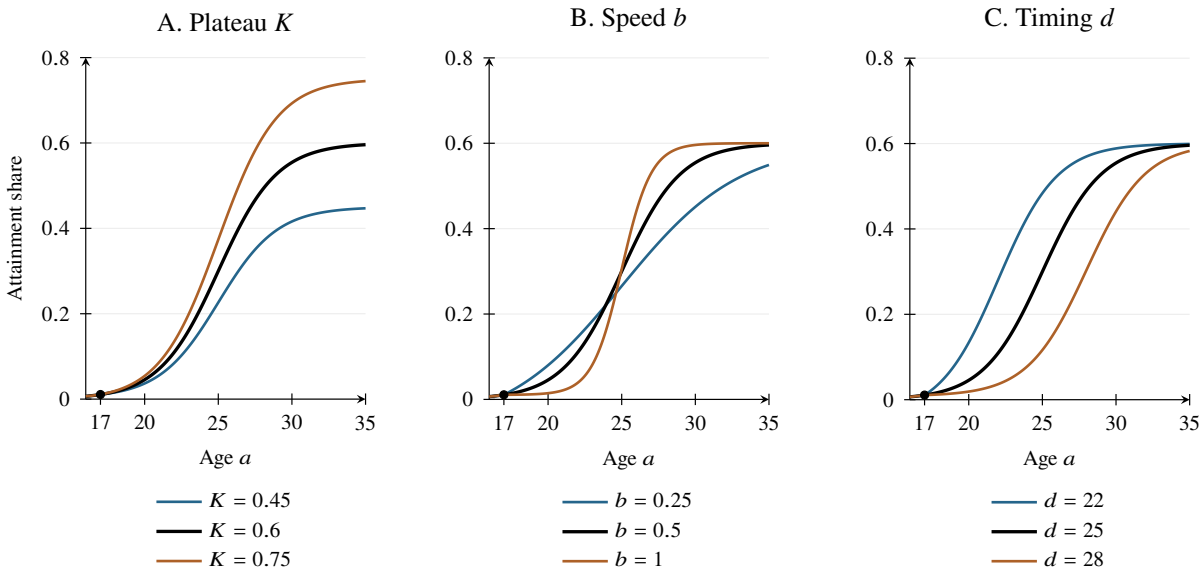
Notes: Bubble area counts papers at each coordinate. Panels separate attained age, individual follow-up, and policy event time, covering 366 distinct papers with overlap across clocks. The coverage bar includes all 459 papers; 93 lack an exact coordinate and receive no imputed endpoint. Correlations are paper-level Spearman correlations. Overall counts and endpoints may summarize different outcomes. Appendix A.2 gives coding rules.

Figure A4. Estimation times and observation endpoints by education outcome



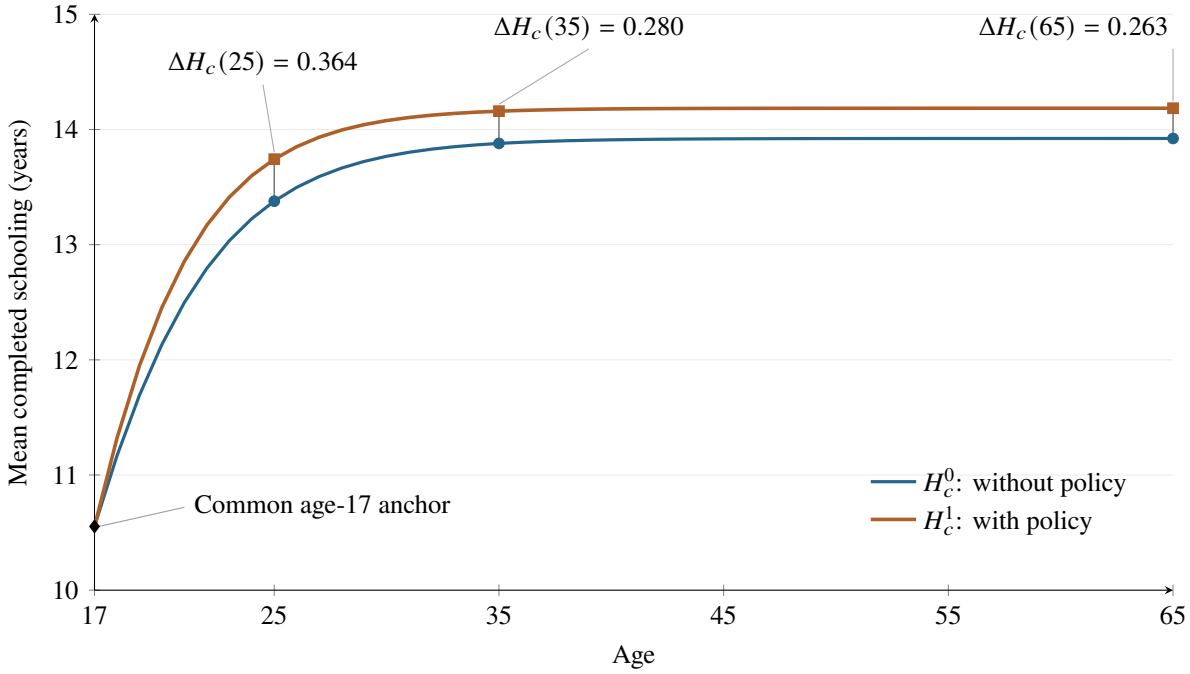
Notes: Rows show college entry, associate-degree outcomes, and BA+ outcomes; columns retain separate clocks. Scales are common within clocks; bubble areas share one count scale. Coverage is 146 CE, 34 AA, and 158 BA+ papers, with overlap. Spearman correlations require at least ten papers and variation in both coordinates. AA preserves each source's exact- or at-least-degree definition.

Figure B1. Parameter changes under a common pre-policy anchor



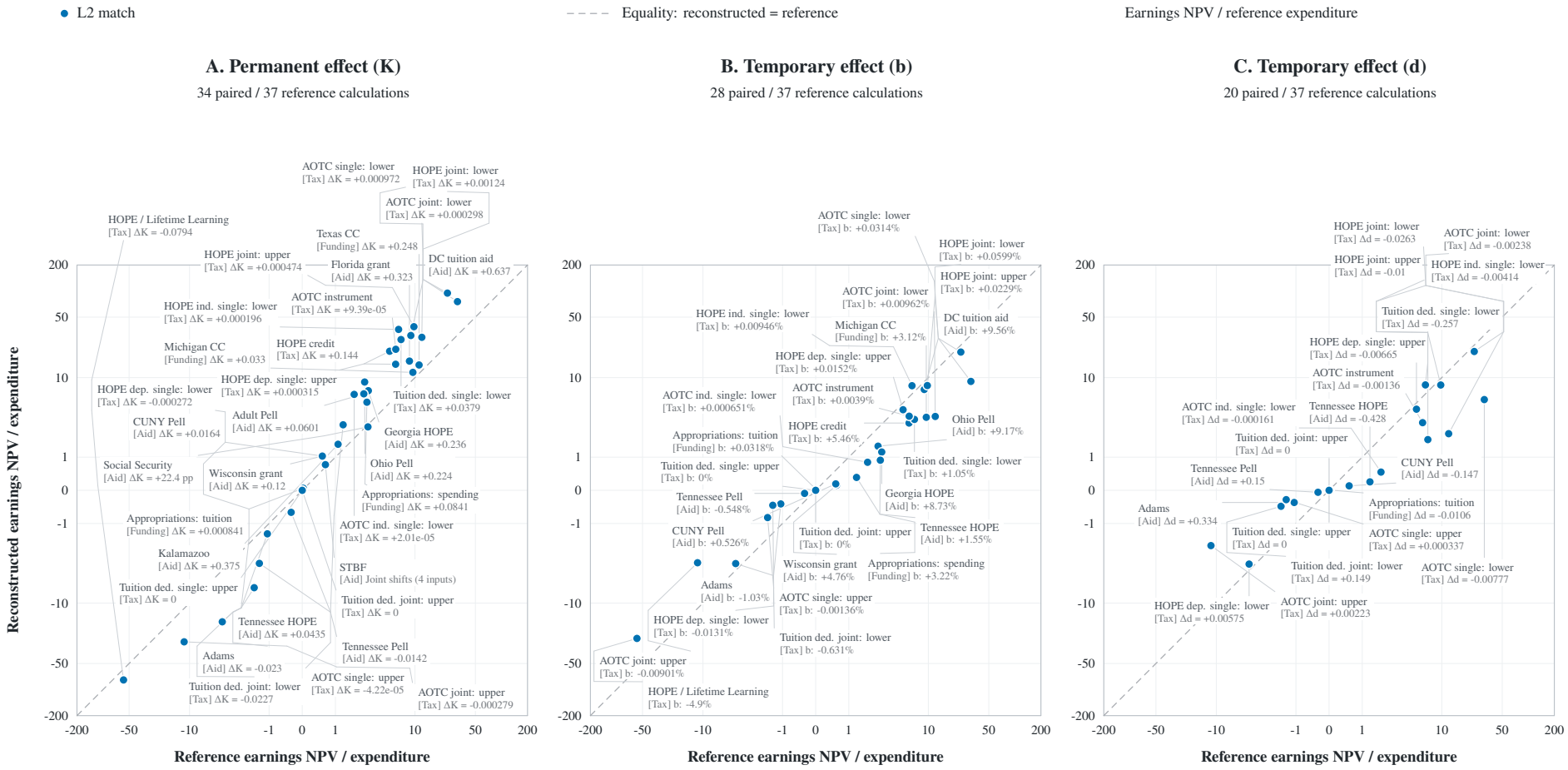
Notes: One parameter varies per panel; the others remain at $(K, b, d) = (0.60, 0.50, 25)$. All paths share the age-17 reference level. Black, blue, and orange denote reference, smaller, and larger parameter values. K governs the plateau; b and d govern speed and timing conditional on the anchor. In this normalized family, d need not mark half the plateau. Axes show hypothetical attainment shares.

Figure B2. Schooling paths underlying Figure 4



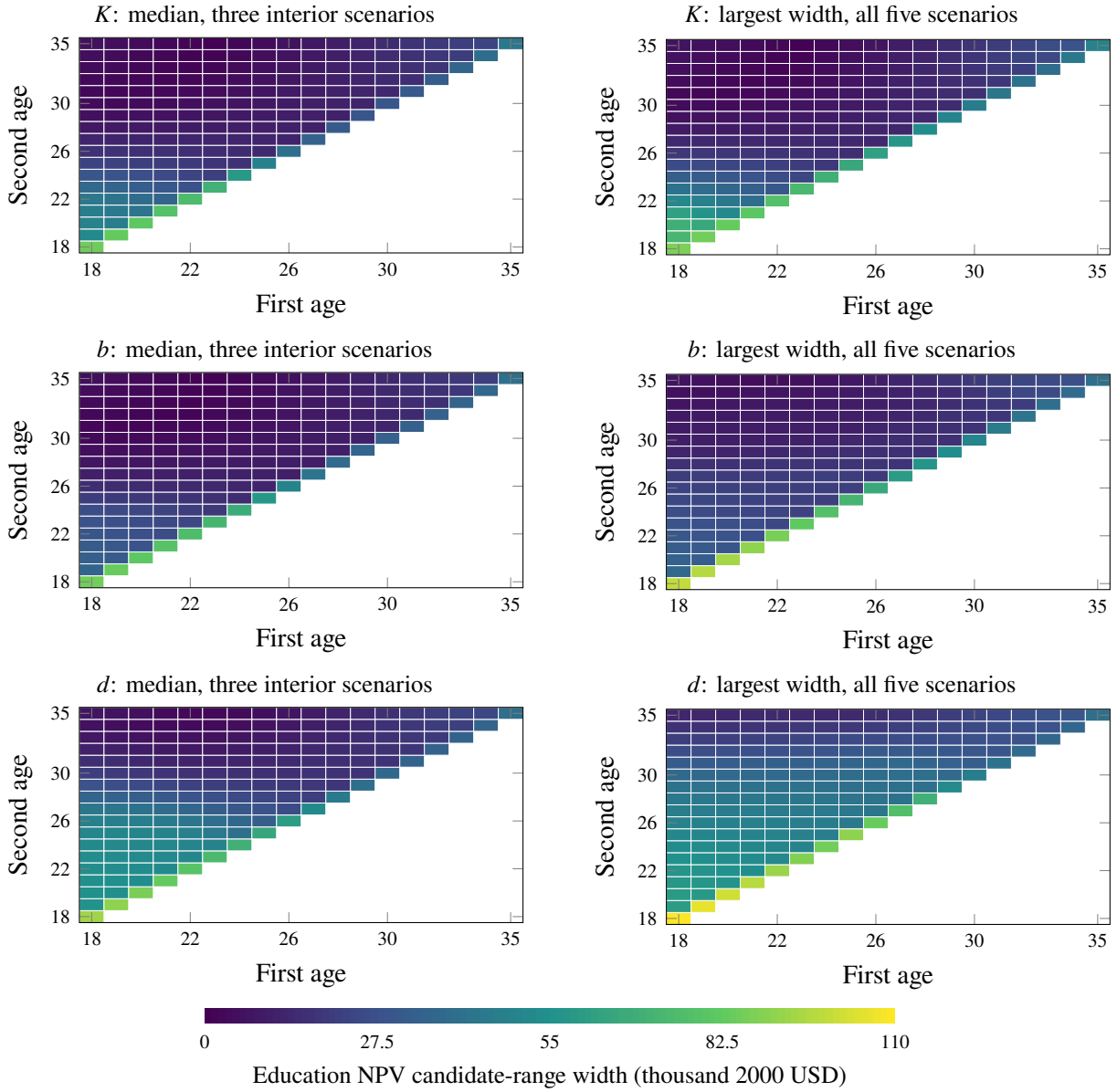
Notes: Blue shows $H_c^0(a)$ and orange shows $H_c^1(a)$ for the same joint-change illustration used in Figure 4. The comparison path averages the fitted ACS profiles for the 1985–1987 birth cohorts with fixed one-third weights. The treated path uses $(\Delta K_c, \Delta b_c, \Delta d_c) = (0.2625, 0.0300, -0.4500)$ and shares the comparison level at age 17. Lines connect the annual values used in the NPV accounts; markers and labels identify the gaps at ages 25, 35, and 65. The early schooling advantage partly narrows as the comparison population catches up, while a positive gap remains at age 65. Baseline values beyond age 35 are extrapolated. The figure displays one stipulated comparison of potential paths.

Figure B3. Reference and reconstructed earnings NPVs under permanent and temporary effects



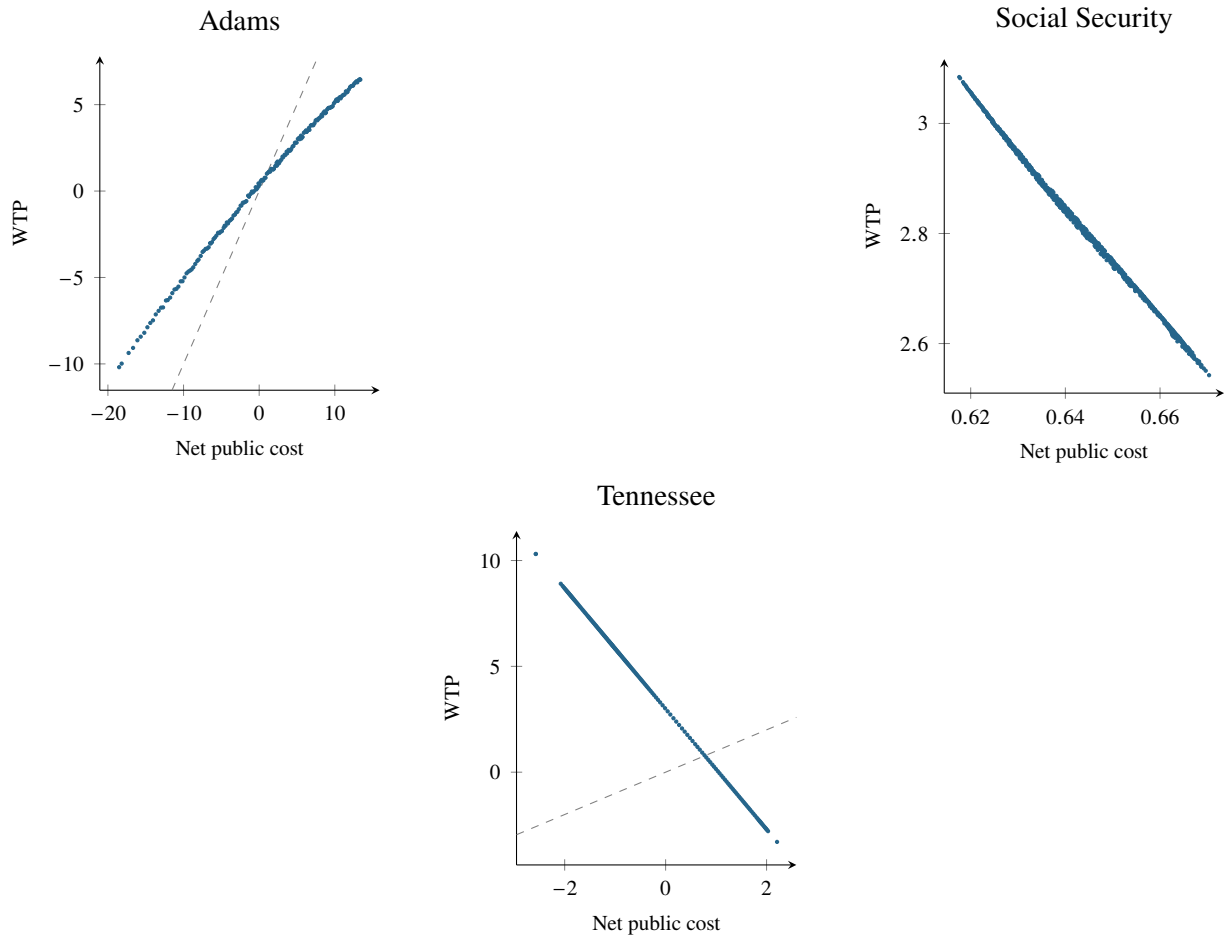
Notes: Both earnings NPVs are divided by the same positive reference expenditure within each calculation, using matched monetary units. Panels use identical asinh scales and square axes; tick labels give actual normalized values. Blue points match L2. None of these single-family solutions also matches every additional point input exactly; the main L3 fits remain separate. Bracketed labels give the policy types defined in Figure 8. Labels report ΔK in schooling years (percentage points for Social Security and Early Head Start), proportional b changes in percent, and Δd in years; the other two parameter families stay fixed. STBF uses four joint time and credential inputs; Tennessee Promise uses three credential inputs. Permanent effects change eventual attainment; temporary effects eventually vanish but can retain lifetime value. Early Head Start and Tennessee Promise lack a verified common price vintage for their public reconstruction and source NPV and are excluded from the paired monetary plot. Table B3 gives the values and accounts for omitted pairs. Self-financing policies can have finite earnings NPVs.

Figure B4. Observation timing and remaining education-NPV ranges



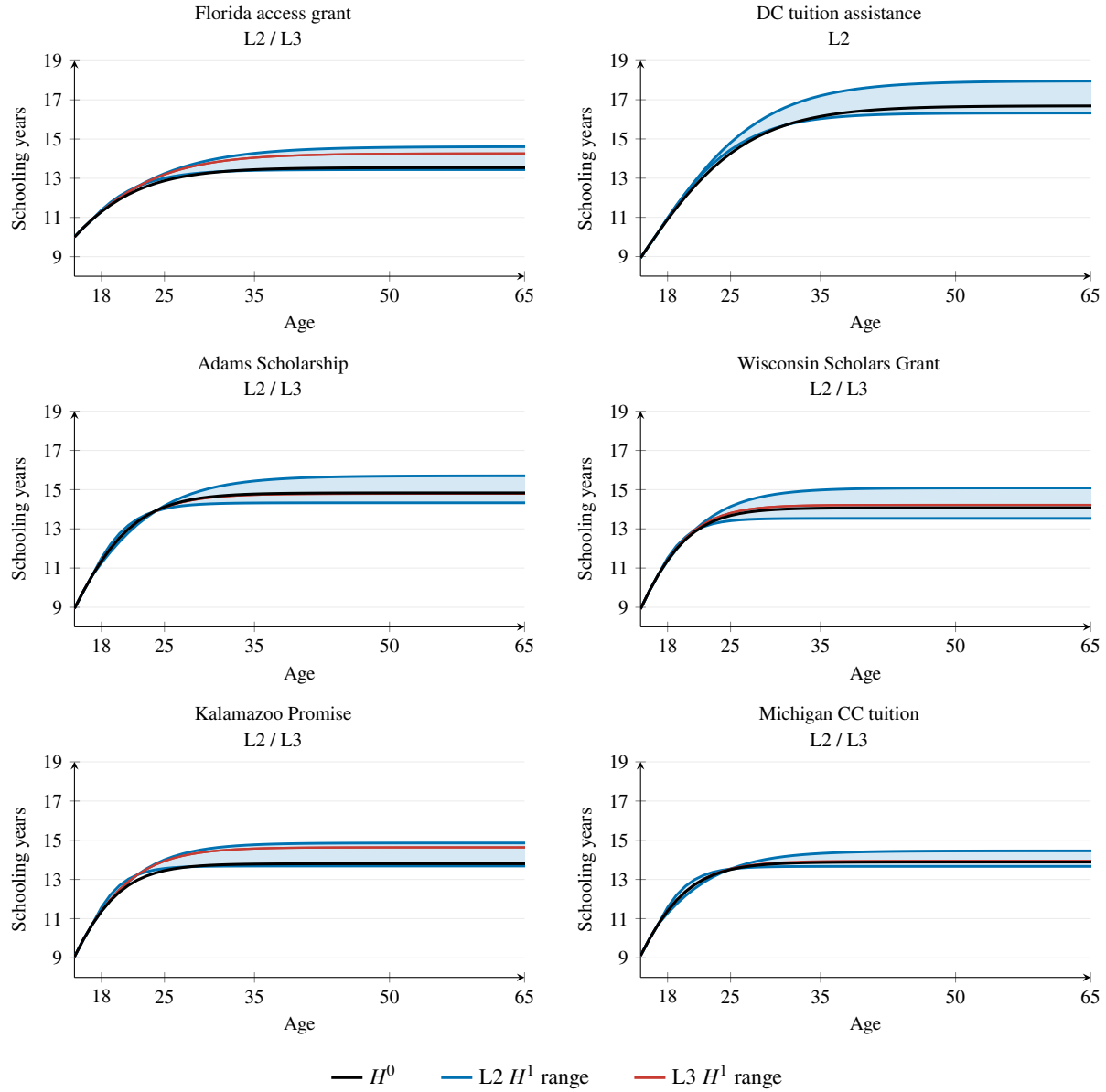
Notes: Diagonal cells represent single-age observations; upper-triangle cells represent two distinct ages. All panels use the same width scale, in thousands of 2000 dollars per person. The widths come from verified numerical candidates under the common baseline, path restrictions, and valuation rule; they are not confidence intervals or certified global bounds. Left panels report medians across the three interior scenarios; right panels report largest widths across all five scenarios.

Figure D1. Paired welfare components under the source-specific path mappings



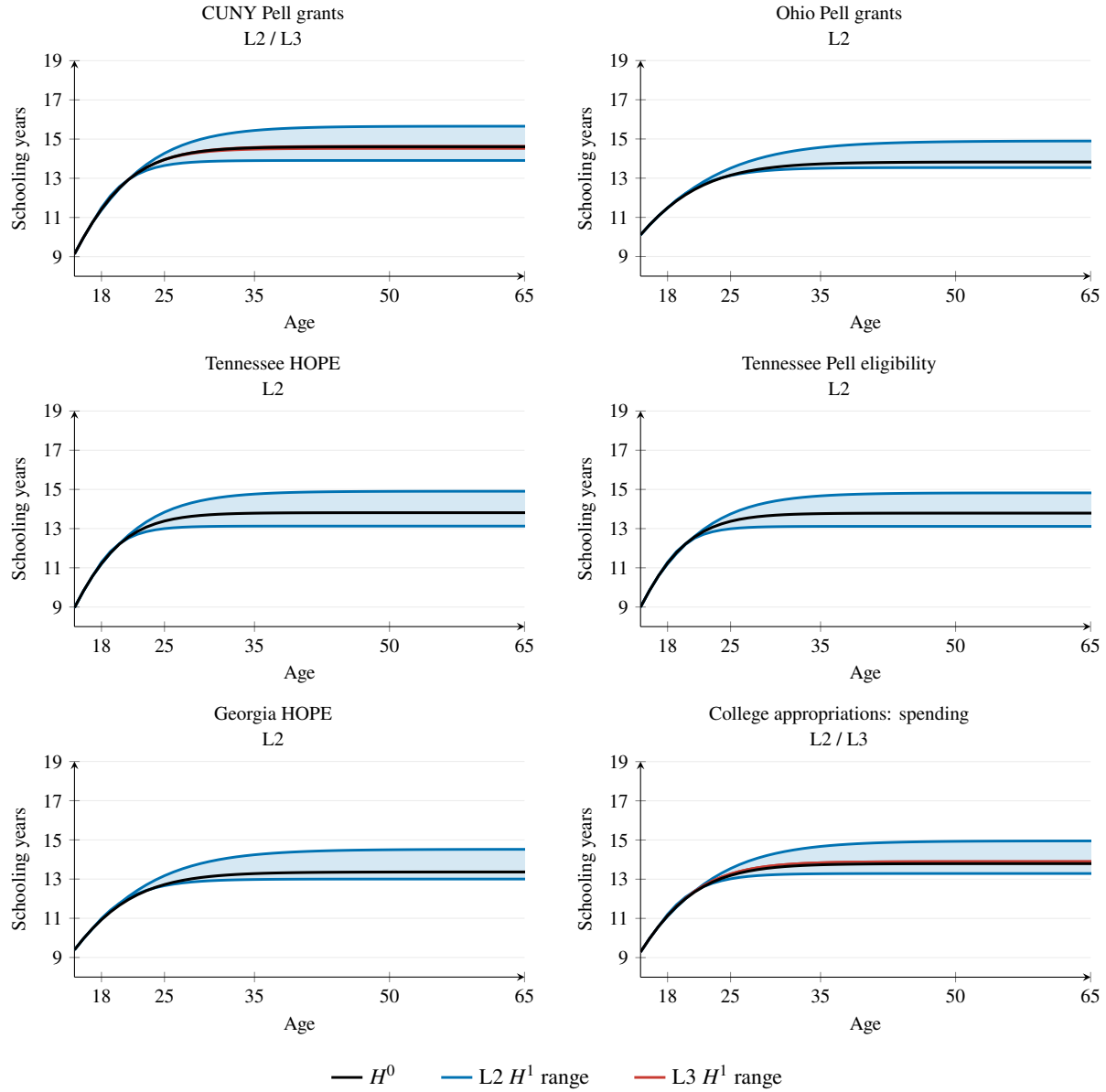
Monetary components are divided by the same fixed reference expenditure within each panel. Points are complete L2 paths, sampled across the earnings range. The dashed line is WTP equal to net public cost. Axis ranges vary by policy; coordinates from different monetary vintages are not dollar comparisons. No separate opportunity cost is deducted from net earnings.

Figure E1. Comparison and treated schooling paths (1 of 6)



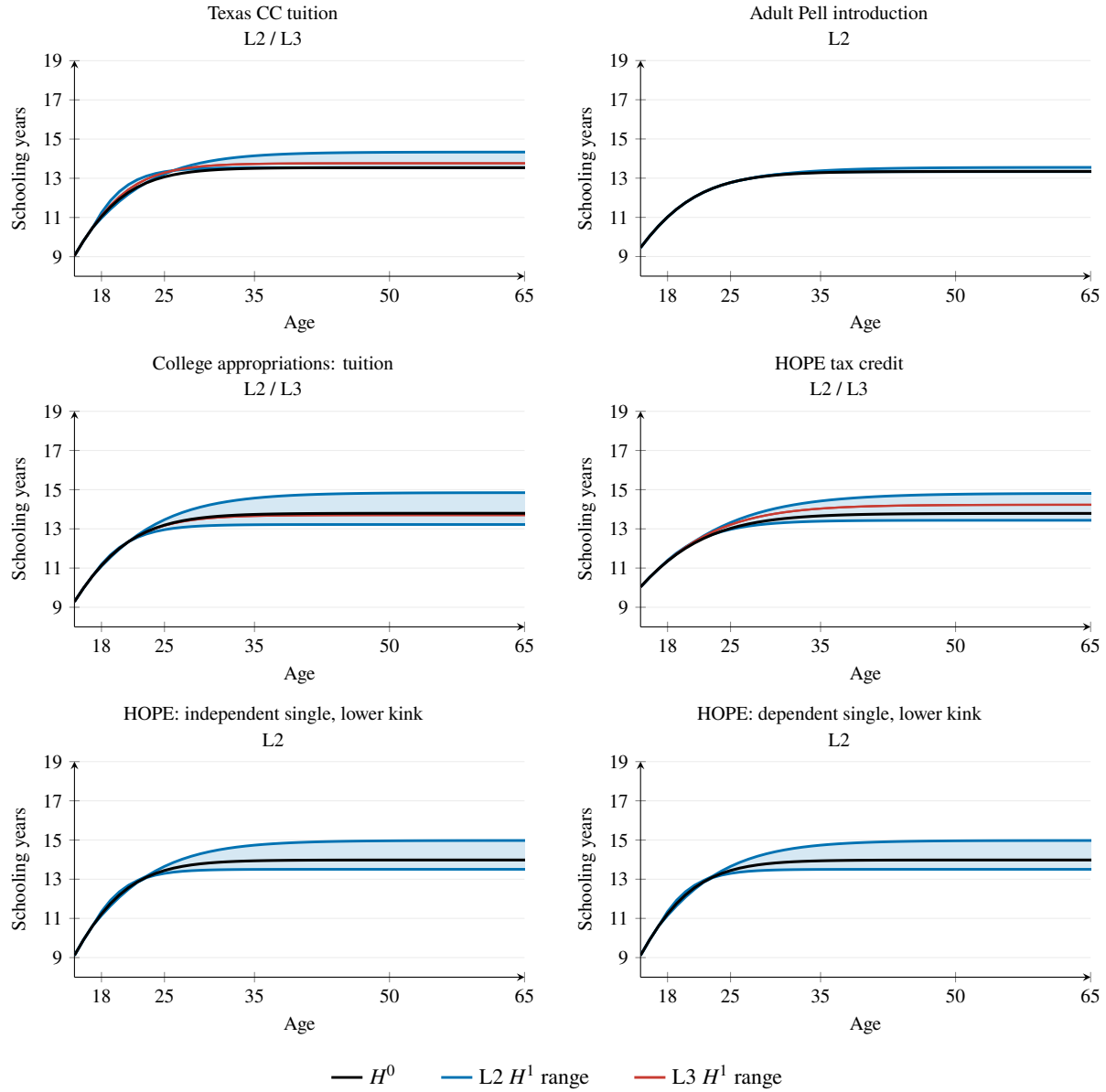
Notes: Black is the matched comparison profile H^0. Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E2. Comparison and treated schooling paths (2 of 6)



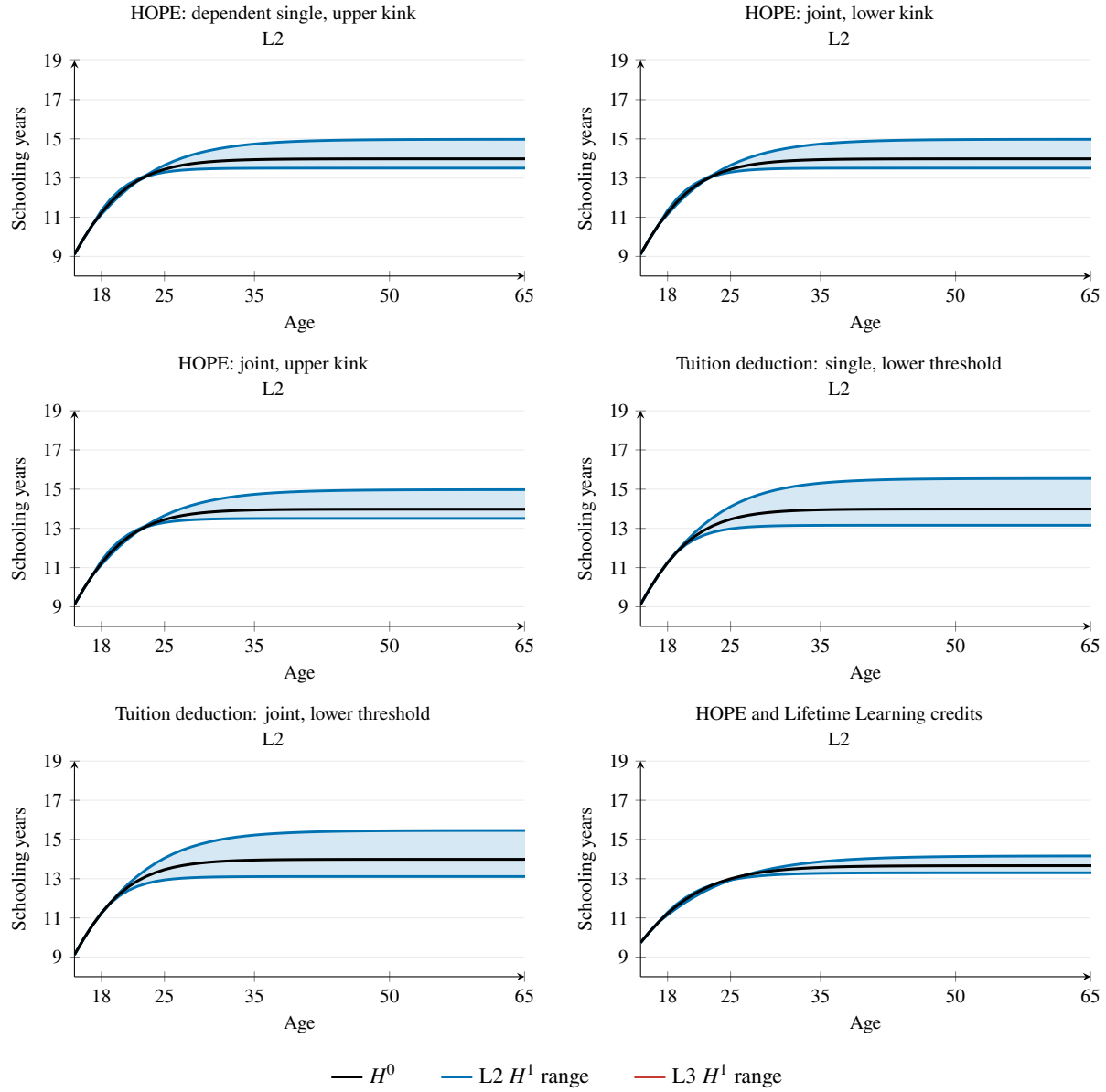
Notes: Black is the matched comparison profile H^0. Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E3. Comparison and treated schooling paths (3 of 6)



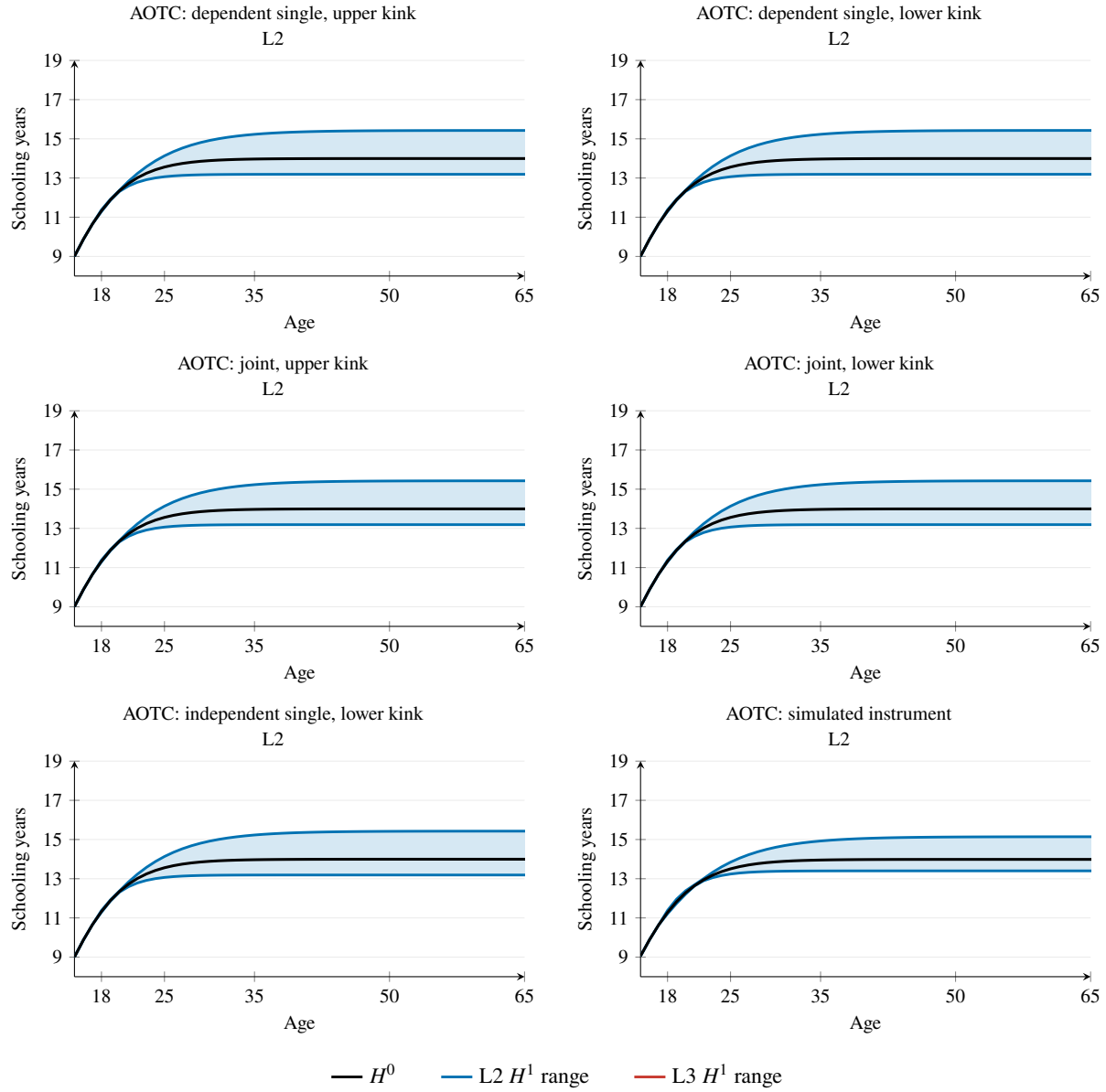
Notes: Black is the matched comparison profile H^0. Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E4. Comparison and treated schooling paths (4 of 6)



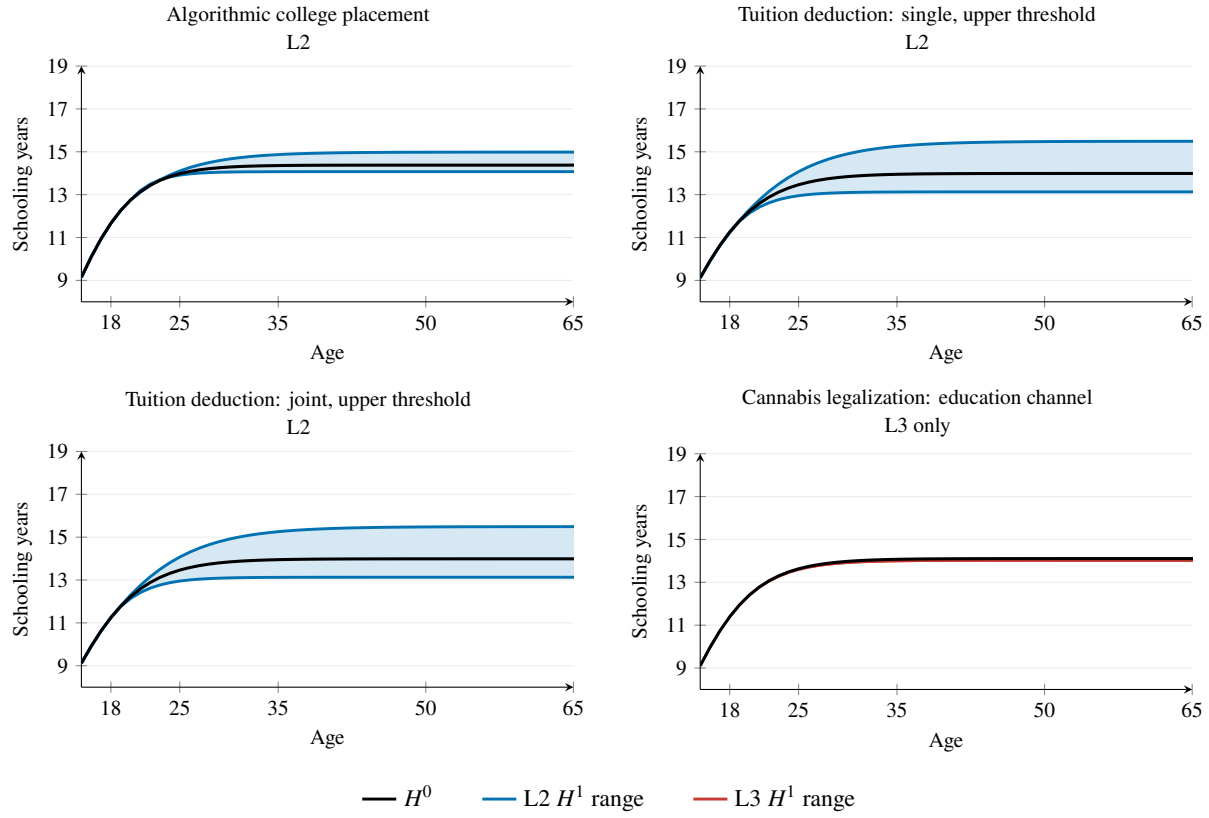
Notes: Black is the matched comparison profile H^0. Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E5. Comparison and treated schooling paths (5 of 6)



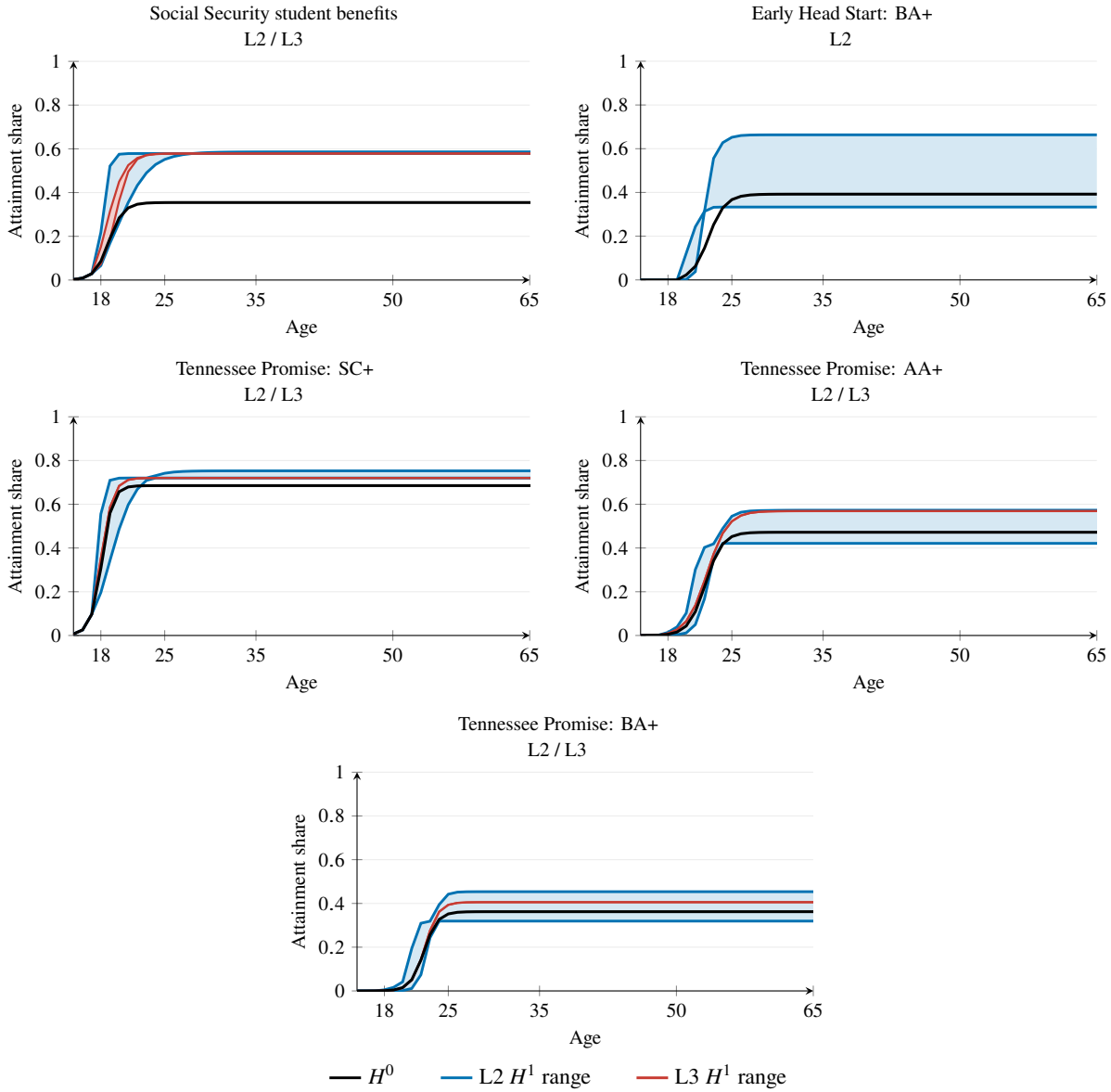
Notes: Black is the matched comparison profile H^0. Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E6. Comparison and treated schooling paths (6 of 6)



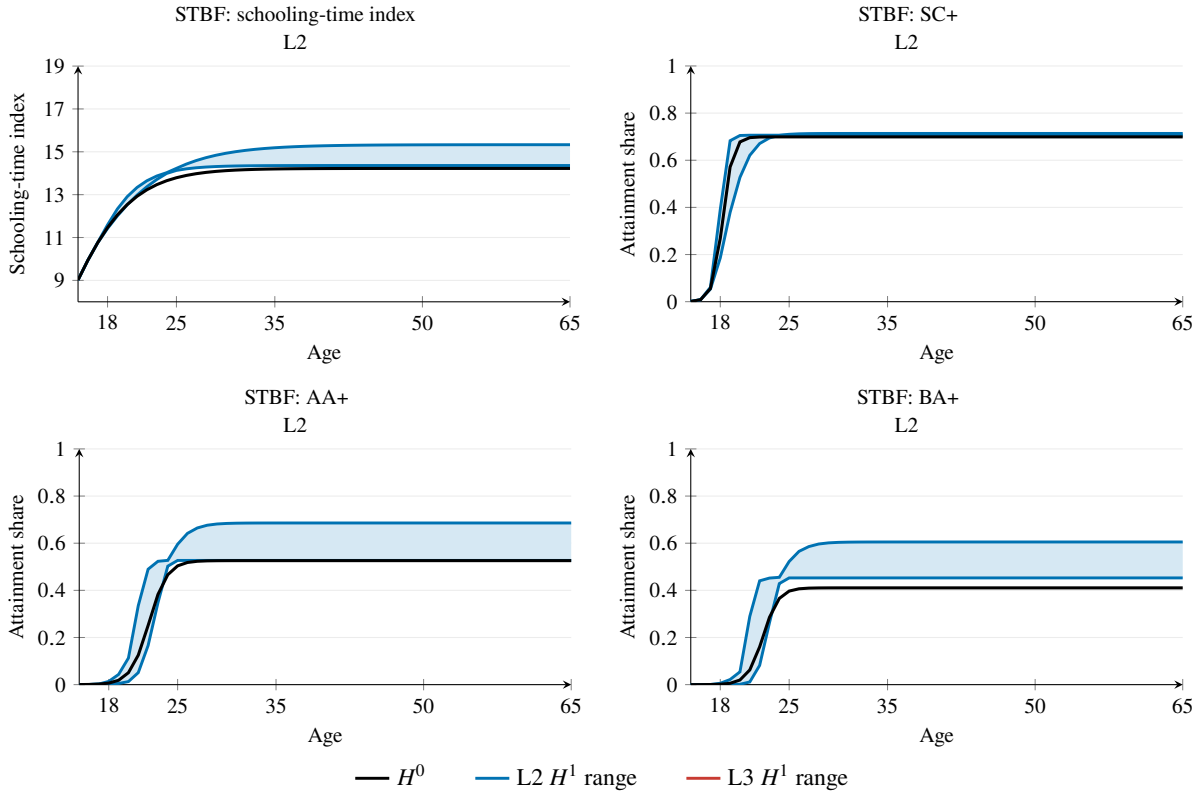
Notes: Black is the matched comparison profile H^0. Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

Figure E7. Comparison and treated credential paths



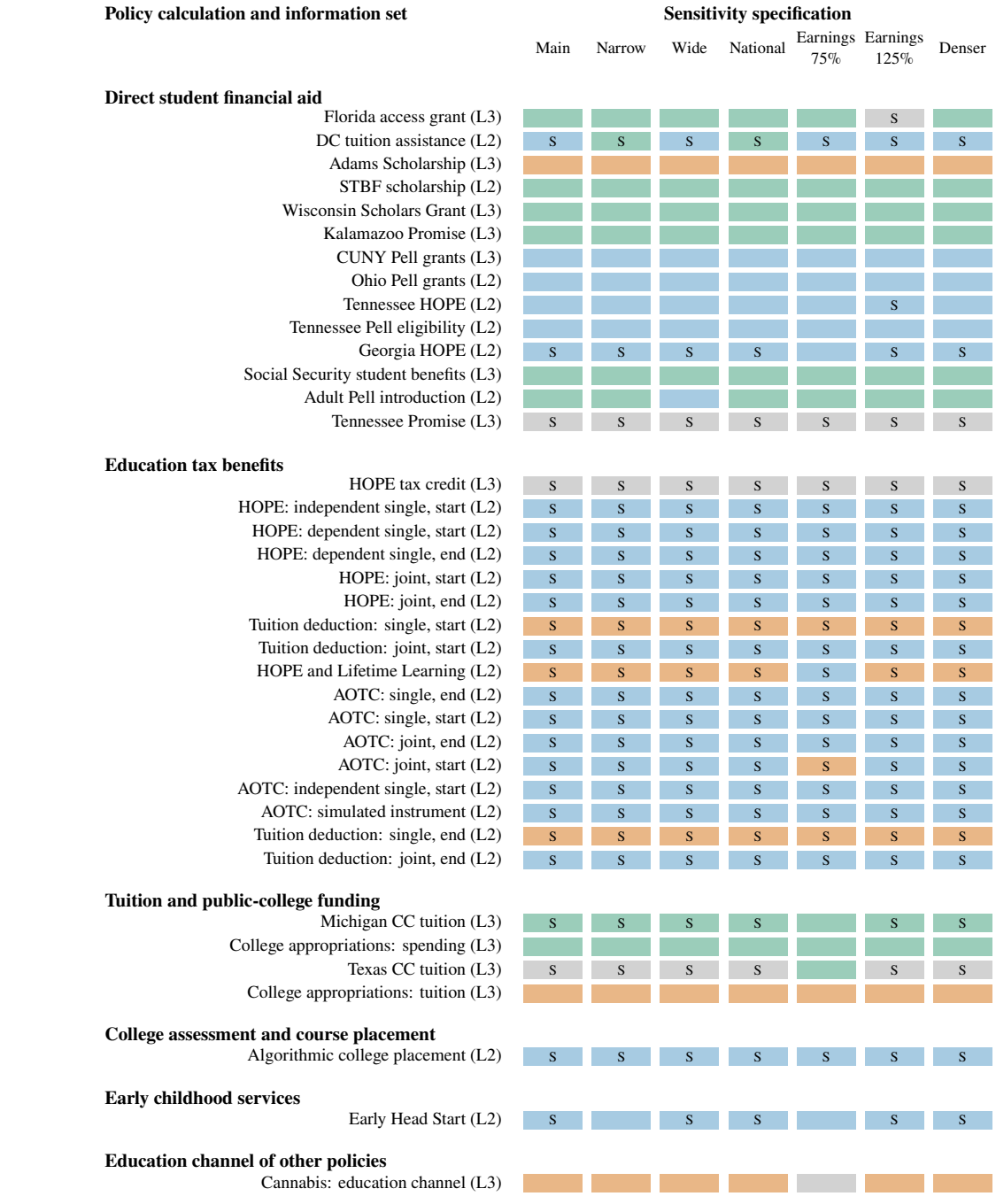
Notes: Black is H^0; blue and red boundaries and shading show the pointwise H^1 ranges for L2 and L3. All panels share age and attainment-share scales. Tennessee ranges come from joint candidates satisfying credential nesting and the restrictions in Appendix D.8. Envelope edges may combine different candidates across ages and credentials; they need not form a feasible joint path. These finite-search ranges are not confidence bands or probability distributions. Targets and calibrations differ across the three applications.

Figure E8. Comparison and treated paths: STBF



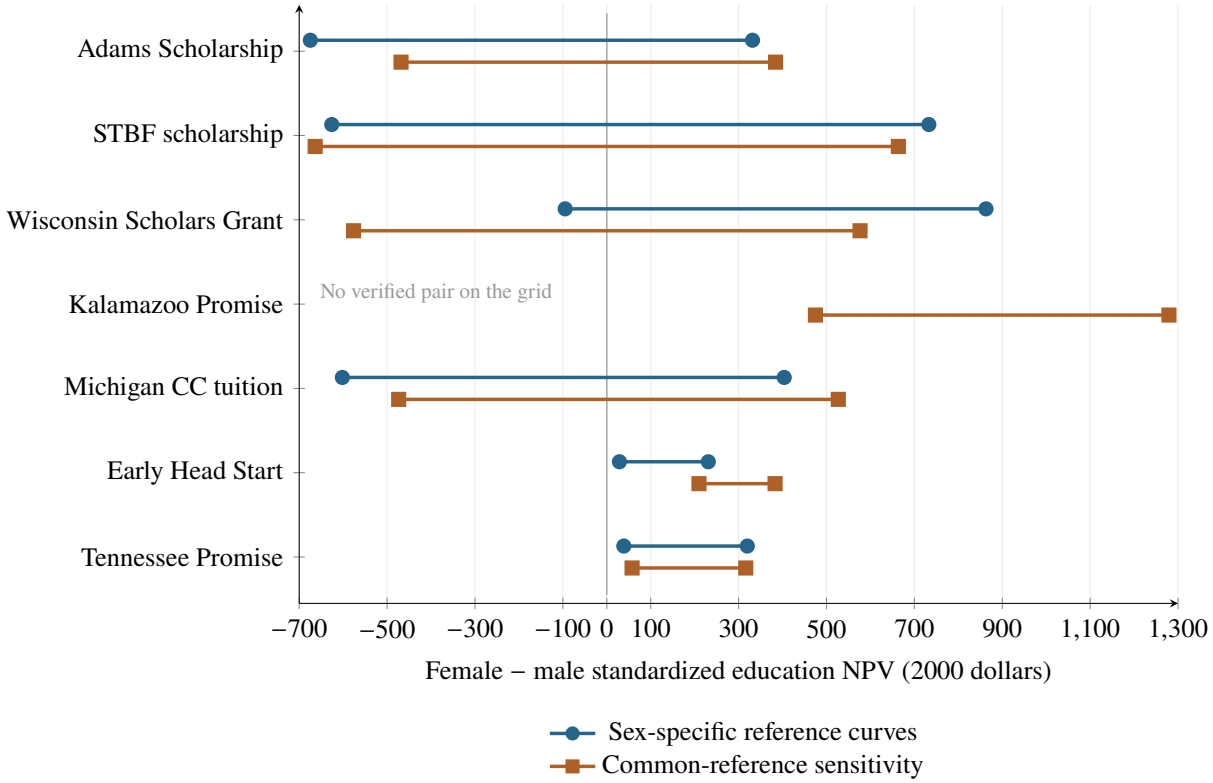
Notes: Black is the comparison profile H^0; blue boundaries and shading show the pointwise L2 treated range H^1. All four outcomes are matched jointly at the same six-year follow-up. The schooling-time index uses the schooling-years calibration scale and is not relabeled as completed schooling in the income model. Its scale differs from the three credential shares. Boundaries can combine different candidates across ages and outcomes; they are not confidence bands. No additional comparable schooling-time or credential window is admitted as L3.

Figure E9. Point-input sensitivity across the policy collection



Notes: Colors refer to positive-cost MVPFs; gray means none was found. S marks a separate self-financing candidate. Equal signs indicate an already national baseline. Each alternative changes one specified assumption, with point inputs fixed. L3 is used where admitted; otherwise L2. Finite searches cannot establish absence of other paths. Table E6 reports values; Appendix E.9 defines the changes.

Figure G1. Sex comparisons after fixing terminal education effects and valuation prices



Notes: Each underlying comparison uses two complete paths that match their own source subgroup effects and the same age-65 BA effect. Blue spans use separate sex-specific ACS reference curves; orange spans use their equally weighted mean. Positive differences favor women in the standardized education account. Endpoints summarize the smallest and largest paired differences across the tested terminal grid: $-15, -10, -5, 0$ percentage points for Adams and $2, 5, 10, 15$ for the other policies. The terminal effect is held fixed within each pair; it can differ across pairs in a row. No span is drawn when the saved grid contains no verified pair. Table G2 reports every grid value and path count. The lines summarize endpoints and do not establish that every intermediate value is attained. They are not sampling intervals or probabilities. All values use the common signed seven-year net-income response, a two-schooling-year equivalent per BA increment, and 3 percent discounting to age 18 in 2000 dollars. These accounts exclude other qualifications, transfers, and fiscal costs.